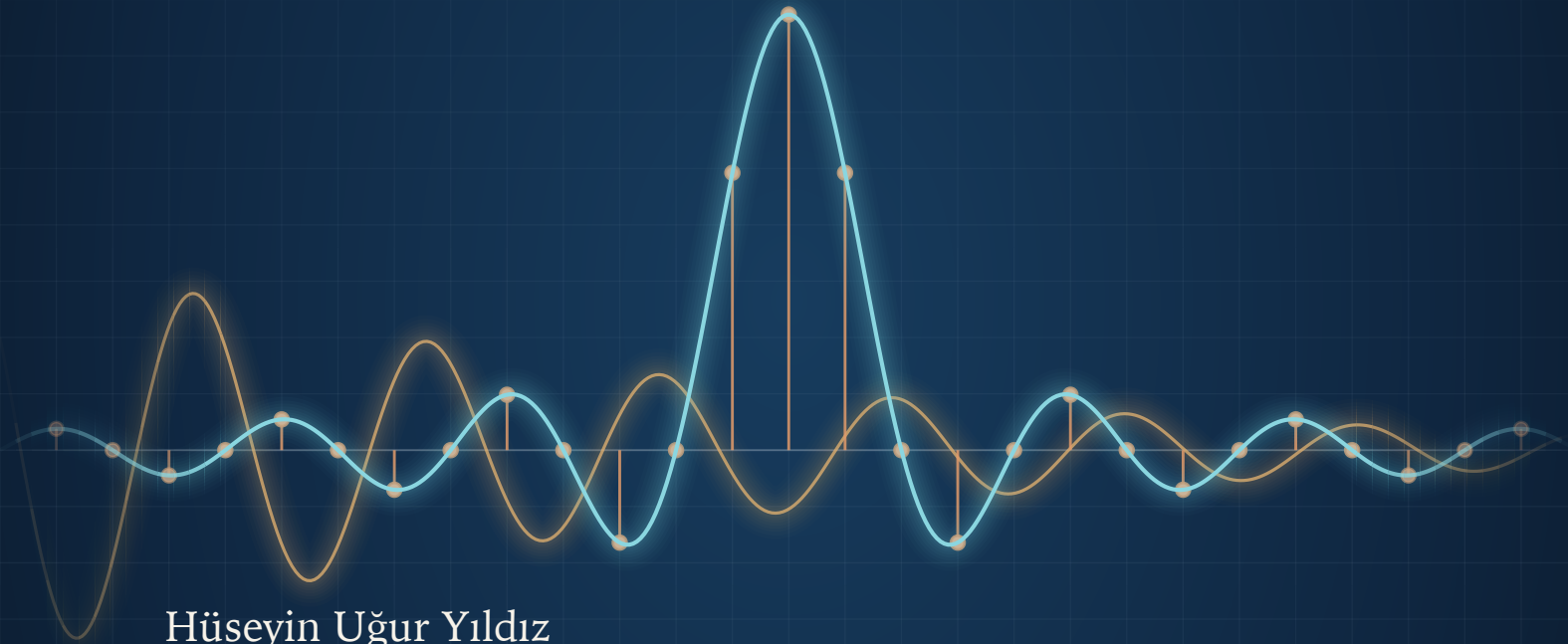




SIGNALS AND SYSTEMS

Signals, Systems and Frequency-Domain Analysis

Lecture Notes



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How to read these notes

These notes teach the full course at undergraduate level. They assume calculus, complex numbers and basic circuits. They define signals, classify systems and show why one function describes a linear time-invariant system. They then represent signals with complex exponentials and explain how sampling changes a signal.

Read the chapters in order because each chapter uses the one before it. First, a signal is defined as a function. Next, a system is defined as a map on functions. A linear time-invariant system is then described by one function. The frequency-domain chapters use this function to turn convolution into multiplication.

Each topic follows the same teaching order. A picture introduces the purpose of the idea. A definition and an equation state it exactly. A short derivation names each step. A worked example applies the method and ends with a check. Each worked example uses five headings: Given, Find, Method, Solution, Check. Try the Check step before you read it.

Two conventions apply throughout the notes. Energy and power are **normalised**: the resistance is $1\ \Omega$, so instantaneous power is $|x|^2$. The imaginary unit is j .

The third contents column gives textbook references. For example, **OW CH1.1–1.4** points to the matching material in Oppenheim and Willsky, *Signals and Systems*, second edition. The **OW** mark separates a textbook address from a course address. The numbering systems differ: these notes introduce the continuous-time Fourier transform in chapter 5, while the textbook introduces it in chapter 4.

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CHAPTER

1

Signals

A signal represents information as a function. This chapter defines continuous-time and discrete-time signals. It then shows how to calculate energy and power, change the time axis, test periodicity and symmetry, and use impulses and complex exponentials. It ends with a catalogue of common signals.

1.1 What a signal is

A signal is a physical quantity that varies and carries information. Mathematically, it is a function of one or more independent variables.

The independent variable in this course is usually time. An image instead uses two space variables. The same signal operations apply to those variables.

Continuous-time and discrete-time signals use different domains. Keep their notation separate.

CONTINUOUS TIME

$$x(t), \quad t \in \mathbb{R}$$

Round brackets. The signal has a value at every real instant.

DISCRETE TIME

$$x[n], \quad n \in \mathbb{Z}$$

Square brackets. The signal has a value only at integer indices. Here n is an integer **index**, not a time in seconds.

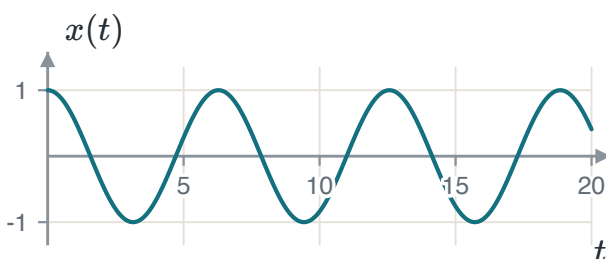


FIGURE 1.1 A continuous-time signal is drawn as an unbroken curve.

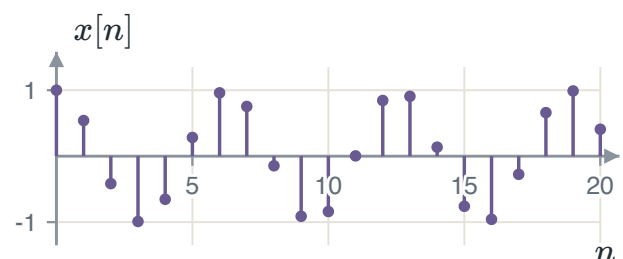


FIGURE 1.2 A discrete-time signal is drawn with stems. The dots are the signal values. No signal value exists between adjacent integer indices.

⊗ NOTATION

Writing $x[t]$ or $x(n)$ states the wrong domain. The domain determines the periodicity test, the convolution limits and the transform properties.

1.2 Signal energy and power

Signal energy and power come from circuit power. If $v(t)$ is the voltage across a resistance R , use Ohm's law to write the instantaneous power as

$$p(t) = v(t) i(t) = v(t) \left(\frac{v(t)}{R} \right) = \frac{1}{R} v^2(t).$$

To find the energy over a time interval, integrate the instantaneous power:

$$E = \int_{t_1}^{t_2} p(t) dt = \int_{t_1}^{t_2} \frac{1}{R} v^2(t) dt.$$

From this point onward, set $R = 1 \Omega$ and use $|x(t)|^2$. This is the normalised convention. Restore the factor $1/R$ when a physical calculation uses a different resistance.

We use the modulus because a signal may be complex valued, and $|x(t)|^2 = x(t) x^*(t)$ is the quantity that is real and never negative. Writing $x^2(t)$ instead is correct only for real signals.

Total energy

TOTAL ENERGY OVER ALL TIME

$$E_{\infty} = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$E_{\infty} = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2 = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

The integral or sum may not converge. Because E_{∞} is a limit, non-convergence means that the signal has no finite total energy. Calculate average power next before classifying the signal.

Average power

Average power measures energy per unit time. Over a finite interval $[t_1, t_2]$, divide the energy by the length of the interval:

$$P = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} p(t) dt.$$

For the average over all time, use the symmetric window $[-T, T]$, as for the energy. Then let the window grow.

AVERAGE POWER OVER ALL TIME

$$P_{\infty} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$P_{\infty} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

In discrete time the divisor is $2N + 1$, because that is the number of samples between $-N$ and $+N$ inclusive. A divisor of $2N$ gives the same limit, but the wrong value at a finite N .

⚠ TAKE THE LIMIT OF THE RATIO

Calculate the energy-to-duration ratio for each finite interval. The ratio may approach any non-negative value, including zero.

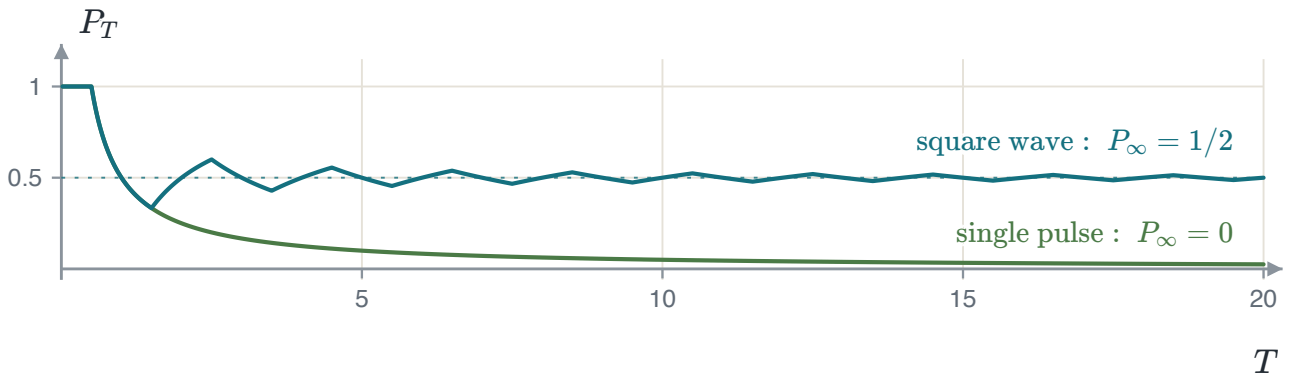


FIGURE 1.3 The running average $P_T = E_T/(2T)$ of two signals. The single pulse is 1 for $|t| < 1/2$ and has energy 1. The square wave repeats that pulse every 2 s. The two averages agree until $T = 3/2$. The square wave keeps adding energy, so its average approaches $1/2$. The pulse adds none, so its average $1/(2T)$ approaches 0.

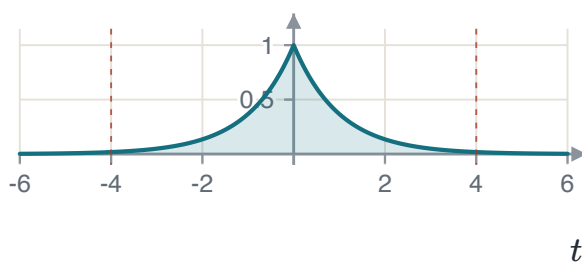


FIGURE 1.4 Converging. As the window grows, the shaded area approaches a finite limit.

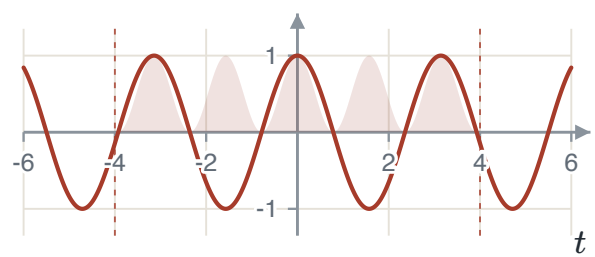


FIGURE 1.5 Diverging. Every period adds the same area, so $E_{\infty} \rightarrow \infty$ and only the average power is meaningful.

Compute the finite-window energy of the diverging signal $x(t) = \cos(2t)$ before taking any limit. Use the identity $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$ and integrate each term:

$$\begin{aligned}
E_T &= \int_{-T}^T \cos^2(2t) dt \\
&= \int_{-T}^T \frac{1 + \cos(4t)}{2} dt \\
&= \left[\frac{t}{2} + \frac{\sin(4t)}{8} \right]_{-T}^T \\
&= T + \frac{\sin(4T)}{4}.
\end{aligned}$$

The sine term stays between $-1/4$ and $1/4$, while T grows without limit. So $E_\infty \rightarrow \infty$. Dividing by the window length gives a finite average power:

$$P_\infty = \lim_{T \rightarrow \infty} \frac{E_T}{2T} = \lim_{T \rightarrow \infty} \left(\frac{1}{2} + \frac{\sin(4T)}{8T} \right) = \frac{1}{2}.$$

1.3 Energy signals, power signals, and neither

TABLE 1.1

TYPE	CONDITION	TYPICAL SIGNALS
Energy signal	$E_\infty < \infty$ and $P_\infty = 0$	Finite pulses and decaying responses
Power signal	$E_\infty \rightarrow \infty$ and $0 < P_\infty < \infty$	Constants, sinusoids and other continuing signals
Neither	$E_\infty \rightarrow \infty$ and $P_\infty \rightarrow \infty$	Signals that grow without bound, such as $t u(t)$

The two energy-signal conditions are not independent. If E_∞ is finite, the energy in every finite window is at most E_∞ . Dividing this bound by $2T$ makes P_∞ approach zero. The power condition follows from the energy condition.

$$\begin{aligned}
E_T &= \int_{-T}^T |x(t)|^2 dt \leq E_\infty \\
0 \leq P_\infty &= \lim_{T \rightarrow \infty} \frac{E_T}{2T} \leq \lim_{T \rightarrow \infty} \frac{E_\infty}{2T} = 0 \\
\therefore P_\infty &= 0.
\end{aligned}$$

The constant $x(t) = 1$ is the simplest power signal. The energy in the window grows with the window, but the window length cancels in the average:

$$\begin{aligned}
E_T &= \int_{-T}^T 1^2 dt = 2T \rightarrow \infty, \\
P_\infty &= \lim_{T \rightarrow \infty} \frac{E_T}{2T} = \lim_{T \rightarrow \infty} \frac{2T}{2T} = 1.
\end{aligned}$$

The ramp $x(t) = t$ for $t \geq 0$, and $x(t) = 0$ for $t < 0$, is in neither class. The lower limit of the energy integral is 0 because the ramp is zero for $t < 0$:

$$E_T = \int_0^T t^2 dt = \left[\frac{t^3}{3} \right]_0^T = \frac{T^3}{3} \rightarrow \infty,$$

$$P_\infty = \lim_{T \rightarrow \infty} \frac{E_T}{2T} = \lim_{T \rightarrow \infty} \frac{T^3/3}{2T} = \lim_{T \rightarrow \infty} \frac{T^2}{6} = \infty.$$

Both limits diverge. Compute both before you assign a class.

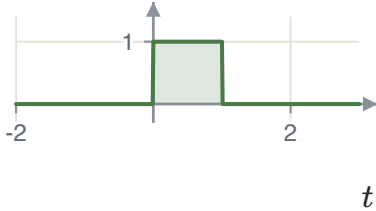


FIGURE 1.6 Energy signal: $E_\infty = 1$, $P_\infty = 0$.

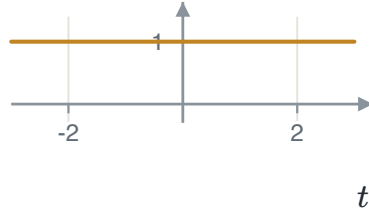


FIGURE 1.7 Power signal: $E_\infty \rightarrow \infty$, $P_\infty = 1$.

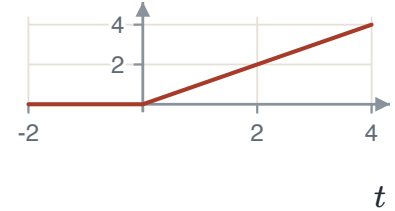


FIGURE 1.8 Neither: both limits diverge.

EXAMPLE 1.1

GIVEN $x(t) = 1$ for $0 \leq t \leq 1$, and $x(t) = 0$ otherwise.

FIND Is this an energy signal or a power signal?

METHOD Find E_∞ first because finite energy implies zero average power. This result then determines the class.

SOLUTION Start with the definitions and then use the support of the signal:

$$\begin{aligned} E_\infty &= \int_{-\infty}^{\infty} |x(t)|^2 dt \\ &= \int_0^1 1^2 dt \\ &= t \Big|_0^1 = 1 < \infty. \end{aligned}$$

For every $T \geq 1$, the window contains the whole pulse. Therefore

$$\begin{aligned} P_\infty &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^1 1 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} = 0. \end{aligned}$$

So $x(t)$ is an **energy signal**, with $E_\infty = 1$ J.

CHECK Halving the amplitude must divide the energy by four. Direct calculation gives $\int_0^1 (1/2)^2 dt = 1/4$, so the result has the required quadratic dependence on amplitude.

EXAMPLE 1.2**GIVEN** $x[n] = 4$ for every integer n .**FIND** Is this an energy signal or a power signal?**METHOD** Each sample contributes the same positive energy, so E_∞ diverges. Calculate P_∞ next and use the exact count $2N + 1$.**SOLUTION** Each sample contributes $|4|^2 = 16$, so

$$\begin{aligned}
 E_\infty &= \sum_{n=-\infty}^{\infty} |x[n]|^2 \\
 &= \sum_{n=-\infty}^{\infty} |4|^2 \\
 &= \sum_{n=-\infty}^{\infty} 16 \rightarrow \infty.
 \end{aligned}$$

The finite window from $-N$ to N contains $2N + 1$ samples. Hence

$$\begin{aligned}
 P_\infty &= \lim_{N \rightarrow \infty} \frac{1}{2N + 1} \sum_{n=-N}^N |x[n]|^2 \\
 &= \lim_{N \rightarrow \infty} \frac{1}{2N + 1} \sum_{n=-N}^N 16 \\
 &= \lim_{N \rightarrow \infty} \frac{(2N + 1)16}{2N + 1} \\
 &= \lim_{N \rightarrow \infty} 16 = 16.
 \end{aligned}$$

So $x[n]$ is a **power signal**, with $P_\infty = 16$.**CHECK** A constant of amplitude A must have $P_\infty = A^2$. The factor $2N + 1$ cancels exactly, which confirms the sample count was right.**⊗ DO NOT INFER POWER FROM ENERGY ALONE**

Infinite energy does not imply infinite average power. For the constant sequence, both the energy sum and the sample count grow at the same rate. Their ratio approaches 16.

1.4 Time shifting**TIME SHIFT**

$$x(t) \longrightarrow x(t - t_0)$$

If $t_0 > 0$ the signal is **delayed** and moves right. If $t_0 < 0$ it is **advanced** and moves left.

Read the argument before moving the graph. At time t , the shifted signal uses the original value at $t - t_0$. For $t_0 > 0$, it reaches each original value t_0 seconds later. The graph therefore moves right.

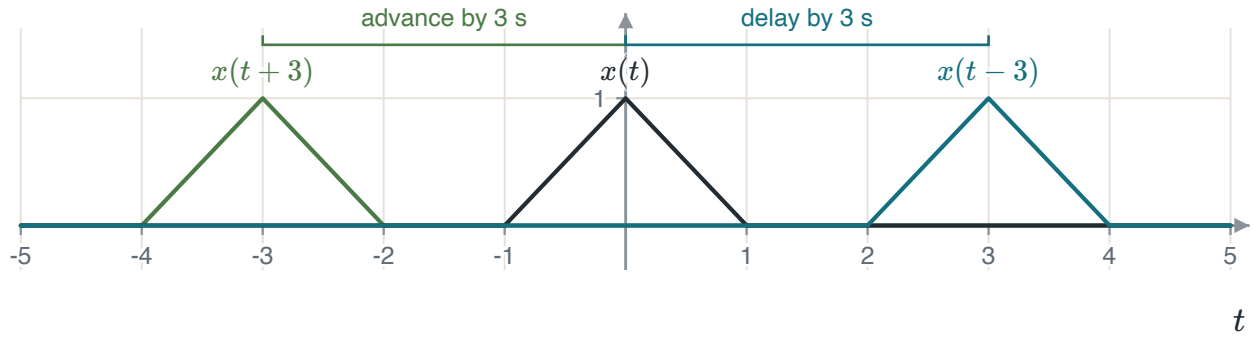


FIGURE 1.9 A time shift changes the signal position but not its shape.

For example, a radar sends a pulse $x(t)$ and the echo returns 0.2 ms later. With t in ms, the echo is $x(t - 0.2)$. It arrives later, so it is a delay with $t_0 = 0.2 > 0$.

TIME SHIFT IN DISCRETE TIME

$$x[n] \longrightarrow x[n - n_0], \quad n_0 \in \mathbb{Z}$$

The sign rule is the same as in continuous time. If $n_0 > 0$ the stems move right. If $n_0 < 0$ they move left.

Every sample moves by the same whole number of steps. The shift must be an integer because a sequence has values only at integer indices. The expression $x[n - \frac{1}{2}]$ is therefore not defined.

Read a shifted sequence one sample at a time. Let $x[n] = 1 - n/4$ for $n = 0, 1, 2, 3$ and $x[n] = 0$ elsewhere, and let $y[n] = x[n - 2]$. Then $y[3] = x[3 - 2] = x[1] = 1 - \frac{1}{4} = 0.75$.

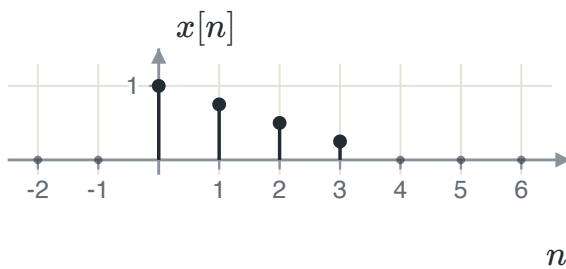


FIGURE 1.10 $x[n] = 1 - n/4$ for $n = 0, 1, 2, 3$.



FIGURE 1.11 $y[n] = x[n - 2]$: every stem moves two samples right. The sample $x[1] = 0.75$ appears at $n = 3$.

A one-sample delay, $x[n - 1]$, is the basic memory element of every difference equation in Chapter 3.

1.5 Time reversal and time scaling

TIME REVERSAL

$$x(t) \longrightarrow x(-t) \quad (x[n] \rightarrow x[-n])$$

Reflect the signal about the vertical axis. The value at time t moves to time $-t$.

Find the new support in the same way as for any change of the argument. If x is non-zero only on $[\alpha, \beta]$, then $x(-t)$ is non-zero where $\alpha \leq -t \leq \beta$, that is on $[-\beta, -\alpha]$. A pulse on $[1, 3]$ moves to $[-3, -1]$, and its width stays 2.

Reversal changes what you hear when the signal is not symmetric in time. A plucked string starts loud and dies away, for example $x(t) = e^{-4t} \sin(2\pi 220 t)$ for $t \geq 0$. Played backwards, $x(-t)$ swells and then stops at once. A steady tone does not change: $\cos(-2\pi 440 t) = \cos(2\pi 440 t)$.

TIME SCALING

$$y(t) = x(at), \quad a > 0$$

If $a > 1$ the signal is compressed and speeded up. If $0 < a < 1$ it is stretched and slowed down.

Write the scaled signal as a new signal $y(t)$. Writing $x(t) = x(at)$ would force $a = 1$.

Find the new support by solving $at \in [\alpha, \beta]$. If x is non-zero only on $[\alpha, \beta]$, then $x(at)$ is non-zero only on $[\alpha/a, \beta/a]$. The width is divided by a .

$$at \in [\alpha, \beta] \iff \alpha \leq at \leq \beta \iff \frac{\alpha}{a} \leq t \leq \frac{\beta}{a}, \quad a > 0.$$

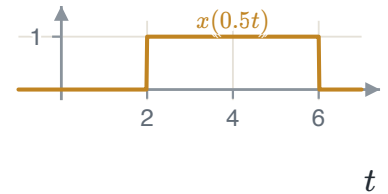
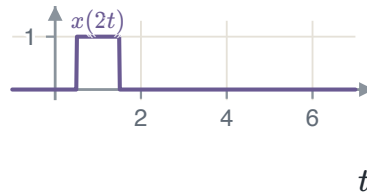
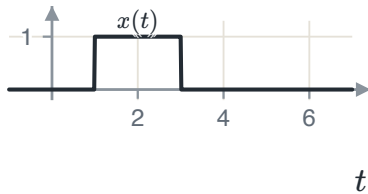


FIGURE 1.12

FIGURE 1.13 Compressed: support $[0.5, 1.5]$.

FIGURE 1.14 Stretched: support $[2, 6]$.

Time scaling also changes frequency. Substitute at for t in one tone:

$$x(t) = \sin(2\pi f_0 t) \implies x(at) = \sin(2\pi f_0(at)) = \sin(2\pi (af_0) t).$$

Every frequency is multiplied by a . A melody that lasts 2 s lasts $2/a$ s after scaling, and each pitch moves with it. At $a = 2$ each frequency doubles, which is one octave up. This is what a voice message sounds like at double speed. At $a = \frac{1}{2}$, a 440 Hz tone becomes $\frac{1}{2} \cdot 440 = 220$ Hz.

⚠ DISCRETE-TIME SCALING NEEDS SEPARATE RULES

$x[2n]$ keeps only the even-indexed samples and discards the odd-indexed samples. This operation is decimation and may lose information. The expression $x[n/2]$ is undefined at odd n unless an interpolation rule supplies those values. Continuous-time scaling does not have this integer-index restriction.

1.6 Combining a shift and a scale

To construct $x(at - b)$ from $x(t)$, use an intermediate signal so that each operation is explicit.

SHIFT, THEN SCALE

$$(1) \quad v(t) = x(t - b) \qquad (2) \quad y(t) = v(at) = x(at - b)$$

Shift by b first. Then scale the result by a .

⊗ WHY THE ORDER MATTERS

Scaling first gives $w(t) = x(at)$. Shifting that by b gives $w(t - b) = x(a(t - b)) = x(at - ab)$. Unless $a = 1$ this is a different signal: it is shifted by b instead of by b/a .

✏ EXAMPLE 1.3

GIVEN $x(t)$ is zero for $t < -2$, equal to 1 on $[-2, 0]$, equal to 2 on $[0, 2]$, and falls linearly from 2 to 0 on $[2, 4]$.

FIND Plot $x(3t - 5)$.

METHOD Here $a = 3$ and $b = 5$. Shift right by 5, then compress by 3.

SOLUTION For the shift, each original corner c moves to $c + 5$, so $-2, 0, 2, 4$ become $3, 5, 7, 9$. For the final signal, solve the argument equation for each original corner:

$$3t - 5 = c \implies t = \frac{c + 5}{3}.$$

Thus

$$c = -2, 0, 2, 4 \implies t = 1, \frac{5}{3}, \frac{7}{3}, 3.$$

These are the four corners of $y(t) = x(3t - 5)$.

CHECK The original support has width 6. Compression by $a = 3$ must give width 2, and the result has support $[1, 3]$. Also set $at - b = 0$. This gives $t = b/a = 5/3$, where $y(t)$ must equal $x(0) = 2$.

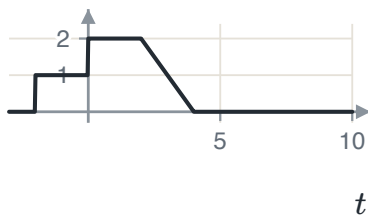


FIGURE 1.15 $x(t)$

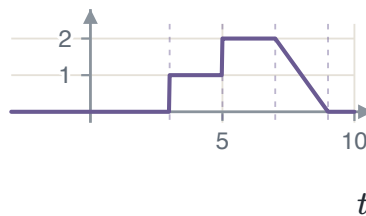


FIGURE 1.16 $v(t) = x(t - 5)$

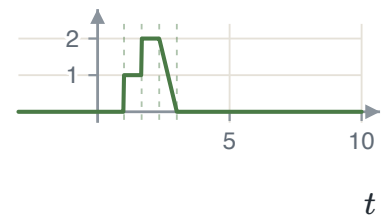


FIGURE 1.17 $y(t) = x(3t - 5)$

1.7 Periodic signals

PERIODICITY

$$x(t) = x(t + T) \quad \text{for all } t \in \mathbb{R}, \text{ for some } T > 0$$

$$x[n] = x[n + N] \quad \text{for all } n \in \mathbb{Z}, \text{ for some integer } N > 0$$

A signal that is not periodic is **aperiodic**.

The definition has two requirements. A positive period must exist, and the equality must hold for every t or n . In continuous time, T may be any positive real number. In discrete time, N must be a positive integer because the sequence exists only at integer indices. A value such as 3.5 samples cannot be a period.

If T is a period then so is $2T$, $3T$, and so on. The **fundamental period** T_0 is the smallest positive period. The same applies to N_0 in discrete time.

FUNDAMENTAL FREQUENCY

$$\omega_0 = \frac{2\pi}{T_0} \quad \text{and} \quad \omega_0 = \frac{2\pi}{N_0}$$

Without the word *smallest*, ω_0 would not be well defined.

Two short cases show the definition at work. For $x(t) = \cos(\pi t/3)$, the angular frequency is $\omega_0 = \pi/3$, so $T_0 = 2\pi/(\pi/3) = 6$. For $x[n] = (-1)^n$, the test $N = 1$ fails because $(-1)^{n+1} = -(-1)^n$. The test $N = 2$ succeeds because $(-1)^{n+2} = (-1)^n$ for every n . So $N_0 = 2$.

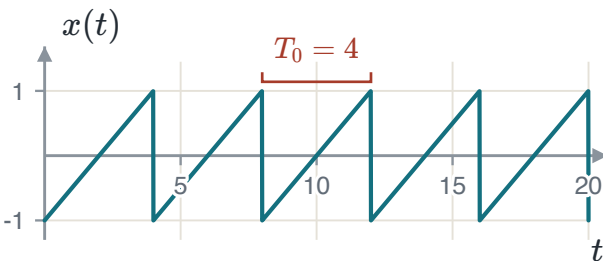


FIGURE 1.18 Periods are 4, 8, 12, ... The fundamental period is $T_0 = 4$.

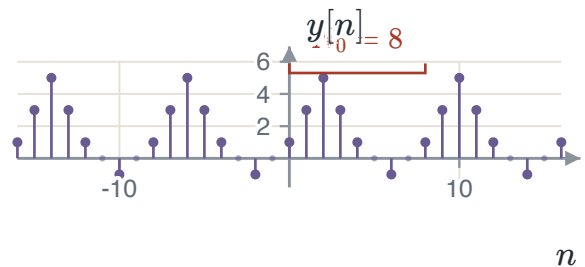


FIGURE 1.19 Periods are 8, 16, 24, ... The fundamental period is $N_0 = 8$.

1.8 Even and odd signals

A signal is **even** if $x(t) = x(-t)$, and **odd** if $x(t) = -x(-t)$. The same definitions apply to $x[n]$ with n in place of t .

To test an odd signal at the origin, set $t = 0$ in the definition. This gives $x(0) = -x(0)$, so $x(0) = 0$. Therefore any signal with $x(0) \neq 0$ is not odd.

Every signal splits into an even part and an odd part, in exactly one way:

EVEN AND ODD PARTS

$$\mathcal{E}\mathbf{v}\{x(t)\} = \frac{1}{2}[x(t) + x(-t)], \quad \mathcal{O}\mathbf{d}\mathbf{d}\{x(t)\} = \frac{1}{2}[x(t) - x(-t)]$$

$$x(t) = \mathcal{E}\mathbf{v}\{x(t)\} + \mathcal{O}\mathbf{d}\mathbf{d}\{x(t)\}$$

Adding the two definitions returns x exactly.

$$\begin{aligned} \mathcal{E}\mathbf{v}\{x(t)\} + \mathcal{O}\mathbf{d}\mathbf{d}\{x(t)\} &= \frac{1}{2}[x(t) + x(-t)] + \frac{1}{2}[x(t) - x(-t)] \\ &= \frac{1}{2}[2x(t)] = x(t). \end{aligned}$$

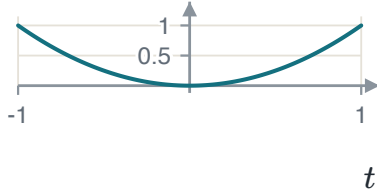


FIGURE 1.20 t^2 is even.

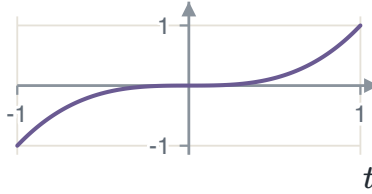


FIGURE 1.21 t^3 is odd.

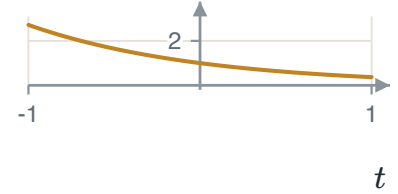


FIGURE 1.22 e^{-t} is neither.

To classify a signal, compute $x(-t)$ and compare it with $x(t)$. Do not rely on the look of the graph alone.

- $x(t) = t \sin t$ gives $x(-t) = (-t) \sin(-t) = t \sin t = x(t)$, so it is even.
- $x(t) = \sin(\pi t)$ gives $x(-t) = -\sin(\pi t) = -x(t)$, so it is odd.
- $x(t) = e^{-2t}u(t)$ has $x(-1) = 0$ but $x(1) = e^{-2}$, so it is not even. It has $x(0) = 1 \neq 0$, so it is not odd. It is neither.
- $x[n] = n$ for $|n| \leq 3$ and 0 otherwise gives $x[-n] = -x[n]$ at every n , so it is odd.

EXAMPLE 1.4

GIVEN The unit pulse $x(t) = 1$ for $0 < t < 1$, and $x(t) = 0$ otherwise.

FIND $\mathcal{E}\mathbf{v}\{x(t)\}$ and $\mathcal{O}\mathbf{d}\mathbf{d}\{x(t)\}$.

METHOD Write down the mirror image $x(-t)$ first. Then apply the two definitions interval by interval.

SOLUTION The mirror image is $x(-t) = 1$ for $-1 < t < 0$, and 0 otherwise. The two pulses do not overlap, so on each interval only one of them is non-zero:

$$\mathcal{E}\mathbf{v}\{x(t)\} = \frac{1}{2}[x(t) + x(-t)] = \begin{cases} \frac{1}{2}, & -1 < t < 0 \\ \frac{1}{2}, & 0 < t < 1 \\ 0, & |t| > 1 \end{cases}$$

$$\mathcal{O}\mathbf{d}\mathbf{d}\{x(t)\} = \frac{1}{2}[x(t) - x(-t)] = \begin{cases} -\frac{1}{2}, & -1 < t < 0 \\ \frac{1}{2}, & 0 < t < 1 \\ 0, & |t| > 1. \end{cases}$$

CHECK Add the two parts. On $0 < t < 1$: $\frac{1}{2} + \frac{1}{2} = 1 = x(t)$. On $-1 < t < 0$: $\frac{1}{2} - \frac{1}{2} = 0 = x(t)$. Outside $[-1, 1]$ both parts are 0. The sum returns $x(t)$ everywhere.

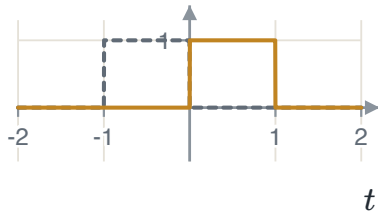


FIGURE 1.23 $x(t)$, with its mirror $x(-t)$ dashed.

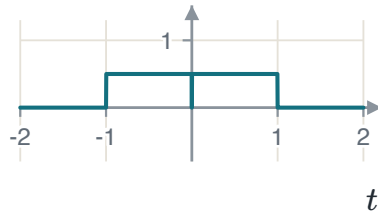


FIGURE 1.24 $\mathcal{E}_v\{x(t)\}$: height $\frac{1}{2}$ on both sides.

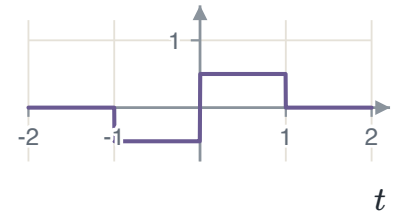


FIGURE 1.25 $\mathcal{O}_{dd}\{x(t)\}$: the left half flips below the axis.

1.9 The unit impulse and the unit step in discrete time

DEFINITIONS

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}$$

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Both are ordinary sequences. Nothing infinite happens here.

A first difference converts the step into the impulse. A running sum converts the impulse back into the step. These operations are the discrete-time forms of differentiation and integration.

FIRST DIFFERENCE AND RUNNING SUM

$$\delta[n] = u[n] - u[n - 1]$$

$$u[n] = \sum_{k=0}^{\infty} \delta[n - k] = \delta[n] + \delta[n - 1] + \delta[n - 2] + \dots$$

The first difference works because the delayed step $u[n - 1]$ equals $u[n]$ at every $n \geq 1$, so the two cancel there. Both are zero for $n < 0$. Only $n = 0$ survives, where $u[0] - u[-1] = 1 - 0 = 1$.

Each term $\delta[n - k]$ of the running sum places one unit sample at $n = k$. To write the sum as an accumulation up to n , substitute $m = n - k$. Then $k = 0$ gives $m = n$, and $k \rightarrow \infty$ gives $m \rightarrow -\infty$:

$$u[n] = \sum_{k=0}^{\infty} \delta[n - k] = \sum_{m=-\infty}^n \delta[m].$$

The two operations undo each other. A first difference reverses a running sum, and a running sum reverses a first difference. For example, $u[n] - u[n - 3]$ keeps the step only until the delayed step starts at $n = 3$. So $u[n] - u[n - 3] = \delta[n] + \delta[n - 1] + \delta[n - 2]$, with three non-zero samples.

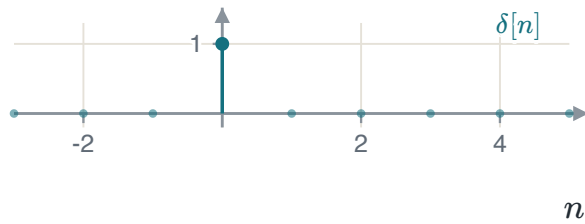


FIGURE 1.26

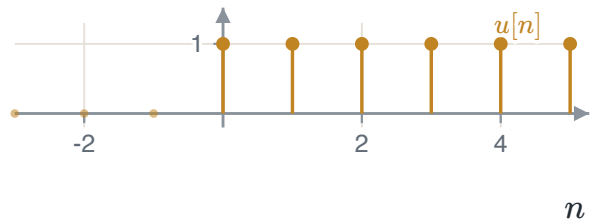


FIGURE 1.27

Sampling and sifting

Sampling and sifting both use a shifted impulse, but they return different types of result.

SAMPLING PROPERTY

$$x[n] \delta[n - n_0] = x[n_0] \delta[n - n_0]$$

Both sides are **sequences**. Multiplying by a shifted impulse keeps one sample and deletes the rest.

SIFTING PROPERTY

$$x[n_0] = \sum_{n=-\infty}^{\infty} x[n] \delta[n - n_0]$$

The right-hand side is a **number**. Sifting is sampling followed by a sum.

EXAMPLE 1.5

GIVEN $x[0] = 1$, $x[1] = 2$, $x[2] = 3$, and $x[n] = 0$ elsewhere. Take $n_0 = 2$.

FIND The sampled sequence and the sifted value.

METHOD Multiply by $\delta[n - 2]$ to keep the sample at $n = 2$. Then sum the sampled sequence to sift out its value.

SOLUTION Sampling: $x[n] \delta[n - 2] = x[2] \delta[n - 2] = 3 \delta[n - 2]$, a single stem of height 3 at $n = 2$.
Sifting: $\sum_n x[n] \delta[n - 2] = x[0] \cdot 0 + x[1] \cdot 0 + x[2] \cdot 1 = 3$.

CHECK The sampled result is a sequence. The sifted result is the number 3.

Apply the sampling property at every integer shift and add the results. This produces the representation used to derive convolution in Chapter 3.

REPRESENTATION PROPERTY

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

Any sequence is a sum of weighted, shifted impulses. The weights are the sample values themselves.

Apply the representation to the step itself. The weight $u[k]$ is 0 for $k < 0$ and 1 for $k \geq 0$, so it switches off every term with $k < 0$:

$$u[n] = \sum_{k=-\infty}^{\infty} u[k] \delta[n-k] = \sum_{k=-\infty}^{-1} 0 \cdot \delta[n-k] + \sum_{k=0}^{\infty} 1 \cdot \delta[n-k] = \sum_{k=0}^{\infty} \delta[n-k].$$

This is the running sum found above.

1.10 The unit impulse and the unit step in continuous time

The continuous-time step is $u(t) = 1$ for $t \geq 0$ and 0 otherwise. The continuous-time impulse is not an ordinary function.

⊗ DEFINE THE IMPULSE BY ITS ACTION

An ordinary function that is zero except at one point has integral 0, not 1. The impulse is therefore a **distribution**, also called a generalized function. A distribution is defined by how it acts inside an integral.

DEFINING PROPERTY (SIFTING)

$$x(t_0) = \int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt$$

This is the definition. Everything else about δ follows from it.

An informal picture is often written as $\delta(t) = \infty$ at $t = 0$, $\delta(t) = 0$ elsewhere, with $\int_{-\infty}^{\infty} \delta(t) dt = 1$. Use it only as a picture. A figure draws $\delta(t)$ as an arrow, and the number beside the arrow is its **weight**. The weight is an area, not a function value. So $\int_{-\infty}^{\infty} 3 \delta(t) dt = 3$.

To picture the impulse, use a rectangle of width ε and height $1/\varepsilon$. Its area is 1. Then let $\varepsilon \rightarrow 0$. The rectangles do not approach an ordinary function at each fixed t . Their integrals against a continuous test function do approach the sifting result.

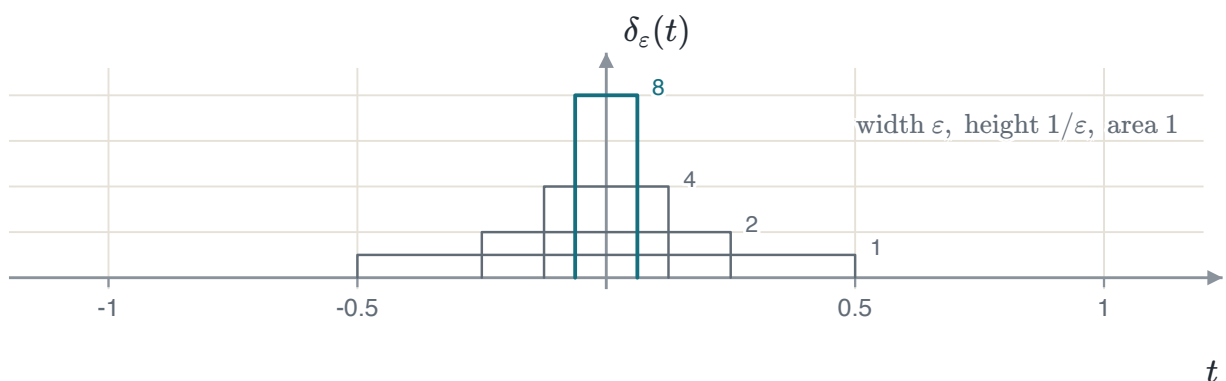


FIGURE 1.28 Unit-area rectangles with $\varepsilon = 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$. The number beside each top is its height $1/\varepsilon$. As $\varepsilon \rightarrow 0$ the rectangle becomes the impulse $\delta(t)$. A rectangle of width 0.1 has height 10.

The impulse and the step are related by differentiation and integration:

$$\delta(t) = \frac{d}{dt}u(t), \quad u(t) = \int_{-\infty}^t \delta(\tau) d\tau.$$

SAMPLING PROPERTY, CONTINUOUS TIME

$$x(t) \delta(t - t_0) = x(t_0) \delta(t - t_0)$$

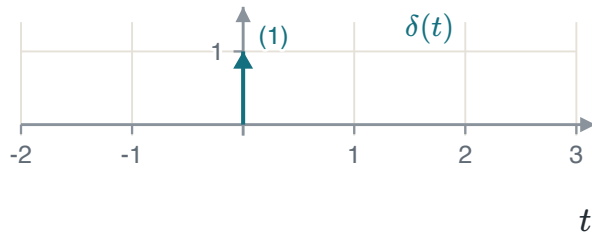


FIGURE 1.29 The arrow height shows the **area**, never a value of a function.

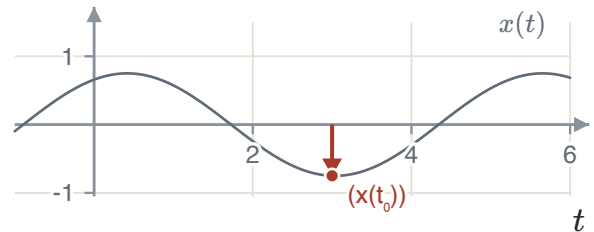


FIGURE 1.30 Sifting: the impulse at t_0 is scaled by the value of x there, and integrating returns that number.

The step and impulse pair $\delta[n] = u[n] - u[n - 1]$ is the discrete-time analogue of $\delta(t) = du/dt$. The discrete-time sum becomes an integral.

EXAMPLE 1.6

GIVEN $x(t) = \cos t$ and $t_0 = \pi$.

FIND $\int_{-\infty}^{\infty} \cos t \delta(t - \pi) dt$.

METHOD Use the sampling property to replace $\cos t$ by its value at the impulse. Then use the unit area of the impulse.

SOLUTION The impulse sits at $t = \pi$, so

$$\begin{aligned} \int_{-\infty}^{\infty} \cos t \delta(t - \pi) dt &= \int_{-\infty}^{\infty} \cos(\pi) \delta(t - \pi) dt \\ &= \cos(\pi) \int_{-\infty}^{\infty} \delta(t - \pi) dt \\ &= (-1)(1) = -1. \end{aligned}$$

CHECK The result is a number, not a signal. It equals $x(t_0) = \cos \pi = -1$, as the sifting property requires.

1.11 Complex exponentials in continuous time

DEFINITION

$$x(t) = C e^{at}, \quad C, a \in \mathbb{C}$$

Complex exponentials describe growth, decay and oscillation in one form. The real and imaginary parts of a determine which behaviour occurs.

Case 1: C and a real

If $a < 0$ the signal decays. If $a > 0$ it grows. If $a = 0$ it is the constant C . A larger $|a|$ makes the decay or growth faster.

Each time t advances by $1/|a|$, the exponent changes by 1 in magnitude. So the signal is multiplied by $e \approx 2.72$ when it grows, or divided by e when it decays. For example, $x(t) = e^{-2t}$ reaches e^{-1} at $t = 1/2$. A signal e^{at} that doubles every second has $e^{a \cdot 1} = 2$, so $a = \ln 2 \approx 0.69$.

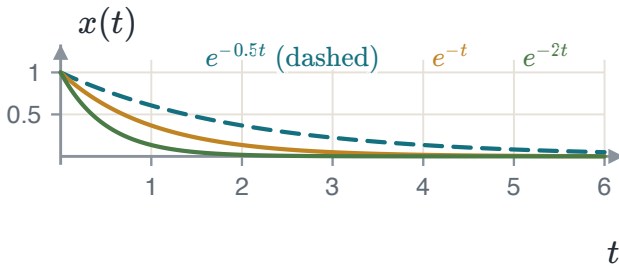


FIGURE 1.31 $a < 0$: decay. A larger $|a|$ decays faster.

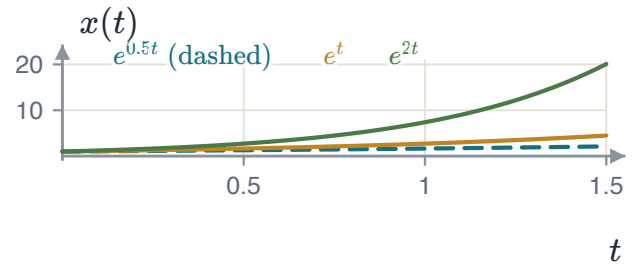


FIGURE 1.32 $a > 0$: growth. Every curve starts at $x(0) = C = 1$.

Case 2: $a = j\omega_0$ purely imaginary

Write $C = Ae^{j\theta}$ and use Euler's relation $e^{jx} = \cos x + j \sin x$:

$$\begin{aligned} x(t) &= Ce^{j\omega_0 t} \\ &= Ae^{j\theta} e^{j\omega_0 t} \\ &= Ae^{j(\omega_0 t + \theta)} \\ &= A \cos(\omega_0 t + \theta) + jA \sin(\omega_0 t + \theta). \end{aligned}$$

Here $A = |C|$ is the amplitude, ω_0 is the angular frequency in rad/s, and $\theta = \angle C$ is the phase in radians. The amplitude is never negative. For $x(t) = -3e^{j2t}$, write $-3 = 3e^{j\pi}$, so $A = 3$ and $\theta = \pi$.

The modulus is constant: $|x(t)| = A |e^{j(\omega_0 t + \theta)}| = A$ for every t . This signal neither grows nor decays. With $\omega_0 = 2\pi f_0$, the real part $\cos(2\pi f_0 t)$ is a pure tone of f_0 hertz. Doubling f_0 halves the period and raises the pitch by one octave.

To test periodicity, impose the condition $x(t) = x(t + T)$ and solve for T :

$$\begin{aligned} x(t + T) &= x(t) \\ Ae^{j[\omega_0(t+T)+\theta]} &= Ae^{j(\omega_0 t + \theta)} \\ e^{j\omega_0 T} &= 1 \\ \omega_0 T &= 2\pi k, \quad k \in \mathbb{Z} \\ T &= \frac{2\pi k}{\omega_0}. \end{aligned}$$

✓ **RESULT**

Taking $k = 1$ gives the fundamental period $T_0 = 2\pi/\omega_0$. Every continuous-time complex exponential with $\omega_0 \neq 0$ is periodic. There is no extra condition.

✂ **EXAMPLE 1.7**

GIVEN $x(t) = e^{j0.5\pi t}$.

FIND The fundamental period.

METHOD Use $T_0 = 2\pi/\omega_0$ because the exponent is purely imaginary and $\omega_0 \neq 0$.

SOLUTION $T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{0.5\pi} = 4$ seconds.

CHECK $0.5\pi \times 4 = 2\pi$, so the phase increases by one full turn over the calculated period.

Case 3: $a = r + j\omega_0$

$$\begin{aligned} x(t) &= Ae^{j\theta} e^{(r+j\omega_0)t} \\ &= Ae^{rt} e^{j(\omega_0 t + \theta)} \\ &= Ae^{rt} \cos(\omega_0 t + \theta) + jAe^{rt} \sin(\omega_0 t + \theta). \end{aligned}$$

The factor e^{rt} is the **envelope** and the factor $e^{j(\omega_0 t + \theta)}$ is the **rotation**. The real part is $\text{Re}\{x(t)\} = Ae^{rt} \cos(\omega_0 t + \theta)$. The curves $\pm Ae^{rt}$ bound the sinusoid. If $r < 0$, the oscillation is damped. If $r > 0$, it grows. If $r = 0$, it is sustained.

Read r and ω_0 directly from the exponent. For $x(t) = e^{(-1+j4)t}$, $r = \text{Re}\{a\} = -1 < 0$, so the real part $e^{-t} \cos 4t$ decays. A plucked string sounds like a tone with $r < 0$.

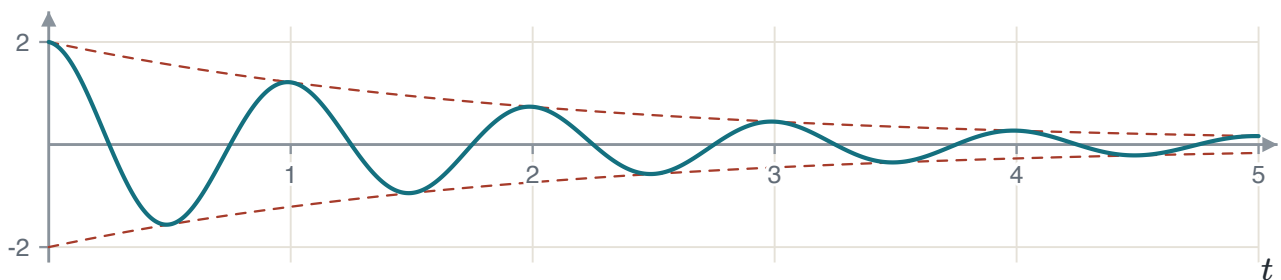


FIGURE 1.33 A damped sinusoid, $\text{Re}\{2e^{-0.5t} e^{j2\pi t}\}$, with its envelope shown dashed.

1.12 Complex exponentials in discrete time

DEFINITION

$$x[n] = C e^{\beta n} = C \alpha^n, \quad \alpha = e^\beta$$

Use the power form in discrete time because solutions of difference equations have this form.

For real C and α : if $0 < \alpha < 1$ the sequence decreases; if $\alpha > 1$ it increases. For complex $C = |C|e^{j\theta}$ and $\alpha = |\alpha|e^{j\omega_0}$:

$$\begin{aligned} x[n] &= C\alpha^n \\ &= |C|e^{j\theta}(|\alpha|e^{j\omega_0})^n \\ &= |C||\alpha|^n e^{j(\omega_0 n + \theta)} \\ &= |C||\alpha|^n \cos(\omega_0 n + \theta) + j|C||\alpha|^n \sin(\omega_0 n + \theta). \end{aligned}$$

Read the modulus $|\alpha|$ as the envelope. If $|\alpha| = 1$ the oscillation is sustained. If $|\alpha| > 1$ it grows. If $|\alpha| < 1$ it decays. The angle ω_0 sets only the oscillation, in radians per sample. For example, $x[n] = (1.1e^{j\pi/3})^n$ grows because $|\alpha| = 1.1 > 1$.

A negative real α is a special case with $\omega_0 = \pi$. For $x[n] = (-0.5)^n$, $n \geq 0$, the samples are $1, -0.5, 0.25, -0.125, \dots$. The sequence decays because $|\alpha| = 0.5 < 1$, and it changes sign at every step.

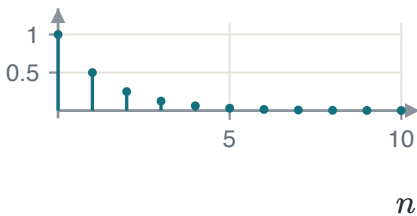


FIGURE 1.34 0.5^n decreases.

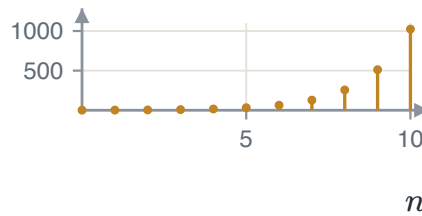


FIGURE 1.35 2^n increases.

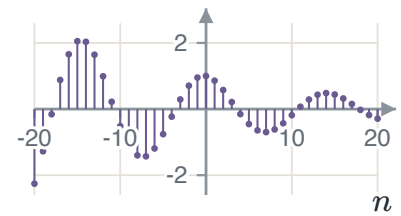


FIGURE 1.36 $0.95^n \cos(0.14\pi n)$: $|\alpha| = 0.95$ decays.

⚠ THE BOUNDARY MOVES

In continuous time the growth-decay boundary is $\text{Re}\{a\} = 0$, the imaginary axis. In discrete time it is $|\alpha| = 1$, the unit circle. The map between them is $\alpha = e^{\beta}$.

1.13 When is a discrete-time exponential periodic?

A discrete-time period must be an integer. Apply $x[n] = x[n + N]$ to $x[n] = Ce^{j\omega_0 n}$ and solve for that integer:

$$\begin{aligned} x[n + N] &= x[n] \\ Ce^{j\omega_0(n+N)} &= Ce^{j\omega_0 n} \\ e^{j\omega_0 N} &= 1 \\ \omega_0 N &= 2\pi k, \quad k \in \mathbb{Z} \\ N &= \frac{2\pi k}{\omega_0}. \end{aligned}$$

PERIODICITY CONDITION

$$\frac{\omega_0}{2\pi} = \frac{k}{N} \in \mathbb{Q}$$

N must be an integer. This is possible only when $\omega_0/2\pi$ is rational. If the ratio is irrational, no integer N satisfies the periodicity condition and the sequence is aperiodic.

For example, $x[n] = e^{j2n}$ has $\omega_0/2\pi = 2/2\pi = 1/\pi$, which is irrational. No integer period exists, so the sequence is aperiodic. The values $N = \pi$ and $N = 2\pi$ are not periods, because a period must be an integer.

TABLE 1.2

	CONTINUOUS TIME, $e^{j\omega_0 t}$	DISCRETE TIME, $e^{j\omega_0 n}$
Periodic?	For every $\omega_0 \neq 0$	Only if $\omega_0/2\pi$ is rational
Distinct frequencies	Each ω_0 gives a different signal	ω_0 and $\omega_0 + 2\pi$ give the same sequence

EXAMPLE 1.8

GIVEN $x[n] = e^{j(3\pi/5)n}$.

FIND The fundamental period N_0 .

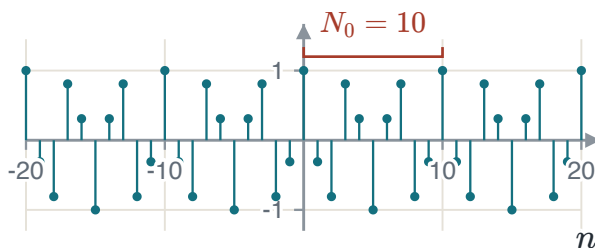
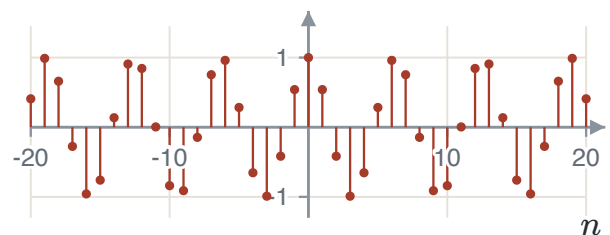
METHOD Use $N = 2\pi k/\omega_0$ and take the smallest k that makes N an integer.

SOLUTION Substitute the frequency before choosing k :

$$\begin{aligned} N &= \frac{2\pi k}{\omega_0} \\ &= \frac{2\pi k}{3\pi/5} \\ &= \frac{10}{3}k. \end{aligned}$$

The smallest positive integer k giving an integer N is $k = 3$, so $N_0 = 10$.

CHECK $\omega_0 N_0 = \frac{3\pi}{5} \times 10 = 6\pi = 2\pi \times 3$. The phase therefore changes by three full turns in ten samples.


 FIGURE 1.37 $\text{Re}\{e^{j3\pi n/5}\}$ repeats every 10 samples.

 FIGURE 1.38 $\cos(n)$ has $\omega_0 = 1$, so $\omega_0/2\pi$ is irrational. The samples never repeat exactly.

△ FREQUENCIES SEPARATED BY 2π GIVE THE SAME SEQUENCE

$e^{j(\omega_0+2\pi)n} = e^{j\omega_0 n}$ for every integer n . Therefore discrete-time frequencies that differ by 2π produce identical samples. This identity makes the discrete-time Fourier transform 2π -periodic and leads to aliasing.

1.14 A catalogue of common signals

A small set of named signals appears again and again in later chapters. This section collects them with their formulas. It introduces no new result. Each one is built from the ideas of this chapter.

Building blocks: step, ramp, sign and exponential

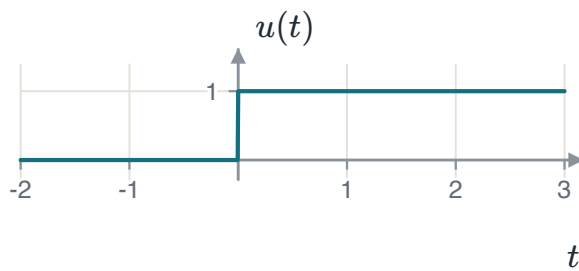


FIGURE 1.39 Unit step: $u(t) = 1$ for $t > 0$ and $u(t) = 0$ for $t < 0$.

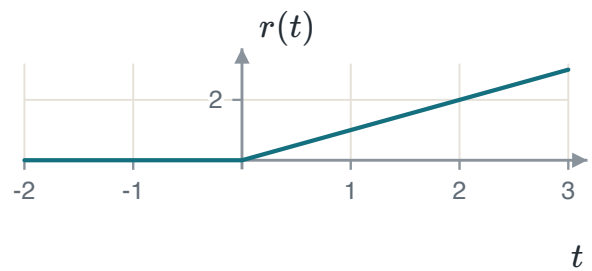


FIGURE 1.40 Unit ramp: $r(t) = t u(t)$. Its slope is 1 for $t > 0$, so $dr/dt = u(t)$.

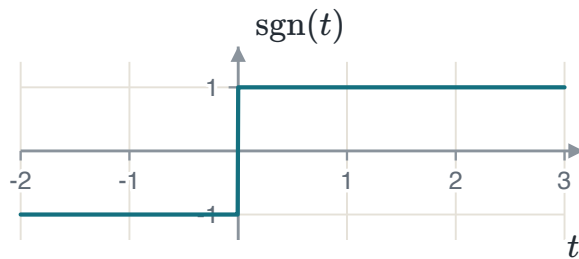


FIGURE 1.41 Sign: $\text{sgn}(t) = 1$ for $t > 0$ and -1 for $t < 0$, so $\text{sgn}(t) = 2u(t) - 1$ for $t \neq 0$.

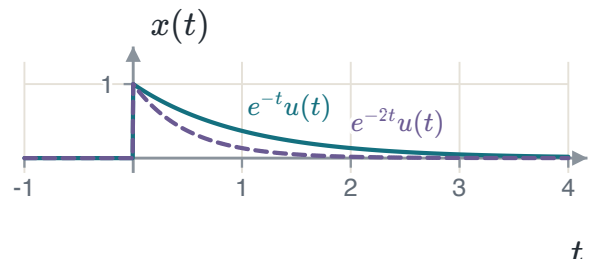


FIGURE 1.42 One-sided exponential: $x(t) = e^{-at}u(t)$ with $a > 0$. Its energy is $1/(2a)$. A larger a decays faster.

Many signals are sums, shifts and products of these four shapes. A voltage V switched on at t_0 is $V u(t - t_0)$. The ramp is the running integral of the step, $r(t) = \int_{-\infty}^t u(\tau) d\tau$. The step is the running integral of $\delta(t)$.

The energy of the one-sided exponential follows from the definition of E_∞ . The signal is zero for $t < 0$, so the lower limit is 0:

$$E_\infty = \int_0^\infty (e^{-at})^2 dt = \int_0^\infty e^{-2at} dt = \left[-\frac{e^{-2at}}{2a} \right]_0^\infty = 0 - \left(-\frac{1}{2a} \right) = \frac{1}{2a}, \quad a > 0.$$

Pulses: rectangle, triangle, sinc and Gaussian

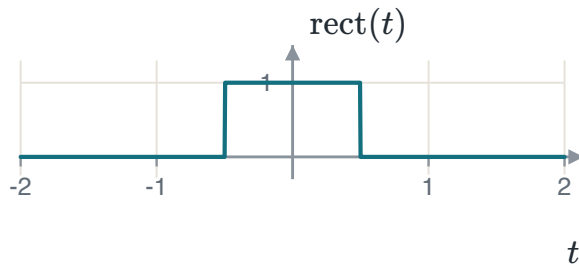


FIGURE 1.43 Rectangle: $\text{rect}(t) = 1$ for $|t| < \frac{1}{2}$ and 0 for $|t| > \frac{1}{2}$. Energy 1.

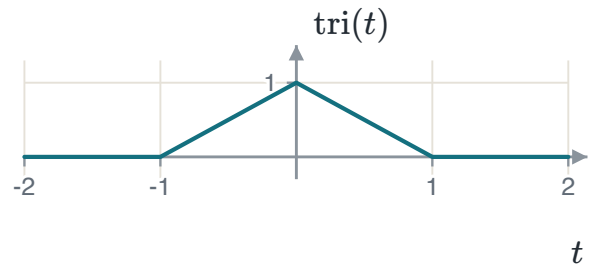


FIGURE 1.44 Triangle: $\text{tri}(t) = 1 - |t|$ for $|t| \leq 1$ and 0 otherwise. Energy $2/3$.

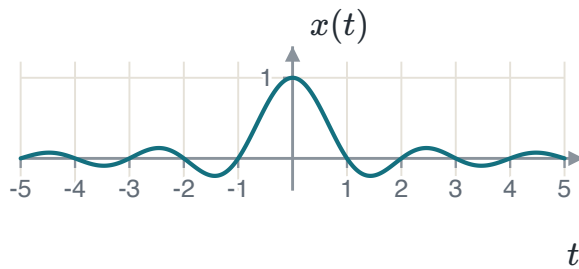


FIGURE 1.45 $x(t) = \text{sinc}(\pi t)$, with the unnormalised $\text{sinc}(\theta) = \sin \theta / \theta$. Peak 1 at $t = 0$; zeros at $t = \pm 1, \pm 2, \dots$

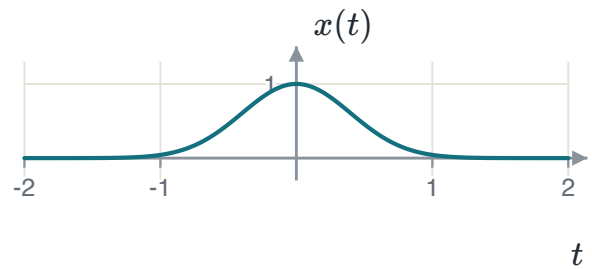


FIGURE 1.46 Gaussian: $x(t) = e^{-\pi t^2}$. It is smooth everywhere and its area is 1.

The triangle energy shows how to use symmetry in an energy integral. The integrand is even, so integrate over $[0, 1]$ and double. Then substitute $s = 1 - t$, $ds = -dt$, which maps $t = 0 \mapsto s = 1$ and $t = 1 \mapsto s = 0$:

$$E_{\infty} = \int_{-1}^1 (1 - |t|)^2 dt = 2 \int_0^1 (1 - t)^2 dt = 2 \int_0^1 s^2 ds = 2 \left[\frac{s^3}{3} \right]_0^1 = \frac{2}{3}.$$

A pulse can shape the loudness of a tone: $p(t - t_0) \cos(2\pi \cdot 440 t)$. The jumps of the rectangle are heard as clicks, while the Gaussian starts and ends softly. The pulses return later. The triangle is the rectangle convolved with itself (Chapter 3). The rectangle and the sinc are a transform pair (Chapter 5).

Periodic waveforms: sine, square, triangle and sawtooth

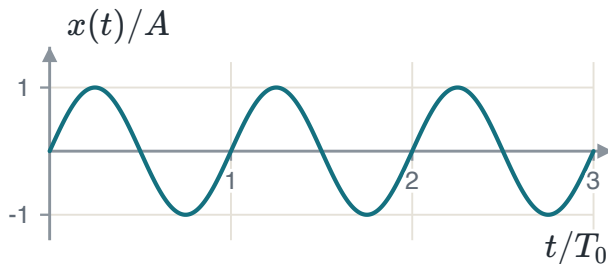


FIGURE 1.47 Sine: $x(t) = A \sin(2\pi t/T_0)$. Average power $A^2/2$.

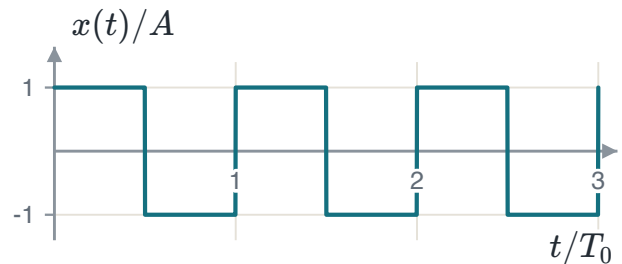


FIGURE 1.48 Square: $x(t) = A$ for $0 \leq t < T_0/2$ and $-A$ for $T_0/2 \leq t < T_0$. Average power A^2 .

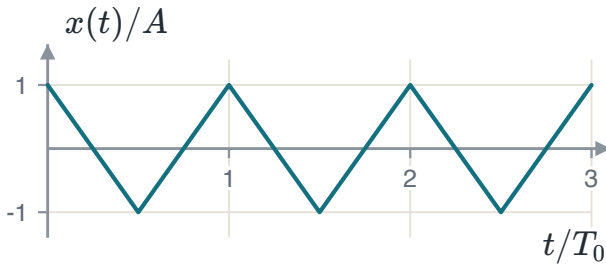


FIGURE 1.49 Triangle: $x(t) = A(1 - 4|t|/T_0)$ for $|t| \leq T_0/2$. Average power $A^2/3$.

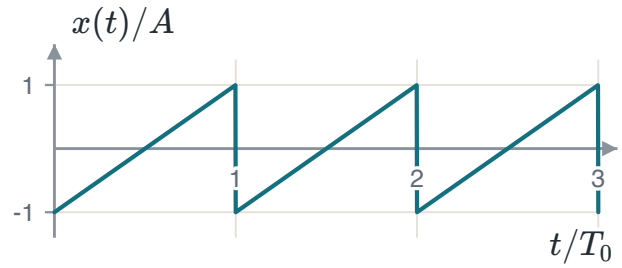


FIGURE 1.50 Sawtooth: $x(t) = A(2t/T_0 - 1)$ for $0 \leq t < T_0$. Average power $A^2/3$.

For a periodic signal, the average over all time equals the average over one period, $P_\infty = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt$. For the sawtooth, substitute $s = 2t/T_0 - 1$, $dt = \frac{T_0}{2} ds$, which maps $t = 0 \mapsto s = -1$ and $t = T_0 \mapsto s = 1$:

$$P_\infty = \frac{1}{T_0} \int_0^{T_0} A^2 \left(\frac{2t}{T_0} - 1 \right)^2 dt = \frac{A^2}{T_0} \cdot \frac{T_0}{2} \int_{-1}^1 s^2 ds = \frac{A^2}{2} \left[\frac{s^3}{3} \right]_{-1}^1 = \frac{A^2}{3}.$$

Played with the same period, $T_0 = 1/220$ s, all four have the same pitch. The shape changes only the colour of the sound, which Chapter 4 explains with harmonics. The triangle and the sawtooth have the same average power, $A^2/3$, and still sound different. Power does not fix the shape.

Signals for the ear: chirp, beats, AM and FM

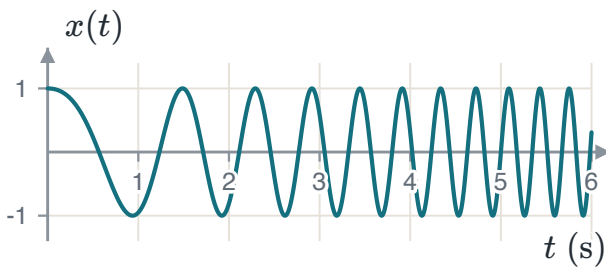


FIGURE 1.51 Chirp: $x(t) = \cos(2\pi(f_0 t + \frac{k}{2} t^2))$. Its frequency $f_0 + kt$ rises with time. Drawn with $f_0 = 0.3$ Hz, $k = 0.5$ Hz/s.

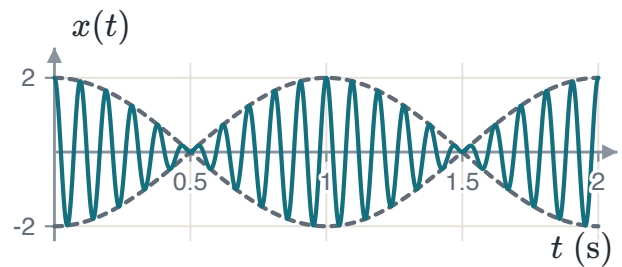


FIGURE 1.52 Beats: $\cos(2\pi f_1 t) + \cos(2\pi f_2 t)$, drawn with $f_1 = 10$ Hz, $f_2 = 11$ Hz. The dashed envelope is $\pm 2 \cos(\pi(f_1 - f_2)t)$.

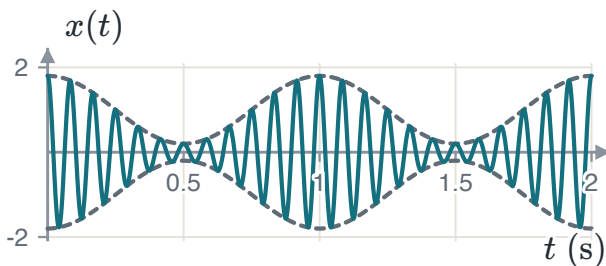


FIGURE 1.53 AM: $x(t) = (1 + m \cos(2\pi f_m t)) \cos(2\pi f_c t)$ with $0 < m \leq 1$. The dashed envelope is $\pm(1 + m \cos(2\pi f_m t))$.

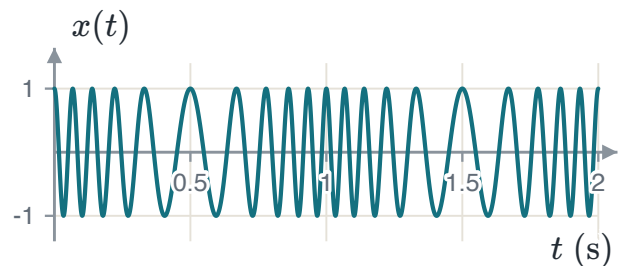


FIGURE 1.54 FM: $x(t) = \cos(2\pi f_c t + \beta \sin(2\pi f_m t))$. Drawn with $f_c = 10$ Hz, $f_m = 1$ Hz, $\beta = 5$.

Beats come from the sum-to-product identity $\cos \theta_1 + \cos \theta_2 = 2 \cos\left(\frac{\theta_1 - \theta_2}{2}\right) \cos\left(\frac{\theta_1 + \theta_2}{2}\right)$. With $\theta_i = 2\pi f_i t$ it gives

$$\cos(2\pi f_1 t) + \cos(2\pi f_2 t) = 2 \cos(\pi(f_1 - f_2)t) \cos(\pi(f_1 + f_2)t).$$

The slow factor sets the loudness, which peaks $|f_1 - f_2|$ times a second. Two strings a few hertz apart beat. A musician turns the peg until the beats slow down and stop, which means $f_1 = f_2$.

The frequency of a signal $\cos \phi(t)$ at time t is $\frac{1}{2\pi} \frac{d\phi}{dt}$. For the chirp this is $f_0 + kt$. For FM it is $f_c + \beta f_m \cos(2\pi f_m t)$: the amplitude stays fixed and the frequency swings about f_c . A chirp sweeps its frequency, as a bird call or a radar pulse does. Beats and AM keep the tone and move its loudness.

Random signals: noise, a noisy tone, a random walk and a telegraph signal

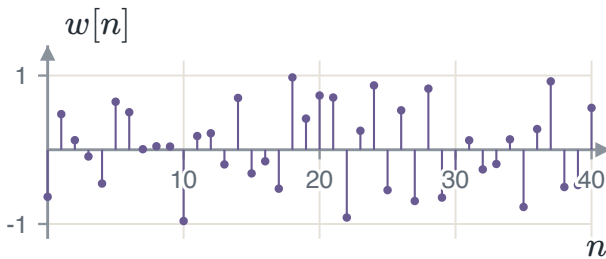


FIGURE 1.55 White noise $w[n]$: the samples are independent, with zero mean and the same variance. No sample predicts the next one.

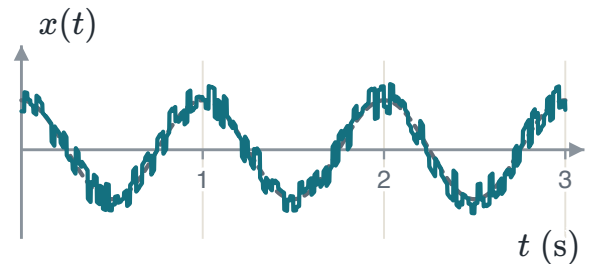


FIGURE 1.56 Signal plus noise: $x(t) = \cos(2\pi f_0 t) + \sigma w(t)$. The dashed curve is the tone alone.

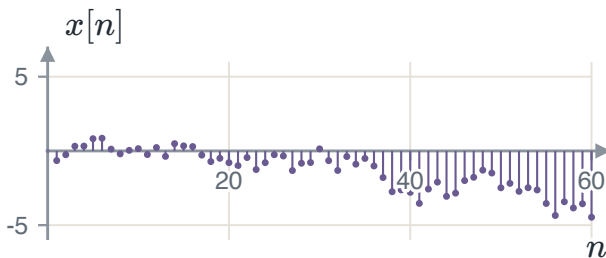


FIGURE 1.57 Random walk: $x[n] = x[n-1] + w[n]$ with $x[0] = 0$. Each step is random, but the sum wanders far from zero.

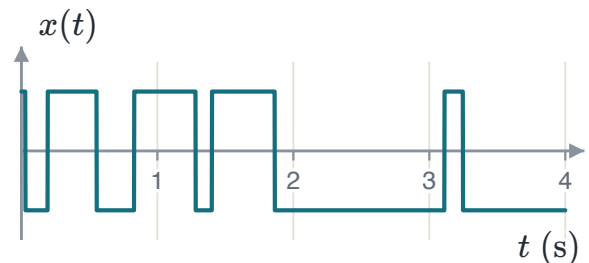


FIGURE 1.58 Random telegraph: $x(t)$ is $+1$ or -1 and flips sign at random times. Only the times of the flips are random.

A random signal has a rule for how it is made, not a formula for each value. Run the rule again and the waveform changes, but its average behaviour does not. A measured signal almost always contains some noise. Heard as sound, white noise is a hiss, the random walk is a low rumble because it changes slowly, and the telegraph signal crackles.

1.15 Chapter summary

1. E_∞ and P_∞ are limits. Finite energy forces zero average power. Finite non-zero average power forces infinite energy. Unbounded growth gives neither.
2. To build $x(at - b)$, shift by b first, then scale by a . The other order gives $x(at - ab)$.
3. T_0 and N_0 are the smallest positive periods, and $\omega_0 = 2\pi/T_0 = 2\pi/N_0$.

4. $\delta[n]$ is an ordinary sequence. $\delta(t)$ is a distribution defined by sifting.
5. Every signal is a sum of weighted, shifted impulses: $x[n] = \sum_k x[k] \delta[n - k]$.
6. Every signal splits in one way into an even and an odd part: $\mathcal{E}\{x(t)\} = \frac{1}{2}[x(t) + x(-t)]$ and $\mathcal{O}\{x(t)\} = \frac{1}{2}[x(t) - x(-t)]$. The odd part is 0 at $t = 0$.
7. $\delta[n] = u[n] - u[n - 1]$ and $u[n] = \sum_{k=-\infty}^n \delta[k]$: a first difference and a running sum.
8. Sifting, $\int x(t) \delta(t - t_0) dt = x(t_0)$, gives a **number**. Sampling, $x(t) \delta(t - t_0) = x(t_0) \delta(t - t_0)$, gives a **signal**.
9. In $e^{(r+j\omega_0)t}$, r sets the envelope e^{rt} and ω_0 sets the oscillation. In discrete time $|\alpha|$ plays the role of the envelope, with the boundary at $|\alpha| = 1$.
10. A discrete-time exponential is periodic only when $\omega_0/2\pi$ is rational. A continuous-time one always is.
11. $e^{j\omega_0 n}$ and $e^{j(\omega_0+2\pi)n}$ are the same sequence, so discrete-time frequencies repeat every 2π .

Exercises

1.1 Classify $x(t) = e^{-3t}u(t)$ as an energy signal, a power signal, or neither. Give the value of whichever quantity is finite.

Answer: Energy signal, $E_\infty = 1/6$.

1.2 For $x[n]$ non-zero only on $-1 \leq n \leq 1$ with values 1, 2, 1, sketch $x[n + 4]$ and $x[n - 5]$ and give the support of each.

Answer: Supports $-5 \leq n \leq -3$ and $4 \leq n \leq 6$.

1.3 A signal $x(t)$ is non-zero only on $[-1, 5]$. On what interval is $x(2t + 3)$ non-zero?

Answer: $[-2, 1]$.

1.4 Find the fundamental period of $x[n] = \cos(4\pi n/7)$, or show that it is aperiodic.

Answer: $N_0 = 7$.

1.5 Show that if $x(t)$ is odd then $\int_{-a}^a x(t) dt = 0$ for every $a > 0$.

1.6 Evaluate $\int_{-\infty}^{\infty} (t^2 + 1) \delta(t - 2) dt$ and $\sum_{n=-\infty}^{\infty} 2^{-n} \delta[n - 3]$.

Answer: 5 and $1/8$.

CHAPTER

2

Systems and their properties

This chapter gives tests for memory, invertibility, causality, stability, time invariance and linearity. Each test describes a system through its input and output signals.

2.1 What a system is

A system is a rule that turns an input signal into an output signal. The rule is deterministic: the same input always gives the same output.

Write the rule as an operator, $y = S\{x\}$. The operator acts on the whole input signal. This form lets us test which input times affect an output and how the system responds to a time shift.

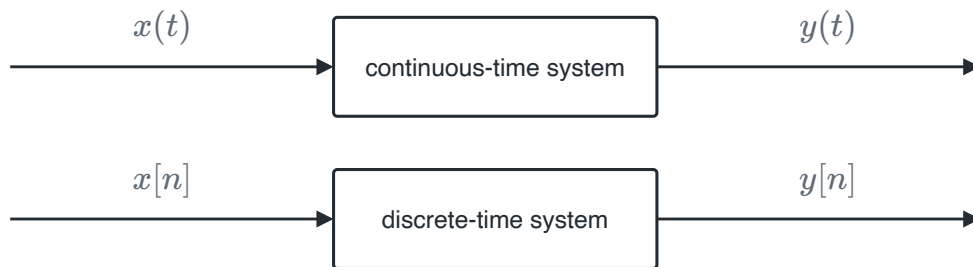


FIGURE 2.1 The operator description applies in continuous time and discrete time.

2.2 Memory

CRITERION

A system is **memoryless** if the output at time t (or n) depends only on the input at that same time.

TABLE 2.1

SYSTEM	VERDICT	REASON
$y(t) = [2x(t) - x^2(t)]^2$	memoryless	Only $x(t)$ appears. There is no $x(t + 1)$ or $x(t - 2)$ term.
$y[n] = x[n]$	memoryless	The identity system.
$y[n] = x[n - 1]$	has memory	The output at n uses the sample at $n - 1$.
$y[n] = x[n] + y[n - 1]$	has memory	See the derivation below.

For the last system, assume initial rest and expose the memory by substituting the feedback relation repeatedly:

$$\begin{aligned}
 y[n] &= x[n] + y[n-1] \\
 &= x[n] + x[n-1] + y[n-2] \\
 &= x[n] + x[n-1] + x[n-2] + y[n-3] \\
 &\vdots \\
 &= x[n] + x[n-1] + x[n-2] + \cdots \\
 &= \sum_{k=0}^{\infty} x[n-k].
 \end{aligned}$$

The result uses $x[n-k]$ for every $k \geq 0$, so the output depends on the whole input history. Output feedback can therefore give a system memory.

✓ CIRCUIT EXAMPLES

A resistor, $v(t) = R i(t)$, is memoryless because the voltage at t uses only the current at t . A capacitor is not memoryless because $v(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau$ uses the current history.

2.3 Invertibility

① CRITERION

A system is **invertible** if distinct inputs always produce distinct outputs. The map must be one-to-one.

To prove invertibility, find a formula that recovers every input from its output. To disprove invertibility, find two distinct inputs that give the same output.

🔗 EXAMPLE 2.1

GIVEN $y(t) = [\cos(t) + 2]x(t)$.

FIND Is the system invertible?

METHOD Solve the system rule for $x(t)$ and verify that the divisor never vanishes.

SOLUTION First bound the gain:

$$-1 \leq \cos(t) \leq 1 \implies 1 \leq \cos(t) + 2 \leq 3.$$

It is never zero. Now solve the system equation for the input:

$$y(t) = [\cos(t) + 2]x(t) \implies x(t) = \frac{y(t)}{\cos(t) + 2}.$$

The input is recovered uniquely, so the system is invertible.

CHECK

Substituting back gives $[\cos t + 2] \frac{y(t)}{\cos t + 2} = y(t)$. Had the gain been $\cos(t)$ alone, the system would fail at every zero of the cosine.

EXAMPLE 2.2**GIVEN** $y(t) = x^2(t)$.**FIND** Is the system invertible?**METHOD** Search for two distinct inputs that lose the same information under squaring.**SOLUTION** No. Take $x_1(t) = 1$ and $x_2(t) = -1$ for all t . Then

$$S\{x_1\} = 1^2 = 1, \quad S\{x_2\} = (-1)^2 = 1.$$

The inputs are different, but their outputs are equal.

CHECK One counterexample is enough. The sign information is destroyed and cannot be recovered from y alone.**⊗ NOT AN INVERSION FORMULA**

Writing $x(t) = \sqrt{y(t)}$ does not invert $y = x^2$. It picks one of two possible inputs by convention. An inversion formula must return the actual input, for every admissible input.

2.4 Causality**ⓘ CRITERION**

A system is **causal** if the output at time t (or n) depends only on inputs at times $\leq t$: the present and the past.

TABLE 2.2

SYSTEM	VERDICT	REASON
$y[n] = x[n - 1]$	causal	Uses only a past sample.
$y[n] = x[n] + x[n + 1]$	not causal	$x[n + 1]$ is a future sample.
$y(t) = \int_{-\infty}^t x(\tau) d\tau$	causal	The upper limit is t , so only $\tau \leq t$ contributes.
$y[n] = x[-n]$	not causal	$y[-1] = x[1]$. The output at $n = -1$ needs a future input.
$y(t) = x(t) \cos(t + 1)$	causal	Uses only $x(t)$. The factor $\cos(t + 1)$ is a known function of t , fixed in advance, not a future input.

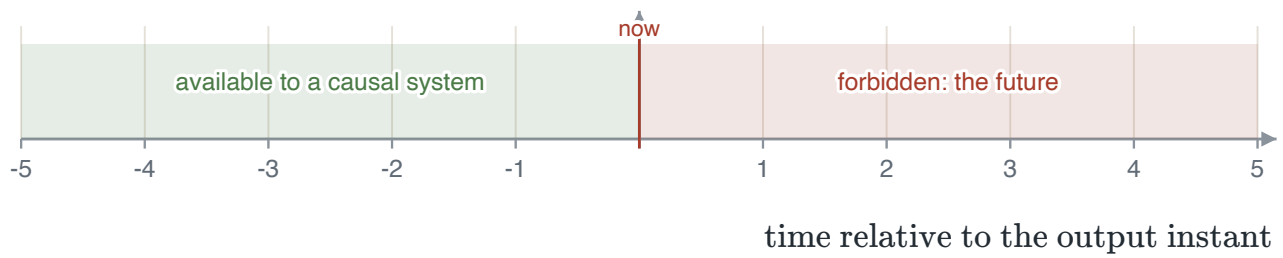


FIGURE 2.2 Causality restricts which part of the input axis the output is allowed to consult.

A real-time system must be causal because future input values are not available. A non-causal system can be used when the complete signal has already been stored, as in offline audio or image processing.

2.5 Stability

A signal is **bounded** if there is a constant $B < \infty$ with $|x(t)| < B$ for all t .

i CRITERION

A system is **BIBO stable** if every bounded input produces a bounded output. Bounded Input, Bounded Output.

Δ PROOF AND COUNTEREXAMPLE

To prove stability, derive an output bound for every bounded input. To disprove stability, find one bounded input that produces an unbounded output. One bounded example cannot prove stability.

P EXAMPLE 2.3

GIVEN $y(t) = 2x^2(t-1) + x(3t)$.

FIND Is the system stable?

METHOD Assume $|x(t)| \leq B < \infty$ and bound $|y|$ with the triangle inequality.

SOLUTION Use the system rule, then apply the triangle inequality, then use the input bound:

$$\begin{aligned} |y(t)| &= |2x^2(t-1) + x(3t)| \\ &\leq |2x^2(t-1)| + |x(3t)| \\ &= 2|x(t-1)|^2 + |x(3t)| \\ &\leq 2B^2 + B < \infty. \end{aligned}$$

The output therefore has the finite bound $2B^2 + B$, so the system is stable.

CHECK Shifting and scaling the time axis cannot change the set of values a signal takes, so $x(t-1)$ and $x(3t)$ are bounded by the same B .

P EXAMPLE 2.4

GIVEN $y[n] = \sum_{k=-\infty}^n x[k]$, the accumulator.

FIND Is the system stable?

METHOD Look for one bounded input with an unbounded output.

SOLUTION Take $x[n] = u[n]$, so $|x[n]| \leq 1$. For $n \geq 0$,

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^n u[k] \\ &= \sum_{k=0}^n 1 \\ &= n + 1 \rightarrow \infty. \end{aligned}$$

The input is bounded and the output is unbounded. The system is **not** stable.

CHECK The input never exceeds 1 and the output grows without bound. That is exactly the failure BIBO stability forbids.

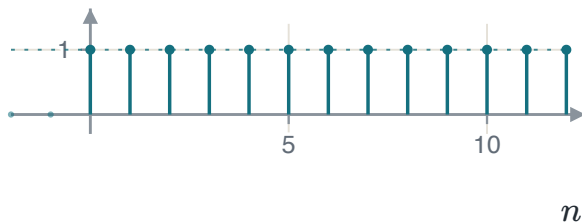


FIGURE 2.3 Bounded input $u[n]$.

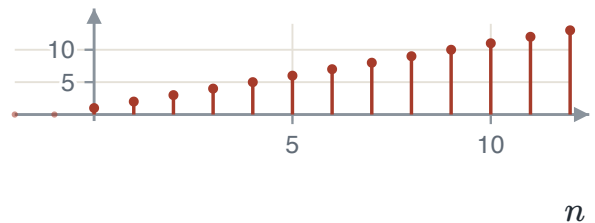


FIGURE 2.4 Unbounded output $n + 1$.

⚠ CAUSAL DOES NOT MEAN STABLE

The accumulator is causal and unstable. The reversal $y[n] = x[-n]$ is stable and not causal. The six properties are independent except where a proof links them.

2.6 Time invariance

① CRITERION

If $x(t) \rightarrow y(t)$, the system is **time invariant** when $x(t - t_0) \rightarrow y(t - t_0)$ for every t_0 .

The test is a comparison of two computed signals.

1. **Path 1.** Shift the input, then run the system: $y_2 = S\{x(t - t_0)\}$.
2. **Path 2.** Run the system, then shift the output: $y_1(t - t_0)$.

The system is time invariant exactly when the two agree, for every input and every t_0 .

🔗 EXAMPLE 2.5

GIVEN $y(t) = \sin(x(t))$.

FIND Is the system time invariant?

METHOD Compute the response to a shifted input and compare it with the shifted response.

SOLUTION Path 1 shifts the input and then applies the system:

$$x_2(t) = x_1(t - t_0) \implies y_2(t) = \sin(x_2(t)) = \sin(x_1(t - t_0)).$$

Path 2 applies the system and then shifts its output:

$$y_1(t) = \sin(x_1(t)) \implies y_1(t - t_0) = \sin(x_1(t - t_0)).$$

Thus $y_2(t) = y_1(t - t_0)$ for every input and every t_0 . The system is time invariant.

CHECK Both paths give the same expression for every input and every t_0 .

EXAMPLE 2.6

GIVEN $y[n] = nx[n]$.

FIND Is the system time invariant?

METHOD Look for a counterexample using an impulse.

SOLUTION Take $x_1[n] = \delta[n]$. The original output and its one-sample delay are

$$y_1[n] = n\delta[n] = 0, \quad y_1[n - 1] = 0.$$

Now delay the input first:

$$\begin{aligned} x_2[n] &= \delta[n - 1], \\ y_2[n] &= n\delta[n - 1] \\ &= 1 \cdot \delta[n - 1] = \delta[n - 1]. \end{aligned}$$

The identity $n\delta[n - 1] = \delta[n - 1]$ follows because the impulse is non-zero only at $n = 1$. Therefore $y_2[n] \neq y_1[n - 1]$, so the system is **not** time invariant.

CHECK The impulse isolates one index, so the unequal outputs form a complete counterexample.

WHEN TO APPLY THE TEST FIRST

An explicit time-dependent coefficient such as n or $\cos(t)$ usually breaks time invariance. A scaled input argument such as $x(3t)$ also changes input shifts. In either case, use the two-path test.

2.7 Linearity

CRITERION

$$a x_1 + b x_2 \longrightarrow a y_1 + b y_2 \quad \text{for all } a, b \in \mathbb{C}$$

where $x_1 \rightarrow y_1$ and $x_2 \rightarrow y_2$. This is additivity and homogeneity in one statement.

EXAMPLE 2.7

GIVEN $y(t) = 2\pi x(t)$.

FIND Is the system linear?

METHOD Apply the system to a weighted sum and compare the result with the same weighted sum of the two outputs.

SOLUTION Let $x_3 = ax_1 + bx_2$. Apply the system and distribute the constant:

$$\begin{aligned} S\{x_3\} &= S\{ax_1 + bx_2\} \\ &= 2\pi(ax_1 + bx_2) \\ &= a(2\pi x_1) + b(2\pi x_2) \\ &= a y_1 + b y_2. \end{aligned}$$

The required superposition equality holds, so the system is linear. It is also time invariant, so it is LTI.

CHECK The final expression is exactly $a S\{x_1\} + b S\{x_2\}$ for arbitrary inputs and scalars.

EXAMPLE 2.8

GIVEN	$y[n] = (x[2n])^2$.
FIND	Is the system linear?
METHOD	Apply superposition and compare the powers and cross term with the required linear result.
SOLUTION	Let $x_3[n] = ax_1[n] + bx_2[n]$. Combining first and then applying the system gives

$$\begin{aligned}
 y_3[n] &= (x_3[2n])^2 \\
 &= (ax_1[2n] + bx_2[2n])^2 \\
 &= a^2 x_1^2[2n] + 2ab x_1[2n]x_2[2n] + b^2 x_2^2[2n].
 \end{aligned}$$

Applying the system first and then combining gives

$$ay_1[n] + by_2[n] = ax_1^2[2n] + bx_2^2[2n].$$

The expressions differ: the first has a cross term and different powers of a and b . The system is **not** linear.

CHECK Scaling the input by a scales this output by a^2 , so homogeneity also fails.

⊗ ZERO-INPUT TEST

$y(t) = x(t) + 5$ is not linear because the zero input does not give the zero output. Every linear system must satisfy $S\{0\} = 0$.

2.8 How to classify an unfamiliar system

1. **Test time invariance.** Apply the two-path test when the rule contains an explicit t or n , a time-dependent gain, or a scaled argument.
2. **Test linearity.** Look for squares, products, $\sin(x)$, absolute values or saturation. Test homogeneity with one scalar first.
3. **Inspect every input argument.** An argument different from t or n gives memory. An argument later than the output time gives non-causality.
4. **Test stability.** Assume $|x| \leq B$ and derive a finite output bound. To disprove stability, choose a bounded input that makes the output unbounded.
5. **Test invertibility.** Find an inverse formula or two distinct inputs with the same output.

✓ ONE GENERAL IMPLICATION

Memoryless implies causal. Establish the other properties independently. A causal system need not be stable, a linear system need not be time invariant, and a stable system need not be memoryless.

TABLE 2.3

SYSTEM	MEMORYLESS	INVERTIBLE	CAUSAL	STABLE	TIME INV.	LINEAR
$y(t) = 2\pi x(t)$	yes	yes	yes	yes	yes	yes
$y[n] = x[n - 1]$	no	yes	yes	yes	yes	yes
$y(t) = x^2(t)$	yes	no	yes	yes	yes	no
$y[n] = n x[n]$	yes	no	yes	no	no	yes
$y[n] = \sum_{k \leq n} x[k]$	no	yes	yes	no	yes	yes
$y[n] = x[-n]$	no	yes	no	yes	no	yes
$y[n] = (x[2n])^2$	no	no	no	yes	no	no

Read each property down its column. The examples show why the six properties require separate tests.

Exercises

2.1 Classify $y(t) = x(t) u(t)$ against all six properties.

Answer: Memoryless, not invertible, causal, stable, not time invariant, linear.

2.2 Show that $y[n] = x[n] - x[n - 1]$ is invertible on inputs that are zero for $n < 0$, and give the inverse.

Answer: $x[n] = \sum_{k=0}^n y[k]$.

2.3 Give a system that is linear and causal but not stable, and one that is stable and time invariant but not linear.

2.4 Is $y(t) = x(t/2)$ time invariant? Prove or give a counterexample.

Answer: No. Path 1 gives $x(t/2 - t_0)$ and path 2 gives $x((t - t_0)/2)$.

CHAPTER

3

Linear time-invariant systems

This chapter develops a direct way to find the output of a linear time-invariant system. Its impulse response describes the system, and convolution uses that response to find the output for any input.

3.1 The impulse response

DEFINITION

The **impulse response** is the output when the input is a unit impulse: $x[n] = \delta[n]$ gives $y[n] = h[n]$. In continuous time, $x(t) = \delta(t)$ gives $y(t) = h(t)$.

For a general system, this experiment gives only one input-output pair. It does not determine the response to another input.

For a linear time-invariant system, one impulse response is enough. Time invariance gives the response to every shifted impulse. Linearity gives the response to every weighted sum of those impulses. Chapter 1 showed that every discrete-time signal can be written as such a sum. Therefore $h[n]$ determines the response to every input.



FIGURE 3.1 For an LTI system, one impulse experiment determines the response to every input.

3.2 Deriving the convolution sum

The purpose of this derivation is to obtain the output from $h[n]$. Start with the representation property from Chapter 1, then apply the system to both sides.

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \quad \longrightarrow \quad S \quad \longrightarrow \quad y[n] = ?$$

1. **Time invariance.** Since $\delta[n] \rightarrow h[n]$, we also have $\delta[n-k] \rightarrow h[n-k]$.
2. **Homogeneity.** The number $x[k]$ does not depend on n , so $x[k]\delta[n-k] \rightarrow x[k]h[n-k]$.
3. **Additivity.** The response to the sum is the sum of the responses.

Write the three moves as one chain. Apply the system to the representation, take the sum outside the system by additivity, take each weight outside by homogeneity, and replace each shifted impulse response by time invariance:

$$\begin{aligned}
 y[n] &= S\left\{\sum_{k=-\infty}^{\infty} x[k] \delta[n-k]\right\} \\
 &= \sum_{k=-\infty}^{\infty} S\{x[k] \delta[n-k]\} \quad (\text{additivity}) \\
 &= \sum_{k=-\infty}^{\infty} x[k] S\{\delta[n-k]\} \quad (\text{homogeneity}) \\
 &= \sum_{k=-\infty}^{\infty} x[k] h[n-k] \quad (\text{time invariance}).
 \end{aligned}$$

CONVOLUTION SUM

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n]$$

Next, replace the summation index by $m = n - k$, so $k = n - m$. As k runs over all integers, m also runs over all integers, only in the opposite direction. The sum has the same terms, so its value is unchanged. This gives an equivalent form and shows that convolution is commutative:

$$\begin{aligned}
 \sum_{k=-\infty}^{\infty} x[k] h[n-k] &= \sum_{m=-\infty}^{\infty} x[n-m] h[n-(n-m)] \\
 &= \sum_{m=-\infty}^{\infty} x[n-m] h[m] = h[n] * x[n].
 \end{aligned}$$

⊗ CHECK THE SYSTEM BEFORE USING CONVOLUTION

Convolution gives the system output only when the system is linear and time invariant. The derivation uses time invariance to shift h and linearity to scale and add the responses. If either property fails, the convolution result is not the output of that system.

3.3 How to compute a convolution

1. **Flip.** Reverse $h[k]$ to get $h[-k]$.
2. **Shift.** Move it by n to get $h[n-k]$.
3. **Multiply and add.** Form $x[k]h[n-k]$ and sum over k .
4. Repeat for every n .

Construct $h[n-k]$ in a fixed order. Treat k as the variable, shift $h[k]$ by n , and then reverse it in k . This is the shift-then-scale procedure from Chapter 1.

⊗ DO NOT OMIT THE REVERSAL

Using an unreversed h gives $\sum_k x[k]h[k - n]$, which is the cross-correlation of x and h , not their convolution. A symmetric h can hide this error because reversal does not change it. Use an asymmetric example to check the construction.

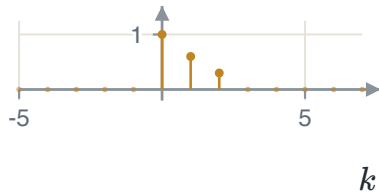


FIGURE 3.2 $h[k]$

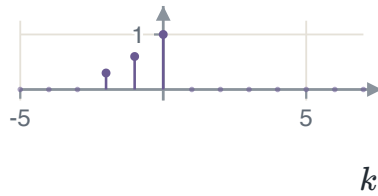


FIGURE 3.3 $h[-k]$: flip

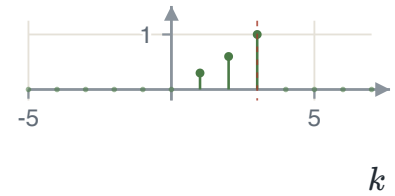


FIGURE 3.4 $h[n-k]$ at $n=3$: shift

✂ EXAMPLE 3.1

GIVEN $x[n] = \{1, 2, 1, 2\}$ for $n = 0, 1, 2, 3$, and $h[n] = \{1, 1\}$ for $n = 0, 1$. Both zero elsewhere.

FIND $y[n] = x[n] * h[n]$.

METHOD Because x has four non-zero samples, expand the convolution sum into four terms. Each term is a shifted copy of h weighted by the corresponding sample of x .

SOLUTION Start from the convolution sum. Only $k = 0, 1, 2, 3$ give non-zero $x[k]$, so the sum has four terms:

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[k] h[n-k] \\ &= x[0]h[n] + x[1]h[n-1] + x[2]h[n-2] + x[3]h[n-3] \\ &= h[n] + 2h[n-1] + h[n-2] + 2h[n-3]. \end{aligned}$$

Now evaluate at each n . Since $h[m] = 1$ only for $m = 0$ and $m = 1$, the term $h[n-k]$ is non-zero only when $n-k = 0$ or $n-k = 1$. Keep only those terms:

$$\begin{aligned} y[0] &= x[0]h[0] = 1 \cdot 1 = 1, \\ y[1] &= x[0]h[1] + x[1]h[0] = 1 \cdot 1 + 2 \cdot 1 = 3, \\ y[2] &= x[1]h[1] + x[2]h[0] = 2 \cdot 1 + 1 \cdot 1 = 3, \\ y[3] &= x[2]h[1] + x[3]h[0] = 1 \cdot 1 + 2 \cdot 1 = 3, \\ y[4] &= x[3]h[1] = 2 \cdot 1 = 2. \end{aligned}$$

For $n < 0$ and $n > 4$ no term survives, so $y[n] = 0$. This gives $y = \{1, 3, 3, 3, 2\}$ on $n = 0, \dots, 4$.

CHECK The support must contain $4 + 2 - 1 = 5$ samples and run from 0 to 4, which the result does. The total must satisfy $\sum_n y[n] = (\sum_n x[n])(\sum_n h[n])$, and $12 = 6 \times 2$. Also, $h = \{1, 1\}$ forms a two-point moving sum, so each output is the sum of two adjacent input samples. This gives the same values.

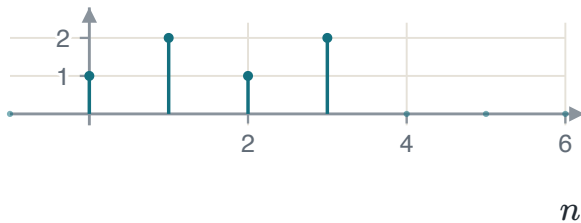


FIGURE 3.5 $x[n]$

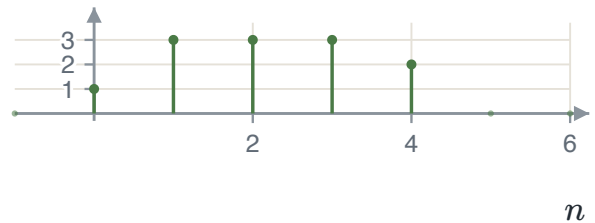


FIGURE 3.6 $y[n] = x[n] * h[n]$

EXAMPLE 3.2

GIVEN $x[n] = \left(\frac{1}{2}\right)^n u[n]$ and $h[n] = u[n]$.

FIND $y[n] = x[n] * h[n]$.

METHOD

Use the support of $h[n - k]$ to set the summation limits: it equals 1 exactly when $k \leq n$. The overlap changes when n changes sign, so treat $n < 0$ and $n \geq 0$ separately.

SOLUTION

Substitute the two signals into the convolution sum:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k] = \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k u[k] u[n - k].$$

The factor $u[k]$ is 1 only for $k \geq 0$. The factor $u[n - k]$ is 1 only for $n - k \geq 0$, that is, for $k \leq n$. Both conditions must hold for a term to survive.

Case 1, $n < 0$. The conditions $k \geq 0$ and $k \leq n < 0$ cannot both hold. No term survives, so $y[n] = 0$.

Case 2, $n \geq 0$. The surviving terms are $0 \leq k \leq n$, and on this range both step factors equal 1:

$$y[n] = \sum_{k=0}^n \left(\frac{1}{2}\right)^k.$$

This is a finite geometric sum with first term 1, ratio $r = \frac{1}{2}$, and $n + 1$ terms. Apply the formula in the box below with $a = 1$, $m = 0$:

$$\begin{aligned} y[n] &= \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}} \\ &= \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{\frac{1}{2}} = 2\left(1 - \left(\frac{1}{2}\right)^{n+1}\right) \\ &= 2 - 2\left(\frac{1}{2}\right)^{n+1} \\ &= 2 - \left(\frac{1}{2}\right)^n. \end{aligned}$$

The last line uses $2\left(\frac{1}{2}\right)^{n+1} = \left(\frac{1}{2}\right)^n$, because multiplying by 2 cancels one factor of $\frac{1}{2}$. Together, $y[n] = \left(2 - \left(\frac{1}{2}\right)^n\right) u[n]$.

CHECK

At $n = 0$, the formula gives $y[0] = 1$, and direct substitution gives $x[0]h[0] = 1$. As $n \rightarrow \infty$, $y[n] \rightarrow 2$, which equals the total sum of the input. This agrees with the interpretation of $h[n] = u[n]$ as an accumulator.

 GEOMETRIC SUM

$\sum_{k=m}^n a r^k = \frac{a(r^m - r^{n+1})}{1 - r}$. The finite sum needs only $r \neq 1$. The condition $|r| < 1$ is needed only when the sum runs to infinity.

3.4 The convolution integral

The continuous-time derivation has the same purpose and uses the same three LTI moves. First, the sifting property represents x as continuously indexed weighted impulses:

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau,$$

This form comes from the sifting definition in Chapter 1, $x(t_0) = \int x(t) \delta(t - t_0) dt$. Rename the integration variable t as τ and the sampling instant t_0 as t . This gives $x(t) = \int x(\tau) \delta(\tau - t) d\tau$. The impulse is even, $\delta(\tau - t) = \delta(t - \tau)$, which gives the form above. Now apply the system to both sides. Additivity takes the integral outside the system, because an integral is a limit of sums. Homogeneity takes the weight $x(\tau) d\tau$ outside, because it does not depend on t . Time invariance replaces $S\{\delta(t - \tau)\}$ by $h(t - \tau)$:

$$\begin{aligned} y(t) &= S\left\{\int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau\right\} \\ &= \int_{-\infty}^{\infty} x(\tau) S\{\delta(t - \tau)\} d\tau \quad (\text{additivity, homogeneity}) \\ &= \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \quad (\text{time invariance}). \end{aligned}$$

CONVOLUTION INTEGRAL

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = x(t) * h(t) \\ y(t) &= \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \end{aligned}$$

The two forms are equal. Choose the form that makes the support conditions easier to write.

To pass from the first form to the second, substitute $\sigma = t - \tau$. Then $\tau = t - \sigma$ and $d\tau = -d\sigma$. As τ goes from $-\infty$ to $+\infty$, σ goes from $+\infty$ to $-\infty$. Swapping the limits back removes the minus sign:

$$\begin{aligned} \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau &= \int_{+\infty}^{-\infty} x(t - \sigma) h(\sigma) (-d\sigma) \\ &= \int_{-\infty}^{\infty} h(\sigma) x(t - \sigma) d\sigma. \end{aligned}$$

The name of the integration variable does not matter, so σ can be written as τ again. This is the second form, and it shows that $x * h = h * x$.

△ BUILD THE REVERSED AND SHIFTED RESPONSE EXPLICITLY

Use τ as the variable and apply the shift before the reversal. For $\delta(-t + 5)$, first form $v(t) = \delta(t + 5)$ and then form $y(t) = v(-t) = \delta(-t + 5)$. The result is an impulse at $t = +5$. Writing both moves prevents a sign error in the integration limits.

EXAMPLE 3.3

GIVEN $x(t) = e^{2t}u(-t)$ and $h(t) = u(t - 3)$.

FIND $y(t) = x(t) * h(t)$.

METHOD

Use the two support conditions to set the integration limit. $x(\tau)$ is non-zero for $\tau \leq 0$, and $h(t - \tau) = u(t - \tau - 3)$ is non-zero for $\tau \leq t - 3$. The overlap ends at $\min(0, t - 3)$, so the active upper limit changes at $t = 3$.

SOLUTION

Substitute the two signals into the convolution integral:

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = \int_{-\infty}^{\infty} e^{2\tau} u(-\tau) u(t - \tau - 3) d\tau.$$

The factor $u(-\tau)$ is 1 only for $\tau \leq 0$. The factor $u(t - \tau - 3)$ is 1 only for $t - \tau - 3 \geq 0$, that is, for $\tau \leq t - 3$. Both must hold, so the integrand is $e^{2\tau}$ for $\tau \leq \min(0, t - 3)$ and zero elsewhere. The antiderivative of $e^{2\tau}$ is $\frac{1}{2}e^{2\tau}$, and $e^{2\tau} \rightarrow 0$ as $\tau \rightarrow -\infty$, so the lower limit contributes zero.

Case 1, $t < 3$. Here $t - 3 < 0$, so the binding limit is $t - 3$:

$$\begin{aligned} y(t) &= \int_{-\infty}^{t-3} e^{2\tau} d\tau \\ &= \left[\frac{1}{2} e^{2\tau} \right]_{-\infty}^{t-3} \\ &= \frac{1}{2} e^{2(t-3)} - 0 = \frac{1}{2} e^{2(t-3)}. \end{aligned}$$

Case 2, $t > 3$. Here $t - 3 > 0$, so the binding limit is 0 and the whole of x is covered:

$$\begin{aligned} y(t) &= \int_{-\infty}^0 e^{2\tau} d\tau \\ &= \left[\frac{1}{2} e^{2\tau} \right]_{-\infty}^0 \\ &= \frac{1}{2} e^0 - 0 = \frac{1}{2}. \end{aligned}$$

CHECK

Both branches give 0.5 at $t = 3$, so they join continuously. Also, $y(\infty)$ equals the total area of x , which is $1/2$. This agrees with the system interpretation: the delayed step makes the output the area of x collected up to three seconds earlier.

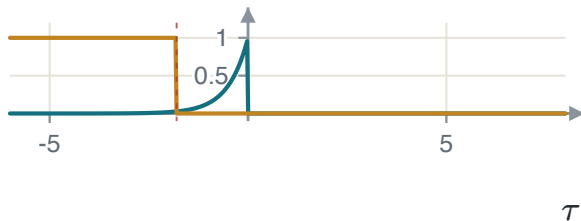


FIGURE 3.7 Case 1: $t < 3$. The shaded area is $y(t)$.

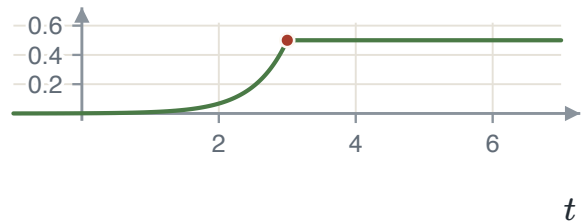


FIGURE 3.8 The result $y(t)$, continuous at $t = 3$.

EXAMPLE 3.4

GIVEN

$x(t) = 1$ on $0 < t < 1$ and zero elsewhere. $h(t) = t$ on $0 < t < 2$ and zero elsewhere.

FIND

$y(t) = x(t) * h(t)$.

METHOD

Use $y(t) = \int h(\tau)x(t - \tau) d\tau$. This form reverses the rectangle, whose constant height makes the overlap limits easy to identify. Its support becomes $[t - 1, t]$, a window of width 1 moving across the ramp on $[0, 2]$.

SOLUTION

Substitute into the chosen form:

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau) d\tau.$$

Here $h(\tau) = \tau$ for $0 < \tau < 2$ and zero elsewhere. The factor $x(t - \tau)$ equals 1 when $0 < t - \tau < 1$. Solve this pair of inequalities for τ : subtract t to get $-t < -\tau < 1 - t$, then multiply by -1 and reverse both inequalities to get $t - 1 < \tau < t$. So the integrand is τ on the overlap of $0 < \tau < 2$ with $t - 1 < \tau < t$, and zero elsewhere.

The moving window has edges $t - 1$ and t , and the fixed ramp has edges 0 and 2. Set each moving edge equal to each fixed edge: $t - 1 = 0$, $t - 1 = 2$, $t = 0$, $t = 2$. This gives $t = 0, 1, 2, 3$, which divides the calculation into five cases. In every case the antiderivative of τ is $\frac{1}{2}\tau^2$.

$t < 0$: the window lies left of the ramp. There is no overlap, so $y = 0$.

$0 < t < 1$: the left edge $t - 1$ is below 0, so the overlap is $0 < \tau < t$:

$$y(t) = \int_0^t \tau d\tau = \left[\frac{1}{2}\tau^2\right]_0^t = \frac{1}{2}t^2 - 0 = \frac{1}{2}t^2.$$

$1 < t < 2$: the whole window lies inside the ramp, so the overlap is $t - 1 < \tau < t$:

$$\begin{aligned} y(t) &= \int_{t-1}^t \tau d\tau = \left[\frac{1}{2}\tau^2\right]_{t-1}^t \\ &= \frac{1}{2}t^2 - \frac{1}{2}(t-1)^2 \\ &= \frac{1}{2}t^2 - \frac{1}{2}(t^2 - 2t + 1) \\ &= t - \frac{1}{2}. \end{aligned}$$

$2 < t < 3$: the right edge t is past 2, so the overlap is $t - 1 < \tau < 2$:

$$\begin{aligned} y(t) &= \int_{t-1}^2 \tau d\tau = \left[\frac{1}{2}\tau^2\right]_{t-1}^2 \\ &= \frac{1}{2} \cdot 4 - \frac{1}{2}(t-1)^2 \\ &= 2 - \frac{1}{2}(t^2 - 2t + 1) \\ &= -\frac{1}{2}t^2 + t + \frac{3}{2}. \end{aligned}$$

$t > 3$: the window lies right of the ramp. There is no overlap, so $y = 0$.

CHECK

Three checks. **Continuity:** at $t = 1$, $\frac{1}{2} \cdot 1^2 = \frac{1}{2}$ and $1 - \frac{1}{2} = \frac{1}{2}$. At $t = 2$, $2 - \frac{1}{2} = \frac{3}{2}$ and $-\frac{1}{2} \cdot 4 + 2 + \frac{3}{2} = \frac{3}{2}$. At $t = 3$, $-\frac{1}{2} \cdot 9 + 3 + \frac{3}{2} = 0$. Every pair of branches agrees.

Support: $[0, 1]$ and $[0, 2]$ give $[0, 3]$, because supports add. **Area:** $\int y = (\int x)(\int h) = 1 \times 2 = 2$. Integrate each branch:

$$\int_0^1 \frac{1}{2}t^2 dt = \left[\frac{1}{6}t^3 \right]_0^1 = \frac{1}{6},$$

$$\int_1^2 \left(t - \frac{1}{2}\right) dt = \left[\frac{1}{2}t^2 - \frac{1}{2}t \right]_1^2 = (2 - 1) - \left(\frac{1}{2} - \frac{1}{2}\right) = 1,$$

$$\int_2^3 \left(-\frac{1}{2}t^2 + t + \frac{3}{2}\right) dt = \left[-\frac{1}{6}t^3 + \frac{1}{2}t^2 + \frac{3}{2}t \right]_2^3$$

$$= \left(-\frac{27}{6} + \frac{9}{2} + \frac{9}{2}\right) - \left(-\frac{8}{6} + 2 + 3\right) = \frac{9}{2} - \frac{11}{3} = \frac{5}{6}.$$

The total is $\frac{1}{6} + 1 + \frac{5}{6} = 2$, as required.

✓ FIND THE BOUNDARIES BEFORE INTEGRATING

List the moving edges and the fixed edges, then set each moving edge equal to each fixed edge. The resulting values divide the calculation into all required cases.

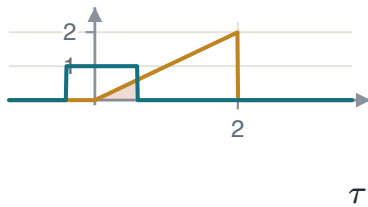


FIGURE 3.9 0

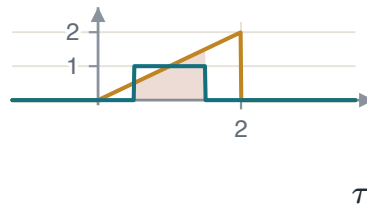


FIGURE 3.10 1

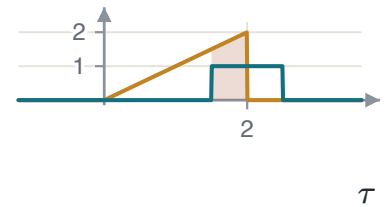


FIGURE 3.11 2

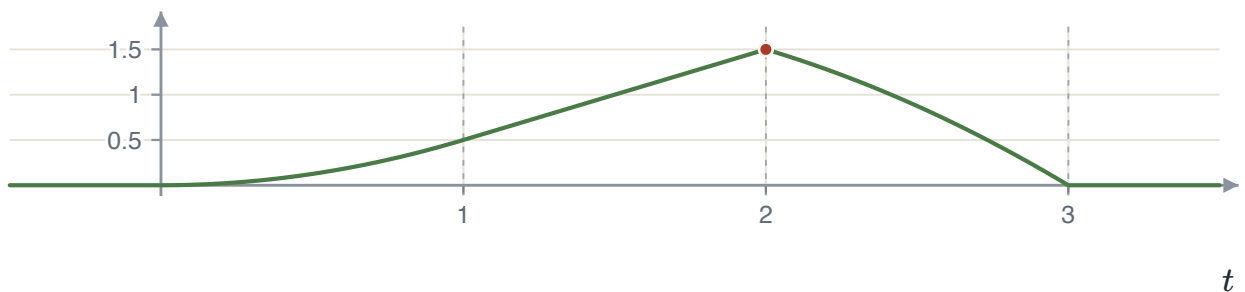


FIGURE 3.12 The complete result. The peak is 1.5 at $t = 2$. A unit-width rectangle acts as a one-second moving sum, so the output rises while the window fills, grows linearly while it is full, and falls as it leaves the ramp.

3.5 Properties of convolution

TABLE 3.1

PROPERTY	STATEMENT	WHAT IT MEANS FOR INTERCONNECTIONS
Commutative	$x * h = h * x$	Input and impulse response play symmetric roles in the algebra.
Distributive	$x * (h_1 + h_2) = x * h_1 + x * h_2$	Two systems in parallel , outputs added, equal one system with impulse response $h_1 + h_2$.
Associative	$x * (h_1 * h_2) = (x * h_1) * h_2$	Two systems in cascade equal one system with impulse response $h_1 * h_2$.

Commutativity was derived in the two sections above by the index change $m = n - k$. The distributive property follows because a sum of two sequences can be split term by term:

$$\begin{aligned} x * (h_1 + h_2)[n] &= \sum_{k=-\infty}^{\infty} x[k] (h_1[n - k] + h_2[n - k]) \\ &= \sum_{k=-\infty}^{\infty} x[k] h_1[n - k] + \sum_{k=-\infty}^{\infty} x[k] h_2[n - k] \\ &= (x * h_1)[n] + (x * h_2)[n]. \end{aligned}$$

For the associative property, write the cascade $(x * h_1) * h_2$ as a double sum, exchange the order of the two sums, and change the inner index. Let $w = x * h_1$ be the output of the first system. The inner sum over m uses the substitution $p = m - k$, so $m = p + k$ and $n - m = (n - k) - p$:

$$\begin{aligned} ((x * h_1) * h_2)[n] &= \sum_{m=-\infty}^{\infty} w[m] h_2[n - m] = \sum_{m=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x[k] h_1[m - k] \right) h_2[n - m] \\ &= \sum_{k=-\infty}^{\infty} x[k] \sum_{m=-\infty}^{\infty} h_1[m - k] h_2[n - m] \\ &= \sum_{k=-\infty}^{\infty} x[k] \sum_{p=-\infty}^{\infty} h_1[p] h_2[(n - k) - p] \\ &= \sum_{k=-\infty}^{\infty} x[k] (h_1 * h_2)[n - k] = (x * (h_1 * h_2))[n]. \end{aligned}$$

The second line moves $x[k]$ outside the sum over m , because $x[k]$ does not depend on m . The fourth line recognises the inner sum as the convolution $h_1 * h_2$ evaluated at $n - k$. Exchanging the two sums is allowed when the double sum converges absolutely, which holds for finite-length or absolutely summable sequences. The same steps with integrals in place of sums give the continuous-time result.

Associativity and commutativity show that the order of cascaded LTI systems does not change their combined impulse response. This conclusion requires both systems to be linear and time invariant. For example, a saturating amplifier followed by a filter is not generally equivalent to the filter followed by the amplifier.

3.6 System properties in terms of h

For an LTI system, each property test from Chapter 2 can be applied directly to the impulse response h .

TABLE 3.2

PROPERTY	CRITERION	REASON
Memoryless	$h(t) = a \delta(t)$, or $h[n] = a \delta[n]$	Then $y[n] = a \sum_k x[k] \delta[n - k] = a x[n]$, a pure gain.
Invertible	$h * g = \delta$ for some g	g is the impulse response of the inverse system, and δ is the identity system.
Causal	$h(t) = 0$ for $t < 0$, or $h[n] = 0$ for $n < 0$	The sum collapses to $y[n] = h[0]x[n] + h[1]x[n-1] + \dots$, which uses only present and past inputs.
BIBO stable	$\sum_k h[k] < \infty$, or $\int h(t) dt < \infty$	See the proof below.

The memoryless row uses the sifting property. Substitute $h[n - k] = a \delta[n - k]$ into the convolution sum. The impulse keeps only the term $k = n$:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] a \delta[n - k] = a \sum_{k=-\infty}^{\infty} x[k] \delta[n - k] = a x[n].$$

The causal row uses the second form of the sum, $y[n] = \sum_k h[k] x[n - k]$. If $h[k] = 0$ for $k < 0$, every term with a negative k vanishes and the sum starts at $k = 0$:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] x[n - k] = \sum_{k=0}^{\infty} h[k] x[n - k] = h[0]x[n] + h[1]x[n-1] + h[2]x[n-2] + \dots$$

Every input index $n - k$ with $k \geq 0$ satisfies $n - k \leq n$, so the output uses only present and past inputs.

To show why absolute summability is sufficient for stability, assume $|x[n]| \leq B$ for all n . Take the modulus of the convolution sum, apply the triangle inequality (the modulus of a sum is at most the sum of the moduli), and then use the input bound on every term:

$$\begin{aligned}
 |y[n]| &= \left| \sum_{k=-\infty}^{\infty} h[k] x[n - k] \right| \\
 &\leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n - k]| \quad (\text{triangle inequality}) \\
 &\leq \sum_{k=-\infty}^{\infty} |h[k]| B \quad (\text{since } |x[n - k]| \leq B) \\
 &= B \sum_{k=-\infty}^{\infty} |h[k]| < \infty.
 \end{aligned}$$

This proves sufficiency: an absolutely summable $h[n]$ maps every bounded input to a bounded output. The bound $B \sum_k |h[k]|$ does not depend on n .

For necessity, suppose instead that $\sum_k |h[k]|$ is infinite. Choose the input $x[n] = \text{sgn } h[-n]$, where sgn is $+1$ for a positive argument, -1 for a negative argument, and 0 at zero. This input is bounded,

because $|x[n]| \leq 1$. Evaluate the output at $n = 0$. Since $x[-k] = \text{sgn } h[k]$, and a number times its own sign is its modulus:

$$y[0] = \sum_{k=-\infty}^{\infty} h[k] x[0-k] = \sum_{k=-\infty}^{\infty} h[k] \text{sgn } h[k] = \sum_{k=-\infty}^{\infty} |h[k]| \rightarrow \infty.$$

A bounded input gives an unbounded output, so the system is not stable. Therefore absolute summability is also necessary.

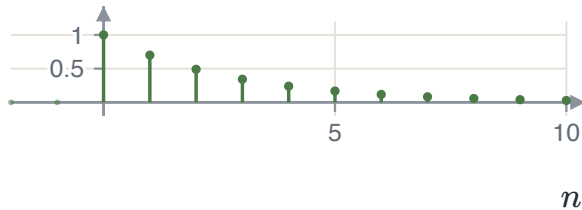


FIGURE 3.13 $h[n] = 0.7^n u[n]$. Here $\sum |h| = \sum_{n=0}^{\infty} 0.7^n = 1/(1-0.7) = 1/0.3 < \infty$: stable and causal.

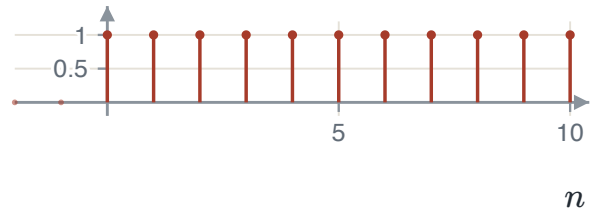


FIGURE 3.14 $h[n] = u[n]$, the accumulator. Here $\sum |h| \rightarrow \infty$: causal but not stable.

3.7 A convolution checklist

1. Confirm that the system is linear and time invariant. Convolution gives the system output only under this condition.
2. Choose the factor whose reversal gives simpler support conditions.
3. Write the support of each factor as an inequality in the dummy variable.
4. Set the moving edges equal to the fixed edges to list every case boundary, before integrating anything.
5. Integrate or sum, case by case.
6. Check continuity at every boundary, check that the supports add, and check that the total area or total sum multiplies.

☑ USE EACH FINAL CHECK FOR A SPECIFIC PURPOSE

A mismatch at a case boundary indicates an incorrect limit. An incorrect output support indicates an error in a shift, reversal, or support condition. An incorrect total area or sum indicates an error in the integrand or summand.

Exercises

3.1 Compute $\{1, 2, 3\} * \{1, -1\}$ with both sequences starting at $n = 0$, and check your answer with the sum rule.

Answer: $\{1, 1, 1, -3\}$; the sums are $6 \times 0 = 0$.

3.2 Let $h[n] = \left(\frac{1}{3}\right)^n u[n]$. Is the system stable? Is it causal? Is it memoryless?

Answer: Stable ($\sum |h| = 3/2$), causal, not memoryless.

3.3 Find $y(t) = u(t) * u(t)$ and sketch it.

Answer: $y(t) = t u(t)$.

3.4 Two systems with $h_1[n] = \delta[n] - \delta[n - 1]$ and $h_2[n] = u[n]$ are cascaded. Find the impulse response of the cascade and identify the resulting system.

Answer: $\delta[n]$: the two systems are inverses.

3.5 $x(t) = 1$ on $0 < t < 3$ and $h(t) = t$ on $0 < t < 2$. List the case boundaries before computing anything, then find $y(t)$.

Answer: Boundaries at $t = 0, 2, 3, 5$; four cases.

CHAPTER

4

Fourier series

Fourier series are used to calculate the response of a linear time-invariant system to a periodic signal. Complex exponentials are eigenfunctions of every LTI system, and periodic signals can be represented as sums of these exponentials. This representation replaces convolution by one multiplication at each harmonic frequency.

4.1 Eigenfunctions of a linear time-invariant system

DEFINITION

A signal is an **eigenfunction** of a system when the output is that same signal multiplied by a constant. The constant is the **eigenvalue**.

Put $x(t) = e^{st}$, with s any complex number, into the convolution integral. Write both limits; they are $-\infty$ and ∞ throughout this chapter. Then split the exponential with $e^{s(t-\tau)} = e^{st}e^{-s\tau}$. The factor e^{st} does not depend on τ , so it moves outside the integral.

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \\ &= \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau \\ &= \int_{-\infty}^{\infty} h(\tau) e^{st} e^{-s\tau} d\tau \\ &= e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau. \end{aligned}$$

The second factor does not depend on t . It is a number attached to the value of s , and it is called the transfer function. The same steps apply in discrete time. Put $x[n] = z^n$ into the convolution sum and use $z^{n-k} = z^n z^{-k}$; the factor z^n does not depend on k , so it moves outside the sum.

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} h[k] x[n - k] = \sum_{k=-\infty}^{\infty} h[k] z^{n-k} \\ &= \sum_{k=-\infty}^{\infty} h[k] z^n z^{-k} = z^n \sum_{k=-\infty}^{\infty} h[k] z^{-k}. \end{aligned}$$

EIGENFUNCTION PROPERTY

$$e^{st} \rightarrow h(t) \rightarrow H(s)e^{st}, \quad H(s) = \int_{-\infty}^{\infty} h(\tau)e^{-s\tau} d\tau$$

$$z^n \rightarrow h[n] \rightarrow H(z)z^n, \quad H(z) = \sum_{k=-\infty}^{\infty} h[k]z^{-k}$$

The discrete-time output is a sequence, $y[n] = H(z)z^n$. Nothing in that line is a function of a continuous variable, and writing the continuous-time result there is the commonest slip in this section.

Setting $s = j\omega$ and $z = e^{j\omega}$ with ω real gives the two **frequency responses**, $H(j\omega)$ and $H(e^{j\omega})$. The discrete-time one repeats every 2π in ω , because $e^{j(\omega+2\pi)k} = e^{j\omega k}$ for integer k . The continuous-time one does not repeat at all.

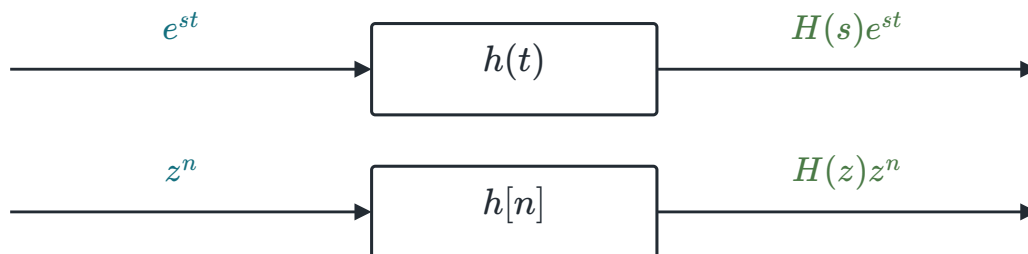


FIGURE 4.1 The two statements, side by side, so that neither can borrow the other's variables.

EXAMPLE 4.1 – A PURE DELAY

GIVEN An LTI system with $y(t) = x(t - 3)$.

FIND The output for $x(t) = e^{j2t}$, and for $x(t) = \cos(4t) + \cos(7t)$, using eigenfunctions only.

METHOD The inputs are complex exponentials, so the eigenfunction method applies. Write h from the delay definition, integrate to obtain $H(s)$, and multiply each exponential by the corresponding value of H .

SOLUTION A delay by 3 is convolution with $\delta(t - 3)$, because $x(t) * \delta(t - 3) = x(t - 3)$. So $h(t) = \delta(t - 3)$. Put this into the definition of $H(s)$ and use the sifting property, which replaces τ by 3 in the exponential:

$$H(s) = \int_{-\infty}^{\infty} \delta(\tau - 3) e^{-s\tau} d\tau = e^{-3s}.$$

For $x(t) = e^{j2t}$ the exponent gives $s = j2$, so

$$y(t) = H(j2) e^{j2t} = e^{-j6} e^{j2t} = e^{j2(t-3)}.$$

For the cosines, first write each one as two exponentials with Euler's relation, then multiply each exponential by its own eigenvalue. For $\cos(4t)$ the two values of s are $\pm j4$, with $H(j4) = e^{-j12}$ and $H(-j4) = e^{+j12}$:

$$\begin{aligned}
 \cos(4t) &= \frac{1}{2}e^{j4t} + \frac{1}{2}e^{-j4t} \\
 &\rightarrow \frac{1}{2}e^{-j12}e^{j4t} + \frac{1}{2}e^{j12}e^{-j4t} \\
 &= \frac{1}{2}e^{j4(t-3)} + \frac{1}{2}e^{-j4(t-3)} \\
 &= \cos(4(t-3)).
 \end{aligned}$$

The same steps with $s = \pm j7$ and $H(\pm j7) = e^{\mp j21}$ give $\cos(7(t-3))$. The system is linear, so the two outputs add:

$$y(t) = \cos(4(t-3)) + \cos(7(t-3)).$$

CHECK

Apply $y(t) = x(t-3)$ directly to both inputs. The answers agree. Note also that $|H(j\omega)| = 1$ and $\angle H(j\omega) = -3\omega$: a delay changes no amplitude and adds a phase proportional to frequency.

This result gives the method used in the rest of the chapter. If $x = \sum_k a_k e^{s_k t}$, multiply each term by its eigenvalue to obtain $y = \sum_k a_k H(s_k) e^{s_k t}$. The next sections show how to write periodic signals in this form.

4.2 Existence: the Dirichlet conditions

Not every signal can be decomposed into complex exponentials. Periodic signals can, provided they satisfy three conditions. Let $x(t) = x(t+T)$ for every t .

1. Over one period, x is absolutely integrable: $\int_T |x(t)| dt < \infty$.
2. In any finite interval, x has only a finite number of maxima and minima.
3. In any finite interval, x has only a finite number of discontinuities, and each is finite.

✓ SUFFICIENT, NOT NECESSARY

A signal meeting all three has a Fourier series that converges to it wherever it is continuous, and to the midpoint of the jump at a discontinuity. A signal failing one of them may still have a series. The conditions guarantee; they do not exclude.

The three are independent. The signal $1/t$ on each period fails the first while the other two hold. The signal $\sin(2\pi/t)$ has finite area but oscillates infinitely often, failing only the second. A staircase whose steps halve in height and width satisfies the first two and fails the third.

4.3 The synthesis and analysis equations

CONTINUOUS-TIME FOURIER SERIES

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}$$

$$a_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

The first equation is synthesis, the second is analysis. One period symbol is used throughout: T_0 , with $\omega_0 = 2\pi/T_0$ stated once here. The sign of the exponent is the only thing that separates the pair.

Proof of the analysis equation

The goal is to isolate one coefficient, a_n . Multiply the synthesis equation by $e^{-jn\omega_0 t}$ and integrate over one period. The sum and the integral can be exchanged, and a_k does not depend on t , so it moves outside the integral. The two exponentials combine because $e^{jk\omega_0 t} e^{-jn\omega_0 t} = e^{j(k-n)\omega_0 t}$.

$$\begin{aligned} \int_{T_0} x(t) e^{-jn\omega_0 t} dt &= \int_{T_0} \left(\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \right) e^{-jn\omega_0 t} dt \\ &= \sum_{k=-\infty}^{\infty} a_k \int_{T_0} e^{j(k-n)\omega_0 t} dt. \end{aligned}$$

Now evaluate the inner integral from $-T_0/2$ to $T_0/2$. Take $k \neq n$ first, so that the antiderivative may divide by $k - n$. At the limits $\omega_0 T_0/2 = \pi$, and the difference of the two exponentials is $2j$ times a sine, by $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$.

$$\begin{aligned} \int_{-T_0/2}^{T_0/2} e^{j(k-n)\omega_0 t} dt &= \frac{e^{j(k-n)\omega_0 t}}{j(k-n)\omega_0} \Big|_{-T_0/2}^{T_0/2} = \frac{e^{j(k-n)\pi} - e^{-j(k-n)\pi}}{j(k-n)\omega_0} \\ &= \frac{2j \sin((k-n)\pi)}{j(k-n)\omega_0} = \frac{2 \sin((k-n)\pi)}{(k-n)\omega_0} \\ &= T_0 \frac{\sin((k-n)\pi)}{(k-n)\pi} = 0, \quad k \neq n. \end{aligned}$$

The last line uses $\omega_0 = 2\pi/T_0$, so $2/\omega_0 = T_0/\pi$. The value is zero because $k - n$ is a non-zero integer, and the sine of an integer multiple of π is zero. For $k = n$ the exponent is zero, the integrand is the constant 1, and the integral is the length of the period: $\int_{-T_0/2}^{T_0/2} 1 dt = T_0$. The same value comes from the limit $(k - n) \rightarrow 0$ of the sine ratio, because $\sin \theta / \theta \rightarrow 1$.

ORTHOGONALITY OF COMPLEX EXPONENTIALS

$$\int_{T_0} e^{j(k-n)\omega_0 t} dt = \begin{cases} T_0, & k = n \\ 0, & k \neq n \end{cases}$$

Only the term with $k = n$ survives in the sum, so $\int_{T_0} x(t) e^{-jn\omega_0 t} dt = a_n T_0$. Dividing both sides by T_0 gives the analysis equation, $a_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jn\omega_0 t} dt$.

① THE DC TERM

Setting $k = 0$ makes the exponential 1, so

$$a_0 = \frac{1}{T_0} \int_{T_0} x(t) dt,$$

the average value of the signal over one period. Use this fact to check a coefficient calculation: calculate or estimate the mean directly from the signal and compare it with a_0 .

Fundamental period of a sum

A sum of periodic signals is periodic when the ratio of every pair of component periods is rational. The fundamental period of the sum is the smallest positive number that is a whole multiple of each component period. Write each period as a fraction in lowest terms:

PERIOD OF A SUM

$$T_0 = \text{LCM}\left(\frac{p_1}{q_1}, \frac{p_2}{q_2}, \dots\right) = \frac{\text{LCM}(p_1, p_2, \dots)}{\text{GCD}(q_1, q_2, \dots)}$$

The result is a least common multiple of fractions, so it need not be an integer. A constant term has no period and is left out.

⊗ WHERE THIS GOES WRONG

Putting every period over one common denominator first, then taking the least common multiple of the numerators, is the same rule in disguise — because for equal denominators $\text{LCM}(d, d) = \text{GCD}(d, d) = d$. Applying LCM to the denominators *without* that rewrite is wrong: for periods $2/9$ and $8/21$ it gives $8/63$, which is a multiple of neither. The correct answer is $\text{LCM}(2, 8) / \text{GCD}(9, 21) = 8/3$. Always divide the answer by each period and confirm both results are whole numbers.

🔗 EXAMPLE 4.2 – COEFFICIENTS OF A SUM OF SINUSOIDS

GIVEN $x(t) = 1 + \frac{1}{2} \cos(2\pi t) + \sin(3\pi t)$.

FIND Every Fourier series coefficient, with magnitude and phase.

METHOD Find T_0 , hence ω_0 . Expand with Euler's relations and match each exponent to $jk\omega_0 t$.

SOLUTION First the period. $\cos(2\pi t)$ repeats every $2\pi/(2\pi) = 1$ s and $\sin(3\pi t)$ every $2\pi/(3\pi) = 2/3$ s. In lowest terms these are $1/1$ and $2/3$, so $T_0 = \text{LCM}(1, 2) / \text{GCD}(1, 3) = 2$ s and $\omega_0 = 2\pi/T_0 = \pi$ rad/s. Check by division: $2/1 = 2$ and $2/(2/3) = 3$, both whole. Now expand each sinusoid with Euler's relations, $\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$ and $\sin \theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$:

$$\begin{aligned} \frac{1}{2} \cos(2\pi t) &= \frac{1}{2} \cdot \frac{1}{2} (e^{j2\pi t} + e^{-j2\pi t}) = \frac{1}{4} e^{j2\pi t} + \frac{1}{4} e^{-j2\pi t}, \\ \sin(3\pi t) &= \frac{1}{2j} (e^{j3\pi t} - e^{-j3\pi t}) = \frac{1}{2j} e^{j3\pi t} - \frac{1}{2j} e^{-j3\pi t}. \end{aligned}$$

Each exponent must be matched to $jk\omega_0 t = jk\pi t$. Here $2\pi t = 2\omega_0 t$, so $k = \pm 2$, and $3\pi t = 3\omega_0 t$, so $k = \pm 3$. The constant is $1 \cdot e^{j0\omega_0 t}$, so $k = 0$. Collecting the terms,

$$x(t) = 1e^{j0\omega_0 t} + \frac{1}{4}e^{j2\omega_0 t} + \frac{1}{4}e^{-j2\omega_0 t} + \frac{1}{2j}e^{j3\omega_0 t} - \frac{1}{2j}e^{-j3\omega_0 t},$$

so $a_0 = 1$, $a_{\pm 2} = 1/4$, $a_3 = 1/(2j)$, $a_{-3} = -1/(2j)$, and every other coefficient is zero.

CHECK

$a_0 = 1$ is also the average of the signal: both sinusoids complete whole cycles in one period and average to zero, leaving the constant. Magnitudes: $|a_0| = 1$, $|a_{\pm 2}| = 1/4$, $|a_{\pm 3}| = 1/2$. Phases: $1/(2j) = -j/2$, so $\angle a_3 = -\pi/2$, and $a_{-3} = +j/2$, so $\angle a_{-3} = +\pi/2$. The signal is real, so magnitudes must be even in k and phases odd, and they are.

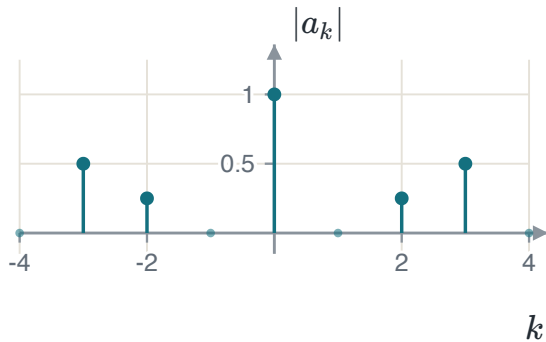


FIGURE 4.2 Magnitudes, even in k .

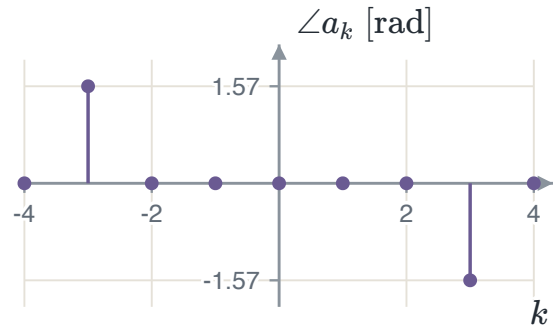


FIGURE 4.3 Phases, odd in k . A value of ± 1.57 rad is $\pm\pi/2$.

4.4 Two standard waveforms

EXAMPLE 4.3 – THE PERIODIC RECTANGULAR WAVE

GIVEN $x(t) = 1$ for $|t| < T_1$ and 0 for $T_1 < |t| < T_0/2$, repeated with period T_0 .

FIND Every coefficient.

METHOD Apply the analysis equation over $-T_0/2$ to $T_0/2$. The signal is 1 only on $[-T_1, T_1]$, so those become the limits. The case $k = 0$ is separate, because the antiderivative divides by k .

SOLUTION Start from the analysis equation with the period from $-T_0/2$ to $T_0/2$. The signal is zero for $T_1 < |t| < T_0/2$, so those parts of the integral drop out and the limits shrink to $\pm T_1$:

$$a_k = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T_0} \int_{-T_1}^{T_1} 1 \cdot e^{-jk\omega_0 t} dt.$$

For $k = 0$ the integrand is 1, so $a_0 = \frac{1}{T_0} \int_{-T_1}^{T_1} 1 dt = \frac{2T_1}{T_0}$. For $k \neq 0$, integrate the exponential, evaluate the two limits, and turn the difference of exponentials into a sine with $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$:

$$\begin{aligned}
 a_k &= \frac{1}{T_0} \left. \frac{e^{-jk\omega_0 t}}{-jk\omega_0} \right|_{-T_1}^{T_1} = \frac{1}{T_0} \cdot \frac{e^{-jk\omega_0 T_1} - e^{jk\omega_0 T_1}}{-jk\omega_0} \\
 &= \frac{1}{T_0} \cdot \frac{e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}}{jk\omega_0} = \frac{1}{T_0} \cdot \frac{2j \sin(k\omega_0 T_1)}{jk\omega_0} \\
 &= \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T_0}.
 \end{aligned}$$

The second line changes the sign of both numerator and denominator, which leaves the fraction unchanged. Finally replace ω_0 by $2\pi/T_0$. The denominator becomes $k\omega_0 T_0 = 2\pi k$, and the argument becomes $k\omega_0 T_1 = 2\pi k T_1/T_0$, so

$$a_k = \frac{2 \sin(2\pi k T_1/T_0)}{2\pi k} = \frac{\sin(2\pi k T_1/T_0)}{\pi k}, \quad k \neq 0.$$

CHECK

As $k \rightarrow 0$ the second branch tends to $2T_1/T_0$, because $\sin \theta/\theta \rightarrow 1$. It therefore agrees with the branch for $k = 0$. The signal is real and even, so the coefficients must be real and even. The formula has this property because $\sin$ and k are both odd. The zeros occur when k is a non-zero multiple of $T_0/(2T_1)$.

The envelope, and the samples taken from it

These coefficients are values of one continuous function of ω , read at equally spaced points. Write the function first:

ENVELOPE OF A SINGLE PULSE

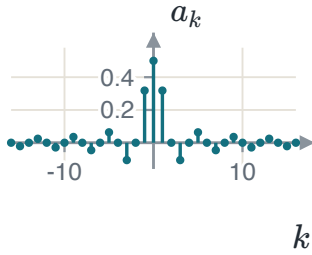
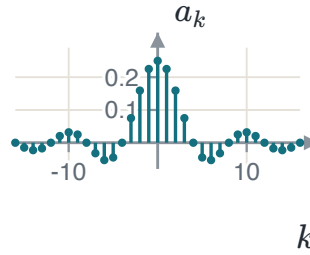
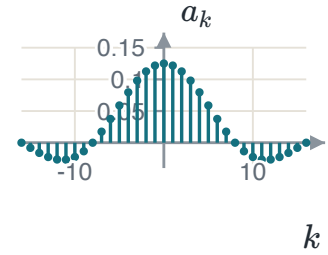
$$\begin{aligned}
 E(\omega) &= \int_{-T_1}^{T_1} e^{-j\omega t} dt = \left. \frac{e^{-j\omega t}}{-j\omega} \right|_{-T_1}^{T_1} = \frac{e^{j\omega T_1} - e^{-j\omega T_1}}{j\omega} = \frac{2 \sin(\omega T_1)}{\omega} \\
 a_k &= \frac{1}{T_0} E(k\omega_0) = \frac{1}{T_0} \cdot \frac{2 \sin(k\omega_0 T_1)}{k\omega_0} = \frac{\sin(2\pi k T_1/T_0)}{\pi k}
 \end{aligned}$$

The first line is the integral of Example 4.3 with $k\omega_0$ replaced by a free variable ω . The second line puts $\omega = k\omega_0$ back and repeats the substitution $\omega_0 = 2\pi/T_0$. E depends on the shape of one pulse and not on how often it repeats. In the unnormalised convention $\text{sinc}(\theta) = \sin \theta/\theta$, it is $E(\omega) = 2T_1 \text{sinc}(\omega T_1)$.

⊗ TWO OBJECTS, NOT ONE

$E(\omega)$ is a function of a continuous variable; a_k is a sequence indexed by an integer. Evaluating the envelope at the single point $\omega = \omega_0$ gives $T_0 a_1$, the first harmonic only. As a function of the harmonic index the coefficient is $T_0 a_k = T_0 \sin(2\pi k T_1/T_0)/(\pi k)$, with k present in both places. Never join the tops of coefficient stems with a line: it claims the signal contains frequencies between the harmonics, which a periodic signal does not.

Increase T_0 while keeping the pulse fixed. The envelope remains the same. The spacing $\omega_0 = 2\pi/T_0$ decreases, and every coefficient decreases as $1/T_0$. The next chapter uses this limiting process to define the Fourier transform.

FIGURE 4.4 $T_0 = 4T_1$.FIGURE 4.5 $T_0 = 8T_1$.FIGURE 4.6 $T_0 = 16T_1$.

EXAMPLE 4.4 – THE SAWTOOTH WAVE

GIVEN $x(t) = t$ for $-T_0/2 < t < T_0/2$, repeated with period T_0 .

FIND Every coefficient.

METHOD The signal is odd, so its DC term is zero; direct integration below confirms this. For $k \neq 0$, the integrand contains t times an exponential, so use integration by parts with $u = t$ and $dv = e^{at}dt$:

$$\int t e^{at} dt = \frac{t e^{at}}{a} - \int \frac{e^{at}}{a} dt = \frac{t e^{at}}{a} - \frac{e^{at}}{a^2} = \frac{(at - 1)e^{at}}{a^2}.$$

SOLUTION The DC term:

$$a_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} t dt = \frac{1}{T_0} \left. \frac{t^2}{2} \right|_{-T_0/2}^{T_0/2} = \frac{1}{T_0} \left(\frac{T_0^2}{8} - \frac{T_0^2}{8} \right) = 0.$$

For $k \neq 0$, substitute the signal into the analysis equation and write $a = -jk\omega_0$, so that the by-parts formula applies:

$$a_k = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} t e^{-jk\omega_0 t} dt = \frac{1}{T_0} \cdot \frac{1}{a^2} \left[(at - 1)e^{at} \right]_{-T_0/2}^{T_0/2}.$$

At the limits, $aT_0/2 = -jk\omega_0 T_0/2 = -jk\pi$, because $\omega_0 T_0/2 = \pi$. Also $e^{\pm jk\pi} = \cos(k\pi) \pm j \sin(k\pi) = \cos(k\pi) = (-1)^k$, since $\sin(k\pi) = 0$. Both exponentials therefore equal $(-1)^k$, and this common factor comes out of the bracket:

$$\begin{aligned} \left[(at - 1)e^{at} \right]_{-T_0/2}^{T_0/2} &= (-jk\pi - 1)e^{-jk\pi} - (jk\pi - 1)e^{jk\pi} \\ &= (-1)^k [(-jk\pi - 1) - (jk\pi - 1)] = (-1)^k (-2jk\pi). \end{aligned}$$

Next, $a^2 = (-jk\omega_0)^2 = j^2 k^2 \omega_0^2 = -k^2 \omega_0^2$. Dividing,

$$a_k = \frac{1}{T_0} \cdot \frac{(-1)^k (-2jk\pi)}{-k^2 \omega_0^2} = \frac{1}{T_0} \cdot \frac{2j\pi (-1)^k}{k \omega_0^2}.$$

Finally $\omega_0^2 = 4\pi^2/T_0^2$, so $\frac{1}{T_0} \cdot \frac{2j\pi}{k \omega_0^2} = \frac{1}{T_0} \cdot \frac{2j\pi T_0^2}{4\pi^2 k} = \frac{jT_0}{2\pi k}$, and

$$a_k = \frac{jT_0(-1)^k}{2k\pi}, \quad k \neq 0.$$

CHECK

The signal is real and odd, so the coefficients must be purely imaginary, and they are. For $T_0 = 1$ the largest is $|a_{\pm 1}| = 1/(2\pi) \approx 0.159$. Every phase is $+\pi/2$ or $-\pi/2$, with the sign alternating in k and flipping again with the sign of k .

EXAMPLE 4.5 – THE PERIODIC IMPULSE TRAIN

GIVEN

$$x(t) = \sum_{m=-\infty}^{\infty} \delta(t - mT_0).$$

FIND

Every coefficient.

METHOD

Take the period from $-T_0/2$ to $+T_0/2$. Exactly one impulse lies inside it, so sifting finishes the integral.

SOLUTION

Substitute the impulse train into the analysis equation. Inside $-T_0/2 < t < T_0/2$ only the term $m = 0$, which is $\delta(t)$, is non-zero; the impulses at $t = \pm T_0, \pm 2T_0, \dots$ lie outside the interval and contribute nothing. Then sift, which sets $t = 0$ in the exponential:

$$\begin{aligned} a_k &= \frac{1}{T_0} \int_{-T_0/2}^{+T_0/2} \sum_{m=-\infty}^{\infty} \delta(t - mT_0) e^{-jk\omega_0 t} dt \\ &= \frac{1}{T_0} \int_{-T_0/2}^{+T_0/2} \delta(t) e^{-jk\omega_0 t} dt = \frac{1}{T_0} e^{-jk\omega_0 \cdot 0} = \frac{1}{T_0} \end{aligned}$$

for every k .

CHECK

Both limits must be written and must differ. An interval with equal endpoints has zero length, so every integral over it is zero. The result is real, positive, and equal at every harmonic. At $t = 0$, every harmonic has value 1 and their contributions add. At other times, the positive and negative contributions cancel.

4.5 How many harmonics are enough

TRUNCATION AND ITS ERROR

$$x_N(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t}, \quad e_N(t) = x(t) - x_N(t)$$

$$\text{MSE} = \frac{1}{T_0} \int_{T_0} |e_N(t)|^2 dt = \sum_{|k| > N} |a_k|^2$$

The second form follows from Parseval's relation in Section 4.7. By linearity, $e_N = x - x_N$ has coefficients

$a_k - a_k = 0$ for $|k| \leq N$ and $a_k - 0 = a_k$ for $|k| > N$. Parseval's relation then gives

$\frac{1}{T_0} \int_{T_0} |e_N(t)|^2 dt = \sum_{|k| > N} |a_k|^2$. It turns a question about a difference of signals into a question about a tail of a series.

⊗ THE GIBBS PHENOMENON

Near a jump the partial sum overshoots, and the overshoot does not shrink as N grows: its height settles at about **8.95% of the size of the jump**. What does shrink is its width. That is how the mean-square error can go to zero while the largest error does not.

TABLE 4.1

N	RECTANGULAR WAVE, $T_0 = 4T_1$	SAWT00TH, $T_0 = 1$
3	0.025	0.0144
9	0.010	0.0053
27	0.004	0.0018
81	0.001	0.0006

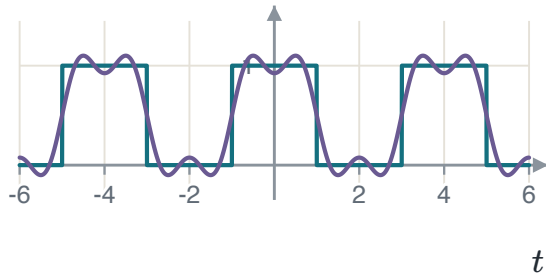


FIGURE 4.7 $N = 3$.

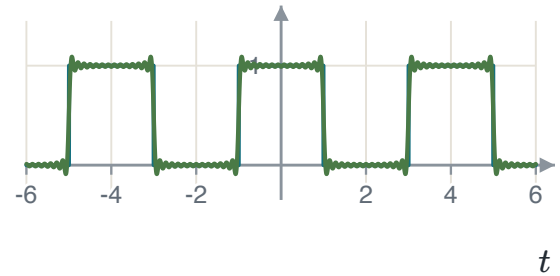


FIGURE 4.8 $N = 27$. The spike beside each jump is no shorter.

The rate at which the error falls is set by the smoothness of the signal. A jump forces the coefficients to decay like $1/k$, so $|a_k|^2$ decays like $1/k^2$. A tail sum of this kind is close to the matching integral, $\sum_{k>N} 1/k^2 \approx \int_N^\infty dk/k^2 = 1/N$, so the tail $\sum_{|k|>N} |a_k|^2$ behaves like $1/N$ and doubling N roughly halves the error. A signal that is continuous but has a corner, such as a triangular wave, has coefficients decaying like $1/k^2$. Then $|a_k|^2$ decays like $1/k^4$, the tail like $\int_N^\infty dk/k^4 = 1/(3N^3)$, and far fewer harmonics are needed.

4.6 The discrete-time Fourier series

DISCRETE-TIME FOURIER SERIES

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}$$

$\langle N \rangle$ means any N consecutive values of the index. Both sums are finite and there are exactly N distinct coefficients.

① THE COEFFICIENTS REPEAT

$a_{k+N} = a_k$. Replace k by $k + N$ in the analysis sum and split the exponential:

$$a_{k+N} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-j(k+N)(2\pi/N)n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n} e^{-j2\pi n} = a_k,$$

because $e^{-j2\pi n} = 1$ for every integer n . The same substitution in continuous time produces $e^{-j2\pi t}$, which is 1 only when t is an integer, and it is not. Continuous-time coefficients therefore do not repeat.

☑ A FINITE, EXACT IDENTITY

A period- N sequence is described by N numbers and its series has N terms. There is no truncation, no limit, no convergence question, and no Gibbs phenomenon anywhere in discrete time.

🔗 EXAMPLE 4.6 – A SUM OF TWO SEQUENCES

GIVEN $x[n] = \sin\left(\frac{5\pi}{6}n\right) + \cos\left(\frac{3\pi}{4}n + \frac{\pi}{5}\right)$.

FIND The fundamental period and every coefficient.

METHOD A discrete-time sinusoid repeats only if $N = (2\pi/\omega)k$ is a positive integer for some integer k . Get each component period, take the least common multiple, then expand.

SOLUTION A sinusoid $\cos(\omega n)$ or $\sin(\omega n)$ repeats with period N when $\omega N = 2\pi k$ for some integer k , that is, $N = 2\pi k/\omega$. For $\omega = 5\pi/6$: $N = \frac{2\pi k}{5\pi/6} = \frac{12k}{5}$, first an integer at $k = 5$, giving $N = 12$. For $\omega = 3\pi/4$: $N = \frac{2\pi k}{3\pi/4} = \frac{8k}{3}$, first an integer at $k = 3$, giving $N = 8$. So $N_0 = \text{LCM}(12, 8) = 24$ and $\omega_0 = 2\pi/24 = \pi/12$. Write each frequency as a multiple of ω_0 :

$$\frac{5\pi}{6} = \frac{20\pi}{24} = 10\omega_0, \quad \frac{3\pi}{4} = \frac{18\pi}{24} = 9\omega_0.$$

Now expand with Euler's relations. The phase $\pi/5$ stays with the coefficient, because $e^{j(9\omega_0 n + \pi/5)} = e^{j\pi/5} e^{j9\omega_0 n}$.

$$\begin{aligned} \sin(10\omega_0 n) &= \frac{1}{2j} e^{j10\omega_0 n} - \frac{1}{2j} e^{-j10\omega_0 n}, \\ \cos\left(9\omega_0 n + \frac{\pi}{5}\right) &= \frac{1}{2} e^{j\pi/5} e^{j9\omega_0 n} + \frac{1}{2} e^{-j\pi/5} e^{-j9\omega_0 n}. \end{aligned}$$

Reading off the coefficient of each $e^{jk\omega_0 n}$,

$$a_{10} = \frac{1}{2j}, \quad a_{-10} = -\frac{1}{2j}, \quad a_9 = \frac{1}{2} e^{j\pi/5}, \quad a_{-9} = \frac{1}{2} e^{-j\pi/5},$$

with every other coefficient in one period equal to zero.

CHECK $|a_{\pm 9}| = |a_{\pm 10}| = 1/2$; $\angle a_9 = \pi/5$, $\angle a_{-9} = -\pi/5$, $\angle a_{10} = -\pi/2$, $\angle a_{-10} = +\pi/2$. The sequence is real, so magnitudes are even and phases odd. Also $a_k = a_{k+24}$, which extends the answer to every k .

EXAMPLE 4.7 – THE DISCRETE-TIME SQUARE WAVE

GIVEN $x[n] = 1$ for $|n| \leq N_1$ and 0 for $N_1 < |n| \leq N/2$, repeated with period N .

FIND Every coefficient.

METHOD Substitute the signal into the analysis sum over the period from $-N/2$ to $N/2$. The signal is 1 only for $-N_1 \leq n \leq N_1$, so the sum reduces to a finite geometric sum with ratio $r = e^{-jk(2\pi/N)}$. For $r \neq 1$ the sum has a closed form. Write $S = \sum_{n=-N_1}^{N_1} r^n$; then $S - rS = r^m - r^{p+1}$, because every other term cancels, so $S = (r^m - r^{p+1})/(1 - r)$. The case $r = 1$ is handled separately.

SOLUTION With $m = -N_1$ and $p = N_1$,

$$a_k = \frac{1}{N} \sum_{n=-N_1}^{N_1} 1 \cdot e^{-jk(2\pi/N)n} = \frac{1}{N} \sum_{n=-N_1}^{N_1} r^n = \frac{1}{N} \cdot \frac{r^{-N_1} - r^{N_1+1}}{1 - r}.$$

The lower limit is $n = -N_1$, so the first numerator term is $r^{-N_1} = e^{+jk(2\pi/N)N_1}$, with a positive exponent. Written out,

$$a_k = \frac{1}{N} \cdot \frac{e^{jk(2\pi/N)N_1} - e^{-jk(2\pi/N)(N_1+1)}}{1 - e^{-jk(2\pi/N)}}.$$

Factor $e^{-jk\pi/N}$ out of the numerator. The exponents that remain are $jk(2\pi/N)N_1 + jk\pi/N = jk(2\pi/N)(N_1 + \frac{1}{2})$ and $-jk(2\pi/N)(N_1 + 1) + jk\pi/N = -jk(2\pi/N)(N_1 + \frac{1}{2})$, so the bracket is $2j$ times a sine:

$$\begin{aligned} e^{jk(2\pi/N)N_1} - e^{-jk(2\pi/N)(N_1+1)} &= e^{-jk\pi/N} \left[e^{jk(2\pi/N)(N_1 + \frac{1}{2})} - e^{-jk(2\pi/N)(N_1 + \frac{1}{2})} \right] \\ &= e^{-jk\pi/N} 2j \sin\left(\frac{2\pi k}{N} \left(N_1 + \frac{1}{2}\right)\right). \end{aligned}$$

Factor the same $e^{-jk\pi/N}$ out of the denominator:

$$1 - e^{-jk(2\pi/N)} = e^{-jk\pi/N} [e^{jk\pi/N} - e^{-jk\pi/N}] = e^{-jk\pi/N} 2j \sin\left(\frac{\pi k}{N}\right).$$

The factors $e^{-jk\pi/N}$ and $2j$ cancel between numerator and denominator, leaving

$$a_k = \frac{1}{N} \frac{\sin\left(\frac{2\pi k}{N} \left(N_1 + \frac{1}{2}\right)\right)}{\sin\left(\frac{\pi k}{N}\right)}, \quad k \neq 0, \pm N, \pm 2N, \dots$$

When k is a multiple of N , $r = e^{-j2\pi(k/N)} = 1$, every term of the sum is 1, and there are $2N_1 + 1$ terms, so

$$a_k = \frac{2N_1 + 1}{N} \text{ when } k = 0, \pm N, \pm 2N, \dots$$

CHECK Here $|r| = 1$ at every k , so a condition $|r| < 1$ would rule the sum out altogether. What matters is $r \neq 1$, and $r = 1$ happens exactly when k is a multiple of N , which is the second

branch. For $N = 10$, $N_1 = 2$ the peak is $(2N_1 + 1)/N = 0.5$, and any plot of these coefficients must be scaled so that 0.5 falls inside the labelled range.

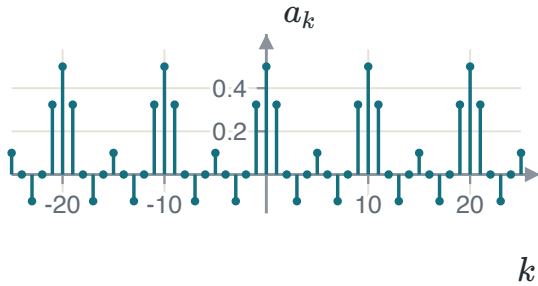


FIGURE 4.9 $N = 10$, $N_1 = 2$: peak 0.5, repeating every 10.

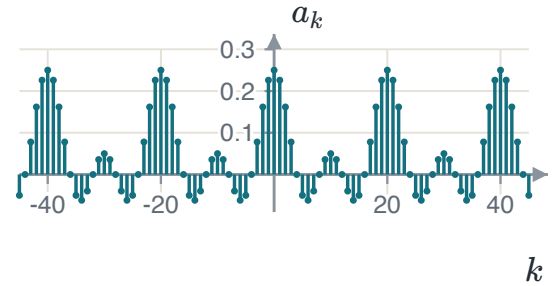


FIGURE 4.10 $N = 20$: peak 0.25, repeating every 20.

EXAMPLE 4.8 – A DISCRETE SAWTOOTH

GIVEN $x[n] = n$ for $-5 \leq n \leq 5$, repeated with period $N = 11$.

FIND Every coefficient.

METHOD Substitute into the analysis sum with $\omega_0 = 2\pi/11$. The sequence is odd, so pair the term at $n = m$ with the term at $n = -m$ before expanding. With $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$,

$$me^{-jk\omega_0 m} + (-m)e^{+jk\omega_0 m} = -m(e^{jk\omega_0 m} - e^{-jk\omega_0 m}) = -2jm \sin(k\omega_0 m).$$

SOLUTION The DC term is the average of the eleven values, and they cancel in pairs:

$$a_0 = \frac{1}{11} \sum_{n=-5}^5 n = \frac{1}{11} [(-5) + (-4) + \cdots + 4 + 5] = 0.$$

For general k , the $n = 0$ term is zero, and the remaining ten terms form five pairs:

$$\begin{aligned} a_k &= \frac{1}{11} \sum_{n=-5}^5 n e^{-jk(2\pi/11)n} \\ &= \frac{1}{11} \sum_{m=1}^5 \left[m e^{-jk(2\pi/11)m} + (-m) e^{+jk(2\pi/11)m} \right] \\ &= \frac{1}{11} \sum_{m=1}^5 (-2jm) \sin\left(\frac{2\pi km}{11}\right) = -\frac{2j}{11} \sum_{m=1}^5 m \sin\left(\frac{2\pi km}{11}\right). \end{aligned}$$

For $k = 1$ the five terms are $1 \sin \frac{2\pi}{11} + 2 \sin \frac{4\pi}{11} + 3 \sin \frac{6\pi}{11} + 4 \sin \frac{8\pi}{11} + 5 \sin \frac{10\pi}{11} \approx 0.5406 + 1.8193 + 2.9695 + 3.0230 + 1.4087 = 9.7610$, so $a_1 \approx -j \frac{2}{11} (9.7610) = -j1.7747$.

CHECK Every coefficient is purely imaginary, as a real odd sequence requires. The largest is $|a_{\pm 1}| = 1.7747$, and $a_k = a_{k+11}$, so the eleven coefficients from $k = -5$ to $k = 5$ are the whole answer.

4.7 Properties

TABLE 4.2

PROPERTY	SIGNAL	COEFFICIENTS
Linearity	$Ax_1 + Bx_2$	$Aa_k + Bb_k$
Time shift, continuous	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0}$
Time shift, discrete	$x[n - n_0]$	$a_k e^{-jk\omega_0 n_0}$
Time reversal	$x(-t)$ or $x[-n]$	a_{-k}
Conjugation	$x^*(t)$ or $x^*[n]$	a_{-k}^*
Frequency shift, continuous	$e^{jM\omega_0 t} x(t)$	a_{k-M} , M an integer
Frequency shift, discrete	$e^{jM(2\pi/N)n} x[n]$	a_{k-M} , M an integer
Time scaling, continuous	$x(\alpha t)$, $\alpha > 0$	a_k , with period T_0/α
Time scaling, discrete	$x_{(m)}[n]$, m a positive integer	$\frac{1}{m} a_k$, with period mN
Periodic convolution, continuous	$\int_{T_0} x(\tau) y(t - \tau) d\tau$	$T_0 a_k b_k$
Periodic convolution, discrete	$\sum_{r=\langle N \rangle} x[r] y[n - r]$	$N a_k b_k$
Multiplication, continuous	$x(t) y(t)$	$\sum_{\ell=-\infty}^{\infty} a_\ell b_{k-\ell}$
Multiplication, discrete	$x[n] y[n]$	$\sum_{\ell=\langle N \rangle} a_\ell b_{k-\ell}$
Differentiation	$dx(t)/dt$	$jk\omega_0 a_k$
Integration	$\int_{-\infty}^t x(\tau) d\tau$	$a_k / (jk\omega_0)$, only if $a_0 = 0$
First difference	$x[n] - x[n - 1]$	$(1 - e^{-jk(2\pi/N)}) a_k$
Running sum	$\sum_{r=-\infty}^n x[r]$	$a_k / (1 - e^{-jk(2\pi/N)})$, only if $a_0 = 0$

Three properties need further explanation. **Time scaling** in continuous time does not change the coefficient values, but it changes their frequencies. In discrete time, $x_{(m)}[n]$ places the original samples m indices apart and inserts $m - 1$ zeros after each sample. The factor $1/m$ appears because the average is taken over m times as many samples. **Periodic convolution** includes the factor T_0 or N before the coefficient product. The analysis equation divides by the period, and the convolution adds an operation over one period. Omitting this factor gives the wrong amplitude. **Integration** and the **running sum** divide by a factor that is zero at $k = 0$. A signal with non-zero mean therefore produces a non-periodic term. Subtract the mean, apply the rule to the zero-mean part, and then add the term $a_0 t$.

The time-shift factor has modulus 1, so $|b_k| = |a_k|$ at every k : a shift never changes a magnitude, only a phase, and it changes that phase by $-k\omega_0 t_0$, proportionally to the harmonic index. Two signals differing only by a delay have identical magnitude plots, which is why a phase plot is not decoration.

⊗ THE DISCRETE-TIME PRODUCT

The multiplication property in discrete time is a **periodic** convolution: the sum runs over one period of ℓ . Discrete-time coefficients repeat, so an infinite sum would add each of the N distinct products over and over and diverge. The product $x[n]y[n]$ is itself periodic with period N and therefore has N coefficients — a number only a sum over one period can deliver.

Symmetry

- x real gives $a_{-k} = a_k^*$, so $|a_k|$ is even in k and $\angle a_k$ is odd.
- x real and even gives a_k real and even.
- x real and odd gives a_k purely imaginary and odd.
- For real x : $\mathcal{E}\mathcal{V}\{x\} \leftrightarrow \text{Re}\{a_k\}$ and $\mathcal{O}\mathcal{D}\mathcal{D}\{x\} \leftrightarrow j \text{Im}\{a_k\}$.

The last line is the general statement and the two before it are its special cases. Write the even part as a combination of $x(t)$ and $x(-t)$, apply linearity and time reversal from the table, and then use $a_{-k} = a_k^*$ for a real signal:

$$\begin{aligned}\mathcal{E}\mathcal{V}\{x(t)\} &= \frac{1}{2}x(t) + \frac{1}{2}x(-t) \leftrightarrow \frac{1}{2}a_k + \frac{1}{2}a_{-k} = \frac{1}{2}(a_k + a_k^*) = \text{Re}\{a_k\}, \\ \mathcal{O}\mathcal{D}\mathcal{D}\{x(t)\} &= \frac{1}{2}x(t) - \frac{1}{2}x(-t) \leftrightarrow \frac{1}{2}a_k - \frac{1}{2}a_{-k} = \frac{1}{2}(a_k - a_k^*) = j \text{Im}\{a_k\}.\end{aligned}$$

The last equality on each line uses $a_k + a_k^* = 2 \text{Re}\{a_k\}$ and $a_k - a_k^* = 2j \text{Im}\{a_k\}$. Splitting a real signal into its even and odd parts splits its coefficients into their real and imaginary parts, and that tells you which half of the coefficients you have to compute.

The third follows in one line. Real gives $a_{-k} = a_k^*$; odd gives $a_{-k} = -a_k^*$; together $a_k^* = -a_k$, so $\text{Re}\{a_k\} = 0$. Both sawtooths of this chapter obey it.

PARSEVAL'S RELATION

$$\begin{aligned}\frac{1}{T_0} \int_{T_0} |x(t)|^2 dt &= \sum_{k=-\infty}^{\infty} |a_k|^2 \\ \frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 &= \sum_{k=\langle N \rangle} |a_k|^2\end{aligned}$$

The left side is the average power over one period, with the normalised convention $R = 1 \Omega$. The right side splits it among the harmonics, and $|a_k|^2$ is the power carried by the k -th harmonic alone.

To derive the continuous-time form, write $|x(t)|^2 = x(t)x^*(t)$. Substitute the series for x , and the conjugated series for x^* with a second index m so that the two sums stay separate. Then integrate over one period. The orthogonality integral of Section 4.3 is T_0 when $m = k$ and zero otherwise, so only the terms with $m = k$ survive:

$$\begin{aligned}\frac{1}{T_0} \int_{T_0} x(t)x^*(t) dt &= \frac{1}{T_0} \int_{T_0} \left(\sum_k a_k e^{jk\omega_0 t} \right) \left(\sum_m a_m^* e^{-jm\omega_0 t} \right) dt \\ &= \sum_k \sum_m a_k a_m^* \frac{1}{T_0} \int_{T_0} e^{j(k-m)\omega_0 t} dt = \sum_k a_k a_k^* = \sum_k |a_k|^2.\end{aligned}$$

The discrete-time form follows the same steps with sums in place of the integral. The orthogonality sum is $\sum_{n=\langle N \rangle} e^{j(k-m)(2\pi/N)n}$. For $k = m$ every term is 1 and the sum is N . For $k \neq m$ within one period, it is a geometric sum with ratio $r = e^{j(k-m)2\pi/N} \neq 1$, and $r^N = e^{j(k-m)2\pi} = 1$ makes the closed form $r^{n_0}(1 - r^N)/(1 - r)$ equal to zero.

4.8 A periodic signal through an LTI system

Apply the eigenfunction property of Section 4.1 to each term of the synthesis equation. The term $a_k e^{jk\omega_0 t}$ is a complex exponential with $s = jk\omega_0$, so the system multiplies it by $H(jk\omega_0)$. The system is linear, so the outputs of all the terms add:

ONE MULTIPLICATION PER HARMONIC

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} \underbrace{a_k H(jk\omega_0)}_{b_k} e^{jk\omega_0 t}$$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} \longrightarrow y[n] = \sum_{k=\langle N \rangle} a_k H(e^{jk\omega_0}) e^{jk\omega_0 n}$$

The frequency response H describes the system. For a periodic input, calculate one product at each harmonic instead of evaluating a convolution.

Putting the output back together

The result above is a sum of complex terms. For real x , $a_{-k} = a_k^*$. For real h , conjugating $H(j\omega) = \int h(\tau) e^{-j\omega\tau} d\tau$ conjugates only the exponential, so $H^*(j\omega) = \int h(\tau) e^{+j\omega\tau} d\tau = H(-j\omega)$. Together, $b_{-k} = a_{-k} H(-jk\omega_0) = a_k^* H^*(jk\omega_0) = b_k^*$. Pair each k with $-k$, write $b_k = |b_k| e^{j\angle b_k}$, and use $e^{j\theta} + e^{-j\theta} = 2 \cos \theta$:

$$\begin{aligned} b_k e^{jk\omega_0 t} + b_{-k} e^{-jk\omega_0 t} &= |b_k| e^{j\angle b_k} e^{jk\omega_0 t} + |b_k| e^{-j\angle b_k} e^{-jk\omega_0 t} \\ &= |b_k| \left[e^{j(k\omega_0 t + \angle b_k)} + e^{-j(k\omega_0 t + \angle b_k)} \right] = 2|b_k| \cos(k\omega_0 t + \angle b_k). \end{aligned}$$

CONJUGATE-PAIR REASSEMBLY

$$b_k e^{jk\omega_0 t} + b_{-k} e^{-jk\omega_0 t} = 2|b_k| \cos(k\omega_0 t + \angle b_k)$$

$$y(t) = b_0 + \sum_{k=1}^{\infty} 2|b_k| \cos(k\omega_0 t + \angle b_k)$$

b_0 has no partner, so it carries no factor of two.

⊗ THE FACTOR OF TWO, AND WHOSE PHASE

The amplitude of the k -th cosine is $2|b_k|$, because both members of the pair contribute. The phase is $\angle b_k$, the one belonging to **positive** k ; taking $\angle b_{-k}$ flips the sign of that cosine's phase. Test any reassembly by setting $H = 1$: it must return the input exactly. A version missing the factor of two returns half the input plus its full average.

EXAMPLE 4.9 – LOW-PASS FILTERING

GIVEN $x(t) = 1 + \cos(\pi t) + \sin(2\pi t) + \cos(3\pi t + \frac{\pi}{3})$ into a system with $h(t) = e^{-t}u(t)$.

FIND The output signal.

METHOD The input is periodic and the system is LTI, so harmonic multiplication applies. Calculate $H(j\omega)$ from h , find T_0 and a_k from x , form $b_k = a_k H(jk\omega_0)$, and combine each conjugate pair.

SOLUTION First the frequency response. Substitute h into the definition; $u(\tau)$ makes the lower limit zero, and the two exponentials combine into one:

$$\begin{aligned} H(j\omega) &= \int_{-\infty}^{\infty} e^{-\tau} u(\tau) e^{-j\omega\tau} d\tau = \int_0^{\infty} e^{-(1+j\omega)\tau} d\tau \\ &= \frac{e^{-(1+j\omega)\tau}}{-(1+j\omega)} \Big|_0^{\infty} = \frac{0 - 1}{-(1+j\omega)} = \frac{1}{1+j\omega}. \end{aligned}$$

The upper limit gives zero because $|e^{-(1+j\omega)\tau}| = e^{-\tau} \rightarrow 0$. So $|H(j\omega)| = 1/\sqrt{1+\omega^2}$ and $\angle H(j\omega) = -\arctan \omega$. Next the input. The component periods are $2\pi/\pi = 2$, $2\pi/(2\pi) = 1$ and $2\pi/(3\pi) = 2/3$ s, so $T_0 = \text{LCM}(2, 1, 2)/\text{GCD}(1, 1, 3) = 2$ s and $\omega_0 = \pi$. The three frequencies are $1\omega_0$, $2\omega_0$ and $3\omega_0$. Euler's relations give the coefficients for positive k :

$$\begin{aligned} \cos(\pi t) &\rightarrow a_1 = \frac{1}{2}, \\ \sin(2\pi t) &= \frac{1}{2j}e^{j2\pi t} - \frac{1}{2j}e^{-j2\pi t} \rightarrow a_2 = \frac{1}{2j} = \frac{1}{2}e^{-j\pi/2}, \\ \cos(3\pi t + \frac{\pi}{3}) &\rightarrow a_3 = \frac{1}{2}e^{j\pi/3} \end{aligned}$$

with $a_0 = 1$ and $a_{-k} = a_k^*$. Now one product per harmonic, $b_k = a_k H(jk\pi)$: magnitudes multiply and phases add. The values of H at the three harmonics are

$$\begin{aligned} |H(j\pi)| &= \frac{1}{\sqrt{1+\pi^2}} = 0.3033, & \angle H(j\pi) &= -\arctan \pi = -1.2626, \\ |H(j2\pi)| &= \frac{1}{\sqrt{1+4\pi^2}} = 0.1572, & \angle H(j2\pi) &= -\arctan 2\pi = -1.4130, \\ |H(j3\pi)| &= \frac{1}{\sqrt{1+9\pi^2}} = 0.1055, & \angle H(j3\pi) &= -\arctan 3\pi = -1.4651. \end{aligned}$$

Then

$$\begin{aligned} b_0 &= a_0 H(j0) = 1 \cdot 1 = 1, \\ b_1 &= \frac{1}{2}(0.3033) e^{-j1.2626} = 0.1517 e^{-j1.2626}, \\ b_2 &= \frac{1}{2}(0.1572) e^{j(-\pi/2-1.4130)} = 0.0786 e^{-j2.9838}, \\ b_3 &= \frac{1}{2}(0.1055) e^{j(\pi/3-1.4651)} = 0.0528 e^{-j0.4179}. \end{aligned}$$

Finally combine each conjugate pair into a cosine of amplitude $2|b_k|$ and phase $\angle b_k$:

$$y(t) = 1 + 0.303 \cos(\pi t - 1.263) + 0.157 \cos(2\pi t - 2.984) + 0.106 \cos(3\pi t - 0.418).$$

CHECK Each amplitude is $2|b_k|$: 0.303, 0.157, 0.106. The output ranges from about 0.615 to 1.417; using half the required amplitudes would give only 0.81 to 1.21. The average is unchanged because $|H(0)| = 1$, and the third-harmonic amplitude is reduced to about one tenth of its input value.

EXAMPLE 4.10 – HIGH-PASS FILTERING

GIVEN The same $x(t)$, now into a system with $H(j\omega) = \frac{j\omega}{1 + j\omega}$.

FIND The output signal.

METHOD The input is periodic and the system is LTI, so use the same harmonic-multiplication method as in Example 4.9. Replace the low-pass response by the given H , calculate each b_k , and combine the conjugate pairs.

SOLUTION $H(j0) = 0/(1 + 0) = 0$, so $b_0 = a_0 H(j0) = 0$ and the output has zero average. For $k \geq 1$, the magnitude of a quotient is the quotient of the magnitudes, and the phase is the difference of the phases. The numerator $jk\pi$ has magnitude $k\pi$ and phase $\pi/2$; the denominator $1 + jk\pi$ has magnitude $\sqrt{1 + k^2\pi^2}$ and phase $\arctan(k\pi)$:

$$|H(jk\pi)| = \frac{k\pi}{\sqrt{1 + k^2\pi^2}}, \quad \angle H(jk\pi) = \frac{\pi}{2} - \arctan(k\pi).$$

With the values of $\sqrt{1 + k^2\pi^2}$ and $\arctan(k\pi)$ from Example 4.9,

$$\begin{aligned} |H(j\pi)| &= \frac{3.1416}{3.2969} = 0.9529, & \angle H(j\pi) &= 1.5708 - 1.2626 = +0.3082, \\ |H(j2\pi)| &= \frac{6.2832}{6.3623} = 0.9876, & \angle H(j2\pi) &= 1.5708 - 1.4130 = +0.1578, \\ |H(j3\pi)| &= \frac{9.4248}{9.4777} = 0.9944, & \angle H(j3\pi) &= 1.5708 - 1.4651 = +0.1057. \end{aligned}$$

The input coefficients are the same as before: $a_1 = \frac{1}{2}$, $a_2 = \frac{1}{2}e^{-j\pi/2}$, $a_3 = \frac{1}{2}e^{j\pi/3}$.

Multiply the magnitudes and add the phases:

$$\begin{aligned} b_1 &= \frac{1}{2}(0.9529)e^{j0.3082} = 0.4764e^{+j0.3082}, \\ b_2 &= \frac{1}{2}(0.9876)e^{j(-\pi/2+0.1578)} = 0.4938e^{-j1.4130}, \\ b_3 &= \frac{1}{2}(0.9944)e^{j(\pi/3+0.1057)} = 0.4972e^{+j1.1529}. \end{aligned}$$

Pairing k with $-k$ as before,

$$y(t) = 0.953 \cos(\pi t + 0.308) + 0.988 \cos(2\pi t - 1.413) + 0.994 \cos(3\pi t + 1.153).$$

CHECK $|H(jk\pi)|$ is 0.953, 0.988, 0.994 for $k = 1, 2, 3$. These values are close to one, so unit-amplitude input harmonics must produce output amplitudes close to one. Values close to 0.5

would show that the factor of two is missing. Use the phases for positive k : $+0.308$, -1.413 , $+1.153$. Using b_{-k} instead changes the signs of the first and third phases.

EXAMPLE 4.11 – FILTERING A DISCRETE IMPULSE TRAIN

GIVEN $x[n] = \sum_{m=-\infty}^{\infty} \delta[n - 4m]$, and the systems $h_1[n] = 0.5\delta[n] - 0.5\delta[n - 1]$ and $h_2[n] = 0.5\delta[n] + 0.5\delta[n - 1]$.

FIND Both output signals.

METHOD The input is periodic and both systems are LTI, so discrete-time harmonic multiplication applies. Find a_k for the impulse train, calculate $b_k = a_k H(e^{jk\omega_0})$ with $\omega_0 = \pi/2$, and then combine the conjugate pairs.

SOLUTION First the input coefficients. One period is $n = 0, 1, 2, 3$ and contains one impulse, at $n = 0$, so

$$a_k = \frac{1}{4} \sum_{n=0}^3 x[n] e^{-jk(\pi/2)n} = \frac{1}{4} \cdot 1 \cdot e^{-jk(\pi/2) \cdot 0} = \frac{1}{4} \quad \text{for every } k.$$

One period of coefficients is $k = -1, 0, 1, 2$. Next the frequency responses, from $H(e^{j\omega}) = \sum_n h[n] e^{-j\omega n}$:

$$H_1(e^{j\omega}) = 0.5 - 0.5e^{-j\omega} = 0.5(1 - e^{-j\omega}), \quad H_2(e^{j\omega}) = 0.5(1 + e^{-j\omega}).$$

Evaluate H_1 at the three harmonics $\omega = 0, \pi/2, \pi$, using $e^{-j\pi/2} = -j$ and $e^{-j\pi} = -1$:

$$\begin{aligned} H_1(e^{j0}) &= 0.5(1 - 1) = 0, \\ H_1(e^{j\pi/2}) &= 0.5(1 + j) = 0.5\sqrt{2}e^{j\pi/4} = 0.7071e^{j\pi/4}, \\ H_1(e^{j\pi}) &= 0.5(1 + 1) = 1. \end{aligned}$$

Multiply by $a_k = 1/4$: $b_0 = 0$, $b_1 = 0.25(0.7071)e^{j\pi/4} = 0.1768e^{j\pi/4}$ and $b_2 = 0.25$. The pair $k = \pm 1$ gives a cosine of amplitude $2|b_1| = 0.3536$ and phase $\pi/4$. The term $k = 2$ has $e^{j2(\pi/2)n} = e^{j\pi n} = (-1)^n$ and no partner, so

$$y[n] = 0.354 \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right) + 0.25(-1)^n.$$

For h_2 the same three evaluations give

$$\begin{aligned} H_2(e^{j0}) &= 0.5(1 + 1) = 1, \\ H_2(e^{j\pi/2}) &= 0.5(1 - j) = 0.7071e^{-j\pi/4}, \\ H_2(e^{j\pi}) &= 0.5(1 - 1) = 0 \end{aligned}$$

so $b_0 = 0.25$, $b_1 = 0.1768e^{-j\pi/4}$ and $b_2 = 0$, and

$$y[n] = 0.25 + 0.354 \cos\left(\frac{\pi}{2}n - \frac{\pi}{4}\right).$$

CHECK

Apply the two systems directly. For h_1 , $y[n] = 0.5x[n] - 0.5x[n-1]$ is 0.5 at $n = 0$, -0.5 at $n = 1$ and 0 at $n = 2, 3$. The series gives the same values: at $n = 0$, $0.354 \cos(\pi/4) + 0.25 = 0.25 + 0.25 = 0.5$; at $n = 1$, $0.354 \cos(3\pi/4) - 0.25 = -0.25 - 0.25 = -0.5$; at $n = 2$, $0.354 \cos(5\pi/4) + 0.25 = 0$; at $n = 3$, $0.354 \cos(7\pi/4) - 0.25 = 0$. The check for h_2 works the same way. Within one coefficient period, the $k = 2$ term has no separate conjugate partner. It therefore appears alone as $b_2(-1)^n$ and has no factor of two. The high-pass system removes the average and retains b_2 ; the low-pass system retains the average and gives $b_2 = 0$.

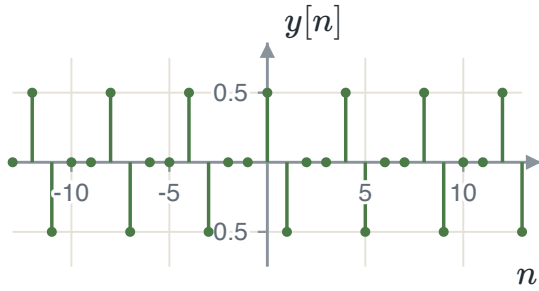


FIGURE 4.11 First difference: zero average.

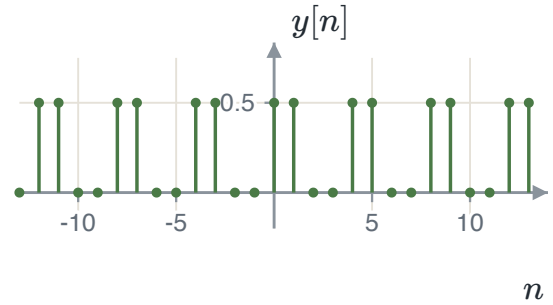


FIGURE 4.12 Two-point average: the average remains.

4.9 Checklist, and what comes next

1. Confirm the signal is periodic and find T_0 or N before anything else.
2. Set $\omega_0 = 2\pi/T_0$ or $2\pi/N$ and write every component frequency as a multiple of it. That multiple is k .
3. If the signal is already a sum of sinusoids, read the coefficients off with Euler's relations. Otherwise integrate or sum, treating $k = 0$ separately.
4. Compare a_0 with the average calculated from the signal, and check the symmetry: a real signal gives $a_{-k} = a_k^*$.
5. Through an LTI system, $b_k = a_k H(jk\omega_0)$ — one product per harmonic.
6. To return to a real signal, pair k with $-k$: amplitude $2|b_k|$, phase $\angle b_k$. The term b_0 stands alone.

Everything in this chapter needed the signal to repeat. A single pulse does not repeat, has no fundamental period, and has no harmonics to carry coefficients. The rectangular wave already showed the way out: its coefficients were samples of one envelope, taken every $\omega_0 = 2\pi/T_0$. Lengthen the period and the envelope stays put while the samples crowd together and shrink as $1/T_0$.

WHERE CHAPTER 5 BEGINS

$$T_0 a_k = E(k\omega_0) \xrightarrow{T_0 \rightarrow \infty} X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

In the limit the stems merge into the curve they were always sampling. That curve is the Fourier transform.

4.10 Exercises

4.1 Two periodic signals have fundamental periods $2/9$ s and $8/21$ s. Find the fundamental period of their sum, and verify it by division.

Answer: $T_0 = \text{LCM}(2, 8) / \text{GCD}(9, 21) = 8/3$ s. Check: $(8/3)/(2/9) = 12$ and $(8/3)/(8/21) = 7$, both whole, and $\text{gcd}(12, 7) = 1$.

4.2 For $x(t) = 1 + \frac{1}{2} \cos(2\pi t) + \sin(3\pi t)$, state a_0 and explain in one sentence why it is not zero.

Answer: $a_0 = 1$: the constant term is the $k = 0$ harmonic, and it is also the average of the signal, since both sinusoids complete whole cycles in one period.

4.3 A rectangular wave has $T_0 = 8T_1$. Find a_0 and a_2 .

Answer: $a_0 = 2T_1/T_0 = 1/4$ and $a_2 = \sin(\pi/2)/(2\pi) = 1/(2\pi) \approx 0.159$.

4.4 Explain why the finite geometric sum used for the discrete-time square wave requires $r \neq 1$ and not $|r| < 1$, and say what happens at the excluded value.

Answer: A finite sum always has a value; only the closed form fails, where its denominator $1 - r$ vanishes. Here $|r| = 1$ at every k , so $|r| < 1$ would exclude the sum entirely. At $r = 1$, that is k a multiple of N , every term is 1 and the sum is $2N_1 + 1$.

4.5 A filtered signal has $b_1 = 0.1517e^{-j1.263}$ and $\omega_0 = \pi$. Write the contribution of the pair $k = \pm 1$ to the real output.

Answer: $2|b_1| \cos(\pi t + \angle b_1) = 0.303 \cos(\pi t - 1.263)$.

4.6 A square wave and a triangular wave are truncated to the same number of harmonics. Which is approximated better, and which shows a Gibbs overshoot?

Answer: The triangular wave: its coefficients decay like $1/k^2$ against $1/k$, so its discarded power falls like $1/N^3$ rather than $1/N$. Only the square wave has a jump, so only it shows a Gibbs overshoot.

CHAPTER

5

The continuous-time Fourier transform

The continuous-time Fourier transform represents an aperiodic signal by a continuous function of frequency. To derive it, construct a periodic extension of a finite-duration signal and increase the repetition period without bound. The harmonic spacing then approaches zero, and the Fourier-series sum approaches an integral.

5.1 From a series to a transform

Let $x(t)$ be zero for $|t| > T_1$. The number T_1 is the half-width of the support and belongs to the signal alone. Build a periodic signal $\tilde{x}$ by repeating the pulse every T seconds, with $T > 2T_1$ so that the copies do not touch:

$$\tilde{x}(t) = \sum_{m=-\infty}^{\infty} x(t - mT), \quad \tilde{x}(t) = \tilde{x}(t + T), \quad T > 2T_1.$$

Inside one period, $\tilde{x}$ equals the original pulse, so its Fourier series can be used. Increase T while keeping the central pulse fixed. The neighbouring copies move to larger values of $|t|$, and $\tilde{x}(t) \rightarrow x(t)$ for every t .

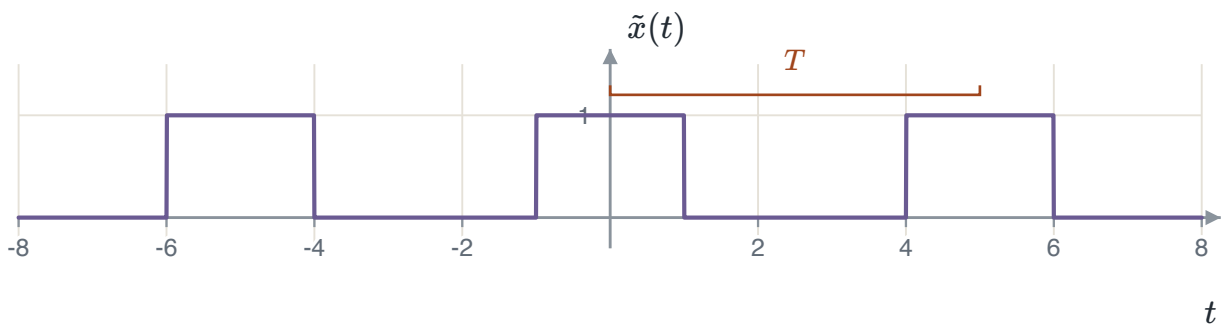


FIGURE 5.1 The periodic extension with $T = 5T_1$. The condition $T > 2T_1$ is what keeps the copies apart.

The coefficients are samples of one curve

Apply the analysis equation of Chapter 4 to $\tilde{x}$ over one period. Inside that period $\tilde{x} = x$, and outside $|t| < T_1$ the integrand is zero, so the limits may be opened to all of time:

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t} dt, \quad \omega_0 = \frac{2\pi}{T}.$$

The right-hand integral has the same form for every k ; only the sampled frequency $k\omega_0$ changes. Replace that frequency by the continuous variable ω and define the transform.

DEFINITION OF THE TRANSFORM

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Then $a_k = \frac{1}{T} X(jk\omega_0)$, so every coefficient of the periodic extension is one point of the curve X , scaled by $1/T$. The curve was built from the pulse alone, so lengthening T cannot move it. What lengthening T does is shrink the spacing $\omega_0 = 2\pi/T$ and shrink each coefficient by the same factor.

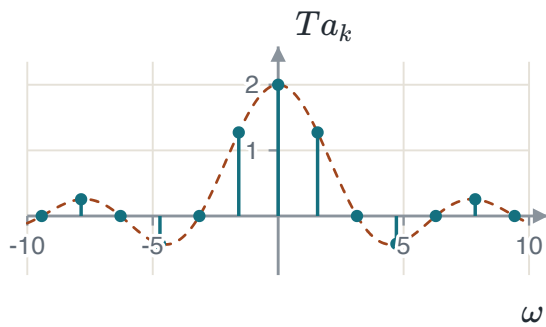


FIGURE 5.2 $T = 4T_1$.

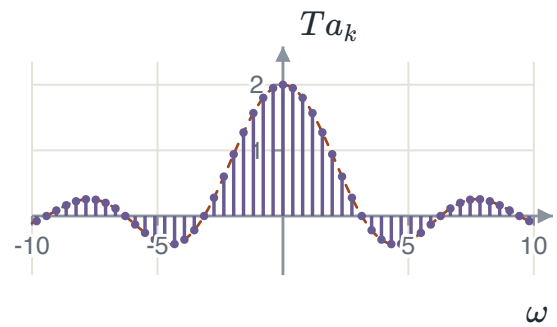


FIGURE 5.3 $T = 16T_1$: the same curve, sampled four times as finely.

The sum becomes an integral

Put $a_k = \frac{1}{T} X(jk\omega_0)$ back into the synthesis equation and replace $1/T$ by $\omega_0/2\pi$. Nothing has been approximated yet.

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(jk\omega_0) e^{jk\omega_0 t} \omega_0.$$

Every term now includes the spacing ω_0 , so the expression is a Riemann sum. Let $T \rightarrow \infty$. Then $\tilde{x}(t) \rightarrow x(t)$, $\omega_0 \rightarrow d\omega$, the sample frequencies cover the frequency axis, and the sum approaches an integral.

THE CONTINUOUS-TIME FOURIER TRANSFORM PAIR

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{analysis})$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad (\text{synthesis})$$

Analysis integrates over t and leaves a function of ω : it takes the signal apart. Synthesis integrates over ω and leaves a function of t : it puts the signal back together. The factor $1/2\pi$ arrived as $\omega_0/2\pi$ when the spacing was substituted, and it stays on the synthesis side throughout. The pair is written $x(t) \leftrightarrow X(j\omega)$, with the argument $j\omega$ and never ω alone.

△ WHICH EQUATION IS WHICH

The variable of integration settles it, and nothing else needs to be remembered. Integrating time away produces a spectrum, and that is analysis. Integrating frequency away produces a signal, and that is synthesis. The two exponents differ by one minus sign, so the pair is as easy to write backwards as forwards, and the names are the only part that can be exchanged by mistake.

5.2 When the transform exists

Two conditions each guarantee that $X(j\omega)$ exists. They are alternatives, joined by "or", and neither implies the other.

1. **Finite energy.** $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$.
2. **The Dirichlet conditions.** x is absolutely integrable, $\int_{-\infty}^{\infty} |x(t)| dt < \infty$, and has finitely many maxima, minima and finite jumps in any finite interval.

That they are independent is shown by one example each way. The signal $\sin(t)/t$ has finite energy, and its modulus decays only like $1/|t|$, so its area diverges. The signal $1/\sqrt{t}$ on $0 < t < 1$ has area 2, and its square is $1/t$, whose integral over the same interval diverges.

Neither condition is necessary. A constant, a complex exponential and every periodic signal fail both and still have spectra, in the limiting sense: the transform is an impulse, defined by what it does inside an integral, exactly as $\delta(t)$ is. Take the pair $e^{-a|t|} \leftrightarrow 2a/(a^2 + \omega^2)$ from Example 5.4 and let $a \rightarrow 0$. The signal tends to the constant 1. The transform grows tall at $\omega = 0$, where it equals $2/a$, and narrow, since it has fallen to half that height at $\omega = \pm a$. Its area does not depend on a at all:

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{2a}{a^2 + \omega^2} d\omega &= \left[2 \tan^{-1} \frac{\omega}{a} \right]_{-\infty}^{\infty} \\ &= 2\left(\frac{\pi}{2}\right) - 2\left(-\frac{\pi}{2}\right) = 2\pi. \end{aligned}$$

A curve of fixed area 2π that becomes tall and narrow is an impulse of weight 2π . So $1 \leftrightarrow 2\pi\delta(\omega)$.

5.3 Basic transform pairs

🔗 EXAMPLE 5.1 – THE IMPULSE AND THE SHIFTED IMPULSE

GIVEN $x(t) = \delta(t)$, and then $x(t) = \delta(t - t_0)$.

FIND $X(j\omega)$ in both cases, with magnitude and phase.

METHOD The signal is an impulse, so the sifting property evaluates the analysis integral directly. Substitute the impulse into the analysis equation.

SOLUTION Substitute $x(t) = \delta(t)$ into the analysis equation. The sifting property $\int \delta(t)g(t)dt = g(0)$ evaluates the exponential at $t = 0$:

$$\begin{aligned}\mathcal{F}\{\delta(t)\} &= \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt \\ &= e^{-j\omega \cdot 0} = 1.\end{aligned}$$

Every frequency is present, equally, with no phase. For the shifted impulse the sifting happens at $t = t_0$, because $\delta(t - t_0)$ is zero everywhere except at that instant:

$$\begin{aligned}\mathcal{F}\{\delta(t - t_0)\} &= \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j\omega t} dt \\ &= e^{-j\omega t_0}.\end{aligned}$$

Write this in polar form: $e^{-j\omega t_0}$ has modulus 1 and angle $-\omega t_0$, so

$$|X(j\omega)| = 1, \quad \angle X(j\omega) = -\omega t_0.$$

CHECK

Push $e^{-j\omega t_0}$ back through the synthesis equation. Combine the two exponentials first:

$$\begin{aligned}x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-j\omega t_0} e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega(t-t_0)} d\omega \\ &= \frac{1}{2\pi} \cdot 2\pi \delta(t - t_0) = \delta(t - t_0).\end{aligned}$$

The last line uses $\int e^{j\omega\tau} d\omega = 2\pi\delta(\tau)$, which is the pair $1 \leftrightarrow 2\pi\delta(\omega)$ of Section 5.2 with the roles of t and ω exchanged. Reading: moving a signal in time never changes the size of any frequency component; it rotates each one by an amount proportional to its frequency.

EXAMPLE 5.2 – ONE IMPULSE IN FREQUENCY

GIVEN $X(j\omega) = 2\pi\delta(\omega - \omega_0)$.

FIND $x(t)$.

METHOD The spectrum is an impulse, so use the synthesis equation and apply sifting in the variable ω .

SOLUTION Substitute the spectrum into the synthesis equation. The impulse sits at $\omega = \omega_0$, so sifting evaluates $e^{j\omega t}$ there:

$$\begin{aligned}x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \delta(\omega - \omega_0) e^{j\omega t} d\omega \\ &= \frac{2\pi}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega \\ &= e^{j\omega_0 t}.\end{aligned}$$

The 2π of the impulse weight and the $1/2\pi$ of the synthesis equation cancel exactly, which is why the weight is written as 2π and not as 1.

CHECK $|x(t)| = 1$ and $\angle x(t) = \omega_0 t$. With an impulse of weight 1 the answer would be $e^{j\omega_0 t}/2\pi$, which is not a unit-amplitude exponential.

TWO CONSEQUENCES OF THE COMPLEX-EXPONENTIAL PAIR

$$e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)$$

$$1 \longleftrightarrow 2\pi\delta(\omega)$$

The second is the first at $\omega_0 = 0$.

EXAMPLE 5.3 – THE ONE-SIDED EXPONENTIAL

GIVEN $x(t) = e^{-at}u(t)$, with a real.

FIND $X(j\omega)$, the condition on a , and the magnitude and phase.

METHOD The unit step makes the signal zero for negative time. Apply the analysis equation from 0 to ∞ .

SOLUTION Substitute the signal into the analysis equation. The step $u(t)$ is zero for $t < 0$ and 1 for $t > 0$, so the lower limit becomes 0 and the step disappears. Then combine the two exponentials into one:

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} e^{-at}u(t) e^{-j\omega t} dt \\ &= \int_0^{\infty} e^{-at} e^{-j\omega t} dt \\ &= \int_0^{\infty} e^{-(a+j\omega)t} dt. \end{aligned}$$

The integrand is an exponential in t with constant coefficient $-(a + j\omega)$, so its antiderivative is the same exponential divided by that coefficient:

$$\begin{aligned} X(j\omega) &= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty} \\ &= \frac{1}{-(a+j\omega)} \left[\lim_{t \rightarrow \infty} e^{-(a+j\omega)t} - e^0 \right] \\ &= \frac{1}{-(a+j\omega)} [0 - 1] = \frac{1}{a+j\omega}. \end{aligned}$$

The limit at the upper end is zero only when $a > 0$: $|e^{-(a+j\omega)t}| = e^{-at}|e^{-j\omega t}| = e^{-at}$, because $|e^{-j\omega t}| = 1$, and $e^{-at} \rightarrow 0$ needs $a > 0$. For $a \leq 0$ the integral does not converge and the transform does not exist. So

$$e^{-at}u(t) \longleftrightarrow \frac{1}{a+j\omega}, \quad a > 0.$$

For the magnitude, the modulus of a quotient is the quotient of the moduli, and $|a + j\omega| = \sqrt{a^2 + \omega^2}$:

$$|X(j\omega)| = \frac{|1|}{|a + j\omega|} = \frac{1}{\sqrt{a^2 + \omega^2}}.$$

For the phase, the angle of a quotient is the angle of the numerator minus the angle of the denominator. The number $a + j\omega$ has real part $a > 0$ and imaginary part ω , so its angle is $\tan^{-1}(\omega/a)$:

$$\angle X(j\omega) = \angle 1 - \angle(a + j\omega) = 0 - \tan^{-1}(\omega/a) = -\tan^{-1}(\omega/a).$$

CHECK

$|X(j0)| = 1/a$ gives 10, 1 and 0.2 for $a = 0.1, 1, 5$. At $a = 1, \omega = 1$ the phase is -0.785398 rad; at $a = 0.1, \omega = 3$ it is -1.537475 ; at $a = 5, \omega = 2$ it is -0.380506 . Every one is negative, and the phase curve falls from $+\pi/2$ to $-\pi/2$. A curve that rises belongs to $+\tan^{-1}(\omega/a)$, which is the angle of the reciprocal.

⚠ WHERE THAT MINUS SIGN IS LOST

The numerator is the real number 1, so its angle is zero and it is tempting to skip that term and copy $\tan^{-1}(\omega/a)$ straight out of the denominator. What is then reported is $\angle(a + j\omega)$, the angle of the reciprocal of the answer. Write the subtraction out, including the term that is zero.

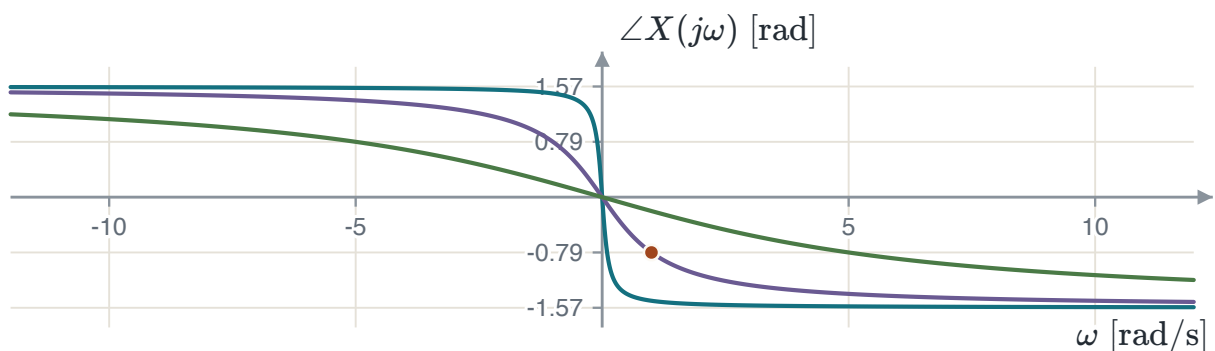


FIGURE 5.4 The phase of $1/(a + j\omega)$ for $a = 0.1, 1$ and 5 , with the marked value -0.785398 at $a = \omega = 1$.

✂ EXAMPLE 5.4 – THE TWO-SIDED EXPONENTIAL

GIVEN $x(t) = e^{-a|t|}$ with $a > 0$.

FIND $X(j\omega)$.

METHOD The absolute value gives different exponential formulas for negative and positive time. Split the analysis integral at $t = 0$; on the negative-time interval the exponent is $+at$.

SOLUTION For $t < 0$ the signal is $e^{-a(-t)} = e^{at}$, and for $t > 0$ it is e^{-at} . Split the analysis integral at $t = 0$ and combine the exponentials in each part:

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^0 e^{at} e^{-j\omega t} dt + \int_0^{\infty} e^{-at} e^{-j\omega t} dt \\ &= \int_{-\infty}^0 e^{(a-j\omega)t} dt + \int_0^{\infty} e^{-(a+j\omega)t} dt. \end{aligned}$$

The second integral is Example 5.3 and equals $1/(a + j\omega)$. Evaluate the first with its antiderivative. At the lower end, $|e^{(a-j\omega)t}| = e^{at} \rightarrow 0$ as $t \rightarrow -\infty$ because $a > 0$:

$$\begin{aligned}\int_{-\infty}^0 e^{(a-j\omega)t} dt &= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0 \\ &= \frac{e^0 - 0}{a-j\omega} = \frac{1}{a-j\omega}.\end{aligned}$$

Add the two fractions over the common denominator $(a-j\omega)(a+j\omega) = a^2 - (j\omega)^2 = a^2 + \omega^2$:

$$\begin{aligned}X(j\omega) &= \frac{1}{a-j\omega} + \frac{1}{a+j\omega} \\ &= \frac{(a+j\omega) + (a-j\omega)}{(a-j\omega)(a+j\omega)} \\ &= \frac{2a}{a^2 + \omega^2}.\end{aligned}$$

The imaginary parts $+j\omega$ and $-j\omega$ cancelled in the numerator, so the transform is real.

CHECK

$X(j0) = 2/a$ gives 4, 2 and 0.4 for $a = 0.5, 1, 5$. The signal is real and even and the transform came out real and even, which is the general rule proved in Section 5.6. Note also that the transform is never zero: at $a = 1$ and $\omega = 10^6$ it is still 2×10^{-12} .

5.4 The rectangular pulse, the sinc, and the inverse relation

EXAMPLE 5.5 – THE RECTANGULAR PULSE

GIVEN $x(t) = 1$ for $|t| < T_1$ and 0 otherwise.

FIND $X(j\omega)$, its value at the origin, and its zeros.

METHOD The signal is non-zero only on $-T_1 < t < T_1$, so restrict the analysis integral to that support and integrate $e^{-j\omega t}$.

SOLUTION Substitute the pulse into the analysis equation. Outside $|t| < T_1$ the signal is zero, so those parts of the integral contribute nothing, and inside the signal equals 1:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-T_1}^{T_1} 1 \cdot e^{-j\omega t} dt.$$

For $\omega \neq 0$ the antiderivative of $e^{-j\omega t}$ is $e^{-j\omega t}/(-j\omega)$. Evaluate at the two limits, then move the minus sign of the denominator into the numerator by swapping the two terms:

$$\begin{aligned}
 X(j\omega) &= \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-T_1}^{T_1} \\
 &= \frac{e^{-j\omega T_1} - e^{j\omega T_1}}{-j\omega} \\
 &= \frac{e^{j\omega T_1} - e^{-j\omega T_1}}{j\omega}.
 \end{aligned}$$

Euler gives $e^{j\theta} - e^{-j\theta} = (\cos \theta + j \sin \theta) - (\cos \theta - j \sin \theta) = 2j \sin \theta$. With $\theta = \omega T_1$:

$$X(j\omega) = \frac{2j \sin(\omega T_1)}{j\omega} = \frac{2 \sin(\omega T_1)}{\omega}.$$

The j cancels, so the answer is real. At $\omega = 0$ the integrand is 1 and the integral is the width $2T_1$ directly.

CHECK

At the origin the formula reads $0/0$. L'Hôpital, differentiating numerator and denominator with respect to ω , gives $\lim_{\omega \rightarrow 0} 2T_1 \cos(\omega T_1)/1 = 2T_1$, which agrees with the direct value and is the area under the pulse: 2, 10 and 20 for $T_1 = 1, 5, 10$. The zeros are where $\sin(\omega T_1) = 0$ with $\omega \neq 0$, that is $\omega T_1 = \pm k\pi$, so $\omega = \pm k\pi/T_1$ for $k = 1, 2, 3, \dots$. The origin is excluded, because there the limit is the peak, not a zero.

① THE SINC CONVENTION USED THROUGHOUT

This course writes $\text{sinc}(\theta) = \frac{\sin \theta}{\theta}$, the **unnormalised** sinc, with $\text{sinc}(0) = 1$ and zeros at $\pm\pi, \pm 2\pi, \dots$. In that convention the result above reads $X(j\omega) = 2T_1 \text{sinc}(\omega T_1)$. Signal-processing software and many communications texts use the **normalised** sinc, $\text{sinc}_n(\theta) = \sin(\pi\theta)/(\pi\theta)$, whose zeros are at the integers; there the same result is $2T_1 \text{sinc}_n(\omega T_1/\pi)$. The two expressions are equal, and the argument is not: it differs by a factor of π . An argument copied between the conventions agrees at the origin and is wrong everywhere else, so the convention is stated at every point of use.

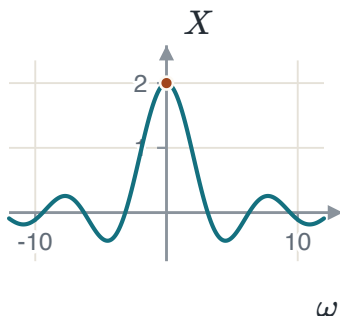


FIGURE 5.5 $T_1 = 1$: peak 2.

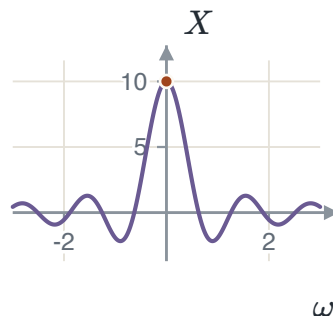


FIGURE 5.6 $T_1 = 5$: peak 10.

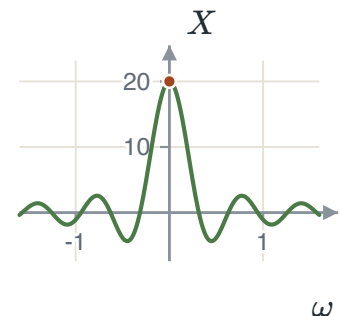


FIGURE 5.7 $T_1 = 10$: peak 20.

🔗 EXAMPLE 5.6 – THE IDEAL LOW-PASS BAND

GIVEN $X(j\omega) = 1$ for $|\omega| < W$ and 0 otherwise.

FIND $x(t)$, its peak, and its zeros.

METHOD The spectrum is given and is non-zero only on $-W < \omega < W$, so apply the synthesis equation over those limits and retain the factor $1/2\pi$.

SOLUTION Substitute the spectrum into the synthesis equation. Outside $|\omega| < W$ it is zero, and inside it equals 1:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-W}^W e^{j\omega t} d\omega.$$

Here t is a fixed parameter and ω is the variable, so for $t \neq 0$ the antiderivative of $e^{j\omega t}$ is $e^{j\omega t}/(jt)$:

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \left[\frac{e^{j\omega t}}{jt} \right]_{-W}^W \\ &= \frac{e^{jWt} - e^{-jWt}}{2\pi jt} \\ &= \frac{2j \sin(Wt)}{2\pi jt} \\ &= \frac{\sin(Wt)}{\pi t}. \end{aligned}$$

The third line is Euler again, $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$ with $\theta = Wt$; the $2j$ then cancels against the $2\pi j$. With the unnormalised sinc, $\text{sinc}(\theta) = \sin \theta / \theta$, multiply and divide by W :

$$x(t) = \frac{W}{\pi} \cdot \frac{\sin(Wt)}{Wt} = \frac{W}{\pi} \text{sinc}(Wt).$$

CHECK At $t = 0$ the synthesis integral is $\frac{1}{2\pi} \int_{-W}^W 1 d\omega = \frac{2W}{2\pi} = \frac{W}{\pi}$, and l'Hôpital on $\sin(Wt)/(\pi t)$ gives $\lim_{t \rightarrow 0} W \cos(Wt)/\pi = W/\pi$ as well: 0.5, 1 and 2 for $W = 0.5\pi, \pi, 2\pi$. Zeros where $\sin(Wt) = 0$ with $t \neq 0$: $t = \pm k\pi/W$, $k = 1, 2, \dots$, again with the origin excluded. This signal is not a pulse: it rings on both sides for ever, alternating in sign.

The two examples are one statement seen twice: a rectangle in either domain is a sinc in the other. Section 5.7 gives that symmetry a name and a proof.

Narrow in time, wide in frequency

The pulse of half-width T_1 has its first zero at $\omega = \pi/T_1$. Halve T_1 and that zero doubles. Taking the duration as the full width $T = 2T_1$ and the bandwidth as the distance to the first null,

$$T \times \text{BW} = 2T_1 \cdot \frac{\pi}{T_1} = 2\pi \quad \text{at every width.}$$

△ THE VALUE DEPENDS ON THE WAVEFORM

The product is invariant for scaled versions of one waveform because time scaling changes duration and frequency width by reciprocal factors. A triangular pulse of the same total duration $2T_1$ has the transform $T_1 \text{sinc}^2(\omega T_1/2)$, with $\text{sinc}(\theta) = \sin \theta / \theta$, because that triangle is the convolution of two rectangles of half-width $T_1/2$ and convolution multiplies transforms (Section 5.9). Its first null is where $\omega T_1/2 = \pi$, that is at $\omega = 2\pi/T_1$, so its product is $2T_1 \cdot 2\pi/T_1 = 4\pi$. In general, narrowing a signal in time widens its spectrum; both widths cannot be made arbitrarily small.

One implication about band limitation is a theorem and its converse is not. A signal of finite duration cannot be band-limited: the pulse above has a sinc transform, which is non-zero on stretches reaching out to every frequency. But infinite duration guarantees nothing. The signal $e^{-a|t|}$ lasts for ever and its transform $2a/(a^2 + \omega^2)$ is strictly positive at every finite frequency. Small is not zero, and Chapter 7 depends on the difference.

5.5 Periodic signals, sinusoids and the impulse train

A periodic signal has a Fourier series, and the series can be transformed term by term. Linearity moves the transform inside the sum and past each constant a_k . Each term is then a complex exponential at frequency $k\omega_0$, and Example 5.2 gives its transform, $e^{jk\omega_0 t} \leftrightarrow 2\pi\delta(\omega - k\omega_0)$:

$$\begin{aligned} X(j\omega) &= \mathcal{F}\left\{ \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \right\} \\ &= \sum_{k=-\infty}^{\infty} a_k \mathcal{F}\{e^{jk\omega_0 t}\} \\ &= \sum_{k=-\infty}^{\infty} a_k \cdot 2\pi \delta(\omega - k\omega_0). \end{aligned}$$

TRANSFORM OF A PERIODIC SIGNAL

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longleftrightarrow X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

The spectrum is a train of impulses at the harmonics, and the impulse at $k\omega_0$ carries weight $2\pi a_k$. The coefficient and the weight are different objects: a_k multiplies a unit exponential, while $2\pi a_k$ is an area under a spectrum. Reporting the coefficients as the transform loses a factor of 2π at every harmonic.

For the rectangular wave of Chapter 4, with value 1 on $|t| < T_1$ in each period T , the coefficients are $a_k = \sin(k\omega_0 T_1)/(k\pi)$ for $k \neq 0$ and $a_0 = 2T_1/T$. Multiply by 2π to get the impulse weights:

$$2\pi a_k = 2\pi \cdot \frac{\sin(k\omega_0 T_1)}{k\pi} = \frac{2 \sin(k\omega_0 T_1)}{k}, \quad 2\pi a_0 = 2\pi \cdot \frac{2T_1}{T} = \frac{4\pi T_1}{T}.$$

Take $T_1 = 1$. Three periods make the sequence visible. $T = 8T_1$ gives $\omega_0 = 2\pi/8 = \pi/4 = 0.392699$ and origin weight $4\pi/8 = \pi/2 = 1.5708$. $T = 16T_1$ gives $\omega_0 = \pi/8 = 0.196350$ and weight $4\pi/16 = 0.7854$. $T = 32T_1$ gives $\omega_0 = \pi/16 = 0.098175$ and weight $4\pi/32 = 0.3927$.

Each doubling of the period halves the spacing and halves every weight, while the envelope $2 \sin(\omega T_1)/\omega$ does not move at all. This is the derivation of Section 5.1, carried out three steps at a time.

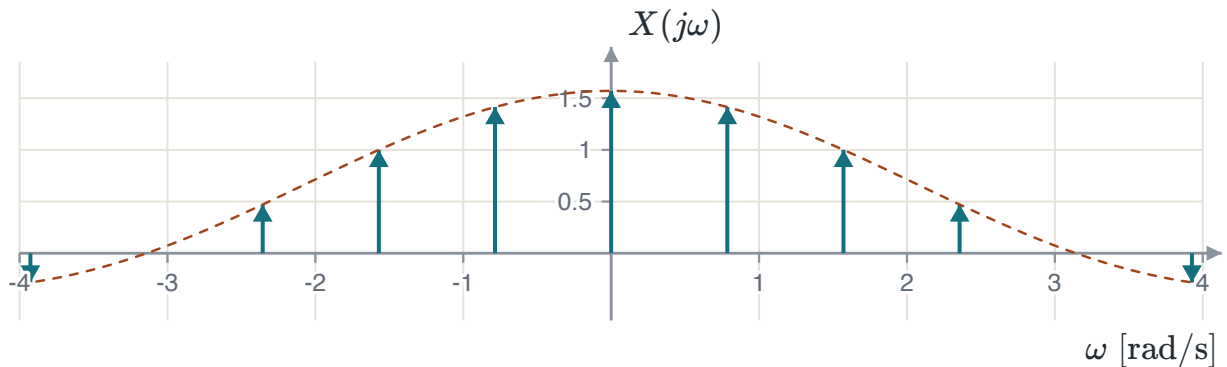


FIGURE 5.8 The square wave with $T = 8T_1$: impulses of weight $2\pi a_k$ sample the dashed envelope. A magnitude plot would not show which weights are negative.

EXAMPLE 5.7 – A CONSTANT, A COSINE AND A SINE

GIVEN $x(t) = 5 + 4 \cos(3\pi t) + 6 \sin(4\pi t)$.

FIND $X(j\omega)$, drawn so that both size and phase can be read.

METHOD Each sinusoid is a sum of complex exponentials, whose transform pair is known. Expand with Euler's relations and apply linearity term by term.

SOLUTION Write each term as complex exponentials with Euler's relations, $\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$ and $\sin \theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$:

$$\begin{aligned} 4 \cos(3\pi t) &= 4 \cdot \frac{1}{2} (e^{j3\pi t} + e^{-j3\pi t}) = 2e^{j3\pi t} + 2e^{-j3\pi t}, \\ 6 \sin(4\pi t) &= 6 \cdot \frac{1}{2j} (e^{j4\pi t} - e^{-j4\pi t}) = \frac{3}{j}e^{j4\pi t} - \frac{3}{j}e^{-j4\pi t}. \end{aligned}$$

Now apply $e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$ to each exponential and multiply by its coefficient. The constant is the case $\omega_0 = 0$:

$$\begin{aligned} 5 &\longleftrightarrow 5 \cdot 2\pi\delta(\omega) = 10\pi\delta(\omega), \\ 2e^{j3\pi t} + 2e^{-j3\pi t} &\longleftrightarrow 2 \cdot 2\pi\delta(\omega - 3\pi) + 2 \cdot 2\pi\delta(\omega + 3\pi) \\ &= 4\pi\delta(\omega - 3\pi) + 4\pi\delta(\omega + 3\pi), \\ \frac{3}{j}e^{j4\pi t} - \frac{3}{j}e^{-j4\pi t} &\longleftrightarrow \frac{3}{j} \cdot 2\pi\delta(\omega - 4\pi) - \frac{3}{j} \cdot 2\pi\delta(\omega + 4\pi) \\ &= \frac{6\pi}{j}\delta(\omega - 4\pi) - \frac{6\pi}{j}\delta(\omega + 4\pi). \end{aligned}$$

The sine gives imaginary weights, since $1/j = -j$, of modulus $6\pi = 18.849556$. Linearity adds the three transforms:

$$\begin{aligned} X(j\omega) &= 10\pi\delta(\omega) + 4\pi\delta(\omega - 3\pi) + 4\pi\delta(\omega + 3\pi) \\ &\quad + \frac{6\pi}{j}\delta(\omega - 4\pi) - \frac{6\pi}{j}\delta(\omega + 4\pi). \end{aligned}$$

CHECK The signal is real, so the magnitude must be even in ω and the phase odd. Magnitudes: 31.4159 at 0, 12.5664 at $\pm 3\pi$, 18.8496 at $\pm 4\pi$. Phases: 0 except at $\pm 4\pi$, where they

are $\mp\pi/2$ because $1/j = -j$. A spectrum with only the positive-frequency impulse violates conjugate symmetry and describes a complex signal.

△ DRAWING A COMPLEX SPECTRUM

Three weights are real and two are imaginary. One vertical axis cannot represent both signed real values and signed imaginary values. Plot magnitude and phase, or plot real part and imaginary part, and identify the chosen pair in the caption.

✂ EXAMPLE 5.8 – THE IMPULSE TRAIN

GIVEN $x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$.

FIND $X(j\omega)$.

METHOD The signal is periodic, so its transform is an impulse at each harmonic. Find its Fourier-series coefficients a_k , then give each impulse the weight $2\pi a_k$.

SOLUTION The signal repeats every T , so $\omega_0 = 2\pi/T$. Apply the analysis equation of the Fourier series over the period $-T/2 < t < T/2$. That interval encloses exactly one impulse, the one at $t = 0$, so inside it the signal is just $\delta(t)$ and sifting evaluates the exponential at $t = 0$:

$$\begin{aligned} a_k &= \frac{1}{T} \int_{-T/2}^{T/2} \sum_m \delta(t - mT) e^{-jk\omega_0 t} dt \\ &= \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t} dt \\ &= \frac{1}{T} e^0 = \frac{1}{T} \quad \text{for every } k. \end{aligned}$$

Now use the transform of a periodic signal: each harmonic $k\omega_0 = 2\pi k/T$ carries an impulse of weight $2\pi a_k = 2\pi/T$:

$$\begin{aligned} X(j\omega) &= \sum_k 2\pi a_k \delta(\omega - k\omega_0) \\ &= \sum_k \frac{2\pi}{T} \delta\left(\omega - \frac{2\pi k}{T}\right). \end{aligned}$$

Hence

$$\sum_k \delta(t - kT) \longleftrightarrow \frac{2\pi}{T} \sum_k \delta\left(\omega - \frac{2\pi k}{T}\right).$$

CHECK Spacing and weight are the same number, $2\pi/T$: 6.2832 for $T = 1$ and 3.1416 for $T = 2$. Crowding the impulses in time spreads them in frequency. This one pair is the mechanism behind sampling, in Chapter 7. Note that the limits must enclose exactly one impulse; writing both as $-T/2$ encloses none.

5.6 Properties

TABLE 5.1

PROPERTY	STATEMENT
Linearity	$ax_1 + bx_2 \leftrightarrow aX_1 + bX_2$
Time shift	$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$
Frequency shift	$e^{j\omega_0 t} x(t) \leftrightarrow X(j(\omega - \omega_0))$
Conjugation	$x^*(t) \leftrightarrow X^*(-j\omega)$
Real signal	$X(-j\omega) = X^*(j\omega)$
Time reversal	$x(-t) \leftrightarrow X(-j\omega)$
Time scaling	$x(at) \leftrightarrow \frac{1}{ a } X(j\omega/a)$
Differentiation in time	$\mathrm{d}^n x / \mathrm{d}t^n \leftrightarrow (j\omega)^n X(j\omega)$
Integration	$\int_{-\infty}^t x(\tau) \mathrm{d}\tau \leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0) \delta(\omega)$
Differentiation in frequency	$t x(t) \leftrightarrow j \mathrm{d}X(j\omega) / \mathrm{d}\omega$
Duality	$X(t) \leftrightarrow 2\pi x(-\omega)$
Convolution	$x * h \leftrightarrow X H$
Multiplication	$x y \leftrightarrow \frac{1}{2\pi} X * Y$
Real and even	$X(j\omega)$ real and even
Real and odd	$X(j\omega)$ purely imaginary and odd
Even-odd parts	$\mathcal{E}\mathcal{V}\{x\} \leftrightarrow \operatorname{Re}\{X\}, \mathcal{O}\mathcal{D}\mathcal{D}\{x\} \leftrightarrow j \operatorname{Im}\{X\}$
Parseval	$\int x ^2 \mathrm{d}t = \frac{1}{2\pi} \int X ^2 \mathrm{d}\omega$

Shifts

For the time shift, put $x(t - t_0)$ into the analysis equation and substitute $\tau = t - t_0$, so $t = \tau + t_0$ and $\mathrm{d}t = \mathrm{d}\tau$. The limits are infinite and do not move, because $\tau \rightarrow \pm\infty$ exactly when $t \rightarrow \pm\infty$. Then split the exponential and take the factor that does not depend on τ outside:

$$\begin{aligned}
 \mathcal{F}\{x(t - t_0)\} &= \int_{-\infty}^{\infty} x(t - t_0) e^{-j\omega t} \mathrm{d}t \\
 &= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega(\tau + t_0)} \mathrm{d}\tau \\
 &= e^{-j\omega t_0} \int_{-\infty}^{\infty} x(\tau) e^{-j\omega \tau} \mathrm{d}\tau \\
 &= e^{-j\omega t_0} X(j\omega).
 \end{aligned}$$

The last integral is the analysis equation with τ as the dummy variable, so it is $X(j\omega)$. Since $|e^{-j\omega t_0}| = 1$, no magnitude changes at any frequency: a delay of t_0 seconds is exactly a linear phase of slope $-t_0$.

For the frequency shift, start from the expression the property states, $e^{+j\omega_0 t}x(t)$, and combine the two exponentials before anything else. The exponent becomes $-j(\omega - \omega_0)t$, which is the analysis integral read at $\omega - \omega_0$, and no substitution is needed:

$$\begin{aligned}\mathcal{F}\{e^{j\omega_0 t}x(t)\} &= \int_{-\infty}^{\infty} e^{j\omega_0 t}x(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} x(t) e^{-j(\omega - \omega_0)t} dt \\ &= X(j(\omega - \omega_0)).\end{aligned}$$

The operand is a complex exponential in **time** with ω_0 fixed; the time-shift kernel $e^{-j\omega t_0}$ has a fixed **time** and the opposite sign, and opening this proof with it proves a different statement.

Conjugation and the symmetry of a real signal

Conjugation. The conjugate of an integral is the integral of the conjugate, and conjugating a product conjugates each factor. The conjugate of $e^{-j\omega t}$ is $e^{+j\omega t}$, which is the analysis kernel evaluated at $-\omega$:

$$\begin{aligned}\mathcal{F}\{x^*(t)\} &= \int_{-\infty}^{\infty} x^*(t) e^{-j\omega t} dt \\ &= \left[\int_{-\infty}^{\infty} x(t) e^{+j\omega t} dt \right]^* \\ &= \left[\int_{-\infty}^{\infty} x(t) e^{-j(-\omega)t} dt \right]^* = [X(-j\omega)]^* = X^*(-j\omega).\end{aligned}$$

Time reversal. Put $x(-t)$ into the analysis equation and substitute $\tau = -t$, so $t = -\tau$ and $dt = -d\tau$. As t runs from $-\infty$ to ∞ , τ runs from ∞ to $-\infty$, so the limits arrive reversed. Swapping them back absorbs the minus sign of $dt = -d\tau$:

$$\begin{aligned}\mathcal{F}\{x(-t)\} &= \int_{-\infty}^{\infty} x(-t) e^{-j\omega t} dt \\ &= \int_{\infty}^{-\infty} x(\tau) e^{-j\omega(-\tau)} (-d\tau) \\ &= \int_{-\infty}^{\infty} x(\tau) e^{-j(-\omega)\tau} d\tau = X(-j\omega).\end{aligned}$$

A real signal. If x is real then $x^*(t) = x(t)$, so the two sides of the conjugation property have the same transform: $X(j\omega) = X^*(-j\omega)$, or, conjugating both sides, $X(-j\omega) = X^*(j\omega)$. Everything follows from that one line. Write $X(j\omega) = R(\omega) + jI(\omega)$ with R and I real. Then $X(-j\omega) = R(-\omega) + jI(-\omega)$ and $X^*(j\omega) = R(\omega) - jI(\omega)$. Equating real parts and imaginary parts gives $R(-\omega) = R(\omega)$ and $I(-\omega) = -I(\omega)$: the real part of X is even and the imaginary part is odd. Taking modulus and angle of the same equation, $|X(-j\omega)| = |X^*(j\omega)| = |X(j\omega)|$ and $\angle X(-j\omega) = \angle X^*(j\omega) = -\angle X(j\omega)$: the magnitude is even and the phase is odd.

A real and even signal. Evenness gives $x(-t) = x(t)$, so by time reversal $X(-j\omega) = X(j\omega)$. Realness gives $X(-j\omega) = X^*(j\omega)$. Together, $X(j\omega) = X^*(j\omega)$, so X is real; and $X(-j\omega) =$

$X(j\omega)$ says it is even. A real and odd signal has $x(-t) = -x(t)$, so $X(-j\omega) = -X(j\omega)$; with $X(-j\omega) = X^*(j\omega)$ this gives $X^*(j\omega) = -X(j\omega)$, so X is purely imaginary, and odd.

⚠ REAL DOES NOT MEAN ZERO PHASE

A real transform can be negative. Where $X(j\omega) < 0$ the magnitude is $-X$ and the phase is π , not 0. Only a real and non-negative transform has zero phase everywhere, and the sinc of the rectangular pulse is the standing counterexample: it is real, and its side lobes are negative.

Differentiation

Differentiate the **synthesis equation**, not the signal. On the right the only factor depending on t is $e^{j\omega t}$, and ω is the variable of integration, so it is held fixed. The derivative of $e^{j\omega t}$ with respect to t is $j\omega e^{j\omega t}$:

$$\begin{aligned}\frac{d}{dt}x(t) &= \frac{d}{dt} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \right] \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \frac{\partial}{\partial t} e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} [j\omega X(j\omega)] e^{j\omega t} d\omega.\end{aligned}$$

The last line is the synthesis equation applied to the function $j\omega X(j\omega)$. So $j\omega X(j\omega)$ is the transform of dx/dt . Differentiating again multiplies by another $j\omega$, and after n derivatives:

$$\frac{d^n x}{dt^n} \longleftrightarrow (j\omega)^n X(j\omega).$$

⊗ DO NOT MIX THE TWO DOMAINS

The equation $dx/dt = j\omega x(t)$ is false. The frequency ω labels the transform output; it is not a constant in the time-domain signal. For $x(t) = e^{-t^2}$, dx/dt at $t = 1$ is -0.735759 , while $j\omega x(t)$ at $t = 1$, $\omega = 3$ is $1.103638j$. The differentiation property is correct only when derived from the transform integral.

Integration, and the impulse it leaves behind

The running integral is a convolution with the unit step. In the convolution integral, the factor $u(t - \tau)$ equals 1 for $\tau < t$ and 0 for $\tau > t$, so it cuts the upper limit at t :

$$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau) u(t - \tau) d\tau = \int_{-\infty}^t x(\tau) d\tau.$$

The transform of the step is needed next. The step fails both existence conditions, so obtain it as a limit. Write $u(t) = \frac{1}{2} + \frac{1}{2} \operatorname{sgn}(t)$, where $\operatorname{sgn}(t)$ is $+1$ for $t > 0$ and -1 for $t < 0$. The constant $\frac{1}{2}$ transforms to $\frac{1}{2} \cdot 2\pi\delta(\omega) = \pi\delta(\omega)$. For the sign function, use $\operatorname{sgn}(t) = \lim_{a \rightarrow 0} [e^{-at}u(t) - e^{at}u(-t)]$ with $a > 0$; the first term is Example 5.3, and the second is its time reversal, so its transform is $1/(a - j\omega)$. Add the fractions and let $a \rightarrow 0$:

$$\begin{aligned}
\mathcal{F}\{\text{sgn}(t)\} &= \lim_{a \rightarrow 0} \left[\frac{1}{a + j\omega} - \frac{1}{a - j\omega} \right] \\
&= \lim_{a \rightarrow 0} \frac{(a - j\omega) - (a + j\omega)}{a^2 + \omega^2} = \lim_{a \rightarrow 0} \frac{-2j\omega}{a^2 + \omega^2} \\
&= \frac{-2j\omega}{\omega^2} = \frac{-2j}{\omega} = \frac{2}{j\omega},
\end{aligned}$$

using $-j = 1/j$ in the last step. So $u(t) \leftrightarrow \pi\delta(\omega) + \frac{1}{2} \cdot \frac{2}{j\omega} = \frac{1}{j\omega} + \pi\delta(\omega)$. Now the convolution property of Section 5.9 multiplies the transforms. In the product, $X(j\omega)\delta(\omega) = X(j0)\delta(\omega)$, because the impulse is zero away from $\omega = 0$ and only the value of X at the origin survives. Writing $X(0)$ for $X(j0)$:

$$\begin{aligned}
\mathcal{F}\left\{\int_{-\infty}^t x(\tau) d\tau\right\} &= X(j\omega) \left[\frac{1}{j\omega} + \pi\delta(\omega) \right] \\
&= \frac{1}{j\omega} X(j\omega) + \pi X(0) \delta(\omega).
\end{aligned}$$

⊗ INCLUDE THE IMPULSE TERM WHEN THE AREA IS NON-ZERO

$X(0) = \int x(t) dt$ is the total signal area. If this area is non-zero, the running integral approaches a non-zero constant, whose transform includes an impulse at $\omega = 0$. Writing only $X(j\omega)/(j\omega)$ omits that constant component. Calculate the area before applying the integration property.

Differentiation in frequency

Differentiate the analysis integral with respect to ω . Now t is the variable of integration and is held fixed, and the derivative of $e^{-j\omega t}$ with respect to ω is $-jt e^{-j\omega t}$:

$$\begin{aligned}
\frac{dX(j\omega)}{d\omega} &= \int_{-\infty}^{\infty} x(t) \frac{\partial}{\partial \omega} e^{-j\omega t} dt \\
&= \int_{-\infty}^{\infty} [-jt x(t)] e^{-j\omega t} dt = \mathcal{F}\{-jt x(t)\}.
\end{aligned}$$

So $dX/d\omega$ is the transform of $-jt x(t)$. Multiply both sides by j and use $j \cdot (-j) = 1$: $j dX/d\omega = \mathcal{F}\{t x(t)\}$, which is the property. It obtains the pair for $te^{-at}u(t)$ from the pair for $e^{-at}u(t)$ without another transform integral. Write $1/(a + j\omega) = (a + j\omega)^{-1}$ and use the chain rule, with $d(a + j\omega)/d\omega = j$:

$$\begin{aligned}
\mathcal{F}\{te^{-at}u(t)\} &= j \frac{d}{d\omega} \left[\frac{1}{a + j\omega} \right] = j \cdot \frac{-1 \cdot j}{(a + j\omega)^2} \\
&= \frac{-j^2}{(a + j\omega)^2} = \frac{1}{(a + j\omega)^2}, \quad a > 0.
\end{aligned}$$

Even and odd parts

For a real signal, transform the even part $\mathcal{E}\{x\} = \frac{1}{2}[x(t) + x(-t)]$ by linearity. Time reversal gives $x(-t) \leftrightarrow X(-j\omega)$, and for a real signal $X(-j\omega) = X^*(j\omega)$. Adding a complex number to its conjugate leaves twice the real part:

$$\begin{aligned}
\mathcal{F}\{\mathcal{E}\{x\}\} &= \frac{1}{2}[X(j\omega) + X(-j\omega)] = \frac{1}{2}[X(j\omega) + X^*(j\omega)] = \text{Re}\{X(j\omega)\}, \\
\mathcal{F}\{\mathcal{O}\{x\}\} &= \frac{1}{2}[X(j\omega) - X(-j\omega)] = \frac{1}{2}[X(j\omega) - X^*(j\omega)] = j \text{Im}\{X(j\omega)\}.
\end{aligned}$$

The second line uses $X - X^* = 2j \operatorname{Im}\{X\}$. Splitting a real signal into even and odd parts splits its spectrum into real and imaginary parts, and the two symmetry rules above are the cases in which one part is absent. Check it on $x(t) = e^{-at}u(t)$. Its even part is $\frac{1}{2}[e^{-at}u(t) + e^{at}u(-t)] = \frac{1}{2}e^{-a|t|}$, and its odd part is $\frac{1}{2}[e^{-at}u(t) - e^{at}u(-t)] = \frac{1}{2}\operatorname{sgn}(t)e^{-a|t|}$. Transform each part from the pairs already found, then compare with the real and imaginary parts of $1/(a + j\omega)$, obtained by multiplying numerator and denominator by $a - j\omega$:

$$\begin{aligned}\frac{1}{2}e^{-a|t|} &\longleftrightarrow \frac{1}{2} \cdot \frac{2a}{a^2 + \omega^2} = \frac{a}{a^2 + \omega^2}, \\ \frac{1}{2}\operatorname{sgn}(t)e^{-a|t|} &\longleftrightarrow \frac{1}{2} \left[\frac{1}{a + j\omega} - \frac{1}{a - j\omega} \right] = \frac{1}{2} \cdot \frac{-2j\omega}{a^2 + \omega^2} = \frac{-j\omega}{a^2 + \omega^2}, \\ \frac{1}{a + j\omega} &= \frac{a - j\omega}{(a + j\omega)(a - j\omega)} = \frac{a}{a^2 + \omega^2} + j \frac{-\omega}{a^2 + \omega^2}.\end{aligned}$$

The even part gives exactly $\operatorname{Re}\{1/(a + j\omega)\}$ and the odd part gives exactly $j \operatorname{Im}\{1/(a + j\omega)\}$, as the property says.

Time scaling

Put $x(at)$ into the analysis equation and substitute $\tau = at$, so $t = \tau/a$ and $dt = d\tau/a$. For $a > 0$ the limits keep their order, since $\tau \rightarrow \pm\infty$ when $t \rightarrow \pm\infty$, and the coefficient is $1/a = 1/|a|$:

$$\begin{aligned}\mathcal{F}\{x(at)\} &= \int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau/a} \frac{d\tau}{a} \\ &= \frac{1}{a} \int_{-\infty}^{\infty} x(\tau) e^{-j(\omega/a)\tau} d\tau = \frac{1}{a} X\left(j\frac{\omega}{a}\right), \quad a > 0.\end{aligned}$$

For $a < 0$ the substitution sends $t \rightarrow -\infty$ to $\tau \rightarrow +\infty$ and $t \rightarrow +\infty$ to $\tau \rightarrow -\infty$, so the limits arrive reversed. Swapping them back costs one minus sign, and $-1/a = 1/|a|$ for negative a . That is the only sign in the calculation:

$$\begin{aligned}\mathcal{F}\{x(at)\} &= \int_{+\infty}^{-\infty} x(\tau) e^{-j(\omega/a)\tau} \frac{d\tau}{a} \\ &= -\frac{1}{a} \int_{-\infty}^{\infty} x(\tau) e^{-j(\omega/a)\tau} d\tau = \frac{1}{|a|} X\left(j\frac{\omega}{a}\right), \quad a < 0.\end{aligned}$$

Both cases read $x(at) \leftrightarrow \frac{1}{|a|} X(j\omega/a)$.

⊗ COUNTING THE FLIP TWICE

Writing the reversed limits **and** an explicit -1 in front applies the same correction twice and gives $-\frac{1}{|a|} X(j\omega/a)$, contradicting the property. Do the flip once: reverse the limits, or write the minus, never both. The case $a = -1$ is time reversal, $x(-t) \leftrightarrow X(-j\omega)$, with coefficient 1.

Worked instance. If $X(j\omega) = 1$ on $|\omega| < 2\pi$, then $x(0.5t) \leftrightarrow 2X(j2\omega)$, which is 2 on $|\omega| < \pi$, and $x(2t) \leftrightarrow 0.5X(j\omega/2)$, which is 0.5 on $|\omega| < 4\pi$. The three areas are $2 \cdot 2\pi$, $1 \cdot 4\pi$ and $0.5 \cdot 8\pi$, all 4π ; since $\int X d\omega = 2\pi x(0)$, the value at $t = 0$ is the same for all three, which is what a time scaling cannot change.

5.7 Duality

The two equations of the pair differ only by a sign and a factor, so any pair can be read a second time with the domains exchanged.

DUALITY

$$x(t) \longleftrightarrow X(j\omega) \implies X(t) \longleftrightarrow 2\pi x(-\omega)$$

The argument on the right is $-\omega$, a real number, and not $-j\omega$: the letter x names a signal and a signal takes a real argument. The j belongs to the frequency-domain function only, and duality is the point in the chapter where that distinction is most easily lost.

Here $X(t)$ means the formula for $X(j\omega)$ with ω replaced by t . Proof. Start from the synthesis equation and multiply by 2π :

$$2\pi x(t) = \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega.$$

The names of the variables carry no meaning, so rename them: call the variable of integration t and the free variable ω . The equation becomes

$$2\pi x(\omega) = \int_{-\infty}^{\infty} X(jt) e^{j\omega t} dt.$$

This holds for every real ω , so it holds with ω replaced by $-\omega$. The exponent $e^{j(-\omega)t} = e^{-j\omega t}$ is now the analysis kernel:

$$2\pi x(-\omega) = \int_{-\infty}^{\infty} X(jt) e^{-j\omega t} dt = \mathcal{F}\{X(t)\}.$$

The right side is the analysis equation applied to the time signal $X(t)$, so its transform is $2\pi x(-\omega)$.

EXAMPLE 5.9 – DUALITY ON THE RECTANGULAR PULSE

GIVEN $x_1(t) = 1$ on $|t| < W$, so $X_1(j\omega) = 2 \sin(W\omega)/\omega$.

FIND The transform of $x_2(t) = 2 \sin(Wt)/t$.

METHOD x_2 has the same formula as X_1 with the independent variable changed, so duality applies. Use duality first, then verify the result with synthesis.

SOLUTION The signal $x_2(t) = 2 \sin(Wt)/t$ is $X_1(j\omega)$ with ω replaced by t , that is $x_2(t) = X_1(t)$ in the notation of the duality box. Duality then gives $X_2(j\omega) = 2\pi x_1(-\omega)$. The pulse x_1 is even, so $x_1(-\omega) = x_1(\omega)$, which is 1 on $|\omega| < W$ and 0 beyond:

$$X_2(j\omega) = 2\pi x_1(-\omega) = 2\pi x_1(\omega) = \begin{cases} 2\pi, & |\omega| < W \\ 0, & |\omega| > W. \end{cases}$$

CHECK The second route works backwards. Put a band of height 2π on $|\omega| < W$ through the synthesis equation; for $t \neq 0$ the antiderivative of $e^{j\omega t}$ in ω is $e^{j\omega t}/(jt)$:

$$\begin{aligned}\frac{1}{2\pi} \int_{-W}^W 2\pi e^{j\omega t} d\omega &= \left[\frac{e^{j\omega t}}{jt} \right]_{-W}^W = \frac{e^{jWt} - e^{-jWt}}{jt} \\ &= \frac{2j \sin(Wt)}{jt} = \frac{2 \sin(Wt)}{t},\end{aligned}$$

which is x_2 . The two routes use different equations, so their agreement is a real check. At the origin, $X_2(j0) = 2\pi = 6.283185$ for any W , and $x_2(0) = \lim_{t \rightarrow 0} 2W \cos(Wt)/1 = 2W$ by l'Hôpital.

Duality also carries properties across: the time-shift rule becomes the frequency-shift rule, and differentiation in time becomes differentiation in frequency. Every pair already derived gives a second one for nothing.

5.8 Parseval's relation

Energy and power in this course are normalised, with $R = 1 \Omega$, so the instantaneous power is $|x(t)|^2$ and the energy is its integral, in joules.

PARSEVAL'S RELATION

$$E_\infty = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

The quantity $|X(j\omega)|^2$ is the energy spectral density: $\frac{1}{2\pi}|X|^2 d\omega$ is the energy in a narrow band.

Proof. Write $|x(t)|^2 = x(t) x^*(t)$. Replace $x^*(t)$ by the conjugate of the synthesis equation; conjugating turns $X(j\omega)$ into $X^*(j\omega)$ and $e^{j\omega t}$ into $e^{-j\omega t}$. Then exchange the order of the two integrals, so that the t integral is done first. The inner integral is the analysis equation and equals $X(j\omega)$:

$$\begin{aligned}\int_{-\infty}^{\infty} |x(t)|^2 dt &= \int_{-\infty}^{\infty} x(t) x^*(t) dt \\ &= \int_{-\infty}^{\infty} x(t) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega) e^{-j\omega t} d\omega \right] dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega) \left[\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right] d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega) X(j\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega.\end{aligned}$$

For a real signal, $|x(t)|^2 = x^2(t)$, and that is the only simplification available. The claim $|x(t)| = x(t)$ is stronger and fails wherever a real signal is negative.

EXAMPLE 5.10 – ENERGY FROM A TWO-BAND SPECTRUM

GIVEN $X_3(j\omega) = 2$ for $|\omega| < 2\pi$, 1 for $2\pi < |\omega| < 4\pi$, and 0 beyond.

FIND The total energy.

METHOD The spectrum is piecewise constant, so the frequency-domain energy integral is simpler than the time-domain integral. Apply Parseval, square each height, and multiply by the

corresponding width.

SOLUTION

Split the frequency integral of Parseval's relation at the points where the height changes. On each piece $|X_3|^2$ is a constant, so the integral is that constant times the width of the piece:

$$\begin{aligned} E_\infty &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |X_3(j\omega)|^2 d\omega \\ &= \frac{1}{2\pi} \left[\int_{-4\pi}^{-2\pi} 1^2 d\omega + \int_{-2\pi}^{2\pi} 2^2 d\omega + \int_{2\pi}^{4\pi} 1^2 d\omega \right] \\ &= \frac{1}{2\pi} [1 \cdot 2\pi + 4 \cdot 4\pi + 1 \cdot 2\pi] \\ &= \frac{1}{2\pi} [2\pi + 16\pi + 2\pi] = \frac{20\pi}{2\pi} = 10 \text{ J}. \end{aligned}$$

The inner band contributes $2^2 \cdot 4\pi = 16\pi$ and the two outer bands $1^2 \cdot 2\pi$ each.

CHECK

Write X_3 as the sum of two ideal low-pass bands of height 1, one on $|\omega| < 4\pi$ and one on $|\omega| < 2\pi$; where they overlap the heights add to 2. Example 5.6 with $W = 4\pi$ and $W = 2\pi$ then gives, by linearity,

$$x_3(t) = \frac{\sin(4\pi t)}{\pi t} + \frac{\sin(2\pi t)}{\pi t} = \frac{\sin(2\pi t) + \sin(4\pi t)}{\pi t}.$$

Integrating x_3^2 over all time returns 10 J as well. Note that $\frac{1}{2\pi} \int X_3 d\omega = \frac{1}{2\pi} [2\pi + 8\pi + 2\pi] = 6$, which is the peak $x_3(0) = 4 + 2$ and not the energy: using the heights unsquared computes the wrong quantity correctly.

The same relation on $e^{-at}u(t)$, $a > 0$, is worth doing both ways. In time, the step cuts the lower limit at 0 and $|e^{-at}|^2 = e^{-2at}$.

$$E_\infty = \int_0^\infty e^{-2at} dt = \left[\frac{e^{-2at}}{-2a} \right]_0^\infty = \frac{0 - 1}{-2a} = \frac{1}{2a}.$$

In frequency, $|X(j\omega)|^2 = 1/(a^2 + \omega^2)$ from Example 5.3. Substitute $\omega = au$, so $d\omega = a du$ and the limits stay infinite; then use the antiderivative $\tan^{-1} u$:

$$\begin{aligned} \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{d\omega}{a^2 + \omega^2} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{a du}{a^2(1 + u^2)} = \frac{1}{2\pi a} [\tan^{-1} u]_{-\infty}^{\infty} \\ &= \frac{1}{2\pi a} \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right] = \frac{1}{2\pi} \cdot \frac{\pi}{a} = \frac{1}{2a}. \end{aligned}$$

The two routes agree. Without the $1/2\pi$ the second route reports 2π times too much.

5.9 Convolution and multiplication

THE TWO DUAL PROPERTIES

$$y(t) = x(t) * h(t) \longleftrightarrow Y(j\omega) = X(j\omega)H(j\omega)$$

$$z(t) = x(t)y(t) \longleftrightarrow Z(j\omega) = \frac{1}{2\pi}X(j\omega) * Y(j\omega)$$

The premise is stated with one set of symbols: $x \leftrightarrow X$ is the input, $h \leftrightarrow H$ the impulse response, y the output. Convolution in time carries no factor; convolution in frequency carries $1/2\pi$. $H(j\omega)$ exists when h is absolutely integrable, which for an LTI system is exactly bounded-input bounded-output stability, so an unstable system has no frequency response to plot.

Proof of the convolution property. Write the convolution integral inside the analysis integral and exchange the order, so that the t integral is done first. The inner bracket is the transform of $h(t - \tau)$ with τ fixed, which the time-shift property gives as $e^{-j\omega\tau}H(j\omega)$. Then $H(j\omega)$ does not depend on τ and leaves the outer integral:

$$\begin{aligned} Y(j\omega) &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \right] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} x(\tau) \left[\int_{-\infty}^{\infty} h(t - \tau) e^{-j\omega t} dt \right] d\tau \\ &= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} H(j\omega) d\tau \\ &= H(j\omega) \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} d\tau = H(j\omega) X(j\omega). \end{aligned}$$

Proof of the multiplication property. Replace $y(t)$ in the analysis integral of $z = xy$ by its synthesis equation, written with the dummy variable θ so that it is not confused with the output frequency ω . Exchange the order of integration. The inner integral is the frequency-shift property, the analysis equation of x read at $\omega - \theta$:

$$\begin{aligned} Z(j\omega) &= \int_{-\infty}^{\infty} x(t) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} Y(j\theta) e^{j\theta t} d\theta \right] e^{-j\omega t} dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} Y(j\theta) \left[\int_{-\infty}^{\infty} x(t) e^{-j(\omega - \theta)t} dt \right] d\theta \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} Y(j\theta) X(j(\omega - \theta)) d\theta = \frac{1}{2\pi} X(j\omega) * Y(j\omega). \end{aligned}$$

The last integral is the convolution of X and Y in the variable ω , and the $1/2\pi$ came in with the synthesis equation of y .

EXAMPLE 5.11 – TWO EXPONENTIALS IN CASCADE

GIVEN $x(t) = e^{-at}u(t)$ and $h(t) = e^{-bt}u(t)$, with $a, b > 0$ and $a \neq b$.

FIND $y(t) = x * h$.

METHOD The output is a convolution in time, so it becomes multiplication in frequency. Multiply the transforms, expand the product into simple fractions, and invert each term.

SOLUTION

Example 5.3 gives $X(j\omega) = 1/(a + j\omega)$ and $H(j\omega) = 1/(b + j\omega)$. Multiply them, then write the product as a sum of two simple fractions with unknown constants A and B :

$$Y(j\omega) = \frac{1}{(a + j\omega)(b + j\omega)} = \frac{A}{a + j\omega} + \frac{B}{b + j\omega}.$$

Multiply through by $(a + j\omega)(b + j\omega)$ to clear the denominators:

$$1 = A(b + j\omega) + B(a + j\omega).$$

This holds for every ω . Choose $j\omega = -a$ to remove the B term: $1 = A(b - a)$, so $A = 1/(b - a)$. Choose $j\omega = -b$ to remove the A term: $1 = B(a - b)$, so $B = 1/(a - b) = -A$. Hence

$$Y(j\omega) = \frac{1}{b - a} \left[\frac{1}{a + j\omega} - \frac{1}{b + j\omega} \right].$$

Each fraction is the pair of Example 5.3, so by linearity

$$y(t) = \frac{1}{b - a} [e^{-at}u(t) - e^{-bt}u(t)] = \frac{e^{-at} - e^{-bt}}{b - a} u(t).$$

CHECK

$y(0) = (1 - 1)/(b - a) = 0$, as a convolution of two causal signals must be. For $a = 1$, $b = 2$, set $dy/dt = -e^{-t} + 2e^{-2t} = 0$, so $e^t = 2$ and the peak is at $t = \ln 2 = 0.693147$, with value $e^{-\ln 2} - e^{-2\ln 2} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$ exactly. At $\omega = 0$: $|X| = 1/a = 1$, $|H| = 1/b = 0.5$ and $|Y| = 1/(ab) = 0.5$, and the product of the first two is the third. That is the property itself at one frequency.

 EXAMPLE 5.12 – IDEAL FILTERS IN CASCADE

GIVEN

$X(j\omega) = 2$ on $|\omega| \leq 4\pi$; an ideal low-pass system with $H(j\omega) = 3$ on $|\omega| \leq 2\pi$.

FIND

$Y(j\omega)$, $y(t)$ and the three time-domain peaks.

METHOD

An LTI system gives $Y = XH$. Multiply the input spectrum and frequency response at each frequency, then apply the inverse transform.

SOLUTION

Multiply the two spectra frequency by frequency. On $|\omega| \leq 2\pi$ both are non-zero: $Y = 2 \cdot 3 = 6$. On $2\pi < |\omega| \leq 4\pi$ the input is 2 but the system is 0: $Y = 2 \cdot 0 = 0$. Beyond 4π both are zero. So $Y = 6$ on $|\omega| \leq 2\pi$ and zero elsewhere; the narrower band decides. Invert with the synthesis equation, which is Example 5.6 with $W = 2\pi$ and an extra factor 6:

$$\begin{aligned} y(t) &= \frac{1}{2\pi} \int_{-2\pi}^{2\pi} 6 e^{j\omega t} d\omega = 6 \cdot \frac{1}{2\pi} \left[\frac{e^{j\omega t}}{jt} \right]_{-2\pi}^{2\pi} \\ &= 6 \cdot \frac{e^{j2\pi t} - e^{-j2\pi t}}{2\pi jt} = 6 \cdot \frac{2j \sin(2\pi t)}{2\pi jt} = \frac{6 \sin(2\pi t)}{\pi t}. \end{aligned}$$

CHECK

At $t = 0$ the synthesis integral is the area of the spectrum divided by 2π , since $e^{j\omega \cdot 0} = 1$. Each peak is therefore the area of its own band over 2π : $x(0) = 2 \cdot 8\pi/2\pi = 8$, $h(0) =$

$3 \cdot 4\pi/2\pi = 6$, $y(0) = 6 \cdot 4\pi/2\pi = 12$. The output peak is the largest because a peak counts area, and $6 \times 4\pi$ exceeds $2 \times 8\pi$.

5.10 Amplitude modulation

A cosine of frequency ω_c is called a carrier. Its transform follows from Euler and the complex-exponential pair: $\cos(\omega_c t) = \frac{1}{2}e^{j\omega_c t} + \frac{1}{2}e^{-j\omega_c t} \leftrightarrow \pi\delta(\omega - \omega_c) + \pi\delta(\omega + \omega_c)$. Multiplying by the carrier therefore convolves the signal spectrum with two impulses. Convolution with a shifted impulse shifts the function, $X(j\omega) * \delta(\omega - \omega_c) = X(j(\omega - \omega_c))$, so:

$$\begin{aligned} Z(j\omega) &= \frac{1}{2\pi} X(j\omega) * [\pi\delta(\omega - \omega_c) + \pi\delta(\omega + \omega_c)] \\ &= \frac{\pi}{2\pi} X(j(\omega - \omega_c)) + \frac{\pi}{2\pi} X(j(\omega + \omega_c)) \\ &= \frac{1}{2} X(j(\omega - \omega_c)) + \frac{1}{2} X(j(\omega + \omega_c)). \end{aligned}$$

The factor $1/2\pi$ in the multiplication property and the weight π of each impulse give a factor $\frac{1}{2}$ for each shifted copy. The frequency-shift property gives the same result term by term: $\frac{1}{2}e^{j\omega_c t}x(t) \leftrightarrow \frac{1}{2}X(j(\omega - \omega_c))$ and $\frac{1}{2}e^{-j\omega_c t}x(t) \leftrightarrow \frac{1}{2}X(j(\omega + \omega_c))$.

DOUBLE-SIDEBAND SUPPRESSED-CARRIER MODULATION

$$z(t) = x(t) \cos(\omega_c t) \longleftrightarrow Z(j\omega) = \frac{1}{2}X(j(\omega - \omega_c)) + \frac{1}{2}X(j(\omega + \omega_c))$$

The spectrum is not moved to ω_c ; it is **uplicated**, one copy at $+\omega_c$ and one at $-\omega_c$, each at half height. A description that says "the signal moved up to the carrier" loses the negative-frequency copy and the factor of one half in the same sentence.

With $x(t) = \cos(\pi t)$ and carrier $\cos(4\pi t)$, the product-to-sum identity $\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha + \beta) + \frac{1}{2} \cos(\alpha - \beta)$ gives $z(t) = \frac{1}{2} \cos(5\pi t) + \frac{1}{2} \cos(3\pi t)$. Each cosine of unit amplitude transforms to two impulses of weight π , so $\frac{1}{2} \cos(3\pi t)$ gives impulses of weight $\frac{1}{2} \cdot \pi = \pi/2 = 1.570796$ at $\pm 3\pi$, and $\frac{1}{2} \cos(5\pi t)$ gives the same weight at $\pm 5\pi$. The property gives the same four positions: $X(j\omega) = \pi\delta(\omega - \pi) + \pi\delta(\omega + \pi)$, and the two half-height copies shifted to $\pm 4\pi$ put weight $\pi/2$ at $\pm(4\pi \pm \pi)$, that is at $\pm 3\pi$ and $\pm 5\pi$. Nothing sits at $\omega_c = 4\pi$ itself, which is what "suppressed carrier" records.

With a band-limited signal the picture is cleaner. If $X = 1$ on $|\omega| < 2\pi$ and the carrier is $\cos(4\pi t)$, the copies occupy $2\pi \leq |\omega| \leq 6\pi$ at height 0.5: the same width as the original band, at half the height, and twice as much of the axis occupied because there are now two of them.

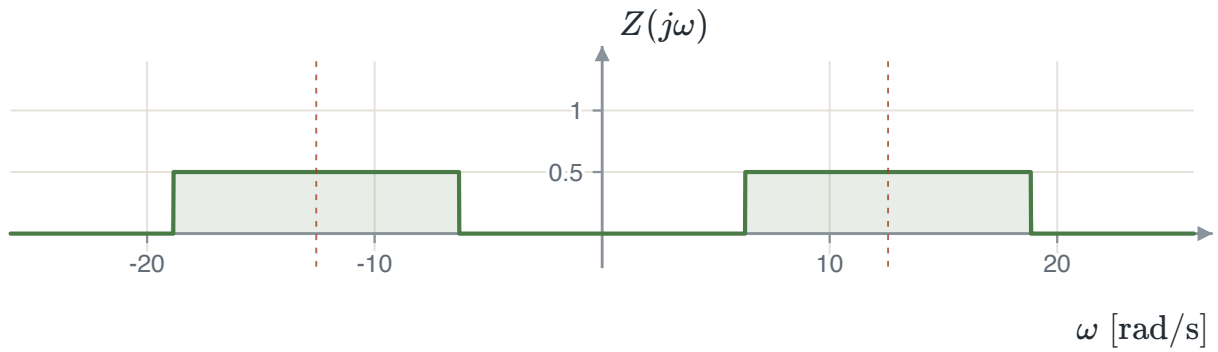


FIGURE 5.9 Two half-height copies on $2\pi \leq |\omega| \leq 6\pi$. The dashed lines mark $\pm\omega_c$, where nothing sits.

⊗ TWO DIFFERENT EVENTS

Copies appear whenever a signal is multiplied by a carrier. That happens at every carrier frequency and it loses nothing. **Copies overlap** only when the carrier is low enough for the shifted bands to reach each other. Overlap is what destroys information, because once two copies have been added there is no way to tell what each contributed. If a band on $\pi \leq |\omega| \leq 3\pi$ is modulated by $\cos(2\pi t)$, the two half-height copies both land on $|\omega| \leq \pi$ and add to 1 there, while only one reaches $3\pi \leq |\omega| \leq 5\pi$, where the height stays 0.5. Chapter 7 asks the same question about the copies that sampling produces.

Products of two band-limited signals follow the same arithmetic. Two rectangles of height A and half-width ω_0 convolve to a triangle: the overlap of the two rectangles is a full $2\omega_0$ wide at zero shift, giving the apex $A \cdot A \cdot 2\omega_0 = 2A^2\omega_0$, and it shrinks linearly to nothing at shift $\pm 2\omega_0$, so the triangle lives on $|\omega| \leq 2\omega_0$. For $x(t) = \sin(2\pi t)/(\pi t)$, Example 5.6 gives $A = 1$ and $\omega_0 = 2\pi$, so $X * X$ is a triangle of apex $2 \cdot 1 \cdot 2\pi = 4\pi$ on $|\omega| \leq 4\pi$; with the $1/2\pi$ of the multiplication property, the transform of $x^2(t)$ is a triangle of apex $4\pi/2\pi = 2$ on $|\omega| \leq 4\pi$. The time-domain peak is $x^2(0) = (2\pi/\pi)^2 = 2^2 = 4$, and the area of that triangle over 2π agrees: $\frac{1}{2} \cdot 8\pi \cdot 2/2\pi = 4$. With unequal half-widths 2π and 4π , the narrow rectangle fits wholly inside the wide one for shifts up to 2π , so the overlap stays at its full width 4π there: the result is a trapezoid of height $4\pi/2\pi = 2$, flat on $|\omega| \leq 2\pi$, falling linearly to zero at $|\omega| = 6\pi$, the sum of the half-widths. Its time-domain peak is the product of the two peaks, $2 \times 4 = 8$. The flat top is as wide as the difference of the two half-widths, and it shrinks to a point when they are equal.

5.11 Systems described by a differential equation

The Fourier transform converts a linear differential equation with constant coefficients into an algebraic equation. Take the transform of both sides. Linearity moves the transform inside each sum and past each constant coefficient, and the differentiation property replaces each derivative d^k/dt^k by the factor $(j\omega)^k$:

$$\sum_{k=0}^N a_k (j\omega)^k Y(j\omega) = \sum_{k=0}^M b_k (j\omega)^k X(j\omega).$$

Both sides now contain a plain product. Factor $Y(j\omega)$ out of the left sum and $X(j\omega)$ out of the right sum, and divide. The ratio Y/X is the frequency response, because for an LTI system $Y = XH$:

FREQUENCY RESPONSE FROM THE COEFFICIENTS

$$\sum_{k=0}^N a_k \frac{d^k y}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x}{dt^k} \implies H(j\omega) = \frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k}$$

The differential-equation coefficients form two polynomials in $j\omega$, and their ratio is the frequency response. Next calculate $Y = XH$, expand it into simple fractions, and invert each term with the transform table. During the algebra, define $s = j\omega$ and substitute back at the end.

 EXAMPLE 5.13 – SIMPLE POLES

GIVEN $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 3y = \frac{dx}{dt} + 2x$, at rest.

FIND $H(j\omega)$ and $h(t)$.

METHOD The differential equation has constant coefficients, so the differentiation property turns it into algebra. Form the frequency-response ratio, factor it, expand it into simple fractions, and invert each term.

SOLUTION Take the transform of both sides. Each derivative becomes a power of $j\omega$, and linearity keeps the coefficients:

$$(j\omega)^2 Y(j\omega) + 4(j\omega)Y(j\omega) + 3Y(j\omega) = (j\omega)X(j\omega) + 2X(j\omega).$$

Write $s = j\omega$ as a shorthand, factor out Y on the left and X on the right, and divide:

$$H = \frac{Y}{X} = \frac{s + 2}{s^2 + 4s + 3}.$$

Factor the denominator. Its roots are the values of s where $s^2 + 4s + 3 = 0$, that is $s = -1$ and $s = -3$, so $s^2 + 4s + 3 = (s + 1)(s + 3)$. Expand into simple fractions:

$$\frac{s + 2}{(s + 1)(s + 3)} = \frac{A}{s + 1} + \frac{B}{s + 3}, \quad s + 2 = A(s + 3) + B(s + 1).$$

Set $s = -1$ so the B term vanishes: $-1 + 2 = A(-1 + 3)$, so $1 = 2A$ and $A = \frac{1}{2}$. Set $s = -3$ so the A term vanishes: $-3 + 2 = B(-3 + 1)$, so $-1 = -2B$ and $B = \frac{1}{2}$.

This is the cover-up rule: cover one factor, and evaluate the rest at the root of that factor. Hence

$$H(j\omega) = \frac{1/2}{1 + j\omega} + \frac{1/2}{3 + j\omega},$$

and each fraction is the pair of Example 5.3, so

$$h(t) = \left[\frac{1}{2}e^{-t} + \frac{1}{2}e^{-3t} \right] u(t).$$

CHECK At $\omega = 0$, $H(j0) = 2/3$ from the ratio. The transform at $\omega = 0$ is the area, and $\int_0^\infty h \, dt = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$, which agrees. Also $h(0^+) = \frac{1}{2} + \frac{1}{2} = 1$. The roots $s = -1$ and $s = -3$ both have negative real part, so both exponentials decay, h is absolutely integrable and H was entitled to exist.

Repeated poles

The cover-up rule assumes each factor of the denominator appears once. When a factor is repeated, one term is needed per power, and covering up reaches only the highest of them. The reason is visible after clearing denominators: with $(s - \lambda)^m$ in the denominator, multiplying by $(s - \lambda)^m$ and setting $s = \lambda$ removes every term except the one with the highest power, which gives c_m . To reach the next coefficient, differentiate once before setting $s = \lambda$; the term c_m is a constant and disappears, the term $c_{m-1}(s - \lambda)$ leaves c_{m-1} , and every other term still carries a factor $(s - \lambda)$ and vanishes. Each further derivative peels off one more coefficient, with a factorial from differentiating the power.

REPEATED-POLE RULE

$$F(s) = \frac{N(s)}{(s - \lambda)^m Q(s)}, \quad c_{m-k} = \frac{1}{k!} \left. \frac{d^k}{ds^k} [(s - \lambda)^m F(s)] \right|_{s=\lambda}$$

Here $c_1, \dots, c_m$ are the coefficients of $1/(s - \lambda), \dots, 1/(s - \lambda)^m$ in the expansion of F , and $k = 0$ is the cover-up rule and gives c_m . The inverse transforms needed are $1/(s + a) \leftrightarrow e^{-at}u(t)$ from Example 5.3 and $1/(s + a)^2 \leftrightarrow t e^{-at}u(t)$ from the differentiation-in-frequency property of Section 5.6: a repeated pole always brings a factor of t into the time domain.

✦ EXAMPLE 5.14 – A REPEATED POLE, AND THE CHECK THAT CATCHES A SIGN

GIVEN The system of Example 5.13, with input $x(t) = e^{-t}u(t)$.

FIND $y(t)$.

METHOD The input passes through an LTI system, so calculate $Y = XH$. Define $s = j\omega$, expand Y into partial fractions, and invert each term.

SOLUTION Example 5.3 gives $X = 1/(s + 1)$ with $s = j\omega$, and Example 5.13 gives H . Multiply:

$$Y = \frac{1}{s + 1} \cdot \frac{s + 2}{(s + 1)(s + 3)} = \frac{s + 2}{(s + 1)^2(s + 3)}.$$

The input pole coincides with a system pole, so $s = -1$ is now double. One term is needed per power of $(s + 1)$:

$$Y = \frac{A}{s + 1} + \frac{B}{(s + 1)^2} + \frac{C}{s + 3}.$$

For B , multiply by $(s + 1)^2$ and set $s = -1$:

$$B = \left[(s + 1)^2 Y \right]_{s=-1} = \left. \frac{s + 2}{s + 3} \right|_{s=-1} = \frac{-1 + 2}{-1 + 3} = \frac{1}{2}.$$

For A , differentiate the same product once before setting $s = -1$. By the quotient rule, $\frac{d}{ds} \frac{s+2}{s+3} = \frac{(s+3) \cdot 1 - (s+2) \cdot 1}{(s+3)^2} = \frac{1}{(s+3)^2}$, so

$$A = \left. \frac{d}{ds} [(s + 1)^2 Y] \right|_{s=-1} = \left. \frac{1}{(s + 3)^2} \right|_{s=-1} = \frac{1}{2^2} = \frac{1}{4}.$$

For C , the factor $(s + 3)$ is simple, so cover it up and set $s = -3$:

$$C = \left[(s + 3) Y \right]_{s=-3} = \left. \frac{s + 2}{(s + 1)^2} \right|_{s=-3} = \frac{-3 + 2}{(-3 + 1)^2} = \frac{-1}{4} = -\frac{1}{4}.$$

Invert term by term with $1/(s + a) \leftrightarrow e^{-at}u(t)$ and $1/(s + a)^2 \leftrightarrow t e^{-at}u(t)$:

$$y(t) = \left[\frac{1}{4}e^{-t} + \frac{1}{2}t e^{-t} - \frac{1}{4}e^{-3t} \right] u(t).$$

CHECK $y(0) = \frac{1}{4} + 0 - \frac{1}{4} = 0$, as a convolution of two causal signals must be. Assembling C as $+\frac{1}{4}$ instead gives $y(0) = \frac{1}{2}$, which no causal convolution can. The two candidates agree to three decimals past $t = 2$, where they are 0.168549 and 0.169789, so only the value at

the origin separates them. Convolution x with h directly, $y(t) = \int_0^t e^{-(t-\tau)} \left[\frac{1}{2}e^{-\tau} + \frac{1}{2}e^{-3\tau} \right] d\tau$, returns the same three terms with the same signs.

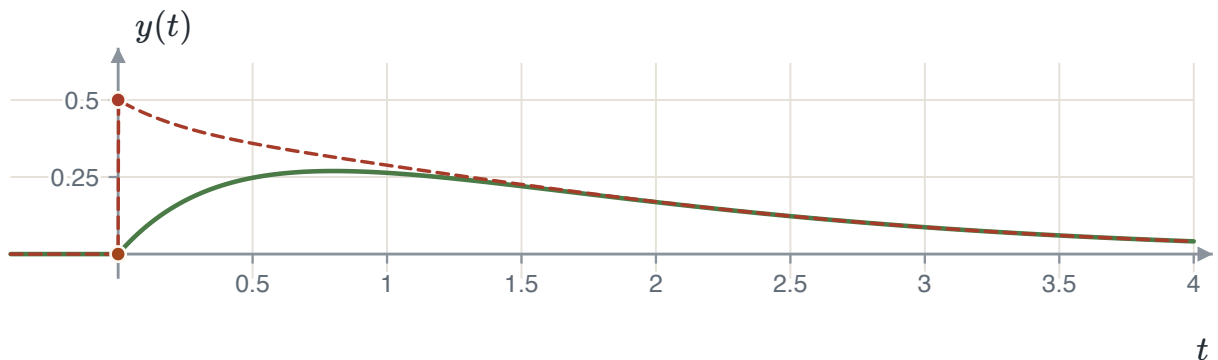


FIGURE 5.10 The output with $C = -\frac{1}{4}$ and, dashed, the version with the sign lost. They are indistinguishable past $t = 2$ and differ by 0.5 at the origin.

5.12 Summary

- Analysis integrates over t and returns $X(j\omega)$; synthesis integrates over ω , carries $1/2\pi$, and returns $x(t)$.
- Finite energy **or** the Dirichlet conditions is enough; neither is necessary, and signals meeting neither have impulse spectra in the limiting sense.
- A periodic signal transforms to impulses of weight $2\pi a_k$ at the harmonics — not to the coefficients themselves.
- The sinc convention here is $\text{sinc}(\theta) = \sin \theta / \theta$, and it is stated at every point of use.
- A real signal has an even magnitude and an odd phase, and both halves of its spectrum are kept.
- Narrowing a signal in time widens its spectrum. Finite duration forbids band limitation; infinite duration guarantees nothing.
- Convolution in time is multiplication in frequency; multiplication in time is convolution in frequency, with $1/2\pi$.
- A carrier makes two copies at half height. Copies always appear; overlap is a separate event and is what loses information.
- A differential equation gives $H(j\omega)$ directly; a repeated pole needs the derivative rule, and a causal convolution must start at zero.

Everything here was continuous time. A computer sees a sequence, not a signal, and the same question has to be asked again for $x[n]$. One thing changes and it changes everything: $e^{-j(\omega+2\pi)n} = e^{-j\omega n}$ for every integer n , so a discrete-time spectrum repeats with period 2π . That is Chapter 6.

CHAPTER

6

The discrete-time Fourier transform

The discrete-time Fourier transform describes the frequency content of a sequence that is not periodic. We derive it in three steps: replicate a finite sequence to make it periodic, apply the series from Chapter 4, and let the period grow without bound. The result is a continuous function of frequency that repeats every 2π . Every figure shows more than one period so that this defining property remains visible.

6.1 From a periodic sequence to an aperiodic one

Let $x[n]$ have finite support: $x[n] = 0$ for $|n| > N_1$. The discrete-time Fourier series applies to periodic sequences only, so build a periodic sequence out of x by laying the same finite sequence down again every N samples.

PERIODIC REPLICATION

$$\tilde{x}[n] = \sum_{r=-\infty}^{\infty} x[n - rN]$$

Periodic replication places a copy of the complete sequence every N samples. It does not insert samples within a copy.

△ THE CONDITION THE CONSTRUCTION NEEDS

The copies must not overlap, so the period must be longer than the support: $N > 2N_1$. Only then is $\tilde{x}[n] = x[n]$ for $-N_1 \leq n \leq N_1$, a band of $2N_1 + 1$ samples centred on the origin. With $N \leq 2N_1$ the tails of neighbouring copies add and no period of $\tilde{x}$ equals x any more. Both ends of the interval point the same way; reversing the second relation would leave an interval with nothing in it.

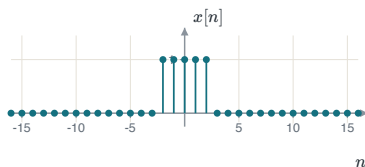


FIGURE 6.1 The finite-support sequence, $N_1 = 2$.

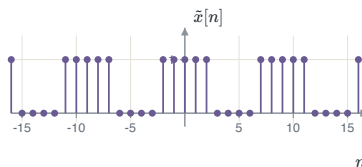


FIGURE 6.2 Replication with $N = 9 > 2N_1$: the copies stand clear.

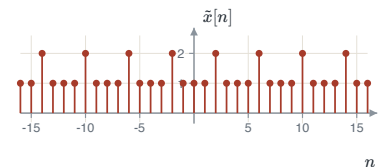


FIGURE 6.3 Replication with $N = 4 \leq 2N_1$: the copies add.

The coefficients are samples of one function

DISCRETE-TIME FOURIER SERIES

$$\tilde{x}[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n}, \quad \omega_0 = \frac{2\pi}{N}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} \tilde{x}[n] e^{-jk\omega_0 n}$$

The analysis exponent is negative. Synthesis builds the sequence up out of exponentials and carries $e^{+jk\omega_0 n}$; analysis takes it apart and carries the conjugate. A plus sign in both places breaks the identity derived immediately below.

Over one period, $\tilde{x}[n] = x[n]$. Because x is zero outside $-N_1 \leq n \leq N_1$, extending the sum to all n adds only zero terms:

$$a_k = \frac{1}{N} \sum_{n=-N_1}^{N_1} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk\omega_0 n}.$$

The last sum has the same form for every k . Define it as a function of the continuous variable ω , and then evaluate it at $\omega = k\omega_0$.

THE ENVELOPE

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \quad \Rightarrow \quad a_k = \frac{1}{N} X(e^{jk\omega_0})$$

The series coefficients are samples of the envelope, spaced $\omega_0 = 2\pi/N$ apart and scaled by $1/N$. The identity holds only because the two exponents match, which is the check on the sign.

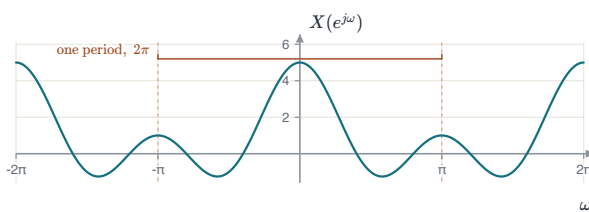


FIGURE 6.4 The envelope of the pulse with $N_1 = 2$, over two periods. Its value at $\omega = 0$ is $2N_1 + 1 = 5$.

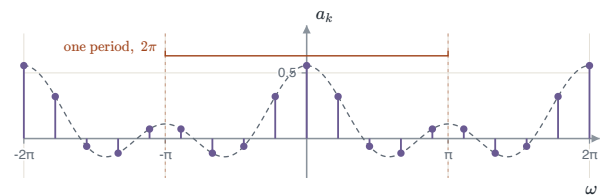


FIGURE 6.5 The coefficients for $N = 9$: the dashed envelope divided by N , sampled at $\omega = k\omega_0$.

Letting the period grow

Put $a_k = \frac{1}{N} X(e^{jk\omega_0})$ back into the synthesis equation and write $\frac{1}{N}$ as $\frac{\omega_0}{2\pi}$:

$$\tilde{x}[n] = \sum_{k=\langle N \rangle} \frac{1}{N} X(e^{jk\omega_0}) e^{jk\omega_0 n} = \frac{1}{2\pi} \sum_{k=\langle N \rangle} X(e^{jk\omega_0}) e^{jk\omega_0 n} \omega_0.$$

Interpret the right-hand side as a sum of rectangles. Each rectangle has height $X(e^{j\omega}) e^{j\omega n}$ at a sampled frequency and width ω_0 . The N rectangles cover an interval of length $N\omega_0 = 2\pi$. As N grows without

bound, the replicated copies separate and $\tilde{x}[n]$ approaches $x[n]$. At the same time, $\omega_0 = 2\pi/N$ approaches zero, so the rectangle sum approaches an integral over an interval of length 2π .

THE DISCRETE-TIME FOURIER TRANSFORM PAIR

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \quad (\text{analysis})$$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \quad (\text{synthesis})$$

Analysis takes the sequence to its spectrum; synthesis rebuilds the sequence from it. The names go by what each equation does, never by which is written first. Three details separate this pair from the continuous-time one: the factor $\frac{1}{2\pi}$ sits on the synthesis side, the integral runs over one period rather than the whole line, and the left-hand side is written $X(e^{j\omega})$ to record that it is a function of $e^{j\omega}$ and therefore repeats.

✓ WHEN THE SUM CONVERGES

A sufficient condition is that x is absolutely summable, $\sum_n |x[n]| < \infty$. Then the analysis sum converges at every ω and $X(e^{j\omega})$ is continuous. Finite-energy sequences also have a transform, in a mean-square sense.

6.2 Why the spectrum repeats

PERIODICITY IN THE FREQUENCY VARIABLE

$$X(e^{j(\omega+2\pi)}) = \sum_n x[n] e^{-j(\omega+2\pi)n} = \sum_n x[n] e^{-j\omega n} e^{-j2\pi n} = X(e^{j\omega})$$

The whole argument is that n is an integer, so $e^{-j2\pi n} = 1$ in every term of the sum and the extra factor disappears term by term.

The same step in continuous time gives $\int x(t) e^{-j\omega t} e^{-j2\pi t} dt$, where t runs over the reals. There $e^{-j2\pi t}$ equals 1 only at integer t , so it cannot leave the integral, and nothing forces $X(j\omega)$ to repeat.

① WHAT FREQUENCY MEANS IN DISCRETE TIME

A sequence $e^{j\omega n}$ is unchanged when ω moves by 2π , because it is only ever sampled at integers. High frequency therefore means ω near $\pm\pi$, not ω large. The fastest sequence a discrete-time signal can carry is $e^{j\pi n} = (-1)^n$.

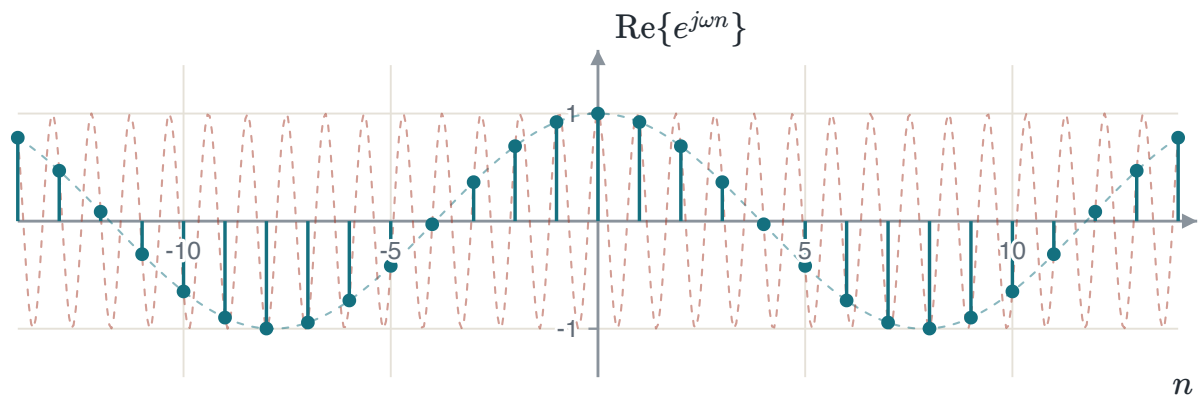


FIGURE 6.6 Two continuous curves, $\omega = 0.4$ and $\omega = 0.4 + 2\pi$, and the one stem sequence they share. At the integers they agree exactly, so no sequence can tell them apart.

⊗ THE HABIT THIS CHAPTER IS BUILT TO PREVENT

A spectrum drawn on $-\pi \leq \omega \leq \pi$ alone is the same information as three periods of it, and it is also the picture a reader later mistakes for a spectrum that stops at $\pm\pi$. Every spectrum here is drawn over more than one period with the period marked.

6.3 Four transforms worked out

EXAMPLE 6.1 – A SHIFTED UNIT SAMPLE

GIVEN $x[n] = \delta[n - n_0]$.

FIND $X(e^{j\omega})$, its magnitude and its phase.

METHOD Use the analysis sum because the sequence is given in time. The unit sample makes every term with $n \neq n_0$ equal to zero, so only one term remains.

SOLUTION Start from the analysis sum and substitute the sequence. The unit sample $\delta[n - n_0]$ is 1 at $n = n_0$ and 0 at every other n , so the sum keeps one term:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} \delta[n - n_0]e^{-j\omega n} \\ &= e^{-j\omega n_0}. \end{aligned}$$

Write the result in polar form. A complex exponential $e^{j\phi}$ with real ϕ has modulus 1 and angle ϕ , so

$$|X(e^{j\omega})| = |e^{-j\omega n_0}| = 1, \quad \angle X(e^{j\omega}) = -n_0\omega.$$

The magnitude is 1 at every frequency and the phase is a straight line of slope $-n_0$.

CHECK At $n_0 = 0$ the sequence is $\delta[n]$ and the transform is the constant 1, which is what the sum gives directly.

READING

A straight line is not periodic, and yet the transform is. What is drawn is the principal value of the phase, which jumps by 2π whenever the line leaves $(-\pi, \pi]$. The sawtooth is that wrapping, and its period is $2\pi/|n_0|$.

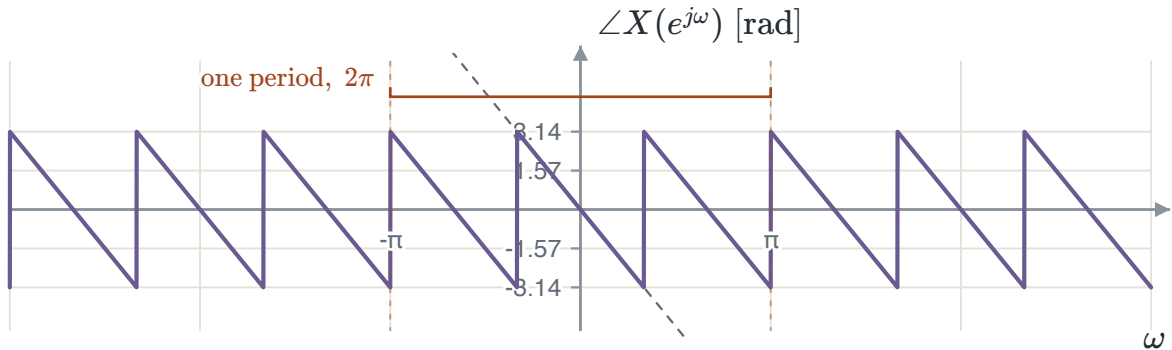


FIGURE 6.7 The phase for $n_0 = 3$. The dashed line is the unwrapped $-n_0\omega$; the sawtooth is its principal value. Both describe the same transform.

EXAMPLE 6.2 – THE ONE-SIDED EXPONENTIAL

GIVEN

$x[n] = a^n u[n]$ with $|a| < 1$.

FIND

$X(e^{j\omega})$, and the extremes of its magnitude and phase.

METHOD

Use the geometric-series formula because the one-sided sequence produces powers of the same ratio $ae^{-j\omega}$. The stated condition makes the infinite sum converge.

SOLUTION

Start from the analysis sum and substitute the sequence. The factor $u[n]$ is 0 for $n < 0$ and 1 for $n \geq 0$, so the lower limit moves to $n = 0$. Then collect the two powers of n into one power:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} a^n u[n] e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} a^n e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} (ae^{-j\omega})^n. \end{aligned}$$

This is an infinite geometric series with first term 1 and ratio $r = ae^{-j\omega}$. The formula $\sum_{n=0}^{\infty} r^n = 1/(1-r)$ holds when $|r| < 1$. Here $|r| = |a| |e^{-j\omega}| = |a| \cdot 1 = |a| < 1$, so the series converges at every ω and

$$X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}.$$

For the magnitude, write the denominator in rectangular form with $e^{-j\omega} = \cos \omega - j \sin \omega$:

$$1 - ae^{-j\omega} = (1 - a \cos \omega) + j a \sin \omega.$$

Its squared modulus is the sum of the squared real and imaginary parts:

$$\begin{aligned}
|1 - ae^{-j\omega}|^2 &= (1 - a \cos \omega)^2 + a^2 \sin^2 \omega \\
&= 1 - 2a \cos \omega + a^2 \cos^2 \omega + a^2 \sin^2 \omega \\
&= 1 - 2a \cos \omega + a^2,
\end{aligned}$$

using $\cos^2 \omega + \sin^2 \omega = 1$. The modulus of a quotient is the quotient of the moduli, and the angle of a quotient is the difference of the angles. The numerator 1 has modulus 1 and angle 0, so

$$\begin{aligned}
|X(e^{j\omega})| &= \frac{1}{\sqrt{1 - 2a \cos \omega + a^2}}, \\
\angle X(e^{j\omega}) &= 0 - \arctan \frac{a \sin \omega}{1 - a \cos \omega} = -\arctan \frac{a \sin \omega}{1 - a \cos \omega}.
\end{aligned}$$

EXTREMES

The magnitude is largest where the denominator $1 - 2a \cos \omega + a^2$ is smallest. For $a > 0$ that is at $\cos \omega = 1$, so $\omega = 0$, where the denominator is $1 - 2a + a^2 = (1 - a)^2$ and $|X|_{\max} = 1/(1 - a)$. It is smallest at $\cos \omega = -1$, so $\omega = \pm\pi$, where the denominator is $(1 + a)^2$ and $|X|_{\min} = 1/(1 + a)$. For negative a the two ends swap, so in general $|X|_{\max} = 1/(1 - |a|)$ and $|X|_{\min} = 1/(1 + |a|)$. For the phase, find where the ratio $r(\omega) = a \sin \omega / (1 - a \cos \omega)$ is stationary. The quotient rule gives

$$r'(\omega) = \frac{a \cos \omega (1 - a \cos \omega) - a \sin \omega \cdot a \sin \omega}{(1 - a \cos \omega)^2} = \frac{a(\cos \omega - a)}{(1 - a \cos \omega)^2},$$

again using $\cos^2 \omega + \sin^2 \omega = 1$ in the numerator. So the extreme is at $\cos \omega = a$, where $\sin \omega = \pm\sqrt{1 - a^2}$ and

$$|r| = \frac{|a|\sqrt{1 - a^2}}{1 - a^2} = \frac{|a|}{\sqrt{1 - a^2}}.$$

A right triangle with opposite side $|a|$ and adjacent side $\sqrt{1 - a^2}$ has hypotenuse 1, so $\arctan |r| = \arcsin |a|$ and $\max |\angle X| = \arcsin |a|$. Geometrically, the denominator $1 - ae^{-j\omega}$ traces a circle of radius $|a|$ about the point 1, and the tangent from the origin to that circle gives the same angle.

NUMBERS

$a = \frac{1}{2}$: magnitude between $\frac{2}{3}$ and 2, and $\max |\angle X| = \arcsin \frac{1}{2} = \pi/6 = 0.5236$ rad exactly. $a = \frac{1}{8}$: magnitude between $\frac{8}{9} = 0.8889$ and $\frac{8}{7} = 1.1429$, and $\max |\angle X| = 0.1253$ rad.

CHECK

Every one of these is a closed form, so none has to be read off the frame of a plot. An extreme that lands exactly on an axis limit is the one place a printed number is likeliest to be the limit rather than the value.

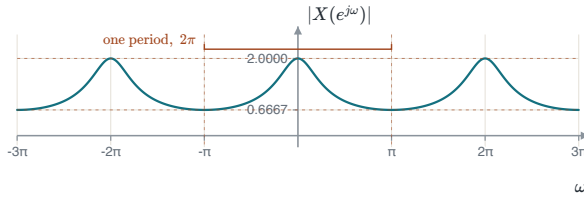


FIGURE 6.8 Magnitude for $a = \frac{1}{2}$, touching $1/(1-a) = 2$ and $1/(1+a) = 2/3$.

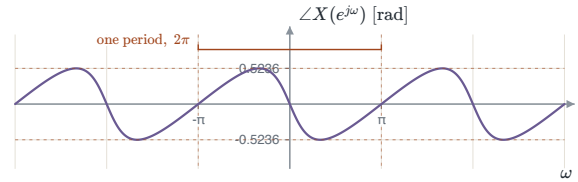


FIGURE 6.9 Phase for $a = \frac{1}{2}$, touching $\pm \arcsin \frac{1}{2} = \pm 0.5236$ rad at $\omega = \mp \pi/3$.

EXAMPLE 6.3 – THE TWO-SIDED EXPONENTIAL

GIVEN $x[n] = a^{|n|}$ with $|a| < 1$.

FIND $X(e^{j\omega})$ and its extremes.

METHOD Split the analysis sum at $n = 0$ because the absolute value gives different exponents on the two sides. Each remaining sum is geometric and converges under the stated condition.

SOLUTION Start from the analysis sum and split it at $n = 0$. For $n \geq 0$, $a^{|n|} = a^n$; for $n \leq -1$, $a^{|n|} = a^{-n}$:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} a^{|n|} e^{-j\omega n} = \sum_{n=0}^{\infty} a^n e^{-j\omega n} + \sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n}.$$

The first sum is the one-sided pair of Example 6.2, equal to $1/(1 - ae^{-j\omega})$ because $|ae^{-j\omega}| = |a| < 1$. In the second sum put $m = -n$. As n runs from $-\infty$ to -1 , m runs from 1 to ∞ :

$$\sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n} = \sum_{m=1}^{\infty} a^m e^{j\omega m} = \sum_{m=1}^{\infty} (ae^{j\omega})^m.$$

This is a geometric series with first term $ae^{j\omega}$ and ratio $ae^{j\omega}$, of modulus $|a| < 1$, so it equals $ae^{j\omega}/(1 - ae^{j\omega})$. Add the two parts over the common denominator $(1 - ae^{-j\omega})(1 - ae^{j\omega})$:

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{1 - ae^{-j\omega}} + \frac{ae^{j\omega}}{1 - ae^{j\omega}} \\ &= \frac{(1 - ae^{j\omega}) + ae^{j\omega}(1 - ae^{-j\omega})}{(1 - ae^{-j\omega})(1 - ae^{j\omega})} \\ &= \frac{1 - ae^{j\omega} + ae^{j\omega} - a^2}{1 - a(e^{j\omega} + e^{-j\omega}) + a^2} \\ &= \frac{1 - a^2}{1 - 2a \cos \omega + a^2}. \end{aligned}$$

In the numerator the two terms $\mp ae^{j\omega}$ cancel and $ae^{j\omega} \cdot ae^{-j\omega} = a^2$. In the denominator, $e^{j\omega} + e^{-j\omega} = 2 \cos \omega$.

EXTREMES At $\omega = 0$ the value is $(1+a)/(1-a)$; at $\omega = \pm\pi$ it is $(1-a)/(1+a)$. For $a = \frac{1}{2}$ these are 3 and $\frac{1}{3}$; for $a = \frac{1}{4}$ they are 1.6667 and 0.6.

READING

The denominator is $|1 - ae^{-j\omega}|^2$, which is positive, and the numerator is positive for $|a| < 1$. So this spectrum is real and strictly positive, and here $|X| = X$ with $\angle X = 0$ throughout. That is a property of this example, not of real spectra in general.

 EXAMPLE 6.4 – THE RECTANGULAR PULSE

GIVEN

$x[n] = 1$ for $-N_1 \leq n \leq N_1$ and zero elsewhere.

FIND

$X(e^{j\omega})$ in closed form and its value at $\omega = 0$.

METHOD

Use the finite geometric-series formula because the pulse has consecutive non-zero samples. Then balance the endpoint exponents so each difference can be written as a sine.

SOLUTION

Start from the analysis sum. The sequence is 1 on $-N_1 \leq n \leq N_1$ and 0 elsewhere, so the limits shrink to the support:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=-N_1}^{N_1} e^{-j\omega n} = \sum_{n=-N_1}^{N_1} (e^{-j\omega})^n.$$

This is a finite geometric series with ratio $r = e^{-j\omega}$, first term $r^{-N_1} = e^{j\omega N_1}$ and $2N_1 + 1$ terms. The finite formula $\sum_{n=p}^q r^n = (r^p - r^{q+1})/(1 - r)$ needs only $r \neq 1$. With $p = -N_1$ and $q = N_1$:

$$X(e^{j\omega}) = \frac{e^{j\omega N_1} - e^{-j\omega(N_1+1)}}{1 - e^{-j\omega}}.$$

Now balance the exponents. Multiply numerator and denominator by $e^{j\omega/2}$, which changes nothing because it is the same factor above and below:

$$\begin{aligned} X(e^{j\omega}) &= \frac{e^{j\omega/2}(e^{j\omega N_1} - e^{-j\omega(N_1+1)})}{e^{j\omega/2}(1 - e^{-j\omega})} \\ &= \frac{e^{j\omega(N_1+\frac{1}{2})} - e^{-j\omega(N_1+\frac{1}{2})}}{e^{j\omega/2} - e^{-j\omega/2}} \\ &= \frac{2j \sin(\omega(N_1 + \frac{1}{2}))}{2j \sin(\omega/2)} \\ &= \frac{\sin(\omega(N_1 + \frac{1}{2}))}{\sin(\omega/2)}. \end{aligned}$$

The third line uses $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$ twice, once with $\theta = \omega(N_1 + \frac{1}{2})$ and once with $\theta = \omega/2$. The minus in each bracket is what makes a sine; a plus would give a cosine and a different function. With $N = 2N_1 + 1$ samples in the pulse, $N_1 + \frac{1}{2} = N/2$, so the same result reads $\sin(\omega N/2)/\sin(\omega/2)$.

EXCLUDED POINTS

$r = e^{-j\omega} = 1$ at $\omega = 0, \pm 2\pi, \dots$, which is exactly where $\sin(\omega/2)$ vanishes. There every term of the sum is 1, so $X = 2N_1 + 1$: five for $N_1 = 2$, nine for $N_1 = 4$.

CHECK

This is not a sinc. The denominator is $\sin(\omega/2)$, not $\omega/2$, and that is what makes the function periodic: a ratio of two sines repeats, while a sine over a straight line decays.

△ WHAT A FINITE GEOMETRIC SUM REQUIRES

$\sum_{n=p}^q r^n = (r^p - r^{q+1})/(1 - r)$ needs $r \neq 1$ and nothing else. There are finitely many terms, so nothing has to converge; only the division can fail. Here $|r| = 1$ exactly, so a condition $|r| < 1$ would exclude the sum altogether, while $|r| \leq 1$ would admit the one value that breaks it. The condition $|a| < 1$ of Example 6.2 belongs to the infinite sum, where it makes the tail vanish.

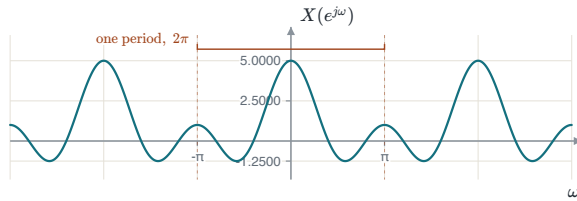


FIGURE 6.10 $N_1 = 2$: peak 5, least value -1.2500 .

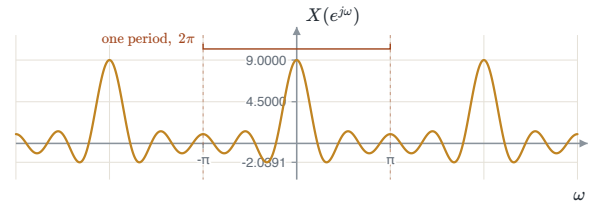


FIGURE 6.11 $N_1 = 4$: peak 9, least value -2.0391 , and a main lobe half as wide.

⊗ REAL IS NOT THE SAME AS NON-NEGATIVE

The kernel above is real at every ω and negative on part of every period. Realness fixes the imaginary part and says nothing about the sign, so it gives $\angle X = 0$ where $X > 0$ and $\angle X = \pi$ where $X < 0$; and $|X| = X$ needs the extra hypothesis $X \geq 0$. A real spectrum has an even magnitude and a phase taking only the values 0 and π . Example 6.3 is the contrasting case: there the spectrum is real **and** strictly positive, so the phase really is zero, because of positivity and not because of realness.

6.4 Inverting an ideal low-pass spectrum

🔗 EXAMPLE 6.5 – THE IDEAL LOW-PASS SEQUENCE

GIVEN $X(e^{j\omega}) = 1$ for $|\omega| \leq W$ and 0 for $W < |\omega| \leq \pi$, repeated with period 2π .

FIND $x[n]$.

METHOD Use the synthesis integral because the spectrum is given and the sequence is required. Choose $-\pi \leq \omega \leq \pi$ so the band where the spectrum is 1 appears once within the integration period.

SOLUTION Start from the synthesis integral over the period $-\pi \leq \omega \leq \pi$. The spectrum is 1 on $|\omega| \leq W$ and 0 on the rest of the period, so the limits shrink to $-W \leq \omega \leq W$:

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-W}^W e^{j\omega n} d\omega.$$

For $n \neq 0$ the antiderivative of $e^{j\omega n}$ in ω is $e^{j\omega n}/(jn)$:

$$\begin{aligned}
 x[n] &= \frac{1}{2\pi} \left[\frac{e^{j\omega n}}{jn} \right]_{-W}^W \\
 &= \frac{1}{2\pi} \cdot \frac{e^{jWn} - e^{-jWn}}{jn} \\
 &= \frac{1}{2\pi} \cdot \frac{2j \sin(Wn)}{jn} \\
 &= \frac{\sin(Wn)}{\pi n}.
 \end{aligned}$$

The third line uses $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$, and then the j and the 2 cancel. At $n = 0$ the antiderivative does not apply because it divides by n . There the integrand is $e^0 = 1$, so the integral is the length $2W$ of the band and

$$x[0] = \frac{1}{2\pi} \cdot 2W = \frac{W}{\pi},$$

which is also the limit of $\sin(Wn)/(\pi n)$ as $n \rightarrow 0$.

NUMBERS $W = \pi/4$: 0.25, 0.225079, 0.159155, 0.075026 at $n = 0, 1, 2, 3$. $W = \pi/2$: 0.5, 0.318310, 0, -0.106103 .

CHECK $x[0]$ must be the fraction W/π of the period that the band occupies. That single number catches both a lost 2π and a wrong prefactor.

THE SINC CONVENTION

This course uses the unnormalised sinc, $\text{sinc } \theta = \sin \theta / \theta$, with no π inside the argument. In that convention the same sequence reads $x[n] = \frac{W}{\pi} \text{sinc}(Wn)$. The other common definition puts a π inside the argument and moves every zero crossing, so the convention is restated at every point of use. The prefactor is the constant W/π : writing W/n makes the expression depend on n twice and is wrong by a factor π at every index, giving 0.707107 at $n = 1$ where the true value is 0.225079.

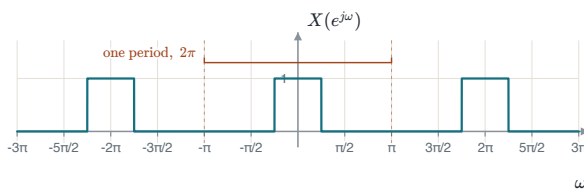


FIGURE 6.12 The spectrum for $W = \pi/4$, over three periods. The band repeats; it does not stop at $\pm\pi$.

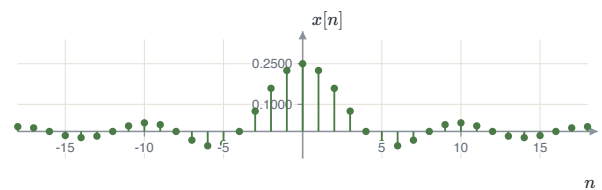


FIGURE 6.13 Its inverse transform, with $x[0] = 0.25$.

6.5 Periodic sequences and impulse spectra

A complex exponential is not absolutely summable, so its analysis sum does not converge in the ordinary sense. Its transform exists as a train of impulses, and the definition is that the synthesis equation works.

THE COMPLEX EXPONENTIAL

$$e^{j\omega_0 n} \longleftrightarrow X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 - 2\pi k)$$

To confirm the pair, put the impulse train into the synthesis equation. Choose the integration period $\omega_0 - \pi < \omega \leq \omega_0 + \pi$, so that it contains exactly one impulse of the train, the one at $\omega = \omega_0$ with $k = 0$. Then the sifting property picks out the integrand at $\omega = \omega_0$:

$$\begin{aligned} x[n] &= \frac{1}{2\pi} \int_{\omega_0 - \pi}^{\omega_0 + \pi} \sum_k 2\pi \delta(\omega - \omega_0 - 2\pi k) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{\omega_0 - \pi}^{\omega_0 + \pi} 2\pi \delta(\omega - \omega_0) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \cdot 2\pi e^{j\omega_0 n} = e^{j\omega_0 n}. \end{aligned}$$

The copies must be there: a single impulse is not 2π -periodic, and a sequence cannot distinguish ω_0 from $\omega_0 + 2\pi$.

① WEIGHT, NOT HEIGHT

An impulse has no value at a point; it has a weight, the number that comes out when it is integrated. Every figure here draws an impulse as an arrow whose height **is** its weight, so the two can be read off the same axis. Arrows drawn to a fixed height with the weight written beside them hide exactly the comparison these pictures are for.

A PERIODIC SEQUENCE

$$\begin{aligned} x[n] = \sum_{k=\langle N \rangle} a_k e^{jk \frac{2\pi}{N} n} &\longleftrightarrow X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta\left(\omega - \frac{2\pi k}{N}\right) \\ \cos \omega_0 n &\longleftrightarrow \pi \sum_k [\delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)] \\ \sin \omega_0 n &\longleftrightarrow \frac{\pi}{j} \sum_k [\delta(\omega - \omega_0 - 2\pi k) - \delta(\omega + \omega_0 - 2\pi k)] \end{aligned}$$

The first line is the complex-exponential pair applied term by term. The series writes $x[n]$ as a sum of N exponentials with frequencies $k \cdot 2\pi/N$ and weights a_k . By linearity each exponential contributes its own impulse train, $2\pi a_k$ at $2\pi k/N$ plus all the 2π copies. Because k runs over one period of N values and each train repeats every 2π , the copies together fill in every integer k , which is why the right-hand sum runs over all k . The spacing is $2\pi/N$, so one period holds exactly N impulses. The cosine and sine lines come from Euler's relations, $\cos \omega_0 n = \frac{1}{2}e^{j\omega_0 n} + \frac{1}{2}e^{-j\omega_0 n}$ and $\sin \omega_0 n = \frac{1}{2j}e^{j\omega_0 n} - \frac{1}{2j}e^{-j\omega_0 n}$. Each exponential brings an impulse train of weight 2π , and $\frac{1}{2} \cdot 2\pi = \pi$ while $\frac{1}{2j} \cdot 2\pi = \pi/j$. A real sequence always produces the pair at $+\omega_0$ and $-\omega_0$; dropping the negative one rebuilds $\frac{1}{2}e^{j\omega_0 n}$, which is complex, instead of the cosine.

For the periodic square wave, 1 on $|n| \leq N_1$ inside each period of length N , take the analysis sum of the series over the period centred on the origin. The sequence is 1 on $-N_1 \leq n \leq N_1$ and 0 on the rest of that period, so the limits shrink to the pulse:

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n} = \frac{1}{N} \sum_{n=-N_1}^{N_1} e^{-j(\frac{2\pi k}{N})n}.$$

The remaining sum is the rectangular-pulse sum of Example 6.4 with ω replaced by $2\pi k/N$. So the coefficients are the same kernel again, sampled at $\omega = 2\pi k/N$ and divided by N :

$$a_k = \frac{1}{N} \frac{\sin(\frac{2\pi k}{N}(N_1 + \frac{1}{2}))}{\sin(\pi k/N)}, \quad a_k = \frac{2N_1 + 1}{N} \text{ when } k \equiv 0 \pmod{N}.$$

The second form covers the excluded points of Example 6.4. When k is a multiple of N , $\sin(\pi k/N) = 0$ and the closed form fails, but every term of the sum is 1, so the sum is $2N_1 + 1$. With $N_1 = 2$ and $N = 10$, $a_0 = 5/10 = 0.5000$ and $a_1 = \frac{1}{10} \sin(\pi/2)/\sin(\pi/10) = \frac{1}{10} \cdot 1/0.3090 = 0.3236$. At $N = 20$ the first two are 0.2500 and 0.2260; at $N = 30$, 0.1667 and 0.1594. The coefficients shrink like $1/N$ while the shape they trace stays put. Some of them are negative. At $N = 10$, $a_3 = \frac{1}{10} \sin(3\pi/2)/\sin(3\pi/10) = \frac{1}{10} \cdot (-1)/0.8090 = -0.1236$, so the impulses of weight $2\pi a_k$ are of unequal size and some point downwards. Sketching them all the same length hides both facts.

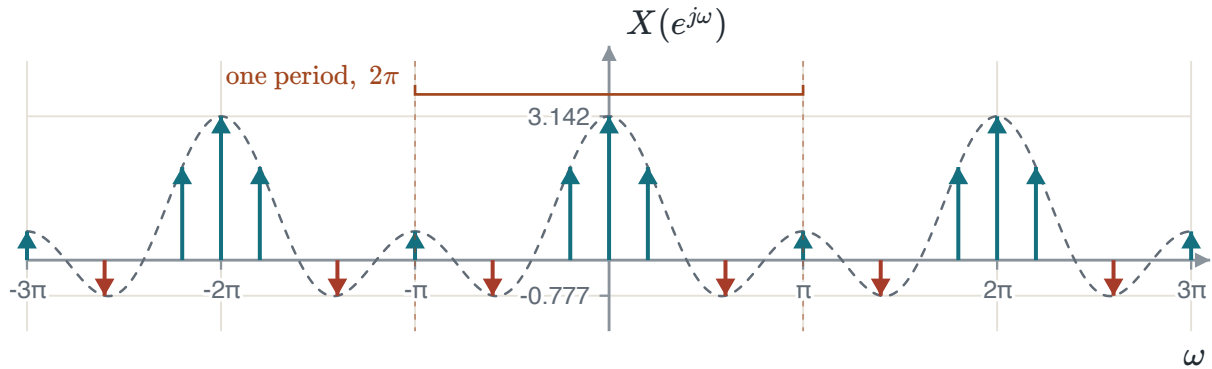


FIGURE 6.14 The square wave with $N = 10$, $N_1 = 2$. The dashed line is the envelope; the red arrows are the harmonics whose weight is negative.

EXAMPLE 6.6 – THE IMPULSE TRAIN

GIVEN $x[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN]$: a unit sample every N indices. Here n is the sequence index and k labels the copies, so the two roles use different letters.

FIND The coefficients and the transform.

METHOD The sequence is periodic with period N , so its transform is an impulse train. Find the series coefficients a_k from the analysis sum of the series, then give each impulse the weight $2\pi a_k$ with the periodic-sequence pair.

SOLUTION Take any run of N consecutive indices that contains $n = 0$. Inside it the only non-zero sample is $\delta[n]$ at $n = 0$, so the analysis sum keeps one term:

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n} = \frac{1}{N} \delta[0] e^{-jk \frac{2\pi}{N} \cdot 0} = \frac{1}{N} \quad \text{for every } k.$$

Put $a_k = 1/N$ into the periodic-sequence pair. Every impulse gets the same weight $2\pi a_k = 2\pi/N$:

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi \cdot \frac{1}{N} \delta\left(\omega - \frac{2\pi k}{N}\right) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right).$$

NUMBERS Weights $2\pi/5 = 1.2566$, $2\pi/10 = 0.6283$, $2\pi/15 = 0.4189$ for $N = 5, 10, 15$.

READING One period always holds N impulses of weight $2\pi/N$, so their weights add to 2π whatever N is. A sparser train in time is a denser train in frequency.

Bringing a frequency into range

EXAMPLE 6.7 – TWO COSINES

GIVEN $x[n] = 2 \cos\left(\frac{5\pi}{3}n\right) + \cos\left(\frac{7\pi}{4}n\right)$.

FIND The spectrum, and the fundamental period of the sequence.

METHOD First reduce each frequency to $-\pi < \omega \leq \pi$ because equivalent discrete-time frequencies describe the same sequence. Then transform each cosine separately and add the results by linearity.

REDUCTION $\frac{5\pi}{3} = 2\pi - \frac{\pi}{3}$, and $2\pi n$ is a whole number of turns at every integer n , so $\cos\left(\frac{5\pi}{3}n\right) = \cos\left(\frac{\pi}{3}n\right)$. Likewise $\cos\left(\frac{7\pi}{4}n\right) = \cos\left(\frac{\pi}{4}n\right)$. These are not merely similar sequences; they take the same value at every n .

SOLUTION After the reduction, $x[n] = 2 \cos\left(\frac{\pi}{3}n\right) + \cos\left(\frac{\pi}{4}n\right)$. Apply the cosine pair to each term and add by linearity:

$$\begin{aligned} X(e^{j\omega}) &= 2\pi \sum_k \left[\delta\left(\omega - \frac{\pi}{3} - 2\pi k\right) + \delta\left(\omega + \frac{\pi}{3} - 2\pi k\right) \right] \\ &\quad + \pi \sum_k \left[\delta\left(\omega - \frac{\pi}{4} - 2\pi k\right) + \delta\left(\omega + \frac{\pi}{4} - 2\pi k\right) \right]. \end{aligned}$$

The amplitude 2 multiplies the weight π of the first cosine pair to $2 \cdot \pi = 2\pi$; the second cosine has amplitude 1 and keeps the weight π . So inside one period, $-\pi < \omega \leq \pi$, there are four impulses: weight 2π at $\pm\pi/3$ and weight π at $\pm\pi/4$. Outside the period the same impulses reappear every 2π (the terms with $k = \pm 1$): weight 2π at $\pm 5\pi/3$ and $\pm 7\pi/3$, weight π at $\pm 7\pi/4$ and $\pm 9\pi/4$.

ORDER On the positive axis $\pi/4 < \pi/3 < 5\pi/3 < 7\pi/4 < 2\pi < 9\pi/4 < 7\pi/3$: the two trains interleave, and the order flips either side of 2π .

PERIOD $N_0 = (2\pi/\omega_0)m$, where m is the smallest positive integer making N_0 an integer. For $\omega_1 = 5\pi/3$: $2\pi/\omega_1 = 6/5$, $m = 5$, $N_0 = 6$. For $\omega_2 = 7\pi/4$: $8/7$, $m = 7$, $N_0 = 8$. The sum repeats after $\text{LCM}(6, 8) = 24$, and $24/6 = 4$, $24/8 = 3$ with $\text{gcd}(4, 3) = 1$, so nothing smaller works.

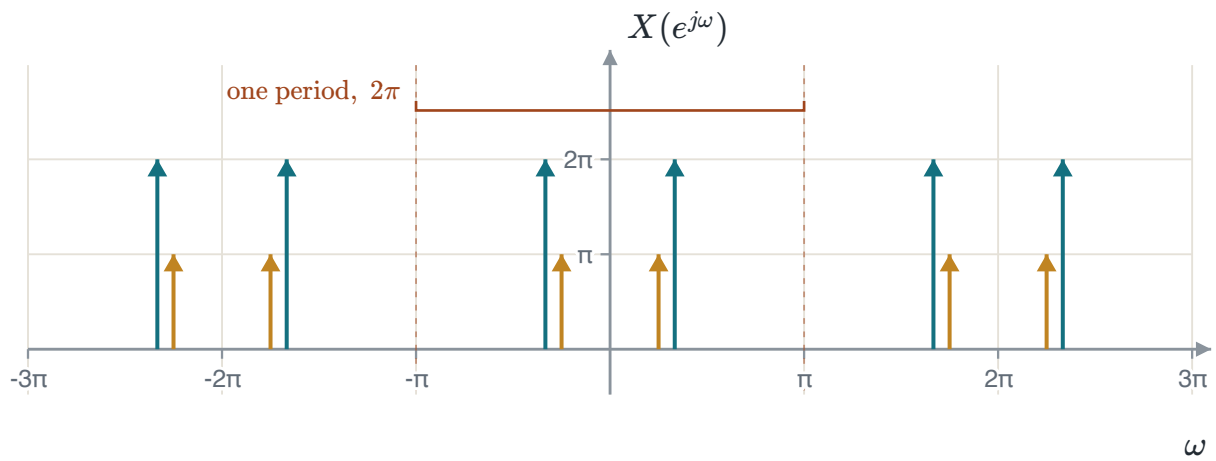


FIGURE 6.15 The spectrum over three periods. Tall arrows carry weight 2π , short ones π , and each pattern repeats every 2π .

6.6 Properties

TABLE 6.1

PROPERTY	PAIR
Linearity	$a x_1[n] + b x_2[n] \leftrightarrow a X_1 + b X_2$
Time shift	$x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(e^{j\omega})$
Frequency shift	$e^{j\omega_0 n} x[n] \leftrightarrow X(e^{j(\omega - \omega_0)})$
Conjugation	$x^*[n] \leftrightarrow X^*(e^{-j\omega})$
Time reversal	$x[-n] \leftrightarrow X(e^{-j\omega})$
Time expansion	$x_{(k)}[n] \leftrightarrow X(e^{jk\omega})$
Differencing in time	$x[n] - x[n - 1] \leftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$
Accumulation	$\sum_{m=-\infty}^n x[m] \leftrightarrow \frac{X(e^{j\omega})}{1 - e^{-j\omega}} + \pi X(e^{j0}) \sum_k \delta(\omega - 2\pi k)$
Differentiation in frequency	$n x[n] \leftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$
Convolution	$x[n] * h[n] \leftrightarrow X(e^{j\omega})H(e^{j\omega})$
Multiplication	$x[n]y[n] \leftrightarrow \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$
Real and even	$X(e^{j\omega})$ real and even
Real and odd	$X(e^{j\omega})$ purely imaginary and odd
Even-odd parts	$\mathcal{E}\{x\} \leftrightarrow \text{Re}\{X\}, \mathcal{O}\{x\} \leftrightarrow j \text{Im}\{X\}$
Parseval	$\sum_n x[n] ^2 = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) ^2 d\omega$

⊗ TWO NAMES THAT MUST NOT BE EXCHANGED

Discrete time has a **difference**, not a derivative: there is nothing between n and $n + 1$ over which to take a limit, and the factor is $1 - e^{-j\omega}$ rather than $j\omega$. The **differentiation** in this chapter is in frequency, and it is a genuine derivative because ω is continuous. Naming the first difference a derivative collapses the continuous and discrete cases on the one page where the distinction is being built.

Where the rows come from

Each row is proved the same way: write the analysis sum for the new sequence, substitute, and rearrange until the sum for x appears. The time shift needs a change of index. Put $m = n - n_0$, so $n = m + n_0$; as n runs over all integers, so does m :

$$\begin{aligned} \sum_{n=-\infty}^{\infty} x[n - n_0] e^{-j\omega n} &= \sum_{m=-\infty}^{\infty} x[m] e^{-j\omega(m+n_0)} \\ &= e^{-j\omega n_0} \sum_{m=-\infty}^{\infty} x[m] e^{-j\omega m} = e^{-j\omega n_0} X(e^{j\omega}). \end{aligned}$$

The factor $e^{-j\omega n_0}$ does not depend on m , so it moves outside the sum. The frequency shift needs no change of index; the two exponentials combine:

$$\sum_n e^{j\omega_0 n} x[n] e^{-j\omega n} = \sum_n x[n] e^{-j(\omega - \omega_0)n} = X(e^{j(\omega - \omega_0)}).$$

Conjugation uses $(ab)^* = a^* b^*$ and $(e^{-j\omega n})^* = e^{j\omega n}$. Time reversal puts $m = -n$, which again runs over all integers:

$$\sum_n x^*[n] e^{-j\omega n} = \left(\sum_n x[n] e^{j\omega n} \right)^* = X^*(e^{j\omega}), \quad \sum_n x[-n] e^{-j\omega n} = \sum_m x[m] e^{-j(-\omega)m} = X(e^{-j\omega}).$$

Differencing is linearity plus the time shift with $n_0 = 1$: $x[n] - x[n - 1]$ transforms to $X(e^{j\omega}) - e^{-j\omega} X(e^{j\omega}) = (1 - e^{-j\omega}) X(e^{j\omega})$. Differentiation in frequency differentiates the analysis sum term by term. Each term gives $\frac{d}{d\omega} e^{-j\omega n} = -jn e^{-j\omega n}$, so

$$\frac{dX(e^{j\omega})}{d\omega} = \sum_n x[n] (-jn) e^{-j\omega n} = -j \sum_n n x[n] e^{-j\omega n} \implies \sum_n n x[n] e^{-j\omega n} = j \frac{dX(e^{j\omega})}{d\omega},$$

where the last step multiplies both sides by j and uses $j \cdot (-j) = 1$. The convolution and multiplication rows are proved in Section 6.7, Parseval at the end of this section, and time expansion just below.

Accumulation is the convolution of x with the unit step $u[n]$, so its row is the convolution row with the transform of $u[n]$ as the second factor; that transform is the stated pair $1/(1 - e^{-j\omega}) + \pi \sum_k \delta(\omega - 2\pi k)$.

For a real sequence, $x^*[n] = x[n]$, so the conjugation row gives $X(e^{-j\omega}) = X^*(e^{j\omega})$. Taking real parts, moduli, imaginary parts and angles of both sides: the real part and the magnitude are even in ω and the imaginary part and the phase are odd. Half a period then determines the whole spectrum, for a real sequence only. A complex sequence needs the full period drawn.

That one line also fixes the last three rows of the table. The even part is $\mathcal{E}_v\{x[n]\} = \frac{1}{2}(x[n] + x[-n])$. By linearity and the time-reversal row it transforms to $\frac{1}{2}[X(e^{j\omega}) + X(e^{-j\omega})]$, and for a real sequence $X(e^{-j\omega}) = X^*(e^{j\omega})$, so this is $\frac{1}{2}[X + X^*] = \text{Re}\{X\}$. The odd part $\frac{1}{2}(x[n] - x[-n])$ transforms in the same way to $\frac{1}{2}[X - X^*] = j \text{Im}\{X\}$. A real and even sequence has no odd part, so its transform is real, and it is even by the symmetry above. A real and odd sequence has no even part, so its transform is purely imaginary and odd.

A worked check. Take $x[n] = a^n u[n]$ with $0 < a < 1$. For $n > 0$, $x[-n] = 0$, so the even part is $\frac{1}{2}a^n$; for $n < 0$ it is $\frac{1}{2}a^{-n}$; at $n = 0$ both $x[0]$ and $x[-0]$ equal 1, so the even part is 1. In one formula, $\mathcal{E}_v\{x[n]\} = \frac{1}{2}a^{|n|} + \frac{1}{2}\delta[n]$, where the δ term supplies the missing $\frac{1}{2}$ at the origin. Transform term by term with Example 6.3 and the unit-sample pair, then put the two terms over the common denominator:

$$\begin{aligned} \frac{1}{2} \cdot \frac{1 - a^2}{1 - 2a \cos \omega + a^2} + \frac{1}{2} &= \frac{(1 - a^2) + (1 - 2a \cos \omega + a^2)}{2(1 - 2a \cos \omega + a^2)} \\ &= \frac{2 - 2a \cos \omega}{2(1 - 2a \cos \omega + a^2)} = \frac{1 - a \cos \omega}{1 - 2a \cos \omega + a^2}. \end{aligned}$$

This is $\text{Re}\{1/(1 - ae^{-j\omega})\}$ exactly: in Example 6.2 the denominator is $(1 - a \cos \omega) + ja \sin \omega$, and multiplying numerator and denominator by its conjugate gives a real part $(1 - a \cos \omega)/(1 - 2a \cos \omega + a^2)$.

Accumulation undoes differencing, so their factors are reciprocals, and the impulse train is the part differencing destroys. A running sum is fixed only up to a constant offset, and the difference cannot see that offset; the weight $\pi X(e^{j0}) = \pi \sum_n x[n]$ is what puts it back. Where the sequence sums to zero the term vanishes on its own. It is the discrete-time counterpart of the $\pi X(0)\delta(\omega)$ in the integration property of Chapter 5, and it fails in the same place for the same reason: $1 - e^{-j\omega}$ is zero at $\omega = 0$.

Time expansion

For a positive integer k , $x_{(k)}[n]$ is $x[n/k]$ when n is a multiple of k and zero otherwise: a stretch with zeros inserted, since discrete time has no operation that stretches a sequence without leaving gaps. In the analysis sum, only the indices $n = rk$ with integer r contribute, because every other term is zero. Put $n = rk$; as n runs over the multiples of k , r runs over all integers:

$$\sum_{n=-\infty}^{\infty} x_{(k)}[n] e^{-j\omega n} = \sum_{r=-\infty}^{\infty} x_{(k)}[rk] e^{-j\omega rk} = \sum_{r=-\infty}^{\infty} x[r] e^{-j(k\omega)r} = X(e^{jk\omega}).$$

The middle step uses $x_{(k)}[rk] = x[rk/k] = x[r]$. The last sum is the analysis sum of x with ω replaced by $k\omega$. Replacing ω by $k\omega$ compresses the frequency axis by k , so one period of length 2π now holds k copies of the old picture.

EXAMPLE 6.8 — EXPANSION OF A FIVE-POINT PULSE

GIVEN $g[n] = 1$ on $|n| \leq 2$, and $x[n] = y_{(2)}[n] + 2y_{(2)}[n - 1]$ where $y[n] = g[n - 2]$.

FIND $X(e^{j\omega})$ and its peak value.

METHOD Build the sequence in three steps, shift, expansion and a shifted sum, and apply the matching property at each step. Each step multiplies or rescales a known transform, so no new sum is needed.

SOLUTION The pulse is Example 6.4 with $N_1 = 2$, so $G(e^{j\omega}) = \sin(5\omega/2)/\sin(\omega/2)$. The shift by $n_0 = 2$ gives

$$Y(e^{j\omega}) = e^{-j2\omega} G(e^{j\omega}) = e^{-j2\omega} \frac{\sin(5\omega/2)}{\sin(\omega/2)}.$$

Expansion by $k = 2$ replaces ω by 2ω everywhere:

$$Y_{(2)}(e^{j\omega}) = Y(e^{j2\omega}) = e^{-j4\omega} \frac{\sin(5\omega)}{\sin(\omega)}.$$

Finally, by linearity and the shift by 1,

$$X(e^{j\omega}) = Y_{(2)}(e^{j\omega}) + 2e^{-j\omega} Y_{(2)}(e^{j\omega}) = (1 + 2e^{-j\omega}) e^{-j4\omega} \frac{\sin(5\omega)}{\sin(\omega)}.$$

At $\omega = 0$ the first factor is $1 + 2 = 3$ and the kernel is its peak value 5, the number of ones in the pulse, so the peak of X is $3 \times 5 = 15$.

CHECK $x[n]$ is $y_{(2)}[n]$ plus twice a copy shifted by one. The expanded sequence has five ones at even indices, the shifted copy has five twos at odd indices, so $\sum_n x[n] = 5 + 10 = 15 = X(e^{j0})$.

WHERE THE ARGUMENT HALVED

G carries $\sin(5\omega/2)/\sin(\omega/2)$ and $Y_{(2)}$ carries $\sin(5\omega)/\sin(\omega)$: the same expression with ω replaced by 2ω . Halving only the numerator argument, $\sin(5\omega/2)/\sin(\omega)$, is wrong for both sequences, and it puts a pole at $\omega = \pi$ where the original has none: it gives 1000.0 at $\omega = \pi - 10^{-3}$, against the true $G(e^{j\pi}) = \sin(5\pi/2)/\sin(\pi/2) = 1$, and -0.7071 at $\pi/2$ against the true $G(e^{j\pi/2}) = \sin(5\pi/4)/\sin(\pi/4) = -1.0000$.

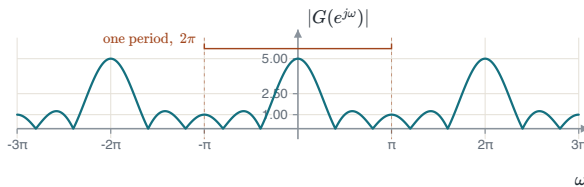


FIGURE 6.16 $|G(e^{j\omega})|$: peak 5, and the finite value 1 at $\omega = \pi$.

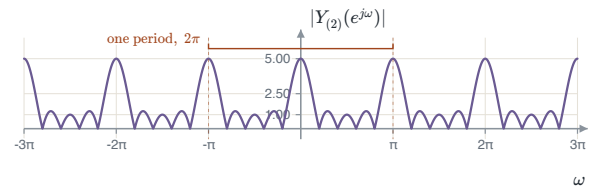


FIGURE 6.17 $|Y_{(2)}(e^{j\omega})|$: the same picture twice inside every 2π .

Parseval

Start from the energy sum, write $|x[n]|^2 = x[n]x^*[n]$, and replace $x^*[n]$ by the conjugate of its synthesis integral. Then swap the sum and the integral and collect the sum over n :

$$\begin{aligned} \sum_n |x[n]|^2 &= \sum_n x[n] x^*[n] = \sum_n x[n] \left[\frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \right]^* \\ &= \frac{1}{2\pi} \int_{2\pi} X^*(e^{j\omega}) \left[\sum_n x[n] e^{-j\omega n} \right] d\omega = \frac{1}{2\pi} \int_{2\pi} X^*(e^{j\omega}) X(e^{j\omega}) d\omega = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega. \end{aligned}$$

Conjugating the integral conjugates X and turns $e^{j\omega n}$ into $e^{-j\omega n}$; the bracket in the second line is then the analysis sum, which is $X(e^{j\omega})$. The quantity $|X(e^{j\omega})|^2$ is the energy-density spectrum. Integrating it over

one period and dividing by 2π gives the total energy. Integrating over part of a period gives the energy in that frequency band. Both the single-period range and the factor $\frac{1}{2\pi}$ are required, although neither is visible from the plot shape.

Two checks. For $a^n u[n]$ with $a = \frac{1}{2}$, the time side is a geometric series with first term 1 and ratio $\frac{1}{4}$: $\sum_{n \geq 0} (\frac{1}{4})^n = 1/(1 - \frac{1}{4}) = \frac{4}{3}$. On the frequency side, $|X|^2 = 1/(1 - 2a \cos \omega + a^2)$ from Example 6.2, and Example 6.3 shows that $(1 - a^2)/(1 - 2a \cos \omega + a^2)$ is the transform of $a^{|n|}$. So $|X|^2$ is that transform divided by $1 - a^2$, and its synthesis integral at $n = 0$ returns $a^{|0|} = 1$:

$$\frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega = \frac{1}{1 - a^2} \cdot \frac{1}{2\pi} \int_{2\pi} \frac{1 - a^2}{1 - 2a \cos \omega + a^2} d\omega = \frac{1}{1 - a^2} \cdot 1 = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3}.$$

For the rectangular pulse with $N_1 = 2$, the time side is five samples of $|1|^2$, so 5. The kernel X is real, so $|X|^2 = X \cdot X$, which by the convolution property is the transform of $x[n] * x[n]$; its synthesis integral at $n = 0$ is $(x * x)[0] = \sum_m x[m]x[-m] = 5$, the same number.

6.7 Convolution and multiplication

CONVOLUTION

$$y[n] = x[n] * h[n] \longleftrightarrow Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

Both pairs $x \leftrightarrow X$ and $h \leftrightarrow H$ are declared before the conclusion is written, and y names the output and nothing else. A product of complex numbers gives two separate statements, $|Y| = |X| \cdot |H|$ and $\angle Y = \angle X + \angle H$. Putting modulus bars around a whole equation is not an operation on anything.

To prove it, put the convolution sum $y[n] = \sum_k x[k]h[n - k]$ into the analysis sum, swap the order of the two sums, and change the index of the inner sum with $m = n - k$. For fixed k , as n runs over all integers so does m :

$$\begin{aligned} Y(e^{j\omega}) &= \sum_n \left[\sum_k x[k]h[n - k] \right] e^{-j\omega n} = \sum_k x[k] \sum_n h[n - k] e^{-j\omega n} \\ &= \sum_k x[k] \sum_m h[m] e^{-j\omega(m+k)} = \left[\sum_k x[k] e^{-j\omega k} \right] \left[\sum_m h[m] e^{-j\omega m} \right] = X(e^{j\omega})H(e^{j\omega}). \end{aligned}$$

In the second line $e^{-j\omega(m+k)} = e^{-j\omega k} e^{-j\omega m}$, and the factor $e^{-j\omega k}$ does not depend on m , so it joins the outer sum. Swapping the sums is allowed when both sequences are absolutely summable.

EXAMPLE 6.9 – CONVOLVING TWO EXPONENTIALS

GIVEN

$x[n] = a^n u[n]$ and $h[n] = b^n u[n]$, with $|a| < 1$, $|b| < 1$ and $a \neq b$. Every one-sided exponential carries its $u[n]$: without it the sequence is defined for negative n too, where it grows without bound.

METHOD

Use the convolution property because the required output is a time convolution. Multiply the transforms, then use partial fractions so each term matches the known one-sided exponential pair. Write $z = e^{-j\omega}$ as an algebraic variable; then $Y = 1/((1 - az)(1 - bz))$ is a rational function of z , and substituting $z = 1/a$ is algebra in z , not a claim that $e^{-j\omega}$ takes that value.

FIND $y[n] = x[n] * h[n]$.

SOLUTION By Example 6.2, $X = 1/(1 - az)$ and $H = 1/(1 - bz)$ with $z = e^{-j\omega}$, so the convolution property gives

$$Y = XH = \frac{1}{(1 - az)(1 - bz)} = \frac{A}{1 - az} + \frac{B}{1 - bz}.$$

Cover-up: to find A , cover the factor $(1 - az)$ and evaluate the rest at $z = 1/a$, the value that makes that factor zero. To find B , cover $(1 - bz)$ and evaluate at $z = 1/b$:

$$A = \left. \frac{1}{1 - bz} \right|_{z=1/a} = \frac{1}{1 - b/a} = \frac{a}{a - b},$$

$$B = \left. \frac{1}{1 - az} \right|_{z=1/b} = \frac{1}{1 - a/b} = \frac{b}{b - a} = -\frac{b}{a - b}.$$

Check by recombining: $A(1 - bz) + B(1 - az) = \frac{a - abz - b + abz}{a - b} = 1$, as required. Each fraction is the one-sided exponential pair, so inverting term by term:

$$y[n] = A a^n u[n] + B b^n u[n] = \frac{a \cdot a^n - b \cdot b^n}{a - b} u[n] = \frac{1}{a - b} [a^{n+1} - b^{n+1}] u[n].$$

CONDITION Both coefficients divide by $a - b$, so the route requires $a \neq b$. At $a = b$ the two poles merge and the answer takes a different form, worked out in Section 6.8.

CHECK $a = \frac{1}{2}$, $b = \frac{1}{4}$: $a - b = \frac{1}{4}$, so $y[n] = 4[(\frac{1}{2})^{n+1} - (\frac{1}{4})^{n+1}]u[n]$, giving $y[0] = 4(\frac{1}{2} - \frac{1}{4}) = 1$, $y[1] = 4(\frac{1}{4} - \frac{1}{16}) = 0.75$, $y[2] = 4(\frac{1}{8} - \frac{1}{64}) = 0.4375$, $y[3] = 4(\frac{1}{16} - \frac{1}{256}) = 0.234375$. Direct convolution gives $y[0] = x[0]h[0] = 1$ and $y[1] = x[0]h[1] + x[1]h[0] = \frac{1}{4} + \frac{1}{2} = 0.75$; the first value must be $x[0]h[0]$ for any two causal sequences. The magnitude $|Y| = |X| |H|$ is largest at $\omega = 0$, $\frac{1}{(1-\frac{1}{2})(1-\frac{1}{4})} = 2.6667$, and smallest at $\omega = \pi$, $\frac{1}{(1+\frac{1}{2})(1+\frac{1}{4})} = 0.5333$.

The product of two ideal low-pass responses is the narrower of the two, so a cascade keeps only the band both filters pass. With cutoffs $\pi/2$ and $\pi/4$, the cascade has cutoff $\pi/4$ and impulse response $\sin(\pi n/4)/(\pi n)$. In writing such a spectrum, close one branch and leave the other open — 1 for $|\omega| \leq \pi/4$ and 0 for $\pi/4 < |\omega| \leq \pi$ — since for an ideal filter the band edge is precisely where a convention is needed rather than assumed.

EXAMPLE 6.10 — A STEPPED SPECTRUM THROUGH A FILTER

GIVEN $X(e^{j\omega}) = 2$ for $|\omega| \leq \pi/4$ and 1 for $\pi/4 < |\omega| \leq 3\pi/4$, zero to π ; H is ideal low-pass with cutoff $\pi/2$.

FIND $Y(e^{j\omega})$ and $y[n]$.

METHOD An LTI system gives $Y = XH$. Multiply the two spectra frequency by frequency, then write the result as a sum of ideal low-pass bands so that Example 6.5 inverts each one.

SOLUTION H is 1 for $|\omega| \leq \pi/2$ and 0 for $\pi/2 < |\omega| \leq \pi$. Multiplying band by band: for $|\omega| \leq \pi/4$, $Y = 2 \cdot 1 = 2$; for $\pi/4 < |\omega| \leq \pi/2$, $Y = 1 \cdot 1 = 1$; for $\pi/2 < |\omega| \leq \pi$, $Y = X \cdot 0 = 0$.

$0 = 0$. Now write Y as two stacked ideal bands, $Y = Y_1 + Y_2$, with $Y_1 = 1$ on $|\omega| \leq \pi/2$ and $Y_2 = 1$ on $|\omega| \leq \pi/4$; the two add to 2 on the inner band and to 1 on the outer band, as required. By linearity and Example 6.5 with $W = \pi/2$ and $W = \pi/4$:

$$\begin{aligned} y[n] &= \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{j\omega n} d\omega + \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} e^{j\omega n} d\omega \\ &= \frac{\sin(\pi n/2)}{\pi n} + \frac{\sin(\pi n/4)}{\pi n}, \\ y[0] &= \frac{\pi/2}{\pi} + \frac{\pi/4}{\pi} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}. \end{aligned}$$

CHECK

$y[0]$ must be the area of one period of Y divided by 2π . That area is $2 \cdot \frac{\pi}{2} + 1 \cdot \frac{\pi}{2} = \frac{3\pi}{2}$ (height 2 over a band of width $\pi/2$, height 1 over two bands of width $\pi/4$), and $\frac{3\pi}{2}/2\pi = \frac{3}{4}$. A sketch of a spectrum is not a solution; the closed form costs one line once the stack is seen.

Multiplication is a periodic convolution

MULTIPLICATION

$$z[n] = x[n]y[n] \longleftrightarrow Z(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) Y(e^{j(\omega-\theta)}) d\theta$$

The integral runs over one period and both factors are 2π -periodic. That operation is a periodic convolution, and it differs from the ordinary convolution of two functions on the line whenever the result is wider than one period.

To prove it, put the product into the analysis sum and replace $x[n]$ by its synthesis integral, with θ as the integration variable so that it is not confused with ω . Then swap the sum and the integral and combine the exponentials:

$$\begin{aligned} Z(e^{j\omega}) &= \sum_n x[n]y[n]e^{-j\omega n} = \sum_n \left[\frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})e^{j\theta n} d\theta \right] y[n]e^{-j\omega n} \\ &= \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) \left[\sum_n y[n]e^{-j(\omega-\theta)n} \right] d\theta = \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) Y(e^{j(\omega-\theta)}) d\theta. \end{aligned}$$

The bracket is the analysis sum of y evaluated at frequency $\omega - \theta$, because $e^{j\theta n}e^{-j\omega n} = e^{-j(\omega-\theta)n}$.

Take $X = 1$ on $|\omega| \leq 3\pi/4$ and $Y = 1$ on $|\omega| \leq \pi/2$, both repeated every 2π . First convolve the two rectangles as functions on the line. The integrand $X(\theta)Y(\omega - \theta)$ is 1 where both factors are 1 and 0 elsewhere, so the integral is the length of the overlap of the interval $|\theta| \leq 3\pi/4$ with the interval $|\omega - \theta| \leq \pi/2$, that is $\omega - \pi/2 \leq \theta \leq \omega + \pi/2$. For $|\omega| \leq \pi/4$ the second interval lies inside the first, so the overlap is its full width π and the value is $\frac{1}{2\pi} \cdot \pi = \frac{1}{2}$. For $\pi/4 \leq |\omega| \leq 5\pi/4$ the overlap runs from $|\omega| - \pi/2$ to $3\pi/4$, of length $5\pi/4 - |\omega|$, so the value is $(5\pi/4 - |\omega|)/(2\pi)$, falling to zero at $|\omega| = 5\pi/4$. The result is a trapezoid of height $\frac{1}{2}$, flat on $|\omega| \leq \pi/4$ and reaching zero at $|\omega| = 5\pi/4$. Since $5\pi/4 > \pi$, the result extends beyond one period. A periodic copy therefore overlaps it near $\omega = \pm\pi$, and the two values add.

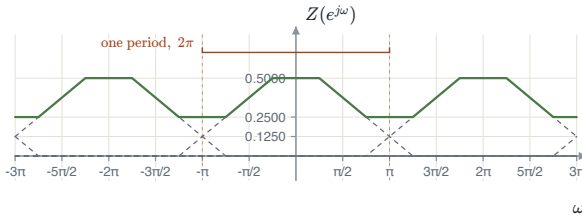


FIGURE 6.18 The dashed trapezoids are the copies; the solid curve is their sum. At $\omega = \pi$ the value doubles from 0.125 to 0.25.

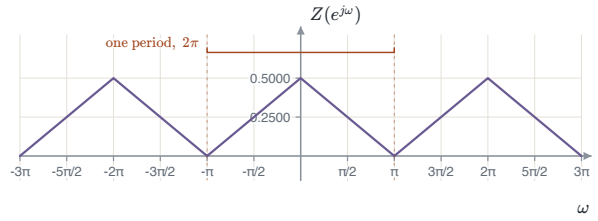


FIGURE 6.19 Two equal bands of half-width $\pi/2$: the triangle fills one period and reaches zero at both boundaries, so repeated copies do not overlap.

The values are $Z(e^{j0}) = Z(e^{j\pi/4}) = 0.5$ on the flat top and $Z(e^{j3\pi/4}) = (5\pi/4 - 3\pi/4)/(2\pi) = (\pi/2)/(2\pi) = 0.25$ on the slope. At $\omega = \pi$ the trapezoid itself gives $(5\pi/4 - \pi)/(2\pi) = (\pi/4)/(2\pi) = \frac{1}{8}$, and the copy centred at 2π is at the same distance from its own centre, so it also gives $\frac{1}{8}$: $Z(e^{j\pi}) = \frac{1}{8} + \frac{1}{8} = 0.25$. The check is the synthesis equation at $n = 0$: $\frac{1}{2\pi} \int_{-\pi}^{\pi} Z d\omega$ must equal $z[0] = x[0]y[0]$. By Example 6.5, $x[0] = \frac{3\pi/4}{\pi} = \frac{3}{4}$ and $y[0] = \frac{\pi/2}{\pi} = \frac{1}{2}$, so $z[0] = \frac{3}{8} = 0.375$. On the frequency side, one period of Z collects the whole area of one trapezoid, because the parts that spill past $\pm\pi$ are exactly the parts that the copies bring in. That area is the height $\frac{1}{2}$ times the mean of the top width $\pi/2$ and the base width $5\pi/2$: $\frac{1}{2} \cdot \frac{3\pi}{2} = \frac{3\pi}{4}$, and $\frac{3\pi/4}{2\pi} = \frac{3}{8}$, as required.

EXAMPLE 6.11 – MULTIPLYING BY A COSINE

GIVEN $X(e^{j\omega}) = 1$ for $|\omega| \leq \pi/4$, repeated every 2π , and $y[n] = \cos(\omega_0 n)$ with $\omega_0 = \pi/3$.

METHOD Write Y as its impulse train, $\pi \sum_l [\delta(\omega - \omega_0 - 2\pi l) + \delta(\omega + \omega_0 - 2\pi l)]$. The periodic convolution integrates over a single period, so choosing $-\pi < \theta \leq \pi$ leaves exactly two impulses, at $\theta = \pm\pi/3$, and only those contribute. Taking the whole train into an ordinary convolution would count every copy and diverge.

FIND $Z(e^{j\omega})$ for $z[n] = x[n]y[n]$, and the band edges.

SOLUTION Periodic convolution is commutative, so put the impulse train in the second factor of the multiplication property and integrate over $-\pi < \theta \leq \pi$, where only the $l = 0$ impulses at $\theta = \pm\omega_0$ lie. The sifting property evaluates X at those two points:

$$\begin{aligned} Z(e^{j\omega}) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} Y(e^{j\theta}) X(e^{j(\omega-\theta)}) d\theta \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \pi [\delta(\theta - \omega_0) + \delta(\theta + \omega_0)] X(e^{j(\omega-\theta)}) d\theta \\ &= \frac{\pi}{2\pi} [X(e^{j(\omega-\omega_0)}) + X(e^{j(\omega+\omega_0)})] \\ &= \frac{1}{2} X(e^{j(\omega-\omega_0)}) + \frac{1}{2} X(e^{j(\omega+\omega_0)}). \end{aligned}$$

The height $\frac{1}{2}$ comes from $\frac{1}{2\pi} \cdot \pi$.

BAND EDGES

The band moves to $\omega_0 \mp \pi/4$, so with $\omega_0 = \pi/3$ the edges are $\pi/12$ and $7\pi/12$, and the mirror image occupies $-7\pi/12 \leq \omega \leq -\pi/12$. Both numbers are computed from ω_0 and the bandwidth, so they cannot drift apart from the drawn figure; with ω_0 below $\pi/4$ the two bands would overlap through the origin instead.

⚠ THIS OVERLAP IS NOT ALIASING

This overlap occurs between periodic copies during frequency-domain convolution. Chapter 7 studies a different overlap between replicas produced by sampling. Only the sampling overlap is called aliasing.

6.8 Difference equations

Transform both sides of a linear constant-coefficient difference equation. Linearity acts term by term, and the time-shift property turns each $y[n - k]$ into $e^{-j\omega k} Y(e^{j\omega})$ and each $x[n - k]$ into $e^{-j\omega k} X(e^{j\omega})$. The transforms Y and X do not depend on k , so they come out of the sums:

FREQUENCY RESPONSE OF A DIFFERENCE EQUATION

$$\sum_{k=0}^N a_k y[n - k] = \sum_{k=0}^M b_k x[n - k]$$

$$\sum_{k=0}^N a_k e^{-j\omega k} Y(e^{j\omega}) = \sum_{k=0}^M b_k e^{-j\omega k} X(e^{j\omega}) \implies Y(e^{j\omega}) \sum_{k=0}^N a_k e^{-j\omega k} = X(e^{j\omega}) \sum_{k=0}^M b_k e^{-j\omega k}$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^M b_k e^{-j\omega k}}{\sum_{k=0}^N a_k e^{-j\omega k}}$$

The last line divides both sides by $X(e^{j\omega})$ and by the denominator sum. The impulse response of the recursion is now available without running the recursion: factor the denominator, split into partial fractions, and invert each piece with the exponential pair. Every step is algebra.

EXAMPLE 6.12 – A SECOND-ORDER SYSTEM

GIVEN $y[n] - \frac{3}{4}y[n - 1] + \frac{1}{8}y[n - 2] = 2x[n]$.

FIND The frequency response and the impulse response.

METHOD The equation has constant coefficients, so the time-shift property turns it into algebra. Read the coefficients into the frequency-response ratio, factor the denominator, expand into partial fractions, and invert each term with the one-sided exponential pair.

SOLUTION Here $a_0 = 1$, $a_1 = -\frac{3}{4}$, $a_2 = \frac{1}{8}$ and $b_0 = 2$. Transforming term by term:

$$Y - \frac{3}{4}e^{-j\omega}Y + \frac{1}{8}e^{-j2\omega}Y = 2X \implies H(e^{j\omega}) = \frac{Y}{X} = \frac{2}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-j2\omega}}.$$

Write $z = e^{-j\omega}$ as an algebraic variable and factor the quadratic $1 - \frac{3}{4}z + \frac{1}{8}z^2$. Two factors $(1 - \alpha z)(1 - \beta z)$ expand to $1 - (\alpha + \beta)z + \alpha\beta z^2$, so we need $\alpha + \beta = \frac{3}{4}$ and $\alpha\beta = \frac{1}{8}$; $\alpha = \frac{1}{2}$ and $\beta = \frac{1}{4}$ satisfy both, since $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ and $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$. Hence

$$H = \frac{2}{(1 - \frac{1}{2}z)(1 - \frac{1}{4}z)} = \frac{A}{1 - \frac{1}{2}z} + \frac{B}{1 - \frac{1}{4}z}.$$

Cover-up: cover $(1 - \frac{1}{2}z)$ and set $z = 2$, then cover $(1 - \frac{1}{4}z)$ and set $z = 4$:

$$A = \frac{2}{1 - \frac{1}{4}z} \Big|_{z=2} = \frac{2}{1 - \frac{1}{2}} = 4, \quad B = \frac{2}{1 - \frac{1}{2}z} \Big|_{z=4} = \frac{2}{1 - 2} = -2.$$

Recombining, $4(1 - \frac{1}{4}z) - 2(1 - \frac{1}{2}z) = 4 - z - 2 + z = 2$, which matches the numerator. Each fraction is the pair $a^n u[n] \leftrightarrow 1/(1 - ae^{-j\omega})$, so

$$h[n] = 4 \left(\frac{1}{2}\right)^n u[n] - 2 \left(\frac{1}{4}\right)^n u[n].$$

CHECK

$h[0] = 4 - 2 = 2$, $h[1] = 4 \cdot \frac{1}{2} - 2 \cdot \frac{1}{4} = 2 - 0.5 = 1.5$, $h[2] = 4 \cdot \frac{1}{4} - 2 \cdot \frac{1}{16} = 1 - 0.125 = 0.875$. Substituting into the equation with $x = \delta$, which is zero for $n \geq 1$: $h[1] - \frac{3}{4}h[0] = 1.5 - 1.5 = 0$ and $h[2] - \frac{3}{4}h[1] + \frac{1}{8}h[0] = 0.875 - 1.125 + 0.25 = 0$, as required. Both poles have modulus below 1, so the system is stable. At $\omega = 0$, $z = 1$ and $|H| = 2/(1 - \frac{3}{4} + \frac{1}{8}) = 2/\frac{3}{8} = 5.3333$; at $\omega = \pi$, $z = -1$ and $|H| = 2/(1 + \frac{3}{4} + \frac{1}{8}) = 2/\frac{15}{8} = 1.0667$. This recursion is a low-pass filter.

The repeated-pole pair

Example 6.9 excluded $a = b$ because its partial fractions divide by $a - b$. When $a = b$, the poles coincide and require a separate pair. Derive it from the geometric pair of Example 6.2, $\sum_{n=0}^{\infty} a^n e^{-j\omega n} = 1/(1 - ae^{-j\omega})$, by differentiating both sides with respect to a while holding ω fixed. On the left, differentiate term by term: $\frac{d}{da} a^n = na^{n-1}$, and the $n = 0$ term is the constant 1, whose derivative is 0, so the sum starts at $n = 1$. On the right, use the chain rule on $(1 - ae^{-j\omega})^{-1}$:

$$\sum_{n=1}^{\infty} n a^{n-1} e^{-j\omega n} = \frac{d}{da} (1 - ae^{-j\omega})^{-1} = -(1 - ae^{-j\omega})^{-2} \cdot (-e^{-j\omega}) = \frac{e^{-j\omega}}{(1 - ae^{-j\omega})^2}.$$

Now put $m = n - 1$ on the left, so $n = m + 1$ and m runs from 0 to ∞ . The exponential splits as $e^{-j\omega(m+1)} = e^{-j\omega} e^{-j\omega m}$, and the factor $e^{-j\omega}$ leaves the sum and cancels the same factor on the right:

$$\begin{aligned} \sum_{m=0}^{\infty} (m+1) a^m e^{-j\omega(m+1)} &= e^{-j\omega} \sum_{m=0}^{\infty} (m+1) a^m e^{-j\omega m} = \frac{e^{-j\omega}}{(1 - ae^{-j\omega})^2} \\ \Rightarrow \sum_{m=0}^{\infty} (m+1) a^m e^{-j\omega m} &= \frac{1}{(1 - ae^{-j\omega})^2}. \end{aligned}$$

REPEATED POLE

$$(n+1)a^n u[n] \longleftrightarrow \frac{1}{(1 - ae^{-j\omega})^2}, \quad |a| < 1$$

The left-hand sum is the analysis sum of $(n+1)a^n u[n]$, with m renamed n . Differentiating a convergent geometric series term by term is allowed for $|a| < 1$, which is why the condition is kept. The exponent 2 comes from the differentiation and the minus sign is inherited unchanged, so neither has to be remembered. One number settles any doubt: at $\omega = 0$ with $a = \frac{1}{4}$ the value is $(1 - \frac{1}{4})^{-2} = \frac{16}{9} = 1.7778$, while a plus sign would give $(\frac{5}{4})^{-2} = 0.6400$.

EXAMPLE 6.13 – THE OUTPUT OF THAT SYSTEM

GIVEN

The system of Example 6.12 driven by $x[n] = (\frac{1}{4})^n u[n]$.

FIND

 $y[n]$.

METHOD

Use $Y = XH$ and partial fractions, because the output transform is rational and each fraction can be inverted with a known pair. The input pole coincides with one system pole, so the expansion must include both a simple and a squared term for that pole.

SOLUTION

With $z = e^{-j\omega}$, the input transform is $X = 1/(1 - \frac{1}{4}z)$ by Example 6.2, and H is the factored form of Example 6.12. Multiply:

$$\begin{aligned} Y = XH &= \frac{1}{1 - \frac{1}{4}z} \cdot \frac{2}{(1 - \frac{1}{2}z)(1 - \frac{1}{4}z)} = \frac{2}{(1 - \frac{1}{2}z)(1 - \frac{1}{4}z)^2} \\ &= \frac{A}{1 - \frac{1}{4}z} + \frac{B}{(1 - \frac{1}{4}z)^2} + \frac{C}{1 - \frac{1}{2}z}. \end{aligned}$$

Coefficients. Cover-up reaches the simple pole and the highest power of the repeated pole. Cover $(1 - \frac{1}{2}z)$ and set $z = 2$:

$$C = \left. \frac{2}{(1 - \frac{1}{4}z)^2} \right|_{z=2} = \frac{2}{(1 - \frac{1}{2})^2} = \frac{2}{\frac{1}{4}} = 8.$$

Cover $(1 - \frac{1}{4}z)^2$ and set $z = 4$:

$$B = \left. \frac{2}{1 - \frac{1}{2}z} \right|_{z=4} = \frac{2}{1 - 2} = -2.$$

Cover-up cannot reach A , so use one more value of z . Multiply the identity through by the full denominator and set $z = 0$; every factor becomes 1, so $2 = A + B + C = A - 2 + 8$ and $A = -4$. Recombine as a check: $-4(1 - \frac{1}{4}z)(1 - \frac{1}{2}z) - 2(1 - \frac{1}{2}z) + 8(1 - \frac{1}{4}z)^2$ expands to $(-4 + 3z - \frac{1}{2}z^2) + (-2 + z) + (8 - 4z + \frac{1}{2}z^2) = 2$, the numerator. Invert term by term. The first and third fractions are the one-sided exponential pair; the middle one is the repeated-pole pair with $a = \frac{1}{4}$:

$$y[n] = -4 \left(\frac{1}{4}\right)^n u[n] - 2(n+1) \left(\frac{1}{4}\right)^n u[n] + 8 \left(\frac{1}{2}\right)^n u[n].$$

CHECK

$y[0] = -4 - 2 + 8 = 2$, $y[1] = -4 \cdot \frac{1}{4} - 2 \cdot 2 \cdot \frac{1}{4} + 8 \cdot \frac{1}{2} = -1 - 1 + 4 = 2$, $y[2] = -\frac{4}{16} - \frac{6}{16} + 2 = 1.375$, $y[3] = -\frac{4}{64} - \frac{8}{64} + 1 = 0.8125$. The first value must equal $h[0]x[0] = 2 \cdot 1 = 2$ for two causal sequences, and it does. Direct convolution, $y[1] = h[0]x[1] + h[1]x[0] = 2 \cdot \frac{1}{4} + 1.5 \cdot 1 = 2$, reproduces the second value as well. Without the repeated-pole pair the middle term could not be inverted at all.

6.9 Duality

There is no duality inside the discrete-time transform pair. The analysis equation is a sum over an integer and the synthesis equation an integral over a continuous variable, so no relabelling turns one into the other. Two dualities do hold.

1. In the discrete-time Fourier series, where both domains are discrete and both objects are periodic with the same period N : if $x[n] \leftrightarrow a_k$, then reading the coefficients as a sequence gives $a[n] \leftrightarrow$

$\frac{1}{N}x[-k]$. The impulse train with $N = 21$ is the example: its coefficients are all $1/21 = 0.0476$, and reading those constants back as a sequence returns an impulse train.

- Between the discrete-time transform and the continuous-time series. $X(e^{j\omega})$ is a 2π -periodic function of a continuous variable, so it has a continuous-time Fourier series, and comparing the two synthesis equations identifies its coefficients as $x[-k]$: the spectrum of a sequence is a periodic signal whose series coefficients are the sequence itself, reversed.

A worked check on the second. Take the 2π -periodic square wave in ω , equal to 1 for $|\omega| \leq \pi/2$ and zero over the rest of the period. Treat ω as the time variable of a continuous-time periodic signal with period 2π , so its fundamental frequency is 1 and its series coefficients are given by the analysis integral over one period. The function is zero outside $|\omega| \leq \pi/2$, so the limits shrink, and for $k \neq 0$ the antiderivative of $e^{-jk\omega}$ is $e^{-jk\omega}/(-jk)$:

$$\begin{aligned} a_k &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{-jk\omega} d\omega = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{-jk\omega} d\omega \\ &= \frac{1}{2\pi} \left[\frac{e^{-jk\omega}}{-jk} \right]_{-\pi/2}^{\pi/2} = \frac{1}{2\pi} \cdot \frac{e^{jk\pi/2} - e^{-jk\pi/2}}{jk} = \frac{1}{2\pi} \cdot \frac{2j \sin(k\pi/2)}{jk} = \frac{\sin(k\pi/2)}{k\pi}. \end{aligned}$$

The third equality reverses the order of the two limit terms to absorb the minus sign, and the fourth uses $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$. At $k = 0$ the integrand is 1, so $a_0 = \frac{1}{2\pi} \cdot \pi = \frac{1}{2}$. The list is 0.5, $1/\pi = 0.3183$, 0, $-1/(3\pi) = -0.1061$ for $k = 0, 1, 2, 3$. The inverse transform of the same square wave is Example 6.5 with $W = \pi/2$: $x[n] = \sin(\pi n/2)/(\pi n)$ with $x[0] = \frac{1}{2}$, which is the same list of numbers, so $a_k = x[-k]$. Here x is even, so $a_k = x[k]$ as well.

6.10 What to carry into Chapter 7

- The pair, with analysis as the sum and synthesis as the integral over one period, carrying $\frac{1}{2\pi}$.
- The 2π -periodicity and its one-line proof, and that the same line fails in continuous time.
- Real means the phase is 0 or π , not that the phase is zero.
- Multiplication in time is a **periodic** convolution over 2π , so anything reaching across a period boundary folds back and adds.
- The unnormalised sinc, $\text{sinc } \theta = \sin \theta / \theta$, restated at every point of use.
- Discrete time has a difference, not a derivative; the derivative in this chapter is in frequency.

A sequence can be formed by reading a continuous-time signal at regular instants. The next chapter studies how this sampling operation changes the spectrum. Sampling produces repeated spectral copies separated by the sampling frequency, so the periodic-overlap ideas from this chapter will be used again.

Exercises

6.1 Prove that $X(e^{j\omega})$ is 2π -periodic, and say precisely where the same argument fails for $X(j\omega)$.

Answer: Substituting $\omega + 2\pi$ into the analysis sum produces a factor $e^{-j2\pi n}$, which is 1 in every term because n is an integer. In continuous time the corresponding factor is $e^{-j2\pi t}$ with t real, which equals 1 only at integer t and cannot leave the integral.

6.2 For $x[n] = a^n u[n]$ with $a = \frac{1}{8}$, give $|X|_{\max}$, $|X|_{\min}$ and $\max |\angle X|$ in closed form and as numbers.

Answer: $1/(1-a) = 8/7 = 1.1429$, $1/(1+a) = 8/9 = 0.8889$, and $\arcsin \frac{1}{8} = 0.1253$ rad.

6.3 Explain why the finite geometric sum behind the Dirichlet kernel needs $r \neq 1$ rather than $|r| < 1$, and give the value of the kernel at the excluded points.

Answer: A finite sum always has a value; only the closed form fails, where its denominator $1-r$ vanishes. Here $|r| = 1$ at every ω , so $|r| < 1$ would exclude the sum entirely. At $r = 1$, that is ω a multiple of 2π , every term is 1 and the sum is $2N_1 + 1$.

6.4 The Dirichlet kernel is real at every ω . Does it follow that $|X| = X$ and $\angle X = 0$?

Answer: No. Realness fixes the imaginary part and not the sign. The kernel is negative on part of every period — least value -1.2500 for $N_1 = 2$ — and there $|X| = -X$ and $\angle X = \pi$. The conclusion needs the extra hypothesis $X \geq 0$, which holds for $a^{|n|}$ but not here.

6.5 Invert the ideal low-pass spectrum with $W = \pi/2$ and give $x[0]$ and $x[2]$. State the sinc convention you use.

Answer: $x[n] = \sin(Wn)/(\pi n)$ with $x[0] = W/\pi = 0.5$, and $x[2] = \sin(\pi)/(2\pi) = 0$. In the unnormalised convention $\text{sinc } \theta = \sin \theta / \theta$, the same sequence is $\frac{W}{\pi} \text{sinc}(Wn)$.

6.6 Where do the impulses of $2 \cos\left(\frac{5\pi}{3}n\right) + \cos\left(\frac{7\pi}{4}n\right)$ lie inside $-\pi < \omega \leq \pi$, and what is the fundamental period of the sequence?

Answer: Weight 2π at $\pm\pi/3$ and weight π at $\pm\pi/4$, after reducing $5\pi/3$ to $\pi/3$ and $7\pi/4$ to $\pi/4$. The two components have periods 6 and 8, so the sum has period $\text{LCM}(6, 8) = 24$.

6.7 Two spectra, 1 on $|\omega| \leq 3\pi/4$ and 1 on $|\omega| \leq \pi/2$, belong to sequences that are multiplied together. Why is $Z(e^{j\pi}) = \frac{1}{4}$ and not $\frac{1}{8}$?

Answer: The multiplication property gives a periodic convolution over one period. The ordinary convolution of the two rectangles is a trapezoid reaching to $|\omega| = 5\pi/4$, which is wider than one period, so near $\omega = \pi$ a second copy arrives from the other side and the two contributions of $\frac{1}{8}$ add.

6.8 Derive the pair for $(n+1)a^n u[n]$ and say what fixes the exponent and the sign.

Answer: Differentiate $\sum_{n \geq 0} a^n e^{-j\omega n} = 1/(1 - ae^{-j\omega})$ with respect to a , shift the index and cancel one factor $e^{-j\omega}$, giving $1/(1 - ae^{-j\omega})^2$. One differentiation of a first power produces the exponent 2, and the minus sign is inherited from the pair being differentiated.

6.9 Find the impulse response of $y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]$ and check it against the equation.

Answer: $h[n] = 4(\frac{1}{2})^n u[n] - 2(\frac{1}{4})^n u[n]$, so $h[0] = 2$, $h[1] = 1.5$, $h[2] = 0.875$. Substituting: $h[1] - \frac{3}{4}h[0] = 0$ and $h[2] - \frac{3}{4}h[1] + \frac{1}{8}h[0] = 0$.

6.10 Which properties of the continuous-time transform carry over unchanged, and which change form?

Answer: Linearity, the time shift, conjugate symmetry, Parseval and the convolution property carry over unchanged. The synthesis equation changes its range to one period; the multiplication property becomes a periodic convolution; the time-domain operator is a difference and not a derivative; and the transform pairs themselves differ, since the discrete-time ones are 2π -periodic and carry their own convergence conditions.

CHAPTER

7

Sampling and aliasing

Sampling keeps the value of a signal every T seconds and discards the values between those instants. This chapter gives the condition for exact recovery, derives the required reconstruction filter, and shows what changes when the condition fails. Sampling always **replicates** the spectrum. **Aliasing** occurs only when those copies overlap.

7.1 Impulse-train sampling

Sampling is modelled as a multiplication. The sampling function is an impulse train of period T , and the sampled signal is the product of the signal with it.

IMPULSE-TRAIN SAMPLING

$$\begin{aligned}
 p(t) &= \sum_{n=-\infty}^{\infty} \delta(t - nT), & x_p(t) &= x(t) p(t) \\
 x_p(t) &= x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) \\
 &= \sum_{n=-\infty}^{\infty} x(t) \delta(t - nT) \\
 &= \sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT)
 \end{aligned}$$

The first step substitutes $p(t)$ into the product. The second step moves $x(t)$ inside the sum, because multiplication distributes over addition. The third step uses the sifting property $x(t)\delta(t - nT) = x(nT)\delta(t - nT)$: an impulse is zero everywhere except at its own instant, so only the value of x at $t = nT$ survives.

HOW AN IMPULSE IS DRAWN

An impulse is drawn as an arrow whose **height is its weight**. In a picture of $x_p(t)$ the arrow at $t = nT$ therefore reaches $x(nT)$, and the outline traced by the arrowheads is the signal itself. The arrows are weights, not values of a function.

The sampler keeps only the numbers $x(nT)$. Written as a sequence, they are $x_p[n] = x(nT)$. These sample values form the discrete-time representation of the signal.

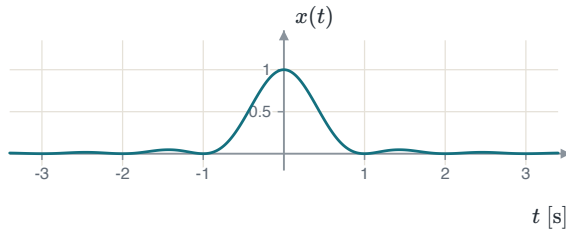


FIGURE 7.1 A band-limited signal, $x(t) = (\sin \pi t / \pi t)^2$.

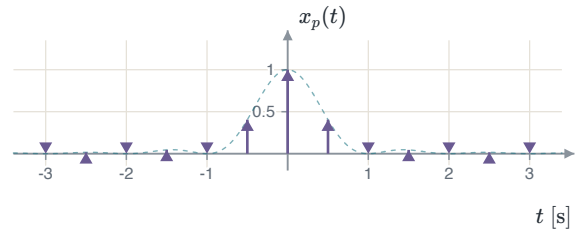


FIGURE 7.2 The same signal sampled with $T = 0.5$ s. Each arrow reaches $x(nT)$.

7.2 The rate, in radians per second and in hertz

A sampling rate can be stated as an angular frequency or in hertz. Define both forms before using them so that their units remain distinct.

SAMPLING RATE, BOTH READINGS

$$\omega_s = \frac{2\pi}{T} \left[\frac{\text{rad}}{\text{s}} \right], \quad f_s = \frac{1}{T} [\text{Hz}], \quad \omega_s = 2\pi f_s$$

ω_s is the sampling **angular** frequency. f_s is the sampling frequency in hertz, the number of samples taken per second. Every spectrum in this chapter is drawn against ω in rad/s, so ω_s is the working symbol; the hertz reading appears only where a physical rate is quoted.

⊗ THE FACTOR OF 2π IS NOT DECORATION

Calling $2\pi/T$ the sampling frequency and then reading it in hertz is the most expensive slip in this chapter. With $T = 0.25 \text{ ms} = 2.5 \times 10^{-4} \text{ s}$ the two readings are $\omega_s = 2\pi/(2.5 \times 10^{-4}) = 8000\pi \text{ rad/s}$ and $f_s = 1/(2.5 \times 10^{-4}) = 4000 \text{ Hz}$. They differ by 2π , and an answer that mixes them is wrong by that factor everywhere it is used. The same applies to the signal: a bandwidth of $\omega_M \text{ rad/s}$ is $f_M = \omega_M/2\pi \text{ hertz}$.

One multiplication tests any pair of values: $\omega_s T = 2\pi$ always, and $f_s T = 1$ always. Every sampling period computed in this chapter is followed by that check.

7.3 The spectrum of a sampled signal

Multiplication in time is convolution in frequency. So the spectrum of $x_p(t)$ needs the transform of the impulse train $p(t)$. Four steps give the result: write $p(t)$ as a Fourier series, transform that series, apply the multiplication property, and evaluate the convolution with the sifting property.

STEP 1: THE FOURIER SERIES OF THE IMPULSE TRAIN

$$\begin{aligned}
 p(t) &= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_s t}, \quad \omega_s = \frac{2\pi}{T} \\
 a_k &= \frac{1}{T} \int_{-T/2}^{T/2} p(t) e^{-jk\omega_s t} dt \\
 &= \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_s t} dt \\
 &= \frac{1}{T} e^{-jk\omega_s \cdot 0} = \frac{1}{T}
 \end{aligned}$$

$p(t)$ is periodic with period T , so it has a Fourier series with fundamental frequency $\omega_s = 2\pi/T$. The coefficient integral runs over one period, $-T/2 < t < T/2$. Only the impulse at $n = 0$ lies inside that window, so the whole sum $p(t)$ reduces to the single term $\delta(t)$ there. The sifting property then evaluates the exponential at $t = 0$, where it equals 1. Every coefficient is the same number, $1/T$.

STEP 2: THE TRANSFORM OF THE IMPULSE TRAIN

$$\begin{aligned}
 e^{jk\omega_s t} &\longleftrightarrow 2\pi \delta(\omega - k\omega_s) \\
 P(j\omega) &= \sum_{k=-\infty}^{\infty} \frac{1}{T} \cdot 2\pi \delta(\omega - k\omega_s) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s)
 \end{aligned}$$

The first line is the transform pair of a complex exponential. The transform is linear, so each term of the series transforms separately and the constant $1/T$ passes through. An impulse train of spacing T in time is therefore an impulse train of spacing ω_s in frequency, with weight $2\pi/T$ on every impulse.

Step 3 is the multiplication property. A product in time is a convolution in frequency, divided by 2π ; the $1/2\pi$ is the same factor that appears in the inverse transform.

$$X_p(j\omega) = \frac{1}{2\pi} [X(j\omega) * P(j\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\theta) P(j(\omega - \theta)) d\theta.$$

Step 4 substitutes P from Step 2 and evaluates the integral.

$$\begin{aligned}
 X_p(j\omega) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\theta) \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - \theta - k\omega_s) d\theta \\
 &= \frac{1}{T} \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} X(j\theta) \delta(\theta - (\omega - k\omega_s)) d\theta \\
 &= \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)).
 \end{aligned}$$

SPECTRUM OF THE SAMPLED SIGNAL

$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))$$

In the second line the constants $1/2\pi$ and $2\pi/T$ multiply to $1/T$, and the sum is taken outside the integral. The impulse is even, so $\delta(\omega - \theta - k\omega_s) = \delta(\theta - (\omega - k\omega_s))$; writing it this way shows where it sits on the θ axis. In the third line the sifting property picks out X at $\theta = \omega - k\omega_s$. Each term is one copy of $X(j\omega)$, moved to $k\omega_s$ and scaled by $1/T$, not by 1. The reconstruction filter later carries gain T , exactly the inverse, so the two must be kept together.

Spacing T in time gives spacing $2\pi/T$ in frequency: the closer the samples, the further apart the copies, and the taller each one is.

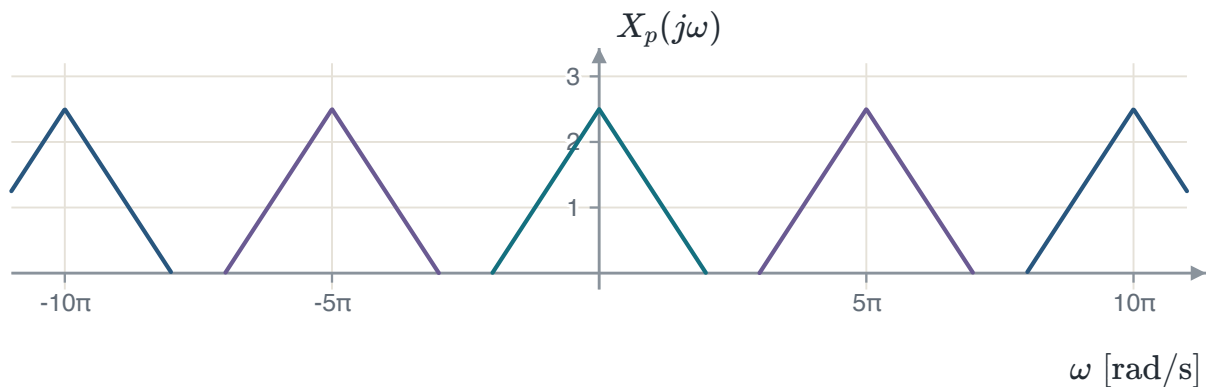


FIGURE 7.3 The sampled spectrum of the signal above at $T = 0.4$ s, so $\omega_s = 5\pi$ rad/s. One copy per multiple of ω_s , each of height $1/T = 2.5$.

7.4 Replication and aliasing are not the same thing

The sum for $X_p(j\omega)$ runs over every integer k for every value of T . There is no rate at which the copies fail to appear and no rate at which they disappear. Two things, and only two, change with the rate: the copies sit ω_s apart, and each is scaled by $1/T$.

① NAMING THE PIECES

The term $k = 0$ is $X(j\omega)/T$, the spectrum of the signal itself, sitting where it always sat. It is the **baseband** and it is not a copy of anything. The terms $k = \pm 1, \pm 2, \dots$ are the **copies**, centred at $\pm\omega_s, \pm 2\omega_s, \dots$. Calling the centre group the first replica shifts every later index by one.

The baseband occupies $|\omega| \leq \omega_M$, and the copy at $k = 1$ occupies $|\omega - \omega_s| \leq \omega_M$, so it begins at $\omega_s - \omega_M$. The gap between them is the quantity that decides everything.

GUARD BAND

$$(\omega_s - \omega_M) - \omega_M = \omega_s - 2\omega_M$$

Positive: the copies stand clear and a filter can cut the baseband out untouched. Zero: they touch. Negative: they have moved into one another and add.

⊗ REPLICATION IS UNCONDITIONAL

It is sometimes said that below the Nyquist rate the spectrum is no longer replicated. That is false. The formula for $X_p(j\omega)$ carries no condition, so the copies are there at every rate; what changes below the Nyquist rate is that they **overlap**. Replication is what sampling does. Aliasing is the overlap of what sampling made.

The overlap cannot be repaired afterwards. Where two copies meet, the sampler stores one number that is the sum of two contributions,

$$X_p(j\omega) = \frac{1}{T} \left[\underbrace{X(j\omega)}_{\text{wanted}} + \underbrace{X(j(\omega - \omega_s))}_{\text{intruder}} \right].$$

A filter multiplies $X_p(j\omega)$ frequency by frequency, including multiplying by zero. It cannot look at one value and report the two numbers that were added to make it. So the only cures are to raise the rate, or to remove the offending frequencies from $x(t)$ before the sampler.

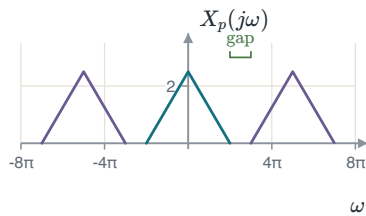


FIGURE 7.4 $\omega_s = 5\pi$: a gap of π rad/s.

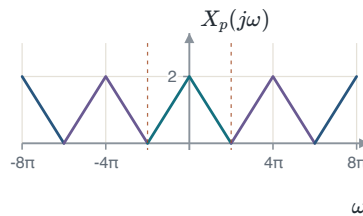


FIGURE 7.5 $\omega_s = 4\pi = 2\omega_M$: the gap is zero.

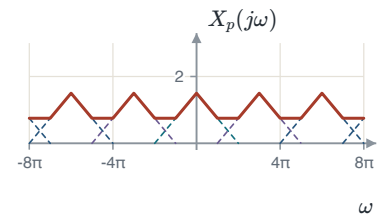


FIGURE 7.6 $\omega_s = 3\pi$: the copies add. The solid curve is the sum.

7.5 The sampling theorem

SAMPLING THEOREM

$$X(j\omega) = 0 \text{ for } |\omega| > \omega_M \text{ and } \omega_s > 2\omega_M$$

$$\implies x(t) \text{ is determined uniquely by } x(nT), n = 0, \pm 1, \pm 2, \dots$$

The signal is recovered by passing $x_p(t)$ through an ideal lowpass filter of gain T and cutoff ω_c with $\omega_M < \omega_c < \omega_s - \omega_M$. The quantity $2\omega_M$ is the **Nyquist rate** of the signal.

⊗ WHY THE INEQUALITY HAS TO BE STRICT

Put $\omega_s = 2\omega_M$ into the cutoff condition. It becomes $\omega_M < \omega_c < \omega_M$, an interval with nothing in it. There is no admissible cutoff at the Nyquist rate, so the theorem cannot be satisfied at its own boundary. Writing $\omega_s \geq 2\omega_M$ promises a filter that the same sentence then rules out. In practice the rate is chosen well above $2\omega_M$, so that the guard band is wide enough for a real filter, whose transition from pass to stop is not vertical.

EXAMPLE 7.1 – A SIGNAL THE NYQUIST RATE THROWS AWAY

GIVEN $x(t) = 1 + \cos(2000\pi t) + \sin(4000\pi t)$, sampled at exactly $\omega_s = 2\omega_M = 8000\pi$ rad/s.

FIND What happens to the term $\sin(4000\pi t)$.

METHOD Use the sampled-spectrum sum because cancellation can occur between the baseband and a copy. Locate every contribution to X_p at $\omega = +4000\pi$, then verify the result from the sample values in time.

SOLUTION The highest frequency present is 4000π rad/s, so $\omega_M = 4000\pi$ rad/s and

$$T = \frac{2\pi}{\omega_s} = \frac{2\pi}{8000\pi} = \frac{1}{4000} \text{ s} = 0.25 \text{ ms}, \quad \frac{1}{T} = 4000.$$

The sine transforms to $\frac{\pi}{j}[\delta(\omega - 4000\pi) - \delta(\omega + 4000\pi)]$; Example 7.3 derives this pair.

Write the sampled-spectrum sum for this term and keep the $k = 0$ and $k = 1$ terms:

$$\begin{aligned} X_p(j\omega) &= \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)) \\ k = 0 : & \quad \frac{1}{T} \cdot \frac{\pi}{j} [\delta(\omega - 4000\pi) - \delta(\omega + 4000\pi)] \\ k = 1 : & \quad \frac{1}{T} \cdot \frac{\pi}{j} [\delta(\omega - 8000\pi - 4000\pi) - \delta(\omega - 8000\pi + 4000\pi)] \\ &= \frac{1}{T} \cdot \frac{\pi}{j} [\delta(\omega - 12000\pi) - \delta(\omega - 4000\pi)]. \end{aligned}$$

The $k = 1$ term replaces ω by $\omega - \omega_s$, which moves every impulse up by 8000π . The impulse that sat at -4000π now sits at $-4000\pi + 8000\pi = +4000\pi$, and it keeps its negative sign. At $\omega = +4000\pi$ the two weights add:

$$\frac{1}{T} \cdot \frac{\pi}{j} - \frac{1}{T} \cdot \frac{\pi}{j} = \frac{4000\pi}{j} - \frac{4000\pi}{j} = 0.$$

The same cancellation happens at -4000π , where the $k = -1$ copy brings the impulse from $+4000\pi$ down by 8000π . In time the sample values confirm it:

$$\sin(4000\pi nT) = \sin\left(4000\pi \cdot \frac{n}{4000}\right) = \sin(\pi n) = 0 \quad \text{for every integer } n.$$

CHECK Every sample of the sine term is taken at a zero crossing, so the sampler never sees it. The component is absent, not merely marginal. The cosine term is not affected: its impulse at -2000π moves to $-2000\pi + 8000\pi = 6000\pi$, which is outside the baseband.

REPAIR Add a guard band. With $\omega_g = 1000\pi$ rad/s the rate becomes $\omega_s = 2\omega_M + \omega_g = 8000\pi + 1000\pi = 9000\pi$ rad/s, so

$$T = \frac{2\pi}{9000\pi} = \frac{1}{4500} \text{ s} \approx 222.2 \text{ } \mu\text{s}.$$

The admissible cutoff interval $\omega_M < \omega_c < \omega_s - \omega_M$ becomes $4000\pi < \omega_c < 9000\pi - 4000\pi = 5000\pi$ rad/s, which is no longer empty.

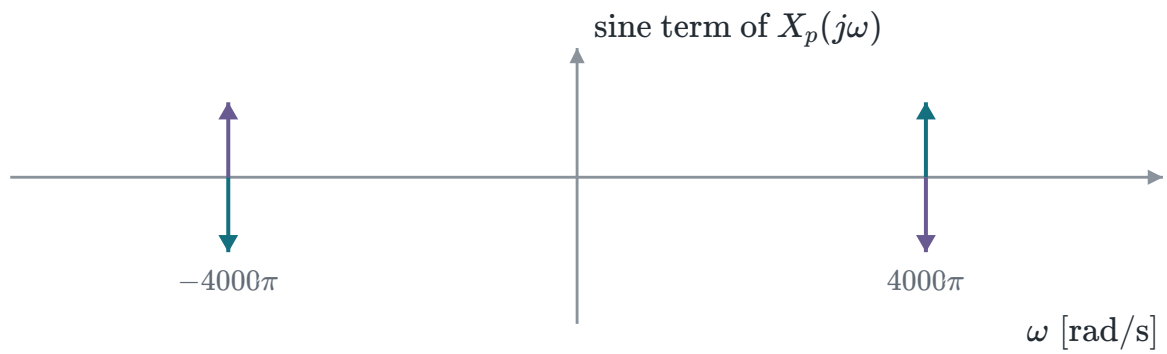


FIGURE 7.7 At $\omega = \pm 4000\pi$ the baseband arrow and the arrow the neighbouring copy brings there have the same length and point in opposite directions.

7.6 Worked examples on the rate

EXAMPLE 7.2 – ONE SIGNAL AT THREE RATES

GIVEN $x(t) = (\sin \pi t / \pi t)^2$, whose transform is a triangle of peak 1 reaching zero at $\pm 2\pi$. So $\omega_M = 2\pi$ rad/s and the Nyquist rate is 4π rad/s.

FIND For $T_1 = 0.40$ s, $T_2 = 0.50$ s and $T_3 = 2/3$ s: the rate, the guard band and the height of each copy.

METHOD Use the sampling definitions because the question asks for the rate, separation, and scale of the copies: $\omega_s = 2\pi/T$, guard band $\omega_s - 2\omega_M$, and copy height $1/T$ times the original peak.

SOLUTION For each period compute the rate from $\omega_s = 2\pi/T$, the guard band from $\omega_s - 2\omega_M$ with $2\omega_M = 4\pi$, and the copy height from $1/T$ times the peak of 1:

$$\begin{aligned}
 T_1 = 0.40 \text{ s} : \quad \omega_s &= \frac{2\pi}{0.40} = 5\pi \text{ rad/s}, \\
 5\pi &> 4\pi, \quad \text{guard} = 5\pi - 4\pi = +\pi, \quad \frac{1}{T_1} = \frac{1}{0.40} = 2.5 \\
 T_2 = 0.50 \text{ s} : \quad \omega_s &= \frac{2\pi}{0.50} = 4\pi \text{ rad/s}, \\
 4\pi &= 4\pi, \quad \text{guard} = 4\pi - 4\pi = 0, \quad \frac{1}{T_2} = \frac{1}{0.50} = 2.0 \\
 T_3 = 2/3 \text{ s} : \quad \omega_s &= \frac{2\pi}{2/3} = 3\pi \text{ rad/s}, \\
 3\pi &< 4\pi, \quad \text{guard} = 3\pi - 4\pi = -\pi, \quad \frac{1}{T_3} = \frac{3}{2} = 1.5
 \end{aligned}$$

Dividing by $2/3$ is multiplying by $3/2$, which gives the last row.

CHECK $\omega_s T = 2\pi$ in each row: $5\pi \times 0.4 = 2\pi$, $4\pi \times 0.5 = 2\pi$, $3\pi \times 2/3 = 2\pi$. Only the first row is safe; the second sits on the boundary and the third overlaps by π rad/s on each side.

EXAMPLE 7.3 – A LINE SPECTRUM, AND ITS SAMPLING PERIOD**GIVEN** $x(t) = 1 + \cos(2000\pi t) + \sin(4000\pi t)$.**FIND** $X(j\omega)$, the bandwidth, and the sampling period at the Nyquist rate.**METHOD** Use linearity because the signal is a sum of standard terms. Transform each term separately, identify the largest frequency, convert the resulting rate to a period, and check the units three ways.**SOLUTION** Start from the pair for one complex exponential, $e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$. It holds because the inverse transform of the right side is $\frac{1}{2\pi} \int 2\pi\delta(\omega - \omega_0)e^{j\omega t} d\omega = e^{j\omega_0 t}$ by sifting. Write each term of $x(t)$ as a sum of exponentials and transform term by term:

$$\begin{aligned}
 1 &= e^{j0t} && \longleftrightarrow 2\pi\delta(\omega) \\
 \cos(2000\pi t) &= \frac{1}{2}e^{j2000\pi t} + \frac{1}{2}e^{-j2000\pi t} && \longleftrightarrow \pi\delta(\omega - 2000\pi) + \pi\delta(\omega + 2000\pi) \\
 \sin(4000\pi t) &= \frac{1}{2j}e^{j4000\pi t} - \frac{1}{2j}e^{-j4000\pi t} && \longleftrightarrow \frac{\pi}{j}\delta(\omega - 4000\pi) - \frac{\pi}{j}\delta(\omega + 4000\pi)
 \end{aligned}$$

Each impulse weight is 2π times the coefficient of its exponential: $2\pi \cdot \frac{1}{2} = \pi$ and $2\pi \cdot \frac{1}{2j} = \pi/j$. Adding the three lines gives

$$\begin{aligned}
 X(j\omega) &= 2\pi\delta(\omega) + \pi\delta(\omega - 2000\pi) + \pi\delta(\omega + 2000\pi) \\
 &\quad + \frac{\pi}{j}\delta(\omega - 4000\pi) - \frac{\pi}{j}\delta(\omega + 4000\pi).
 \end{aligned}$$

Five impulses, each written as a function of ω so that its position can be read. The furthest is at 4000π , so $\omega_M = 4000\pi$ rad/s, that is $f_M = \omega_M/2\pi = 2000$ Hz. The Nyquist rate is $2\omega_M = 8000\pi$ rad/s, that is $2f_M = 4000$ Hz. Then

$$T = \frac{2\pi}{\omega_s} = \frac{2\pi}{8000\pi} = \frac{1}{4000} \text{ s} = 2.5 \times 10^{-4} \text{ s} = 0.25 \text{ ms}.$$

CHECK Three independent tests. $\omega_s T = 8000\pi \times 2.5 \times 10^{-4} = 2\pi$. The copies are scaled by $1/T = 4000$. And doubling the rate must halve the period: at $\omega_s = 16000\pi$ the period is $125 \mu\text{s}$, and 0.25 ms divided by $125 \mu\text{s}$ is exactly 2.**WARNING** The answer is a quarter of a millisecond. Cancelling the π but not the thousand gives 0.25 s , and every later number in the problem is then wrong by a factor of 1000. A quarter of a second between samples would be four samples per second for a signal carrying 2000 Hz.**EXAMPLE 7.4 – AREA, PEAK, AND THE HEIGHT OF A COPY****GIVEN** $x(t) = (\sin(4000\pi t)/\pi t)^2$, the square of a signal whose transform is a rectangle of height 1 on $|\omega| \leq 4000\pi$.**FIND** The peak of $X(j\omega)$, the bandwidth, the Nyquist period, and the height of one copy after sampling.**METHOD** Use the multiplication property because the signal is squared in time. The spectrum therefore convolves with itself and divides by 2π .

SOLUTION

Write $W = 4000\pi$ and $g(t) = \sin(Wt)/\pi t$, so that $x(t) = g(t)^2$. First confirm the transform of g by inverting the rectangle $R(j\omega)$ of height 1 on $|\omega| \leq W$:

$$\begin{aligned} g(t) &= \frac{1}{2\pi} \int_{-W}^W 1 \cdot e^{j\omega t} d\omega = \frac{1}{2\pi} \left[\frac{e^{j\omega t}}{jt} \right]_{-W}^W \\ &= \frac{1}{2\pi} \cdot \frac{e^{jWt} - e^{-jWt}}{jt} = \frac{1}{2\pi} \cdot \frac{2j \sin(Wt)}{jt} = \frac{\sin(Wt)}{\pi t}. \end{aligned}$$

The bracket is the antiderivative of $e^{j\omega t}$ with respect to ω , and $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$ removes the j . Squaring in time is convolution in frequency divided by 2π :

$$X(j\omega) = \frac{1}{2\pi} [R * R](\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} R(j\theta) R(j(\omega - \theta)) d\theta.$$

The integrand is 1 where both rectangles are 1, that is where $-W \leq \theta \leq W$ and $\omega - W \leq \theta \leq \omega + W$ at the same time, and 0 elsewhere. For $0 \leq \omega \leq 2W$ the overlap runs from $\omega - W$ to W :

$$[R * R](\omega) = \int_{\omega-W}^W 1 d\theta = W - (\omega - W) = 2W - \omega.$$

For negative ω the same argument gives $2W + \omega$, so $[R * R](\omega) = 2W - |\omega|$ on $|\omega| \leq 2W$ and 0 outside: a triangle. Its value at the origin, A , is the area of the full rectangle, and the peak of X is A divided by 2π :

$$\begin{aligned} A &= [R * R](0) = \int_{-W}^W (1)(1) d\theta = 2W = 8000\pi, \\ X_{\max} &= X(0) = \frac{A}{2\pi} = \frac{8000\pi}{2\pi} = 4000. \end{aligned}$$

The triangle reaches zero at $2W$, twice the rectangle half-width, so $\omega_M = 8000\pi$ rad/s. The Nyquist rate is $2\omega_M = 16000\pi$ rad/s, and

$$T = \frac{2\pi}{16000\pi} = \frac{1}{8000} \text{ s} = 1.25 \times 10^{-4} \text{ s} = 125 \mu\text{s}.$$

The copy height is $X_{\max}/T = 4000 \times 8000 = 3.2 \times 10^7$.

CHECK

This bandwidth is twice the one in Example 7.3, so this period is half of it. And $\omega_s T = 16000\pi \times 1.25 \times 10^{-4} = 2\pi$.

WARNING

A is an area and $X_{\max}$ is the peak of a spectrum. They differ by 2π , so substituting one where the other belongs inflates every later height by that factor. Give the peak its own symbol.

7.7 Reconstruction and band-limited interpolation

If the copies stand clear, recovering $x(t)$ means keeping the one at the origin and discarding the rest.

IDEAL RECONSTRUCTION FILTER

$$H_r(j\omega) = \begin{cases} T, & |\omega| < \omega_c \\ 0, & |\omega| > \omega_c \end{cases} \quad \omega_M < \omega_c < \omega_s - \omega_M$$

$$X_r(j\omega) = H_r(j\omega) X_p(j\omega) = T \cdot \frac{1}{T} X(j\omega) = X(j\omega) \implies x_r(t) = x(t)$$

Filtering multiplies the spectra. Inside $|\omega| < \omega_c$ only the $k = 0$ term of the sum for $X_p(j\omega)$ is non-zero, and that term is $X(j\omega)/T$; every other copy starts at $\omega_s - \omega_M$ or beyond, which is outside the passband. The gain is T because the baseband is $X(j\omega)/T$. A filter of gain 1 returns a signal T times too small, and since T changes with the rate the error changes with it. The second line is true only when the copies do not overlap.

Filtering is convolution, and x_p is a train of impulses, so the output is a sum of shifted copies of the filter's impulse response. Each sample is replaced by a curve of its own height, and the curves are added.

BAND-LIMITED INTERPOLATION

$$\begin{aligned} h_{LP}(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} H_r(j\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} T e^{j\omega t} d\omega \\ &= \frac{T}{2\pi} \left[\frac{e^{j\omega t}}{jt} \right]_{-\omega_c}^{\omega_c} = \frac{T}{2\pi} \cdot \frac{e^{j\omega_c t} - e^{-j\omega_c t}}{jt} \\ &= \frac{T}{2\pi} \cdot \frac{2j \sin(\omega_c t)}{jt} = \frac{T \sin(\omega_c t)}{\pi t} \end{aligned}$$

$$\begin{aligned} x_r(t) &= x_p(t) * h_{LP}(t) = \int_{-\infty}^{\infty} \left[\sum_{n=-\infty}^{\infty} x(nT) \delta(\tau - nT) \right] h_{LP}(t - \tau) d\tau \\ &= \sum_{n=-\infty}^{\infty} x(nT) h_{LP}(t - nT) = \sum_{n=-\infty}^{\infty} x(nT) \frac{T}{\pi} \cdot \frac{\sin(\omega_c(t - nT))}{t - nT} \end{aligned}$$

$$\omega_c = \frac{\pi}{T} : \quad h_{LP}(t) = \frac{T \sin(\pi t/T)}{\pi t} = \frac{\sin(\pi t/T)}{\pi t/T} = \text{sinc}\left(\frac{\pi t}{T}\right)$$

The first chain is the inverse transform of H_r . The limits shrink to $\pm\omega_c$ because H_r is zero outside them, and T comes out as a constant. The bracket is the antiderivative of $e^{j\omega t}$ with respect to ω ; then $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$ removes the j , and $2/2\pi = 1/\pi$. The second chain is the convolution definition with the impulse train substituted for x_p ; the sifting property evaluates h_{LP} at $\tau = nT$ for each n . Throughout this course $\text{sinc}(\theta) = \sin \theta / \theta$, **unnormalised**. In the third line the second equality divides numerator and denominator by T . At $t = 0$ the ratio $\sin \theta / \theta$ tends to 1, and at $t = mT$ with $m \neq 0$ it is $\sin(\pi m) / (\pi m) = 0$. Anything written as $\text{sinc}(t/T)$ belongs to the other convention and is a different function.

⊗ KEEP THE FACTOR OF π INSIDE THE SINE

The kernel must equal 1 at $t = 0$ and 0 at every other sample instant. Then each sample sets its own value without changing the others. Removing π breaks this condition: with $T = 1$ the correct kernel at $t = 1$ is $\sin(\pi)/\pi = 0$, while $\sin(1)/\pi = 0.267849$. Also, $\omega_c = \pi/T$ equals $\omega_s/2$, which is an endpoint of the admissible interval rather than an interior point. State this cutoff as a choice.

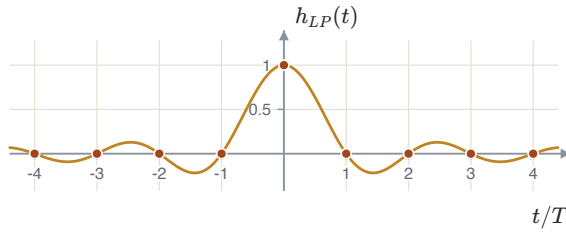


FIGURE 7.8 The kernel with $\omega_c = \pi/T$: one at its own instant, zero at every other.

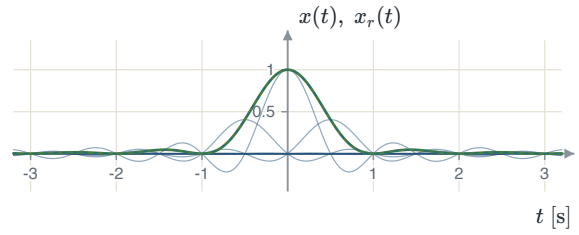


FIGURE 7.9 One kernel per sample, in grey, and their sum. It lands on the dashed original.

7.8 The zero-order and first-order holds

The ideal filter is not buildable: its impulse response starts before $t = 0$ and never ends. What hardware does instead is hold.

ZERO-ORDER HOLD

$$h_0(t) = \begin{cases} 1, & 0 \leq t < T \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} H_0(j\omega) &= \int_{-\infty}^{\infty} h_0(t) e^{-j\omega t} dt = \int_0^T e^{-j\omega t} dt = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^T = \frac{1 - e^{-j\omega T}}{j\omega} \\ &= \frac{e^{-j\omega T/2} (e^{j\omega T/2} - e^{-j\omega T/2})}{j\omega} = \frac{e^{-j\omega T/2} \cdot 2j \sin(\omega T/2)}{j\omega} \\ &= e^{-j\omega T/2} \frac{2 \sin(\omega T/2)}{\omega} = T e^{-j\omega T/2} \frac{\sin(\omega T/2)}{\omega T/2} \end{aligned}$$

The limits shrink to 0 and T because h_0 is zero elsewhere, and the integrand there is $1 \cdot e^{-j\omega t}$. The bracket is the antiderivative of $e^{-j\omega t}$; evaluating it at T and at 0 gives $(e^{-j\omega T} - 1)/(-j\omega)$, which is the fraction shown. The second line factors $e^{-j\omega T/2}$ out of $1 - e^{-j\omega T}$, which leaves $e^{j\omega T/2} - e^{-j\omega T/2} = 2j \sin(\omega T/2)$; the j then cancels. The last form multiplies and divides by $T/2$ to show the sinc, with $\text{sinc}(\theta) = \sin \theta / \theta$ unnormalised: $H_0(j\omega) = T e^{-j\omega T/2} \text{sinc}(\omega T/2)$. The magnitude is $|2 \sin(\omega T/2)/\omega|$, and where the sine is positive the phase is $-\omega T/2$, a delay of half a sampling period. At the origin $\sin \theta / \theta \rightarrow 1$, so $H_0(0) = T$: the hold already supplies the gain the ideal filter needed.

The zero-order hold differs from the ideal filter in three ways. Its magnitude decreases below T as ω grows, so it changes the wanted spectrum. Its response is non-zero beyond the cutoff, so parts of neighbouring copies remain. Its phase also adds a delay. Correcting these effects would require the compensator

$$H_r(j\omega) = \frac{H(j\omega)}{H_0(j\omega)} = e^{j\omega T/2} H(j\omega) \frac{\omega}{2 \sin(\omega T/2)},$$

which has to advance in time and to grow without bound where $\sin(\omega T/2)$ vanishes. It is approximated in practice, not built.

FIRST-ORDER HOLD

$$h_1(t) = \frac{1}{T} [g * g](t), \quad g(t) = \begin{cases} 1, & |t| \leq T/2 \\ 0, & \text{otherwise} \end{cases}$$

$$G(j\omega) = \int_{-T/2}^{T/2} e^{-j\omega t} dt = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-T/2}^{T/2} = \frac{e^{j\omega T/2} - e^{-j\omega T/2}}{j\omega} = \frac{2 \sin(\omega T/2)}{\omega} = \frac{\sin(\omega T/2)}{\omega/2}$$

$$H_1(j\omega) = \frac{1}{T} G(j\omega) G(j\omega) = \frac{1}{T} \left[\frac{\sin(\omega T/2)}{\omega/2} \right]^2$$

G is the transform of the centred rectangle; the limits are $\pm T/2$ because g is zero outside them, and $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$ cancels the j as before. Convolution in time is multiplication in frequency, so $g * g$ transforms to $G \cdot G$, and the constant $1/T$ passes through by linearity. The convolution of the rectangle with itself is $[g * g](t) = \int_{|t|-T/2}^{T/2} 1 d\tau = T - |t|$ for $|t| \leq T$ and 0 outside: a triangle of peak T . So h_1 is a triangle of peak 1 on the same interval and the output joins consecutive samples by straight lines. Every one of these is a function of t ; the independent variable of all three sketches is time. H_1 is real and non-negative because it is a square. At $\omega = 0$, $\sin(\omega T/2)/(\omega/2) \rightarrow T$, so $H_1(0) = T^2/T = T$, and it falls off as $1/\omega^2$ instead of $1/\omega$.

⊗ A HOLD IS NOT A RECONSTRUCTION

Both holds are filters with a real, non-flat, non-band-limited response. The staircase is a different signal from $x(t)$, and no argument about small steps makes them equal: raising the rate makes the error smaller but never zero at any finite rate. The sampling theorem guarantees that the samples *determine* $x(t)$ and names the filter that recovers it. Feeding the same samples to a different filter gives a different output.

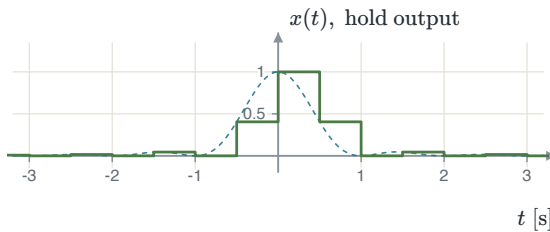


FIGURE 7.10 The zero-order hold: each sample held for one period.

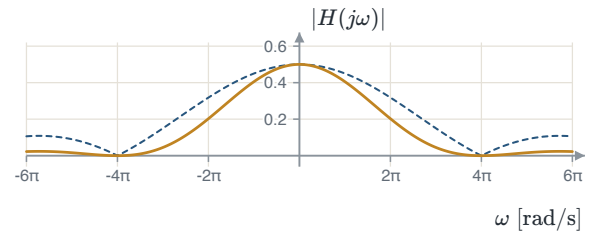


FIGURE 7.11 With $T = 0.5$ s: the first-order hold, solid, against the zero-order hold, dashed. Both start at T .

7.9 What comes back, and where from

Take $x(t) = \cos(\omega_0 t)$, so $\omega_M = \omega_0$, and reconstruct with the cutoff at $\omega_c = \omega_s/2$. The transform is $X(j\omega) = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$. Substitute it into the sampled-spectrum sum:

$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)) = \frac{\pi}{T} \sum_{k=-\infty}^{\infty} [\delta(\omega - k\omega_s - \omega_0) + \delta(\omega - k\omega_s + \omega_0)].$$

So sampling puts a line at every $k\omega_s \pm \omega_0$, each of weight π/T . The filter multiplies by T inside $|\omega| < \omega_c$ and by 0 outside, so each surviving line returns to weight π . List the lines for $k = 0$ and $k = 1$ and

test each against the cutoff; the $k = -1$ lines are the mirror images at negative frequencies and give the same verdict.

$$\begin{aligned}
 \omega_s = 6\omega_0, \omega_c = 3\omega_0 : \quad & k = 0 : \pm\omega_0 \text{ inside;} \\
 & k = 1 : 6\omega_0 \pm \omega_0 = 5\omega_0, 7\omega_0 \text{ outside} \\
 & \Rightarrow x_r(t) = \cos(\omega_0 t) \\
 \omega_s = 3\omega_0, \omega_c = 1.5\omega_0 : \quad & k = 0 : \pm\omega_0 \text{ inside;} \\
 & k = 1 : 3\omega_0 \pm \omega_0 = 2\omega_0, 4\omega_0 \text{ outside} \\
 & \Rightarrow x_r(t) = \cos(\omega_0 t) \\
 \omega_s = 1.5\omega_0, \omega_c = 0.75\omega_0 : \quad & k = 0 : \pm\omega_0 \text{ outside;} \\
 & k = 1 : 1.5\omega_0 - \omega_0 = 0.5\omega_0 \text{ inside, } 1.5\omega_0 + \omega_0 = 2.5\omega_0 \text{ outside} \\
 & \Rightarrow x_r(t) = \cos(0.5\omega_0 t)
 \end{aligned}$$

THE ALIAS FREQUENCY

$$X_r(j\omega) = \pi\delta(\omega - (\omega_s - \omega_0)) + \pi\delta(\omega + (\omega_s - \omega_0)) \xrightarrow{\mathcal{F}^{-1}} x_r(t) = \cos((\omega_s - \omega_0)t)$$

In the third case the line at ω_0 lies outside the filter and is discarded, while the line the $k = 1$ copy contributes at $\omega_s - \omega_0$ lies inside and is kept. The signal takes the identity of a lower frequency, $|\omega_s - \omega_0|$. Each impulse is written as a function of ω ; an expression such as $\pi\delta(\omega_s - \omega_0)$ is a constant and locates nothing. The arrow is the inverse transform, going from a spectrum back to a signal.

EXAMPLE 7.5 – THREE PERIODS FOR ONE COSINE

- GIVEN** $x(t) = \cos(2\pi t)$, so $\omega_M = 2\pi$ rad/s and the Nyquist rate is 4π rad/s. Take $\omega_c = \omega_s/2$.
- FIND** Whether $T_1 = 1/4$ s, $T_2 = 1/3$ s and $T_3 = 2/3$ s recover the signal, and what is recovered when one does not.
- METHOD** Use the sampled line spectrum because the filter selects individual lines. Compute $\omega_s = 2\pi/T$, compare it with 4π , list all line positions, and keep only those inside the cutoff.
- SOLUTION** The lines sit at $k\omega_s \pm 2\pi$. First the rates:

$$\begin{aligned}
 T_1 = 1/4 \text{ s} : \quad \omega_s &= \frac{2\pi}{1/4} = 8\pi > 4\pi, \quad \omega_c = 4\pi \\
 T_2 = 1/3 \text{ s} : \quad \omega_s &= \frac{2\pi}{1/3} = 6\pi > 4\pi, \quad \omega_c = 3\pi \\
 T_3 = 2/3 \text{ s} : \quad \omega_s &= \frac{2\pi}{2/3} = 3\pi < 4\pi, \quad \omega_c = 1.5\pi
 \end{aligned}$$

In the first two rows the baseband lines $\pm 2\pi$ are inside the cutoff, and the nearest copy lines, $8\pi - 2\pi = 6\pi$ and $6\pi - 2\pi = 4\pi$, are outside, so $x_r(t) = \cos(2\pi t)$. In the third row the baseband lines $\pm 2\pi$ lie outside $|\omega| < 1.5\pi$ and are discarded. The $k = 1$ copy puts lines at $3\pi \pm 2\pi$, that is at 5π and at π ; the one at π is inside. The $k = -1$ copy gives the mirror line at $-\pi$. So

$$X_r(j\omega) = \pi\delta(\omega - \pi) + \pi\delta(\omega + \pi) \implies x_r(t) = \cos(\pi t).$$

The alias frequency is $|\omega_s - \omega_0| = |3\pi - 2\pi| = \pi$ rad/s, with the integer $k = 1$ chosen.

CHECK

The sample values of the two signals must agree. With $T_3 = 2/3$:

$$\cos(2\pi n T_3) = \cos\left(\frac{4\pi n}{3}\right), \quad \cos(\pi n T_3) = \cos\left(\frac{2\pi n}{3}\right).$$

Since $4\pi n/3 = 2\pi n - 2\pi n/3$, the cosine has period 2π , and the cosine is even, $\cos(4\pi n/3) = \cos(-2\pi n/3) = \cos(2\pi n/3)$. The two sample sequences are identical, so no reconstruction could prefer one over the other.

WARNING

The whole verdict rests on the number the rate is compared with, and for this signal that number is $2\omega_M = 4\pi$. Comparing against a neighbouring problem's bandwidth of 6π makes the middle row read $6\pi \geq 6\pi$, an equality, so a comfortable margin is turned into the boundary case the theorem excludes.

 EXAMPLE 7.6 – TWO COMPONENTS, ONE RATE

GIVEN

$x(t) = \cos(\pi t) + \cos(3\pi t)$, sampled with $T = 2/5$ s.

FIND

$x_r(t)$, with $\omega_c = \omega_s/2$.

METHOD

Treat the components separately because one can alias while the other remains unchanged. First test the rate against this signal's Nyquist rate, then track each baseband and copy line through the filter.

SOLUTION

$\omega_M = 3\pi$ rad/s, so the Nyquist rate is $2\omega_M = 6\pi$ rad/s, while

$$\omega_s = \frac{2\pi}{T} = \frac{2\pi}{2/5} = 5\pi \text{ rad/s}, \quad \omega_c = \frac{\omega_s}{2} = 2.5\pi \text{ rad/s}.$$

Since $5\pi < 6\pi$, aliasing is expected. Each component transforms to a pair of impulses of weight π , and sampling puts lines at $k\omega_s \pm \omega_0$ for each of the two values of ω_0 . Test each line against $|\omega| < 2.5\pi$:

$$\begin{aligned} \omega_0 = \pi : \quad & k = 0 : \pm\pi \text{ inside;} \\ & k = 1 : 5\pi - \pi = 4\pi, 5\pi + \pi = 6\pi \text{ outside} \\ \omega_0 = 3\pi : \quad & k = 0 : \pm 3\pi \text{ outside;} \\ & k = 1 : 5\pi - 3\pi = 2\pi \text{ inside, } 5\pi + 3\pi = 8\pi \text{ outside} \end{aligned}$$

Larger $|k|$ moves the lines further out, and $k = -1$ gives the mirror lines at -2π and -4π . The filter multiplies the surviving lines by T , which cancels the $1/T$ of sampling and returns the weight π . Hence

$$X_r(j\omega) = \pi [\delta(\omega - \pi) + \delta(\omega + \pi)] + \pi [\delta(\omega - 2\pi) + \delta(\omega + 2\pi)]$$

and $x_r(t) = \cos(\pi t) + \cos(2\pi t)$. The alias frequency of the second component is $|\omega_s - \omega_0| = |5\pi - 3\pi| = 2\pi$ rad/s, with the integer $k = 1$ chosen.

CHECK

The aliased component and its replacement must give the same samples. With $T = 2/5$:

$$\cos(3\pi n T) = \cos\left(\frac{6\pi n}{5}\right), \quad \cos(2\pi n T) = \cos\left(\frac{4\pi n}{5}\right).$$

Since $6\pi n/5 = 2\pi n - 4\pi n/5$ and the cosine is even with period 2π , $\cos(6\pi n/5) = \cos(4\pi n/5)$, so the two sequences agree.

INTERPRETATION The component at π remains unchanged. The component at 3π appears at 2π , a frequency that was not present in the original signal.

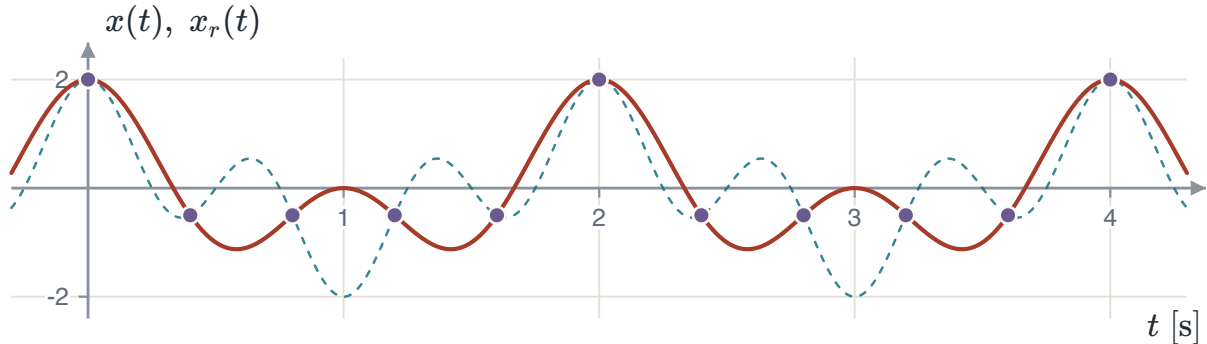


FIGURE 7.12 Both curves pass through every sample. The solid one is what the filter returns.

7.10 The anti-aliasing filter

A real signal is not band-limited, and a sampler at a fixed rate cannot be helped after the fact. The repair is a lowpass filter placed **before** the sampler, which removes everything above $\omega_s/2$ while it can still be removed cleanly.

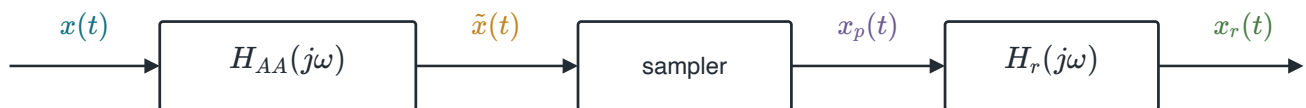


FIGURE 7.13 The chain in the only order that works. The sampler output is $x_p(t)$; the name $x_r(t)$ belongs to the output of the reconstruction filter and to nothing before it.

⊗ PLACE THE FILTER BEFORE THE SAMPLER

After sampling, aliased high-frequency content has already been added to wanted content at the same frequency. A later filter cannot separate those contributions. The anti-aliasing filter must remove the high frequencies before the sampler.

The comparison is quantitative. Take $x(t) = \cos(\pi t) + \cos(3\pi t)$ at $T = 2/5$ s, as in Example 7.6. Without the filter, $x_r(t) = \cos(\pi t) + \cos(2\pi t)$ and the error is $e(t) = \cos(3\pi t) - \cos(2\pi t)$. With a lowpass at $\omega_c = 2.5\pi$ ahead of the sampler, the component at 3π is removed first, what is sampled is $\cos(\pi t)$ alone, it is recovered exactly, and the error is $e(t) = \cos(3\pi t)$.

MEAN-SQUARE ERROR, AVERAGED OVER ONE PERIOD OF 2 S

$$e^2(t) = [\cos(3\pi t) - \cos(2\pi t)]^2 = \cos^2(3\pi t) - 2\cos(3\pi t)\cos(2\pi t) + \cos^2(2\pi t) \\ = \cos^2(3\pi t) - \cos(\pi t) - \cos(5\pi t) + \cos^2(2\pi t)$$

$$\overline{e^2}\Big|_{\text{no filter}} = \frac{1}{2} \int_0^2 e^2(t) dt = \frac{1}{2} - 0 - 0 + \frac{1}{2} = 1, \quad \overline{e^2}\Big|_{\text{filtered}} = \frac{1}{2} \int_0^2 \cos^2(3\pi t) dt = \frac{1}{2}$$

The cross term uses $2\cos a \cos b = \cos(a - b) + \cos(a + b)$ with $a = 3\pi t$ and $b = 2\pi t$. The window of 2 s holds a whole number of periods of every term: $\cos(\pi t)$ has period 2 s, $\cos(5\pi t)$ has period 0.4 s, and $\cos^2(3\pi t) = \frac{1}{2} + \frac{1}{2}\cos(6\pi t)$ has period $1/3$ s. A cosine averages to zero over a whole number of periods, so each cross term contributes 0 and each squared cosine contributes $\frac{1}{2}$. Filtering first halves the error power. It costs the component at 3π , which was lost either way; what it buys is the removal of an impostor at 2π , a frequency the signal never contained. Losing content is a known, bounded error. Adding content that was never there is not, because nothing downstream can tell it apart from the signal.

❶ NO FINITE-DURATION SIGNAL IS BAND-LIMITED

A rectangular pulse of width $2T_1$ has transform $2\sin(\omega T_1)/\omega$, which crosses zero again and again but is never zero on a whole interval. There is no ω_M beyond which it vanishes, so no rate makes the copies disjoint and some aliasing is present at every rate. What is done in practice is to impose the band limit with a filter, and then to reconstruct the *filtered* signal exactly. The difference from the original is a design decision with a known size, not an accident.

7.11 Aliasing in time and in space

A camera is a sampler with $T = 1/f_s$, and a strobe lamp is the same device built from light. Nothing about the rule changes.

EXAMPLE 7.7 – A WHEEL THAT TURNS BACKWARDS

GIVEN A marked spoke turns at 9 revolutions per second. A camera records 10 frames per second.

FIND The rotation the recording appears to show.

METHOD Model the spoke angle as a sinusoid because the camera records it at regular time instants. Convert both rates to rad/s, compare them, and locate the surviving line with the cutoff at $\omega_c = \omega_s/2$.

SOLUTION The spoke turns at $f_0 = 9$ Hz, so $\omega_0 = 2\pi f_0 = 18\pi$ rad/s. The camera samples at $f_s = 10$ Hz, so $\omega_s = 2\pi f_s = 20\pi$ rad/s and $\omega_c = 10\pi$ rad/s. The Nyquist rate is $2\omega_0 = 36\pi$ rad/s and $\omega_s = 20\pi < 36\pi$, so the rate is below it. The lines sit at $k\omega_s \pm \omega_0$:

$$k = 0 : \quad \pm 18\pi \text{ outside } |\omega| < 10\pi$$

$$k = 1 : \quad 20\pi - 18\pi = 2\pi \text{ inside}, \quad 20\pi + 18\pi = 38\pi \text{ outside}$$

The $k = -1$ copy gives the mirror line at -2π . So the surviving line sits at $|\omega_s - \omega_0| = 2\pi$ rad/s, with $k = 1$ chosen, that is $2\pi/2\pi = 1$ revolution per second. In hertz the same subtraction reads $f_s - f_0 = 10 - 9 = 1$ Hz. The line at $+2\pi$ comes from the impulse at

-18π , the negative-frequency component, moved up by ω_s . The positive and negative components have exchanged roles, so the apparent rotation is in the opposite direction.

CHECK

Frame n is taken at $t = n/10$ s. The recorded angle is

$$\cos\left(18\pi \cdot \frac{n}{10}\right) = \cos\left(2\pi n - \frac{2\pi n}{10}\right) = \cos\left(\frac{2\pi n}{10}\right),$$

using $18\pi n/10 = 2\pi n - 2\pi n/10$, the period 2π of the cosine, and its evenness. These are exactly the frames a 1 Hz rotation would produce. Between frames the spoke advances $9/10$ of a turn, which the eye reads as $1/10$ of a turn backwards. At 20 frames per second, $\omega_s = 40\pi > 36\pi$ and the recording is correct.

Position takes the place of time without any change to the arithmetic. A striped pattern of 9 cycles per millimetre recorded on a grid of 10 sample points per millimetre satisfies $2 \times 9 > 10$, so it is recorded as a pattern of $|10 - 9| = 1$ cycle per millimetre. The broad bands the grid produces are smooth, regular and convincing, and nothing in the samples marks them as false. Two regular patterns laid over one another give the same effect without any sampler: stripes at 1.0 mm and at 1.1 mm have spatial frequencies 1.000 and 0.909 cycles per millimetre, so the pattern they make together repeats every 11 mm.

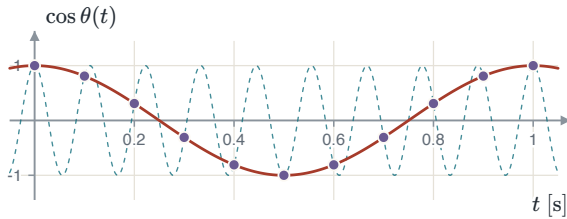


FIGURE 7.14 Nine revolutions per second, dashed, ten frames per second as dots, and the one revolution per second the frames describe.

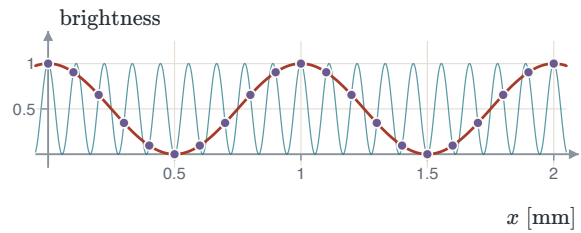


FIGURE 7.15 The same arithmetic in space: nine cycles per millimetre recorded as one.

7.12 Summary

1. Find ω_M , the highest angular frequency the signal carries. If there is none, the signal is not band-limited and must be filtered before anything else.
2. Choose ω_s strictly above $2\omega_M$, leaving a guard band $\omega_s - 2\omega_M$ wide enough for a real filter.
3. Compute $T = 2\pi/\omega_s$ and check $\omega_s T = 2\pi$. Quote $f_s = 1/T$ in hertz only when hertz was asked for.
4. Draw the copies at every multiple of ω_s , each $1/T$ times the original height, and mark the baseband $k = 0$.
5. Place the reconstruction filter: gain T , cutoff strictly inside $\omega_M < \omega_c < \omega_s - \omega_M$.
6. If the copies overlap, say which line came from which copy and give the alias frequency $|\omega_s - \omega_0|$ for each component that moved.

⊗ THE FOUR TRAPS, IN ONE PLACE

Saying the copies disappear below the Nyquist rate — they never do, they overlap. Treating $\omega_s = 2\omega_M$ as safe — the admissible cutoff interval is empty there. Reading $2\pi/T$ in hertz — that is rad/s, and the error is a factor of 2π . Calling a hold output the reconstructed signal — a hold is an approximation with a measurable error.

When the signal is band-limited and the sampling rate satisfies the strict condition, its samples determine it uniquely. This result allows the earlier continuous-time operations to be represented and processed with discrete samples.

7.13 Exercises

7.1 A signal is sampled at $\omega_s = 8000\pi$ rad/s. Give T and f_s , and verify the period with a unit check.

Answer: $T = 2\pi/\omega_s = 1/4000$ s = 0.25 ms and $f_s = 1/T = 4000$ Hz. Check: $\omega_s T = 8000\pi \times 2.5 \times 10^{-4} = 2\pi$. Note that 0.25 s fails the same check by a factor of 1000.

7.2 State in one sentence each what sampling does at every rate, and what happens only below the Nyquist rate.

Answer: At every rate sampling puts a copy of $X(j\omega)$ at every multiple of ω_s , scaled by $1/T$. Below the Nyquist rate those copies overlap and add, and that overlap is aliasing.

7.3 Show that the sampling theorem cannot be satisfied at $\omega_s = 2\omega_M$, and give a signal that demonstrates it.

Answer: The cutoff condition $\omega_M < \omega_c < \omega_s - \omega_M$ becomes $\omega_M < \omega_c < \omega_M$, which is empty. For $x(t) = \sin(\omega_M t)$ sampled at $\omega_s = 2\omega_M$ every sample is $\sin(\pi n) = 0$: the baseband and the $k = 1$ copy contribute equal and opposite weights at $\pm\omega_M$.

7.4 A triangular spectrum has peak $X_{\max} = 4000$ and is sampled with $T = 125$ μ s. Find the height of one copy, and say why the area $A = 8000\pi$ is not the number to use.

Answer: $X_{\max}/T = 4000 \times 8000 = 3.2 \times 10^7$. The area of the convolution and the peak of the spectrum differ by 2π , since squaring in time convolves in frequency and divides by 2π ; using A inflates every height by that factor.

7.5 $x(t) = \cos(\pi t) + \cos(3\pi t)$ is sampled with $T = 2/5$ s and reconstructed with $\omega_c = \omega_s/2$. Find $x_r(t)$ and name the component that moved.

Answer: $\omega_s = 5\pi < 6\pi$, so aliasing occurs. Surviving lines: π from the baseband and $5\pi - 3\pi = 2\pi$ from the $k = 1$ copy, giving $x_r(t) = \cos(\pi t) + \cos(2\pi t)$. The component at 3π moved to 2π ; the one at π passed untouched.

7.6 Explain why the anti-aliasing filter must precede the sampler, and state what the output of the whole chain is equal to.

Answer: After the sampler the intruding content has been added to the wanted content at the same frequency, and a filter cannot recover two numbers from their sum. Placed first, the filter removes that content while it is still at its own frequency. The output then equals the filtered signal exactly, not the original, and the difference is set by the chosen cutoff.

APPENDIX

A

Summary of formulas

A.1 Energy and power

TABLE A.1 Energy and power in continuous and discrete time.

QUANTITY	CONTINUOUS TIME	DISCRETE TIME
Total energy	$E_{\infty} = \int_{-\infty}^{\infty} x(t) ^2 dt$	$E_{\infty} = \sum_{n=-\infty}^{\infty} x[n] ^2$
Average power	$P_{\infty} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) ^2 dt$	$P_{\infty} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x[n] ^2$
Energy signal	$E_{\infty} < \infty, P_{\infty} = 0$	same
Power signal	$E_{\infty} \rightarrow \infty, 0 < P_{\infty} < \infty$	same

A.2 Operations and periodicity

TABLE A.2 Operations on the time axis, and the test for periodicity.

ITEM	STATEMENT
Shift	$x(t - t_0)$: delay if $t_0 > 0$, advance if $t_0 < 0$
Reversal	$x(-t), x[-n]$
Scaling	$y(t) = x(at)$: compressed if $a > 1$, stretched if $0 < a < 1$; support divided by a
Combination	$x(at - b)$: shift by b first, then scale by a
Periodicity	$x(t) = x(t + T); x[n] = x[n + N]$ with N an integer
Fundamental frequency	$\omega_0 = 2\pi/T_0 = 2\pi/N_0$
Even and odd	$\mathcal{E}\{x\} = \frac{1}{2}[x(t) + x(-t)], \mathcal{O}\{x\} = \frac{1}{2}[x(t) - x(-t)]$

A.3 Impulses and steps

TABLE A.3 The unit impulse and the unit step in continuous and discrete time.

ITEM	CONTINUOUS TIME	DISCRETE TIME
Step and impulse	$\delta(t) = \frac{d}{dt}u(t), u(t) = \int_{-\infty}^t \delta(\tau) d\tau$	$\delta[n] = u[n] - u[n-1], u[n] = \sum_{k=0}^{\infty} \delta[n-k]$
Sampling	$x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$	$x[n]\delta[n-n_0] = x[n_0]\delta[n-n_0]$
Sifting	$x(t_0) = \int_{-\infty}^{\infty} x(t)\delta(t-t_0)dt$	$x[n_0] = \sum_n x[n]\delta[n-n_0]$
Representation	$x(t) = \int x(\tau)\delta(t-\tau)d\tau$	$x[n] = \sum_k x[k]\delta[n-k]$

A.4 Complex exponentials

TABLE A.4 Complex exponentials in continuous and discrete time, and their periods.

ITEM	STATEMENT
Continuous time	$x(t) = Ce^{at}$, with $C = Ae^{j\theta}$ and $a = r + j\omega_0$: $x(t) = Ae^{rt}e^{j(\omega_0 t + \theta)}$
Period	$T_0 = 2\pi/\omega_0$, always periodic for $\omega_0 \neq 0$
Discrete time	$x[n] = C\alpha^n$ with $\alpha = e^{\beta}$; growth boundary at $ \alpha = 1$
Period	$N = \frac{2\pi}{\omega_0}k$; periodic only if $\omega_0/2\pi \in \mathbb{Q}$
Frequency wrap-around	$e^{j(\omega_0+2\pi)n} = e^{j\omega_0 n}$

A.5 Systems and convolution

TABLE A.5 System properties, in general and for an LTI system in terms of h .

PROPERTY	GENERAL CRITERION	LTI CRITERION IN TERMS OF h
Memoryless	output at t uses only input at t	$h(t) = a\delta(t)$ or $h[n] = a\delta[n]$
Invertible	distinct inputs give distinct outputs	$h * g = \delta$ for some g
Causal	output uses only $\tau \leq t$	$h(t) = 0$ for $t < 0$; $h[n] = 0$ for $n < 0$
BIBO stable	bounded input gives bounded output	$\int h dt < \infty$; $\sum_k h[k] < \infty$
Time invariant	$x(t-t_0) \rightarrow y(t-t_0)$	—
Linear	$ax_1 + bx_2 \rightarrow ay_1 + by_2$	—

CONVOLUTION

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) \, d\tau$$

Convolution is commutative, distributive, and associative. The output support is the sum of the input supports. The total areas, or the total sums in discrete time, multiply.

A.7 Fourier series properties

A periodic signal of period T_0 with $\omega_0 = 2\pi/T_0$, or a periodic sequence of period N with $\omega_0 = 2\pi/N$. In both columns $x \leftrightarrow a_k$ and $y \leftrightarrow b_k$.

TABLE A.6 Fourier series properties in continuous and discrete time.

PROPERTY	CONTINUOUS TIME	DISCRETE TIME
Linearity	$Ax(t) + By(t) \leftrightarrow Aa_k + Bb_k$	$Ax[n] + By[n] \leftrightarrow Aa_k + Bb_k$
Time shift	$x(t - t_0) \leftrightarrow a_k e^{-jk\omega_0 t_0}$	$x[n - n_0] \leftrightarrow a_k e^{-jk(2\pi/N)n_0}$
Frequency shift	$e^{jM\omega_0 t} x(t) \leftrightarrow a_{k-M}$	$e^{jM(2\pi/N)n} x[n] \leftrightarrow a_{k-M}$
Conjugation	$x^*(t) \leftrightarrow a_{-k}^*$	$x^*[n] \leftrightarrow a_{-k}^*$
Time reversal	$x(-t) \leftrightarrow a_{-k}$	$x[-n] \leftrightarrow a_{-k}$
Time scaling	$x(\alpha t), \alpha > 0: a_k, \text{ period } T_0/\alpha$	$x_{(m)}[n]: \frac{1}{m} a_k, \text{ period } mN$
Periodic convolution	$\int_{T_0} x(\tau)y(t - \tau)d\tau \leftrightarrow T_0 a_k b_k$	$\sum_{r=\langle N \rangle} x[r]y[n - r] \leftrightarrow N a_k b_k$
Multiplication	$x(t)y(t) \leftrightarrow \sum_{\ell=-\infty}^{\infty} a_\ell b_{k-\ell}$	$x[n]y[n] \leftrightarrow \sum_{\ell=\langle N \rangle} a_\ell b_{k-\ell}$
Derivative or difference	$dx/dt \leftrightarrow jk\omega_0 a_k$	$x[n] - x[n - 1] \leftrightarrow (1 - e^{-jk(2\pi/N)})a_k$
Integral or running sum	$\int_{-\infty}^t x d\tau \leftrightarrow \frac{a_k}{jk\omega_0}, \text{ only if } a_0 = 0$	$\sum_{r=-\infty}^n x[r] \leftrightarrow \frac{a_k}{1 - e^{-jk(2\pi/N)}}, \text{ only if } a_0 = 0$
Real signal	$a_k = a_{-k}^*; a_k \text{ even, } \angle a_k \text{ odd}$	same
Real and even	a_k real and even	same
Real and odd	a_k purely imaginary and odd	same
Even-odd parts	$\mathcal{E}\{x\} \leftrightarrow \text{Re}\{a_k\}, \mathcal{O}\{x\} \leftrightarrow j \text{Im}\{a_k\}$	same
Parseval	$\frac{1}{T_0} \int_{T_0} x ^2 dt = \sum_k a_k ^2$	$\frac{1}{N} \sum_{n=\langle N \rangle} x ^2 = \sum_{k=\langle N \rangle} a_k ^2$

The two columns differ in three rows because discrete-time coefficients repeat, $a_k = a_{k+N}$. For multiplication, the discrete-time sum therefore covers one period instead of all integers. Time expansion carries the factor $\frac{1}{m}$ because the average uses m times as many samples. Discrete time also uses a first difference where continuous time uses a derivative. At $k = 0$, integration or accumulation requires zero mean; otherwise the result grows without bound.

A.8 Continuous-time Fourier transform

TABLE A.7 Properties of the continuous-time Fourier transform.

PROPERTY	STATEMENT
Linearity	$ax_1(t) + bx_2(t) \leftrightarrow aX_1(j\omega) + bX_2(j\omega)$
Time shift	$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$
Frequency shift	$e^{j\omega_0 t} x(t) \leftrightarrow X(j(\omega - \omega_0))$
Conjugation	$x^*(t) \leftrightarrow X^*(-j\omega)$
Time reversal	$x(-t) \leftrightarrow X(-j\omega)$
Time and frequency scaling	$x(at) \leftrightarrow \frac{1}{ a } X(j\omega/a)$
Convolution	$x(t) * h(t) \leftrightarrow X(j\omega)H(j\omega)$
Multiplication	$x(t)y(t) \leftrightarrow \frac{1}{2\pi} X(j\omega) * Y(j\omega)$
Differentiation in time	$\frac{d^n x}{dt^n} \leftrightarrow (j\omega)^n X(j\omega)$
Integration	$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
Differentiation in frequency	$t x(t) \leftrightarrow j \frac{dX(j\omega)}{d\omega}$
Duality	$X(t) \leftrightarrow 2\pi x(-\omega)$
Real signal	$X(-j\omega) = X^*(j\omega); X \text{ even, } \angle X \text{ odd}$
Real and even	$X(j\omega) \text{ real and even}$
Real and odd	$X(j\omega) \text{ purely imaginary and odd}$
Even-odd parts	$\mathcal{E}\{x\} \leftrightarrow \text{Re}\{X\}, \mathcal{O}\{x\} \leftrightarrow j \text{Im}\{X\}$
Parseval	$\int_{-\infty}^{\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) ^2 d\omega$

TABLE A.8 Continuous-time Fourier transform pairs.

SIGNAL	TRANSFORM
$\delta(t)$	1
$\delta(t - t_0)$	$e^{-j\omega t_0}$
$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$
$e^{-at}u(t), a > 0$	$\frac{1}{a + j\omega}$
$te^{-at}u(t), a > 0$	$\frac{1}{(a + j\omega)^2}$
$\frac{t^{n-1}}{(n-1)!}e^{-at}u(t), a > 0$	$\frac{1}{(a + j\omega)^n}$
$e^{-a t }, a > 0$	$\frac{2a}{a^2 + \omega^2}$
1 on $ t < T_1$, 0 beyond	$\frac{2 \sin(\omega T_1)}{\omega} = 2T_1 \text{sinc}(\omega T_1)$
$\frac{\sin(Wt)}{\pi t} = \frac{W}{\pi} \text{sinc}(Wt)$	1 on $ \omega < W$, 0 beyond
1	$2\pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos \omega_0 t$	$\pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$
$\sin \omega_0 t$	$\frac{\pi}{j}\delta(\omega - \omega_0) - \frac{\pi}{j}\delta(\omega + \omega_0)$
$\sum_k a_k e^{jk\omega_0 t}$	$\sum_k 2\pi a_k \delta(\omega - k\omega_0)$
Square wave: 1 on $ t < T_1$, 0 on $T_1 < t \leq T/2$	$\sum_k \frac{2 \sin(k\omega_0 T_1)}{k} \delta(\omega - k\omega_0)$
$\sum_k \delta(t - kT)$	$\frac{2\pi}{T} \sum_k \delta(\omega - \frac{2\pi k}{T})$

This table uses the unnormalised definition $\text{sinc}(\theta) = \sin \theta / \theta$. The normalised convention divides the argument by π . Convert the argument when moving between conventions; otherwise the result loses a factor of π . The exponential pairs require $a > 0$.

A.9 Discrete-time Fourier transform

TABLE A.9 Properties of the discrete-time Fourier transform.

PROPERTY	STATEMENT
Linearity	$ax_1[n] + bx_2[n] \leftrightarrow aX_1(e^{j\omega}) + bX_2(e^{j\omega})$
Time shift	$x[n - n_0] \leftrightarrow e^{-j\omega n_0} X(e^{j\omega})$
Frequency shift	$e^{j\omega_0 n} x[n] \leftrightarrow X(e^{j(\omega - \omega_0)})$
Conjugation	$x^*[n] \leftrightarrow X^*(e^{-j\omega})$
Time reversal	$x[-n] \leftrightarrow X(e^{-j\omega})$
Time expansion	$x_{(k)}[n] \leftrightarrow X(e^{jk\omega})$
Convolution	$x[n] * h[n] \leftrightarrow X(e^{j\omega})H(e^{j\omega})$
Multiplication	$x[n]y[n] \leftrightarrow \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$
Differencing in time	$x[n] - x[n - 1] \leftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$
Accumulation	$\sum_{m=-\infty}^n x[m] \leftrightarrow \frac{X(e^{j\omega})}{1 - e^{-j\omega}} + \pi X(e^{j0}) \sum_k \delta(\omega - 2\pi k)$
Differentiation in frequency	$n x[n] \leftrightarrow j dX(e^{j\omega})/d\omega$
Real sequence	$X(e^{-j\omega}) = X^*(e^{j\omega}); X \text{ even, } \angle X \text{ odd}$
Real and even	$X(e^{j\omega}) \text{ real and even}$
Real and odd	$X(e^{j\omega}) \text{ purely imaginary and odd}$
Even-odd parts	$\mathcal{E}\{x\} \leftrightarrow \text{Re}\{X\}, \mathcal{O}\{x\} \leftrightarrow j \text{Im}\{X\}$
Parseval	$\sum_n x[n] ^2 = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) ^2 d\omega$

TABLE A.10 Discrete-time Fourier transform pairs.

SEQUENCE	TRANSFORM
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
$u[n]$	$\frac{1}{1-e^{-j\omega}} + \pi \sum_k \delta(\omega - 2\pi k)$
$a^n u[n], a < 1$	$\frac{1}{1-ae^{-j\omega}}$
$(n+1)a^n u[n], a < 1$	$\frac{1}{(1-ae^{-j\omega})^2}$
$\frac{(n+r-1)!}{n!(r-1)!} a^n u[n], a < 1$	$\frac{1}{(1-ae^{-j\omega})^r}$
$a^{ n }, a < 1$	$\frac{1-a^2}{1-2a \cos \omega + a^2}$
1 on $ n \leq N_1$, 0 beyond	$\frac{\sin(\omega(N_1 + \frac{1}{2}))}{\sin(\omega/2)}$
$\frac{\sin(Wn)}{\pi n}, 0 < W < \pi$	1 on $ \omega \leq W$, 0 on $W < \omega \leq \pi$
1	$2\pi \sum_k \delta(\omega - 2\pi k)$
$e^{j\omega_0 n}$	$2\pi \sum_k \delta(\omega - \omega_0 - 2\pi k)$
$\cos \omega_0 n$	$\pi \sum_k [\delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)]$
$\sin \omega_0 n$	$\frac{\pi}{j} \sum_k [\delta(\omega - \omega_0 - 2\pi k) - \delta(\omega + \omega_0 - 2\pi k)]$
$\sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$	$2\pi \sum_k a_k \delta(\omega - \frac{2\pi k}{N})$
$\sum_k \delta[n - kN]$	$\frac{2\pi}{N} \sum_k \delta(\omega - \frac{2\pi k}{N})$

Every transform in this section is periodic in ω with period 2π because the time index is discrete. A periodic sequence has discrete spectral coefficients, while an aperiodic sequence has a continuous spectrum. In each case, discreteness in one domain produces periodicity in the other.

A.10 Symbols

TABLE A.11 Symbols and their meanings.

SYMBOL	MEANING
$x(t), x[n]$	input signal, continuous and discrete time
$y(t), y[n]$	output signal
$h(t), h[n]$	impulse response of a linear time-invariant system
$\delta(t), \delta[n]$	unit impulse
$u(t), u[n]$	unit step
E_∞, P_∞	total energy and average power over all time
T_0, N_0	fundamental period
ω_0	fundamental angular frequency, rad/s or rad/sample
$*$	convolution
j	imaginary unit, $j^2 = -1$



Signals and Systems · Lecture Notes

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